TSB-FCST: A Billion-Scale Time-Series Forecasting Benchmark with Taxonomy-Specific Evaluation

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Abstract

Time series, defined as sequences of ordered observations, have become ubiquitous across domains such as nature, industry, healthcare, and social processes. Among the core tasks in this field, time-series forecasting—using historical patterns to predict future outcomes—has gained significant attention due to its broad practical impact. Accordingly, numerous forecasting methods and evaluation benchmarks have been developed to track and drive progress. Despite decades of advances, existing evaluation frameworks still exhibit critical limitations that hinder a comprehensive understanding of model efficacy: (i) many empirical studies rely on homogeneous or imbalanced datasets that lack sufficient diversity or quality assurance to evaluate forecasting methods across dynamic environments and domains; (ii) many benchmarks focus on only a limited subset of forecasting methodologies; and (iii) existing studies often provide limited statistical and taxonomy-specific analysis, leading to potentially superficial interpretations of model performance based on aggregated metrics. To address these limitations, we introduce TSB-FCST, a billion-scale time-series forecasting benchmark curated with rigorous data cleaning and quality assurance for comprehensive and reliable evaluation. TSB-FCST provides a standardized evaluation of 100 algorithms, ranging from classical statistical and machine learning methods to deep learning-based and recently emerged foundation models. Our extensive evaluation reveals: (i) although Foundation Models (FMs) demonstrate strong generalist capabilities, classical statistical and machine learning methods continue to excel in specific application domains, underscoring the importance of traditional baselines and taxonomy-specific evaluation; (ii) the effectiveness of Channel-Dependent (CD) versus Channel-Independent (CI) architectures is closely tied to dataset characteristics, highlighting the need for appropriate and diverse large-scale benchmarks to verify algorithmic progress; and (iii) forecasting difficulty is jointly shaped by horizon length and temporal data patterns, suggesting that evaluation should move beyond the excessive pursuit of long-horizon accuracy and explicitly account for data-driven challenges. Finally, we release all source code and datasets to support reproducible comparison studies. In summary, TSB-FCST offers a rigorous roadmap for understanding recent developments and advancing reliable time-series forecasting research.

1 Introduction

The recent explosion in sensor networks and data acquisition frameworks has led to an unprecedented accumulation of high-fidelity signal data. These sequential measurements, characterized as ordered, real-valued observations, are formally referred to as *time series* in both scientific research and industrial applications [1]. Effectively mining such temporal information is essential for revealing hidden patterns and supporting strategic decisions. Consequently, the field has

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Table 1: **Comparison with existing time-series forecasting benchmarks**. TSB-FCST provides 1) the largest data scale and composition, 2) broader model and metric coverage, and 3) stronger data quality assurance and evaluation protocol than prior benchmarks.

Benchmark	**Dataset**				**Model**					**Quality Assurance**		**Eval Protocol**		
	#Obs.	UTS	MTS	Tot.	Stat	ML	DL	FM	Tot.	Clean	Inspect	HPO Set	Sig.Test	Metric
Monash [27]	182M	34	16	50	5	2	5	0	12	✗	✗	✗	✗	4
TSLib [28]	30M	0	14	14	0	0	13	0	13	✗	✗	✗	✗	2
TFB [29]	198M	28	25	53	3	3	15	0	21	✗	✗	✗	✗	8
GIFT-Eval [30]	157M	35	20	55	5	0	8	4	17	✗	✗	✗	✗	2
fev-bench [31]	761M	65	35	100	6	0	0	8	14	✗	✗	✓	✗	2
TSFM-bench [32]	40M	0	21	21	0	0	7	14	21	✗	✗	✗	✗	2
TSB-FCST (Ours)	**1B**	**64**	**64**	**128**	**19**	**15**	**47**	**19**	**100**	✓	✓	✓	✓	**8**

seen a surge in downstream tasks, ranging from classification [2, 3, 4, 5] and clustering [6, 7, 8] to anomaly detection [9, 10, 11, 12], similarity discovery [13, 14, 15, 16, 17], and forecasting [18, 19, 20, 21, 22, 23, 24, 25, 26].

Among these tasks, time-series forecasting holds a unique position by leveraging historical patterns to anticipate future observations. It serves not only as a critical tool for direct prediction, but also as a foundation for broader analytical goals. For example, weather forecasting supports planning in agriculture and aviation [33], while demand forecasting helps enterprises align inventory with market needs, reducing operational costs and preventing stock-outs [34]. Over the past two decades, forecasting methodologies have rapidly evolved. Traditional statistical methods, such as ARIMA [33], model temporal dependencies through structured assumptions and often provide efficient training and inference. Machine learning methods, especially tree-based ensembles, formulate forecasting as supervised regression over historical features [35]. More recently, deep learning models have advanced forecasting by capturing nonlinear temporal dynamics through diverse architectures [18, 36, 37], while foundation models have demonstrated promising zero-shot capabilities by learning from large-scale, cross-domain time-series corpora [38, 39, 40]. In parallel, substantial efforts have been made to build forecasting benchmarks for comparing models across different scales and domains [29, 30, 31, 27, 32], which have established important foundations for standardized evaluation. However, existing practices still exhibit several limitations that hinder a comprehensive understanding of model progress: (i) many studies rely on constrained or imbalanced data collections without quality assurance that do not fully reflect the complexity of multi-domain forecasting scenarios; (ii) some benchmarks focus on only a limited subset of model families, overlooking holistic comparisons between traditional models and newly emerged architectures; and (iii) many evaluations rely primarily on aggregate average metrics, with limited statistical and taxonomy-specific analysis, leading to potentially superficial interpretations of model performance.

To address these challenges, we introduce `TSB-FCST`, a large-scale benchmark for comprehensive and rigorous evaluation of time-series forecasting models. The benchmark contains 128 datasets across eight diverse domains, balanced between 64 univariate (UTS) and 64 multivariate (MTS) datasets. To the best of our knowledge, `TSB-FCST` is the first *billion*-scale archive specifically curated for systematic forecasting evaluation. To ensure data quality, we design a multi-stage curation pipeline that combines automated screening with human inspection to identify and correct anomalies, artifacts, and low-quality observations. Additionally, we provide a standardized evaluation of 100 forecasting models, ranging from classical statistical and machine learning methods to modern deep learning architectures and recent foundation models. As a practical contribution, we also release an integrated library with standardized pipelines and open-source implementations of all evaluated methods [41].

Our experimental study yields three key insights (Section 5): (i) although Foundation Models (FMs) demonstrate strong generalist capabilities, traditional models continue to excel in specific application domains, underscoring the importance of traditional baselines and taxonomy-specific evaluation; (ii) the effectiveness of Channel-Dependent (CD) versus Channel-Independent (CI) architectures is closely tied to dataset characteristics, highlighting the need for appropriate and diverse large-scale benchmarks to verify algorithmic progress; and (iii) long-term forecasting challenges arises from both horizon length and temporal data patterns, suggesting that evaluation should move beyond the excessive pursuit of long-horizon accuracy and explicitly account for data-driven challenges.

We start with the preliminaries and related work in the field of time-series forecasting (Section 2). Then we highlight our contributions of this work, detailed as follows:

- We construct TSB-FCST, a 1B-scale time-series forecasting benchmark curated with rigorous cleaning and quality assurance. Notably, it ensures balanced representation across eight diverse domains to provide an unbiased and comprehensive evaluation platform.
- We evaluate over 100 representative algorithms across a broad class, spanning traditional statistical and machine learning methods to state-of-the-art deep learning and foundation models.
- We implement a rigorous training and evaluation framework. To ensure both efficiency and reliability, we utilize a dedicated TSB-FCST-50M subset for systematic parameter tuning.
- We provide a systematic study uncovering the domain-specific nature of forecasting performance across different categories, establishing a roadmap for future taxonomy-specific research.

2 Preliminaries and Related Work

In this section, we start with an introduction of notation and concepts, as well as related studies on time-series forecasting models and the limitations of current benchmark studies.

2.1 Notation and Definitions

Definition 1 (Time Series)**.** *Let $\mathcal{X} = \{X_1, X_2, \dots, X_n\}$ denote a repository of n time-series instances. Each instance $X_i \in \mathbb{R}^{T \times d}$ consists of T temporal observations across d channels.*

Based on the dimensionality of the observation space, a time-series is classified as *univariate* (UTS) if $d = 1$ and *multivariate* (MTS) if $d > 1$. While UTS analysis primarily focuses on capturing temporal dependencies, MTS introduces additional complexity by requiring models to account for inter-channel correlations and high-dimensional dynamics.

Definition 2 (Subsequence)**.** *Given a time-series $X \in \mathbb{R}^{T \times d}$, a subsequence $X_{s,e}$ is defined as a contiguous segment of observations starting from offset s and ending at offset e. Formally, $X_{s,e} = (\vec{x}_s, \vec{x}_{s+1}, \dots, \vec{x}_e)$, where $1 \leq s \leq e \leq T$. Each $\vec{x}_t \in \mathbb{R}^d$ represents the d-dimensional observation at time t. The length of the subsequence is given by $|X_{s,e}| = e - s + 1$.*

Definition 3 (Sliding Window)**.** *Based on the definition of subsequences, a sliding window of width w is a specific subsequence $X_{i,i+w-1}$. By traversing the series X with a fixed stride s, we extract a collection of such windows: $\mathcal{W} = \{X_{1+(k-1)s, w+(k-1)s}\}_{k=1}^{K}$. Here, $K = \lfloor \frac{T-w}{s} \rfloor + 1$ represents the total number of extracted windows, where each window is a local segment of shape (w, d).*

Definition 4 (Time Series Forecasting)**.** *Given a time-series $X \in \mathbb{R}^{T \times d}$ and a look-back window of length L, the task of forecasting is to learn a mapping $f : \mathbb{R}^{L \times d} \to \mathbb{R}^{H \times d}$ that generates a forecast $\hat{X}_{t+1,t+H} = f(X_{t-L+1,t})$ of a horizon H. The model is optimized to minimize the discrepancy between the prediction and the ground truth $X_{t+1,t+H}$, formally defined by a objective function $\mathcal{L}$:*

$$\mathcal{L} = \psi(X_{t+1,t+H}, \hat{X}_{t+1,t+H})$$

where $\psi(\cdot)$ denotes a distance metric or loss function (e.g., ℓ_1 or ℓ_2 norm) that quantifies the prediction error over the forecast horizon H. In our study, we employ a diverse set of evaluation metrics to provide a comprehensive assessment of this discrepancy (Section 4.2).

We also characterize several fundamental time-series patterns including trend, seasonality, stationarity, transition, and shift. Detailed formulations and definitions are provided in the Appendix B.

2.2 Related Work

In this section, we briefly review representative forecasting models as well as existing benchmark and evaluation studies that motivate the design of our TSB-FCST benchmark.

Time-Series Forecasting Models. Time series forecasting, as one of the most critical tasks in time series analysis, has been continuously evolving over the past few decades [42]. Traditional time series forecasting methods can broadly be categorized into statistical and machine learning approaches. Statistical models[43, 44, 45, 46, 47, 48] are usually contingent on explicit, structured, and principled probabilistic assumptions; Machine learning methods [49, 50, 51] formulate time

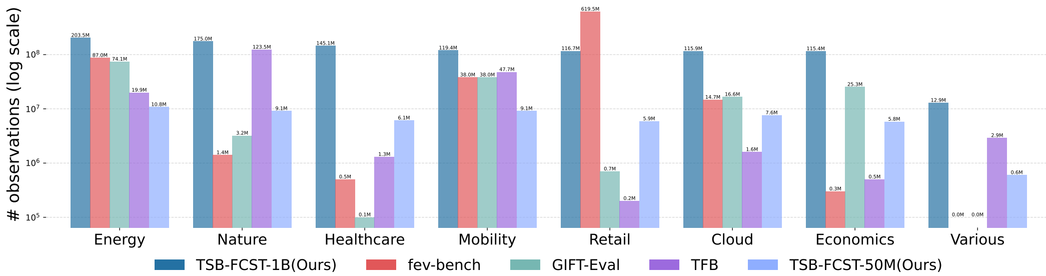

Figure 1: Domain-wise distribution of observations across time series forecasting benchmarks. Our proposed benchmarks demonstrate a more balanced and comprehensive coverage across domains.

series forecasting as a supervised learning problem. More recently, the past decade has witnessed that deep learning methodologies and emergent foundation models have been dominating the majority of breakthrough [52, 53, 54, 55, 18, 56, 57, 58, 39].

Time-Series Forecasting Benchmark and Evaluation Studies. The continual advances in developing new time series forecasting algorithms have been accompanied by a fast-growing community of benchmarking studies [27, 28, 29, 30, 31]. Monash [27], the first time series forecasting benchmark, collects 20 publicly available time series datasets. To account for advances in deep learning architectures [59, 54, 60, 61], Time Series Library (TSLib) [28] evaluates 13 advanced deep time series. TFB [29] proposes to perform a thorough evaluation of 21 forecasting algorithms spanning statistical learning, machine learning, and deep learning categories. GIFT-Eval [30] encompasses 44,000 time series and 177 million data points to improve the evaluation comprehensiveness. fev-bench [31] further offers a collection of 46 tasks with covariates.

3 Common Limitations in Existing Time-Series Forecasting Benchmark

In this section, we summarize common limitations from two perspectives: dataset construction and model evaluation. These limitations motivate the design of `TSB-FCST`, which aims to provide a large-scale, high-quality, and taxonomy-aware evaluation platform.

Limitations in Benchmark Datasets. Public time-series datasets often suffer from data quality issues such as missing values, abnormal spikes, and long inactive regions, all of which can distort evaluation and lead to misleading predictability. For instance, abnormal spikes in the test region may disproportionately penalize otherwise robust models, while inactive regions can artificially lower forecasting errors because even naive predictors perform well on nearly constant signals. Likewise, purely stochastic series provide limited meaningful signal for evaluating forecasting capability. Beyond data quality, existing benchmarks also exhibit substantial domain imbalance, with observations concentrated in only a few application areas while others remain underrepresented (Figure 1). Such imbalance can bias evaluation results, since models effective in domains like energy or mobility may not generalize to healthcare, cloud, economics, or retail forecasting. Therefore, a reliable forecasting benchmark should incorporate explicit quality assurance procedures while also providing broad and balanced domain coverage to ensure robust and representative evaluation.

Limitations in Model Collection and Evaluation Protocols. Existing forecasting benchmarks may also provide incomplete coverage of forecasting methodologies, potentially leading to oversimplified conclusions about model progress. Although modern architectures have achieved strong performance, traditional statistical and machine learning methods remain highly competitive in many forecasting settings. Without representative coverage across statistical, machine learning, deep learning, and foundation model families, it becomes difficult to determine whether improvements stem from genuinely stronger forecasting ability or merely from comparisons against limited baselines. Moreover, fair evaluation remains challenging because forecasting models differ substantially in their sensitivity to hyperparameters, normalization strategies, and training procedures. While default settings may underestimate certain models, exhaustive per-dataset tuning is often computationally impractical; therefore, standardized yet efficient tuning protocols, such as representative Hyperparameter Optimization (HPO) subsets, are necessary to balance fairness and scalability. Finally, heavy reliance on aggregate average scores can obscure substantial heterogeneity across domains, forecasting horizons, temporal

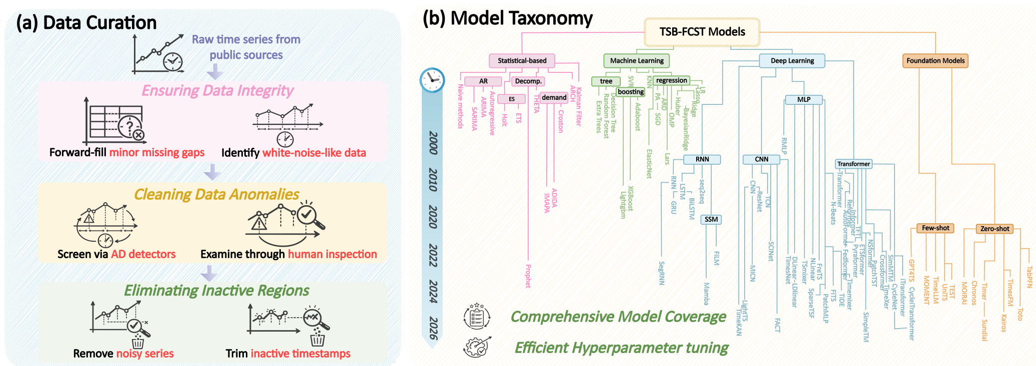

Figure 2: Overview of TSB-FCST. (a) Data curation pipeline: raw public time series are processed through integrity checking, anomaly cleaning, and inactive-region elimination. (b) Model taxonomy: TSB-FCST provides comprehensive coverage of forecasting paradigms of 100 models.

patterns, and multivariate dependency structures [62]. In particular, the relative effectiveness of Channel-Dependent (CD) and Channel-Independent (CI) modeling in multivariate forecasting often depends strongly on dataset-specific correlations, highlighting the importance of taxonomy-aware and fine-grained evaluation analyses.

4 TSB-FCST: Our Proposed Benchmark

We propose TSB-FCST, a large-scale time-series forecasting benchmark designed for comprehensive and rigorous model evaluation. In the following, we go through the details including (i) dataset acquisition and standization; (ii) data curation and quality control; (iii) model search space and optimization protocol; and (iv) evaluation pipeline and metrics. More technical details of each curation process and the model pipeline can be found in Appendix (Section E).

4.1 Data Preparation and Quality Assurance

Data Acquisition and Standardization. To capture diverse real-world forecasting scenarios, we collect public time-series datasets spanning multiple domains, including energy, nature, healthcare, mobility, retail, cloud computing, and economics. The datasets cover heterogeneous temporal resolutions, from high-frequency measurements to yearly observations, and include both univariate and multivariate forecasting settings. Such diversity is important for evaluating model generalization across varying sampling frequencies, seasonal structures, noise levels, and domain-specific dynamics.

Data Curation and Quality Control. To ensure that TSB-FCST contains meaningful forecasting tasks rather than artifacts of raw data collection, we apply a multi-stage curation pipeline combining automated screening with human inspection. We first handle missing values and temporal inconsistencies through forward-fill imputation, temporal regularity verification, and cross-channel alignment checks. We then detect and inspect abnormal spikes to remove clear recording artifacts while preserving legitimate domain events. In addition, we filter or trim sequences dominated by stochastic behavior, random-walk-like dynamics, or excessively long inactive regions, as these patterns either provide little predictive signal or artificially simplify forecasting evaluation. Additionally, we recompute dataset statistics after curation to ensure a balanced distribution across domains. Through this process, we construct a reliable benchmark archive focused on active and predictable time-series behaviors. Representative examples of each temporal pattern are visualized in the Appendix, Figure 9.

4.2 Model and Evaluation

Model Search Space and Optimization Protocol. TSB-FCST evaluates 100 representative forecasting models across five major methodological families: naive methods, statistical methods, machine learning models, deep learning architectures, and time-series foundation models. This coverage spans classical baselines, feature-based regressors, modern neural architectures, and recent pre-trained foundation models[1], enabling unified comparison between traditional and modern forecasting

[1] For foundation models, we report zero-shot performance when supported and few-shot performance otherwise, while all other task-specific models are evaluated in the full-shot setting.

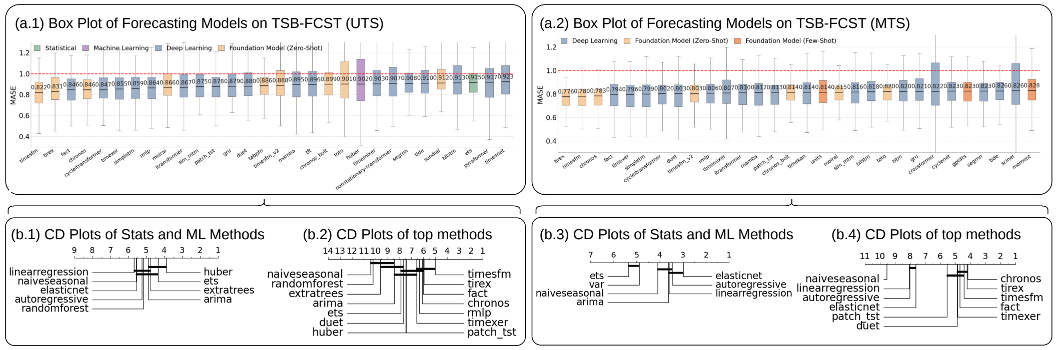

Figure 3: Global performance on TSB-FCST across model categories. (a) Box plots detailing the forecasting error of top 30 models across UTS and MTS. (b) Critical Difference (CD) diagrams illustrating the average model rankings with the Friedman-Nemenyi test. Models connected by a solid horizontal bar show no significant performance difference.

paradigms. Detailed model descriptions are provided in Section C. Such broad coverage avoids narrow interpretations of forecasting progress, since traditional statistical and machine learning methods remain competitive across many domains and temporal patterns despite recent advances in deep learning and foundation models. To ensure fair and scalable evaluation, we further adopt an efficient hyperparameter optimization protocol. Rather than exhaustively tuning every model on every dataset, which is computationally infeasible at a billion scale, we use a representative HPO subset such as TSB-FCST-50M to select competitive configurations (see Appendix Section E). This strategy balances fairness, scalability, and reproducibility while reducing excessive tuning.

Evaluation Pipeline and Metrics. We implement a standardized evaluation pipeline to ensure consistent comparison across heterogeneous datasets and model families. The pipeline covers data partitioning, normalization, forecasting schemes, horizon selection, and evaluation metrics. Following common practice [29, 36], we adopt both fixed and rolling forecasting setups, where the former evaluates prediction accuracy on a designated future window and the latter measures robustness across multiple forecast windows. All models follow consistent preprocessing and normalization procedures to ensure comparable training and inference conditions. For univariate forecasting, we use scale-aware metrics such as Mean Absolute Scaled Error (MASE) and Mean Symmetric Mean Absolute Percentage Error (MSMAPE), while for multivariate forecasting, we report standard metrics including Mean Squared Error (MSE) and Mean Absolute Error (MAE) (details in Appendix Section D). In addition, we define frequency-aware forecasting horizons corresponding to short-, medium-, and long-term forecasting tasks aligned with each dataset's sampling frequency and practical context. Together with taxonomy-specific analysis and statistical significance testing, this evaluation pipeline provides a reliable framework for identifying model strengths, weaknesses, and remaining challenges.

5 Experimental Results

In this section, we comprehensively evaluate the forecasting performance of diverse model categories across the 128 datasets within the TSB-FCST archive. Our analyses are structured around three core dimensions: (i) a macroscopic evaluation highlighting the top-performing methods across model categories under both UTS and MTS settings; (ii) a granular, taxonomy-specific comparison exploring domain-level variations, intrinsic feature impacts, multivariate (MTS) dynamics, and varying forecasting horizons; and (iii) accuracy versus efficiency with retrospect and prospect.

5.1 Global Benchmarking: The Overall Performance Landscape

In this section, we present a global overview of model performance across the TSB-FCST benchmark. Following established benchmarking protocols, we first establish an overall performance landscape to evaluate the general efficacy and robustness of various model categories across diverse datasets.

Our analysis begins with Univariate Time Series (UTS) forecasting, providing a granular comparison of traditional and modern methodologies. A distinguishing feature of our study is the inclusion of a vast array of statistical and machine learning (ML) methods, which are commonly overlooked in contemporary forecasting literature. As illustrated in the Critical Difference (CD) diagram for

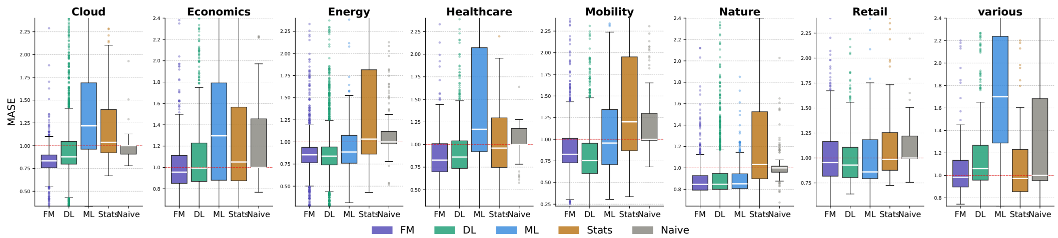

Figure 4: Performance comparison across diverse domains (measured in MASE), where the top performing methods are reported for each category. The result underscores the necessity of taxonomy-specific evaluation, as no single model category consistently dominates all scenarios.

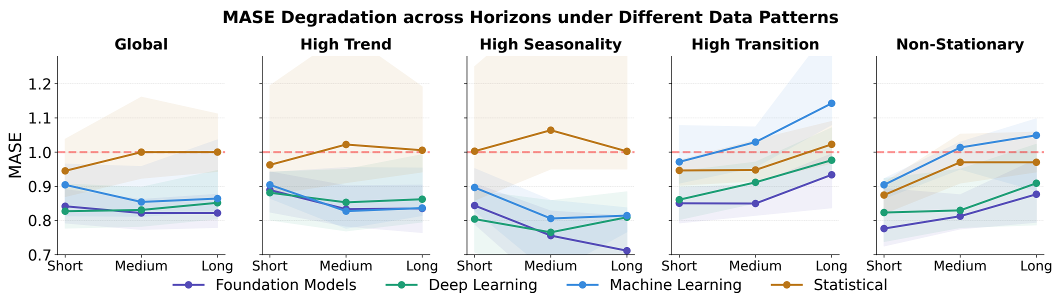

Figure 5: Performance degradation across varying forecasting horizons under distinct data patterns and model categories (measured by MASE). Each panel corresponds to a different data pattern.

the UTS top-performing statistical/ML methods (Figure 3, b.1), the majority of these classical approaches significantly outperform the `NaiveSeasonal` baseline across the 64 UTS datasets. Notably, regression-based methods and robust estimators such as `Huber` exhibit a statistically significant performance gain over the naive baseline. This observation is further corroborated by the performance distribution in Figure 3(a.1), where two statistical and ML models successfully secure positions within the top 30 out of the 100 evaluated algorithms on our global leaderboard, establishing a strong baseline in forecasting scenarios.

When evaluating all categories (Figure 3(b.2)), Foundation Models (FMs) and Deep Learning (DL) architectures—such as `TimesFM` and `TiReX`, followed by `FACT` and `Chronos`—occupy the leading positions, though many top-tier modern models do not significantly outperform the best classical ML methods (e.g., `Huber`) in univariate settings. Conversely, the landscape shifts in Multivariate Time Series (MTS) forecasting (Figure 3(b.3, b.4)), where FMs and DL models exhibit a clear, statistically significant advantage over classical baselines[2]. This performance gap stems from modern architectures' superior ability to handle high-dimensional and complex temporal dependencies. Notably, these top-performing FMs achieve their rankings in a strictly zero-shot setting, indicating substantial room for future improvement via domain-specific fine-tuning. In summary, while foundation models are highly effective on complex multivariate forecasting tasks, traditional methods remain highly competitive and should be included in a reliable forecasting benchmark. In the following section, we move beyond the global winner and conduct a more fine-grained, taxonomy-specific analysis.

5.2 Taxonomy-Specific Evaluation: Domains, Dynamics, and Dependencies

Domain-Specific Efficacy and Feature Sensitivity. To provide a more granular understanding of model capabilities and move beyond global averages, we dissect performance across distinct application domains (Figure 4) and intrinsic data characteristics (Figure 6). Figure 4 reveals that model dominance is highly domain-dependent. While Deep Learning (DL) architectures and Foundation Models (FMs) demonstrate strong superiority in the Cloud and Mobility domains, traditional Statistical and ML methods remain highly competitive in Energy, Nature, and Retail domains. Notably, in the Retail domain, ML methods surpass modern architectures with a top-ranked median MASE.

[2]We select top-performing and scalable statistical and ML methods on MTS tasks. As most natively support only univariate forecasting, we apply a Channel-Independent (CI) strategy to evaluate them on multivariate tasks as baselines.

To further interpret these dynamics, Figure 6 examines model sensitivity under different data characteristics. The ridge plots show that forecasting errors can shift substantially between low- and high-regime datasets, but the direction and magnitude of these shifts vary across model families. In particular, high transition and non-stationary regimes tend to induce broader or more right-skewed error distributions for several model categories, indicating increased risk of large forecasting errors (see Figure 9 and 10 in Appendix for more representative examples). This effect is especially visible for some machine learning methods, while foundation models and deep learning models generally exhibit more concentrated error distributions but are still affected by challenging dynamics. Seasonality and trend also lead to heterogeneous behavior: they do not uniformly benefit or degrade all models, suggesting that their impact depends on both the data regime and the modeling assumptions. Overall, these results reinforce that model performance is strongly conditioned on temporal characteristics, and such feature-level sensitivities can be obscured by aggregate benchmark scores.

When Horizons Meet Data Dynamics. Figure 5 further shows that forecasting degradation across horizons is strongly shaped by temporal data patterns and model-family characteristics. Under high trend and seasonality, foundation models maintain strong robustness and even improve as the horizon increases, suggesting their ability to exploit recurring structures over longer prediction windows. However, this advantage does not transfer uniformly to all regimes. In high-transition and non-stationary datasets, most model families exhibit increasing forecasting errors as the horizon becomes longer. This degradation is more pronounced for machine learning and statistical methods, which deteriorate further over longer prediction windows given complex data patterns. This pattern helps explain their lower average rankings across varying datasets versus horizons from the global evaluation perspective. Foundation models and deep learning models are generally more stable, but they also suffer from clear long-horizon degradation under challenging dynamics. More broadly, future forecasting evaluation should not focus solely on increasingly long horizons or aggregated accuracy scores. Instead, benchmarks are suggested to account for the intrinsic challenges induced by data characteristics, such as seasonality, trend, and non-stationary behaviors.

Identifying Datasets Suitable for Channel-Dependency Evaluation. Next, we revisit the long-standing discussion on Channel-Independent (CI) versus Channel-Dependent (CD) modeling strategies for multivariate time-series forecasting, as shown in Figure 7. Building on prior studies that relate the effectiveness of cross-channel modeling to dataset characteristics [29, 26], our benchmark extends this analysis to a larger and more diverse evaluation space, providing a clearer picture across domains and data characteristics regimes. Figure 7(a) shows that the relative advantage of CD models is not uniform across application domains. For example, mobility datasets exhibit a more visible CD advantage, suggesting that explicitly modeling inter-channel interactions can be beneficial when channels are strongly coupled. In contrast, other domains show much smaller differences between CI and CD strategies, indicating that cross-channel modeling is not always necessary.

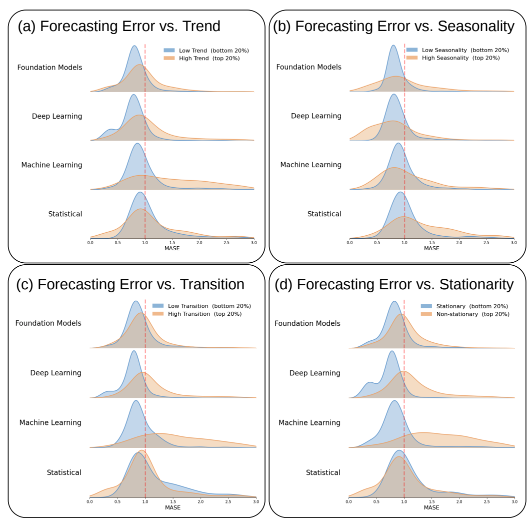

Figure 6: Performance sensitivity to data characteristics, visualized by ridge plots.

Figure 7(b) further connects this observation to inter-channel correlation. The CI–CD MASE difference increases with channel correlation, suggesting that CD models become more advantageous when the channels contain stronger shared structure. Meanwhile, several commonly used MTS benchmark datasets, such as ETTh and ETTm variants, are highlighted in the plot and lie in regions where the CI–CD performance gap is negligible. This implies that repeatedly debating channel dependency on a narrow set may provide limited diagnostic insight. To solve this problem, `TSB-FCST` covers a broader spectrum of datasets with diverse dependency structures, enabling a more systematic evaluation of when channel-dependent modeling is truly beneficial. These results suggest that reliable MTS benchmarks should explicitly account for dependency structure rather than treating all multivariate datasets as a homogeneous evaluation setting.

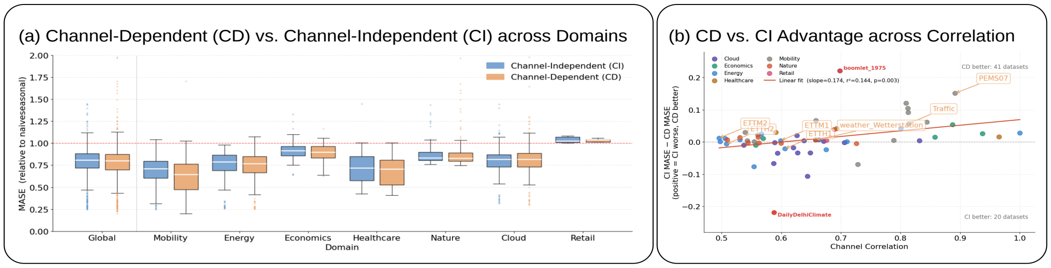

Figure 7: Channel-Dependent (CD) vs. Channel-Independent (CI) model performance on MTS datasets. (a) MASE distribution across domains, sorted in descending order of CD advantage. (b) Scatter plot of CI-CD MASE difference against channel correlation. Widely used MTS datasets (e.g., ETT datasets) are also highlighted for further discussion.

5.3 Accuracy vs. Efficiency: Retrospect and Prospect

To better understand the practical trade-offs among forecasting model families, Figure 8 compares the forecasting error and computational efficiency of top-performing models across categories on TSB-FCST, where the y-axis reports MASE, the x-axis shows cumulative benchmark runtime on a log scale, and the bubble size indicates the number of model parameters.

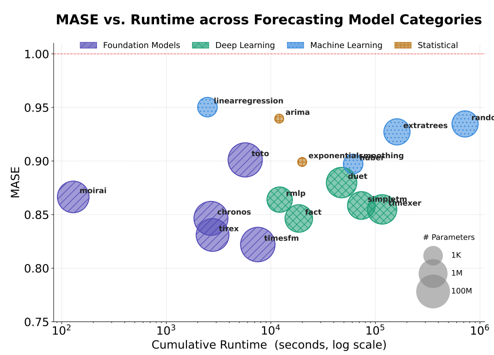

Figure 8: Accuracy vs. Runtime comparison across forecasting model categories on TSB-FCST (evaluated by MASE). Each bubble represents a model, where the bubble size indicates the number of parameters (log scale) and the color and hatch pattern denote the model category.

Notably, Foundation models provide strong forecasting capabilities while demonstrating competitive runtime performance through the zero-shot inference. However, as shown in our taxonomy-specific analyses in Section 5.2, foundation models are not *universally dominant*; they still exhibit weaknesses under certain domains and temporal patterns. On the other hand, traditional statistical and machine learning methods provide a complementary perspective: these methods remain highly competitive in several regimes but many require retraining on each dataset. Deep learning models occupy an intermediate position, balancing strong representation capacity with moderate computational cost.

Overall, the experimental analysis suggests that forecasting evaluation should move beyond a single aggregate accuracy ranking. Reliable benchmarking should jointly consider accuracy, model scale, and computing resource consumption, while also accounting for the domain-specific knowledge and data characteristics that shape model behavior.

6 Conclusion

In this work, we present TSB-FCST, a billion-scale time-series forecasting benchmark designed for comprehensive and reliable evaluation. We curate 128 high-quality datasets across eight domains and evaluate 100 forecasting models spanning statistical, machine learning, deep learning, and foundation-model paradigms. Our results show that no single model family consistently dominates all forecasting scenarios: although foundation models demonstrate strong overall and zero-shot performance, traditional methods remain highly competitive in many domains. We further show that forecasting performance is heavily influenced by application domains, temporal dynamics, forecasting horizons, and multivariate dependency structures, highlighting the limitations of relying solely on aggregate metrics. Nevertheless, we acknowledge several limitations. This work primarily focuses on point forecasting, while probabilistic forecasting may be more suitable for certain application scenarios. In addition, forecasting with exogenous variables represents another important direction that is not fully explored in the current benchmark. We leave these extensions for future work. Overall, TSB-FCST provides a rigorous dataset benchmark and evaluation framework for future taxonomy-aware time-series forecasting research.

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Dataset	Domain	Freq	#Series	#Channels	#Observations	Length Range	ShortH	MedH	LongH	Seasonality	Trend	Shift	Stationarity	Transition	Demand Pattern
Extended_Wikipedia_Web_Traffic_Daily_Dataset	Cloud	Daily	29984	1	75138457	[289, 2557]	30	-	-	0.728	0.13	0.508	0.0	0.015	Erratic
WebTraffic	Cloud	Daily	29737	1	23038505	[271, 803]	30	-	-	0.515	0.259	0.308	0.0	0.015	Erratic
Wike2000	Cloud	Daily	9535	1	7551720	[792, 792]	14	-	-	0.611	0.494	0.216	0.004	0.028	Smooth
Bitcoin	Economics	Daily	18	1	74925	[2659, 4581]	14	30	60	0.674	0.853	0.867	0.564	0.074	Erratic
CIF_2016	Economics	Monthly	62	1	6733	[57, 120]	12	-	-	0.628	0.943	0.679	0.9	0.111	Smooth
NN5	Economics	Daily	111	1	4916856	[791, 87801]	14	30	60	0.806	0.749	0.112	0.0	0.002	Smooth
NN5(Weekly)	Economics	Weekly	111	1	12543	[113, 113]	13	-	-	0.516	0.383	0.336	0.0	0.021	Smooth
Tourism(Monthly)	Economics	Monthly	366	1	109280	[91, 333]	18	-	-	0.873	0.934	0.703	0.89	0.074	Erratic
Tourism(Quarterly)	Economics	Quarterly	426	1	42514	[45, 130]	8	-	-	0.876	0.974	0.73	0.915	0.074	Smooth
Tourism(Yearly)	Economics	Yearly	342	1	9545	[26, 47]	6	-	-	0.663	0.982	0.743	0.949	0.167	Smooth
ausbeer	Economics	Quarterly	1	1	211	[211, 211]	8	-	-	0.96	0.985	0.081	0.096	0.08	Smooth
AustralianElectricity	Energy	half hourly	5	1	1155264	[230736, 232272]	48	192	-	0.959	0.932	0.076	0.0	0.001	Smooth
Elecdemand	Energy	half hourly	1	1	17520	[17520, 17520]	48	192	336	0.928	0.907	0.046	0.001	0.001	Smooth
Solar(5Min.)	Energy	every 5 min	2000	1	83766083	[8726, 104952]	144	-	-	0.913	0.438	0.06	0.0	0.023	Lumpy
SolarPower	Energy	every 4 seconds	1	1	500000	[500000, 500000]	30	75	150	0.934	0.496	0.033	0.0	0.033	Lumpy
Wind_Farms	Energy	every minute	29	1	10036864	[2975, 500000]	48	144	288	0.695	0.415	0.228	0.0	0.051	Lumpy
kdd_cup_2022_10T	Energy	every 10 minutes	134	1	4727177	[35194, 35279]	14	-	-	0.474	0.007	0.089	0.0	0.001	Lumpy
kdd_cup_2022_1D	Energy	Daily	134	1	32562	[243, 243]	14	30	-	0.546	0.365	0.086	0.0	0.001	Erratic
kdd_cup_2022_30T	Energy	every 30 minutes	134	1	1575525	[11731, 11758]	48	192	-	0.475	0.01	0.088	0.0	0.001	Lumpy
us_gasoline	Energy	Weekly	1	1	1578	[1578, 1578]	13	-	-	0.554	0.874	0.316	0.001	0.074	Smooth
CoronaVirus_2019_confirmed	Healthcare	Daily	252	1	112866	[187, 494]	14	30	-	0.756	0.995	0.775	0.946	0.167	Erratic
CoronaVirus_2019_deaths	Healthcare	Daily	184	1	75414	[146, 483]	14	30	-	0.654	0.996	0.775	0.928	0.167	Erratic
CoronaVirus_2019_recovered	Healthcare	Daily	236	1	99133	[141, 494]	14	30	-	0.665	0.995	0.786	0.966	0.167	Erratic
Hospital	Healthcare	Monthly	767	1	64428	[84, 84]	18	-	-	0.678	0.558	0.393	0.061	0.04	Smooth
ecdc_ili	Healthcare	Weekly	25	1	4650	[89, 216]	13	-	-	0.814	0.387	0.269	0.003	0.015	Erratic
heart_rate	Healthcare	every 5 seconds	1	1	1800	[1800, 1800]	12	60	180	0.518	0.326	0.104	0.0	0.03	Smooth
physionet_2012	Healthcare	irregular	3238	1	296587	[80, 189]	6	-	-	0.656	0.549	0.383	0.006	0.037	Smooth
AirPassengers	Mobility	Monthly	1	1	144	[144, 144]	12	24	-	0.987	0.998	0.785	0.992	0.167	Smooth
Pedestrian_Counts	Mobility	Hourly	65	1	3131770	[1176, 96424]	48	168	-	0.948	0.516	0.153	0.0	0.008	Erratic
Taxi(Half_Hourly)	Mobility	half hourly	4742	1	4096924	[221, 1488]	48	-	-	0.806	0.194	0.14	0.0	0.005	Erratic
Taxi(Hourly)	Mobility	Hourly	5000	1	4114178	[149, 1488]	12	-	-	0.806	0.199	0.137	0.0	0.005	Erratic
UberTLC_daily	Mobility	Daily	1	1	181	[181, 181]	14	-	-	0.825	0.863	0.591	0.624	0.033	Smooth
UberTLC_hourly	Mobility	Hourly	1	1	4345	[4345, 4345]	48	168	-	0.956	0.573	0.27	0.0	0.017	Smooth
VehicleTrips	Mobility	Daily	306	1	38370	[60, 243]	12	-	-	0.749	0.593	0.232	0.113	0.018	Smooth
BrazilianCitiesTemperature	Nature	Monthly	12	1	9029	[481, 1320]	12	30	-	0.607	0.196	0.314	0.0	0.012	Erratic
Flood_Modeling_Dataset_1	Nature	Hourly	667	1	34931367	[926, 125139]	48	168	-	0.383	0.041	0.031	0.0	0.069	Lumpy
Flood_Modeling_Dataset_2	Nature	Hourly	661	1	34254046	[926, 123796]	48	168	-	0.378	0.031	0.039	0.0	0.072	Lumpy
Flood_Modeling_Dataset_3	Nature	Hourly	607	1	28979647	[862, 113882]	48	168	-	0.394	0.05	0.034	0.0	0.07	Lumpy
SaugeenRiverFlow	Nature	Daily	1	1	23741	[23741, 23741]	14	30	60	0.544	0.029	0.053	0.0	0.002	Erratic
Sunspot	Nature	Daily	1	1	73924	[73924, 73924]	14	30	60	0.687	0.269	0.28	0.0	0.001	Erratic
USBirths	Nature	Daily	1	1	7305	[7305, 7305]	14	30	-	0.93	0.928	0.67	0.013	0.052	Smooth
Weather	Nature	Daily	2994	1	42172024	[989, 65981]	14	30	-	0.72	0.015	0.058	0.0	0.004	Smooth
Car_Parts	Retail	Monthly	267	1	1861289	[166, 13770]	12	30	-	0.505	0.001	0.224	0.0	0.256	Lumpy
M5	Retail	Daily	29897	1	46279060	[271, 1968]	60	-	-	0.554	0.087	0.251	0.0	0.062	Lumpy
favorita_stores_1D	Retail	Daily	1570	1	2367756	[114, 1688]	14	-	-	0.671	0.431	0.288	0.001	0.021	Smooth
favorita_stores_1M	Retail	Monthly	1466	1	73643	[37, 54]	8	-	-	0.709	0.764	0.407	0.211	0.068	Smooth
favorita_stores_1W	Retail	Weekly	1569	1	337858	[79, 240]	13	-	-	0.7	0.65	0.4	0.047	0.052	Smooth
favorita_transactions_1D	Retail	Daily	51	1	79389	[118, 1688]	14	-	-	0.738	0.597	0.193	0.0	0.006	Smooth
favorita_transactions_1M	Retail	Monthly	45	1	2415	[39, 54]	8	-	-	0.94	0.804	0.148	0.656	0.021	Smooth
favorita_transactions_1W	Retail	Weekly	50	1	11277	[97, 240]	13	-	-	0.829	0.676	0.273	0.013	0.052	Smooth
rossmann_1D	Retail	Daily	1115	1	1049806	[757, 942]	14	-	-	0.808	0.088	0.13	0.0	0.009	Smooth
rossmann_1W	Retail	Weekly	1115	1	148232	[108, 133]	13	-	-	0.537	0.275	0.226	0.0	0.015	Smooth
M1(Monthly)	various	Monthly	617	1	55998	[48, 150]	8	-	-	0.782	0.901	0.561	0.542	0.074	Smooth
M1(Quarterly)	various	Quarterly	172	1	9282	[36, 114]	8	-	-	0.743	0.983	0.728	0.843	0.167	Smooth
M1(Yearly)	various	Yearly	66	1	2242	[26, 58]	6	-	-	0.668	0.977	0.809	0.832	0.167	Smooth
M3(Monthly)	various	Monthly	1088	1	143988	[78, 144]	18	-	-	0.637	0.927	0.635	0.47	0.074	Smooth
M3(Quarterly)	various	Quarterly	697	1	35522	[35, 72]	8	-	-	0.665	0.984	0.673	0.749	0.167	Smooth
M3(Yearly)	various	Yearly	198	1	8329	[26, 47]	6	-	-	0.623	0.954	0.651	0.537	0.093	Smooth
M4(Daily)	various	Daily	4225	1	10022640	[76, 9933]	14	-	-	0.618	0.922	0.609	0.521	0.074	Smooth
M4(Hourly)	various	Hourly	414	1	373372	[748, 1008]	48	-	-	0.986	0.91	0.251	0.108	0.009	Smooth
M4(Monthly)	various	Monthly	4996	1	1164417	[78, 2812]	18	-	-	0.71	0.958	0.633	0.516	0.074	Smooth
M4(Quarterly)	various	Quarterly	4906	1	497181	[35, 874]	8	-	-	0.65	0.977	0.716	0.75	0.167	Smooth
M4(Weekly)	various	Weekly	359	1	371579	[93, 2610]	13	-	-	0.656	0.944	0.501	0.42	0.074	Smooth
M4(Yearly)	various	Yearly	4681	1	195371	[26, 585]	6	-	-	0.631	0.991	0.761	0.936	0.167	Smooth

Table 2: **Dataset statistics (Univariate, UTS)**. We report the dataset domain, frequency, the number of time series, channels and observations, the length range of time series, and forecasting horizons (short-term, medium-term, long-term); We also report the median of a variety of time series patterns such as seasonality, trend, shift, stationarity, transition, and the most common demand pattern.

A Dataset Task Details

Table 2 and Table 3 present the full statistical summary of all forecasting datasets included in our TSB-FCST across both univariate and multivariate settings. For each dataset, we report its application domain, temporal frequency, number of series, dimensionality, total observations, sequence length range, forecasting horizon categories, and a collection of representative temporal characteristics, including seasonality, trend, shift, stationarity, transition behavior, correlation (only for multivariate datasets), and dominant demand pattern. These statistics highlight the substantial diversity of TSB-FCST in terms of scale, domain coverage, and forecasting difficulty.

B Time-series Data Patterns

Definition 5 (Trend). *The trend of a given time series represents the long-term patterns that occur over time. Formally, a Seasonal-Trend Decomposition Procedure Based on Loess (STL) [63] is used to decompose a time series X into three terms:*

$$X = X_S + X_T + X_R$$

where X_T, X_S, X_R refer to trend, seasonality, and the remainder, respectively. The strength of trend is then computed as follows.

$$Trend = max\left(0, 1 - \frac{var\left(X_R\right)}{var\left(X - X_S\right)}\right) \tag{1}$$

Definition 6 (Seasonality). *Seasonality is defined as the recurring pattern in a time series that repeats at regular intervals. Similar to trend, the strength of seasonality in a time series X can be calculated*

Dataset	Domain	Freq	#Series	#Channels	#Observations	Length Range	ShortH	MedH	LongH	Correlation	Seasonality	Trend	Shift	Stationarity	Transition	Demand Pattern
boomlet_1062	Cloud	every 1minute	1	21	344064	[16384, 16384]	60	240	720	0.663	0.567	0.294	0.094	0.037	0.04	Lumpy
boomlet_1209	Cloud	every 1minute	1	53	868352	[16384, 16384]	24	-	-	0.589	0.524	0.494	0.389	0.022	0.054	Lumpy
boomlet_1225	Cloud	every 1minute	1	49	802816	[16384, 16384]	60	240	720	0.649	0.499	0.392	0.238	0.006	0.029	Lumpy
boomlet_1230	Cloud	every 1minute	1	23	376832	[16384, 16384]	24	-	-	0.537	0.523	0.392	0.355	0.002	0.03	Lumpy
boomlet_1282	Cloud	every 1minute	1	35	573440	[16384, 16384]	24	-	-	0.671	0.469	0.346	0.375	0.073	0.027	Lumpy
boomlet_1487	Cloud	every 1minute	1	54	884736	[16384, 16384]	60	240	720	0.626	0.374	0.251	0.063	0.0	0.038	Lumpy
boomlet_1631	Cloud	every 1minute	1	40	418520	[10463, 10463]	60	240	720	0.594	0.512	0.663	0.418	0.014	0.068	Lumpy
boomlet_1676	Cloud	every 1minute	1	100	1046300	[10463, 10463]	60	240	720	0.644	0.473	0.541	0.313	0.068	0.04	Lumpy
boomlet_1855	Cloud	every 1minute	1	52	272012	[5231, 5231]	60	240	720	0.587	0.698	0.489	0.548	0.047	0.047	Lumpy
boomlet_1975	Cloud	every 1minute	1	75	392325	[5231, 5231]	24	-	-	0.698	0.771	0.554	0.294	0.019	0.015	Lumpy
boomlet_2187	Cloud	every 1minute	1	100	523100	[5231, 5231]	24	-	-	0.638	0.665	0.236	0.378	0.004	0.037	Lumpy
boomlet_285	Cloud	every 1minute	1	75	1228800	[16384, 16384]	60	240	720	0.706	0.628	0.084	0.225	0.0	0.155	Lumpy
boomlet_619	Cloud	every 1minute	1	52	851968	[16384, 16384]	60	240	720	0.832	0.78	0.178	0.146	0.0	0.032	Lumpy
boomlet_772	Cloud	every 1minute	1	67	1097728	[16384, 16384]	60	240	720	0.627	0.502	0.367	0.24	0.008	0.036	Lumpy
boomlet_963	Cloud	every 1minute	1	28	458752	[16384, 16384]	60	240	720	0.62	0.501	0.41	0.259	0.002	0.034	Lumpy
FRED-MD	Economics	Monthly	1	107	77896	[728, 728]	12	30	60	0.657	0.567	0.906	0.592	0.574	0.114	Erratic
NASDAQ	Economics	Daily	4818	6	105874440	[136, 14664]	14	30	-	0.914	0.674	0.806	0.653	0.351	0.069	Erratic
NYSE	Economics	Daily	500	5	4255690	[251, 1762]	14	30	-	0.94	0.576	0.809	0.575	0.486	0.081	Smooth
National_Stock_Exchange	Economics	Daily	3	12	7440	[248, 248]	14	30	-	0.828	0.547	0.666	0.416	0.182	0.036	Smooth
australian_tourism	Economics	Monthly	1	96	3456	[36, 36]	6	-	-	0.557	0.715	0.447	0.167	0.224	0.02	Smooth
gvar	Economics	Quarterly	26	6	27768	[178, 178]	8	-	-	0.563	0.678	0.892	0.674	0.416	0.127	Intermittent
Appliances_Energy_Prediction	Energy	every 10 minutes	1	28	552580	[19735, 19735]	48	144	576	0.71	0.649	0.697	0.532	0.03	0.067	Smooth
ETTH1	Energy	Hourly	1	7	121940	[17420, 17420]	48	168	336	0.606	0.731	0.853	0.205	0.001	0.02	Erratic
ETTH2	Energy	Hourly	1	7	121940	[17420, 17420]	48	168	336	0.508	0.615	0.906	0.404	0.022	0.042	Lumpy
ETTM1	Energy	every 15 min	1	7	487760	[69680, 69680]	32	96	384	0.599	0.76	0.614	0.203	0.0	0.027	Erratic
ETTM2	Energy	every 15 min	1	7	487760	[69680, 69680]	32	96	384	0.496	0.615	0.906	0.406	0.003	0.038	Lumpy
Power_Consumption_of_Tetouan_City	Energy	every 10 minutes	1	8	419328	[52416, 52416]	48	144	576	0.567	0.859	0.801	0.151	0.0	0.026	Erratic
Residential_Power_and_Battery_Data	Energy	every 1 minute	41	3	37647018	[30367, 500000]	120	240	-	0.518	0.597	0.24	0.234	0.0	0.002	Erratic
Spanish_energy	Energy	Hourly	1	26	911560	[35060, 35060]	48	168	336		0.335	0.245	0.283	0.0	0.03	Intermittent
Spanish_weather	Energy	Hourly	1	13	455832	[35064, 35064]	48	168	336	0.561	0.733	0.504	0.431	0.0	0.058	Lumpy
WIND	Energy	every 15 minutes	1	7	340711	[48673, 48673]	32	96	384	0.506	0.557	0.639	0.243	0.007	0.028	Smooth
electricity_2016	Energy	Hourly	1	321	8443584	[26304, 26304]	48	168	-	0.801	0.945	0.795	0.194	0.005	0.01	Smooth
electricityloaddiagrams20112014	Energy	every 15 min	1	370	51894720	[140256, 140256]	48	96	192	0.675	0.912	0.885	0.361	0.009	0.05	Lumpy
gasrate_co2	Energy	Daily	1	2	592	[296, 296]	14	30	-	1.0	0.826	0.554	0.495	0.037	0.044	Lumpy
gefcom2012_load	Energy	Daily	21	24	31752	[63, 63]	14	-	-	0.551	0.519	0.598	0.255	0.012	0.022	Smooth
gefcom2012_wind	Energy	Hourly	1	7	185983	[26569, 26569]	48	168	336	0.498	0.713	0.022	0.072	0.0	0.01	Erratic
COVID-19_Deaths	Healthcare	Daily	1	266	56392	[212, 212]	14	-	-		0.331	0.731	0.638	0.662	0.13	Lumpy
ILY	Healthcare	Hourly	1	7	6944	[992, 992]	12	24	60	0.693	0.748	0.646	0.698	0.169	0.038	Erratic
MIT-BIH	Healthcare	microsecond	20	2	4000000	[100000, 100000]	90	360	720	1.0	0.515	0.275	0.096	0.0	0.008	Lumpy
Sleep_EDFX	Healthcare	microsecond	171	6	96405040	[6937, 100000]	50	250	500	0.588	0.495	0.393	0.251	0.159	0.011	Lumpy
Bike_Sharing	Mobility	Hourly	1	11	191169	[17379, 17379]	48	168	336	0.539	0.727	0.489	0.186	0.0	0.074	Lumpy
METR-LA	Mobility	every 5 min	1	207	7094304	[34272, 34272]	48	144	288	0.789	0.49	0.107	0.03	0.0	0.004	Smooth
Metro_Interstate_Traffic_Volume	Mobility	Hourly	1	5	202875	[40575, 40575]	48	168	336	0.561	0.588	0.127	0.122	0.0	0.125	Lumpy
MexicoCityBikes	Mobility	every 1 minute	10	4	3360088	[21024, 141860]	60	240	720	0.554	0.38	0.145	0.013	0.0	0.001	Smooth
PEMS-BAY	Mobility	every 5 min	1	325	16941600	[52128, 52128]	48	144	-	0.844	0.618	0.1	0.091	0.0	0.005	Smooth
PEMS03	Mobility	every 5 min	1	358	9382464	[26208, 26208]	48	144	-	0.813	0.87	0.096	0.072	0.0	0.006	Smooth
PEMS04	Mobility	every 5 min	1	307	5216544	[16992, 16992]	48	144	-	0.809	0.854	0.077	0.085	0.0	0.007	Smooth
PEMS07	Mobility	every 5 min	1	883	24921792	[28224, 28224]	48	144	-	0.891	0.854	0.098	0.087	0.0	0.005	Smooth
PEMS08	Mobility	every 5 min	1	170	3035520	[17856, 17856]	48	144	-	0.812	0.85	0.112	0.095	0.0	0.005	Smooth
Rideshare	Mobility	Hourly	156	15	1246464	[541, 541]	12	24	60	0.732	0.7	0.816	0.443	0.263	0.073	Lumpy
Traffic	Mobility	Hourly	1	862	15122928	[17544, 17544]	48	96	168	0.814	0.88	0.263	0.171	0.0	0.011	Erratic
Traffic_Flow_Forecasting	Mobility	every 15 minutes	36	11	831996	[2101, 2101]	32	96	-	0.655	0.955	0.333	0.101	0.0	0.012	Smooth
vessel_ais	Mobility	every 7 minutes	40	8	20476928	[63885, 64000]	48	144	288	0.539	0.483	0.182	0.128	0.0	0.005	Smooth
Czelan	Nature	every 10 minutes	1	11	578171	[52561, 52561]	48	192	336	0.689	0.742	0.761	0.143	0.135	0.056	Lumpy
DailyDelhiClimate	Nature	Daily	1	4	6300	[1575, 1575]	14	30	60	0.588	0.8	0.149	0.07	0.039	0.017	Smooth
KDDCup2018	Nature	Hourly	54	6	2817230	[3056, 10905]	48	168	336	0.506	0.605	0.387	0.614	0.0	0.018	Erratic
Oikolab	Nature	Hourly	1	8	800456	[100057, 100057]	48	168	336	0.53	0.642	0.547	0.029	0.0	0.002	Smooth
Temperature-Rain	Nature	Daily	422	76	23252200	[725, 725]	14	30	60	0.73	0.552	0.131	0.238	0.013	0.066	Lumpy
Zafnoo_env	Nature	half hourly	1	11	211475	[19225, 19225]	48	192	336	0.599	0.765	0.686	0.147	0.002	0.03	Lumpy
Zafnoo_sapf	Nature	half hourly	1	6	112470	[18745, 18745]	48	192	336	0.509	0.803	0.637	0.26	0.0	0.041	Lumpy
beijing_multisite_air_quality	Nature	Hourly	12	12	5049216	[35064, 35064]	48	168	336		0.585	0.036	0.183	0.0	0.032	Lumpy
italy_air_quality	Nature	Hourly	1	13	121641	[9357, 9357]	48	168	336	0.562	0.599	0.611	0.296	0.0	0.029	Lumpy
weather_Wetterstation	Nature	every 10 minutes	1	21	1106784	[52704, 52704]	48	144	576	0.658	0.652	0.674	0.23	0.0	0.053	Lumpy
Dominick	Retail	Weekly	3889	86	64502697	[59, 399]	13	-	-	0.676	0.6	0.425	0.337	0.099	0.067	Lumpy

Table 3: **Dataset statistics (Multivariate, MTS)**. We report the dataset domain, frequency, the number of time series, channels and observations, the length range of time series, and forecasting horizons (short-term, medium-term, long-term); We also report the median of a variety of time series patterns such as correlation across different channels, seasonality, trend, shift, stationarity, transition, and the most common demand pattern.

as follows.

$$Seasonality = \max\left(0, 1 - \frac{var\left(X_R\right)}{var\left(X - X_T\right)}\right) \tag{2}$$

Definition 7 (Stationarity)**.** *Stationarity of a time series X is defined by three key properties: the mean of any observation is constant; for all observations, the variance is finite; the distance between any two observations exclusively determines the covariance. Augmented Dick-Fuller (ADF) test statistic [64] is employed to measure stationarity, computed as follows.*

$$Stationarity = \begin{cases} True, & ADF(X) \leq 0.05 \\ False, & Otherwise \end{cases} \tag{3}$$

Definition 8 (Shifting)**.** *Shifting is known as temporal changes in the probability distribution of the given time series. The shifting value $\delta \in (0, 1)$ is calculated as follows. Formally, given a Z-normalized time series X', Let $\{s_i\}_{i=1}^m$ be a set of uniformly sampled thresholds over the interval $(\min(X'), \max(X'))$. Define*

$$M_i = \text{median}(t \mid X'_t > s_i). \tag{4}$$

Let $M'_i = \text{min\_max}(M_i)$. The final shifting value is

$$\delta = \text{median}_{1 \leq i \leq m} M'_i \tag{5}$$

Definition 9 (Transition)**.** *The transition of a time series X is defined as the trace of the covariance matrix of its symbol-to-symbol transition matrix, obtained after discretizing the series into a 3-letter alphabet [65]. Formally, given a time series X of length T, let the first zero crossing of the autocorrelation function $\tau = \text{firstzero\_ac}(X)$ and define the downsampled series $Y = \text{Downsample}(X, \tau)$ of length T'. Let $I = \text{argsort}(Y)$, then the 3-symbol representation is* *defined as* $Z_j = \left\lfloor \frac{I_j}{T'/3} \right\rfloor$, $j = 1, \ldots, T'$. *We construct the transition matrix* $M \in \mathbb{R}^{3\times 3}$ *by counting symbol transitions such that we can compute the transition value* Δ*:*

$$\varDelta = \operatorname{trace}\Bigg(\operatorname{Cov}\bigg(\frac{1}{T'}\Big[\#\{j \mid Z_j = a,\ Z_{j+1} = b\}\Big]_{a,b=1}^{3}\bigg)\Bigg) \tag{6}$$

Definition 10 (Correlation)**.** *Correlation measures the degree to which different variables (i.e., channels) within a time-series instance share similar structural patterns. Let* $X_i \in \mathbb{R}^{T\times d}$ *denote a time-series instance with* d *variables. For each variable* $j \in \{1, \ldots, d\}$*, extract its Catch22 [66] feature vector* $F_i^j = \mathrm{Catch22}(X_i^{(:,j)})$*, and collect all feature vectors into*

$$F_i = [F_i^1, \ldots, F_i^d] \in \mathbb{R}^{22\times d}. \tag{7}$$

We compute all pairwise Pearson correlation coefficients between variables:

$$P_i = \{r(F_i^j, F_i^k) \mid 1 \le j < k \le d\}. \tag{8}$$

The correlation of instance X_i *is defined as*

$$\mathrm{Corr}(X_i) = \mathrm{mean}(P_i) + \frac{1}{1 + \mathrm{var}(P_i)}. \tag{9}$$

Definition 11 (ADI)**.** *The average demand interval (ADI) measures how frequently non-zero demand occurs in a channel. Let* $X_i \in \mathbb{R}^{T\times d}$ *and consider channel* j*. Let*

$$N_{ij} = \sum_{t=1}^{T} \mathbf{1}\Big(X_i^{(t,j)} > 0\Big) \tag{10}$$

be the number of non-zero observations. The ADI of channel j *is*

$$\mathrm{ADI}(X_i^{(:,j)}) = \frac{T}{N_{ij}}. \tag{11}$$

Definition 12 (CV^2)**.** *The squared coefficient of variation,* CV^2*, quantifies the relative variability of positive demand values. For channel* j*, let* μ_{ij} *and* σ_{ij} *denote the mean and standard deviation of its strictly positive observations. Then* CV^2 *is calculated as follows:*

$$\mathrm{CV}^2(X_i^{(:,j)}) = \left(\frac{\sigma_{ij}}{\mu_{ij}}\right)^2. \tag{12}$$

Definition 13 (Demand Pattern)**.** *Demand patterns classify each time series channel based on its frequency and variability. Formally, given* ADI *and* CV^2*, channel* j *is categorized as*

$$\begin{cases} Smooth, & \mathrm{ADI} \le 1.32,\ \mathrm{CV}^2 \le 0.49, \\ & \textit{(frequent and stable demand)} \\ Erratic, & \mathrm{ADI} \le 1.32,\ \mathrm{CV}^2 > 0.49, \\ & \textit{(frequent but highly variable demand)} \\ Intermittent, & \mathrm{ADI} > 1.32,\ \mathrm{CV}^2 \le 0.49, \\ & \textit{(infrequent but stable demand sizes)} \\ Lumpy, & \mathrm{ADI} > 1.32,\ \mathrm{CV}^2 > 0.49, \\ & \textit{(infrequent and highly variable demand).} \end{cases} \tag{13}$$

C Details about Forecasting Models

Tables 5, 4, 6, and 7 summarize the complete forecasting model zoo included in TSB-FCST, covering statistical methods, machine-learning-based approaches, deep learning architectures, and recent time-series foundation models. For each model, we report its architectural category, implementation source, and representative hyperparameter configuration. For multivariate forecasting models, we additionally indicate whether the method follows a channel-dependent (CD) design, which explicitly models cross-channel dependencies, or a channel-independent setting (CI) that processes each channel separately. Overall, the model suite is designed to provide the most comprehensive coverage of forecasting methodologies.

Table 4: **19 Statistical forecasting models included in the benchmark.** We illustrate the type of model, the source of implementation, and key hyperparameters.

Model	Type	Source	Key Hyperparameters
naivemovingaverage	naive	Darts [67]	`input_chunk_length=12`
naiveseasonal	naive	Darts [67]	`K=12`
naivemean	naive	Darts [67]	–
naivedrift	naive	Darts [67]	–
autoregressive	auto-regressive	Statsmodels [68]	`lags=7`
arima	auto-regressive	Statsmodels [68]	`order=(1, 1, 1)`
sarima	auto-regressive	Statsmodels [68]	`seasonal_order=(1, 1, 1, 12)`
theta	decomposition	Statsmodels [68]	`period=12`
prophet	decomposition	Meta [69]	`growth="linear"`
ets	exponential-smoothing	Statsmodels [68]	`error="add"`
simpleexpsmoothing	exponential-smoothing	Statsmodels [68]	`smoothing_level=0.6`
exponentialsmoothing	exponential-smoothing	Statsmodels [68]	`trend="add"`
holt	exponential-smoothing	Statsmodels [68]	`initialization_method="estimated"`
croston	intermittent-demand	statsforecast [70]	`n_windows=4`
adida	intermittent-demand	statsforecast [70]	`n_windows=4`
imapa	intermittent-demand	statsforecast [70]	`n_windows=4`
arch	others	statsforecast [70]	–
garch	others	statsforecast [70]	–
kalmanfilter	others	KalmanFilter[43]	–

Table 5: **15 machine-learning-based forecasting models included in the benchmark.** We illustrate the type of model, the source of implementation, and key hyperparameters.

Model	Type	Source	Key Hyperparameters
decisiontree	tree-based	Sklearn [71]	–
randomforest	tree-based	Sklearn [71]	`n_estimators=100`
extratrees	tree-based	Sklearn [71]	`n_estimators=100`
xgboost	boosting	Sklearn [71]	`max_depth=6`, `learning_rate=0.1`
adaboost	boosting	Sklearn [71]	`n_estimators=50`, `learning_rate=1.0`
lightgbm	boosting	Sklearn [71]	`lag=10`
linearregression	regression	Sklearn [71]	–
lasso	regression	Sklearn [71]	`alpha=0.1`
ridge	regression	Sklearn [71]	`alpha=1.0`
passiveaggressive	regression	Sklearn [71]	`C=1.0`, `max_iter=500`
elasticnet	regression	Sklearn [71]	`alpha=0.1`
sgdregressor	regression	Sklearn [71]	–
bayesianridge	regression	Sklearn [71]	–
ardregressor	regression	Sklearn [71]	`max_iter=300`
huber	regression	Sklearn [71]	`lag=10`
orthogonalmatchingpursuit	regression	Sklearn [71]	–
lars	regression	Sklearn [71]	`nonzero_coefs=10`
kneighbors	similarity	Sklearn [71]	`n_neighbors=5`, `lag=10`
svr	kernel-based	Sklearn [71]	`C=1.0`, `epsilon=0.1`

D Evaluation Metrics

We evaluate the forecasting performance using a diverse set of deterministic and probabilistic evaluation metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (SMAPE), Weighted Absolute Percent Error (WAPE), Modified Symmetric Mean Absolute Percentage Error (MSMAPE), and Mean Absolute Scaled Error (MASE).

Mean Absolute Error (MAE). MAE measures the average absolute difference between the forecast and the ground truth. It is easy to interpret because it has the same unit as the original time series. A lower MAE indicates that the model's predictions are globally closer to the true values.

Table 6: **47 deep learning models included in the benchmark.** 'CD' indicates channel dependence during multivariate time series forecasting.

Model	Type	CD	Source	Key Hyperparameters
autoformer	Transformer-based	✓	THUML [28]	`e_layers=2`, `d_model=512`
patch_tst	Transformer-based	✗	THUML [28]	`e_layers=1`, `d_model=512`
crossformer	Transformer-based	✓	THUML [28]	`e_layers=2`, `factor=3`
itransformer	Transformer-based	✓	THUML [28]	`e_layers=2`, `des="Exp"`
pyraformer	Transformer-based	✓	THUML [28]	`e_layers=2`, `factor=3`
nonstationary-transformer	Transformer-based	✓	THUML [28]	`p_hidden_layers=2`
reformer	Transformer-based	✗	THUML [28]	`e_layers=2`, `factor=3`
informer	Transformer-based	✓	THUML [28]	`e_layers=2`, `factor=3`
fedformer	Transformer-based	✓	THUML [28]	`e_layers=2`, `moving_avg=25`
etsformer	Transformer-based	✓	THUML [28]	`e_layers=2`, `top_k=5`
transformer	Transformer-based	✓	THUML [28]	`e_layers=2`, `n_heads=8`
cyclenet	Transformer-based	✗	CycleNet [72]	`e_layers=3`, `use_revin=True`
cycleitransformer	Transformer-based	✓	CycleNet [72]	`e_layers=3`, `use_revin=True`
tft	Transformer-based	✓	THUML [28]	`e_layers=1`, `moving_avg=3`
timexer	Transformer-based	✓	THUML [28]	`e_layers=1`, `moving_avg=25`
sim_mtm	Transformer-based	✗	SimMTM [73]	`e_layers=2`, `factor=5`
simpletm	Transformer-based	✗	SimpleTM [74]	`wv="db2"`, `factor=5`
duet	Transformer-based	✓	DUET [37]	`noisy_gating=True`, `moving_avg=25`
dlinear	MLP-based	✗	THUML [28]	`e_layers=2`, `moving_avg=25`
tsmixer	MLP-based	✓	THUML [28]	`e_layers=2`, `d_model=512`
nlinear	MLP-based	✗	NLinear [75]	`enc_in=1`, `individual=True`
ldlinear	MLP-based	✗	CycleNet [72]	`enc_in=1`
rmlp	MLP-based	✗	CycleNet [72]	–
frets	MLP-based	✗	THUML [28]	–
patchmlp	MLP-based	✗	PatchMLP [76]	`e_layers=4`, `d_model=512`
fits	MLP-based	✗	FITS [77]	`H_order=6`, `loss="MSE"`
tide	MLP-based	✓	THUML [28]	`downsampling_layers=3`
nbeats	MLP-based	✗	N-Beats [78]	`nb_blocks_per_stack=3`
sparse_tsf	MLP-based	✗	SparseTSF [79]	`e_layers=2`, `factor=3`
timemixer	MLP-based	✓	THUML [28]	`downsampling_window=2`
cnn	CNN-based	✓	MLC [80]	–
lightts	CNN-based	✗	THUML [28]	`chunk_size=24`
tcn	CNN-based	✓	TCN [81]	`kernel_size=3`, `num_filters=3`
micn	CNN-based	✓	THUML [28]	`e_layers=2`
scinet	CNN-based	✓	THUML [28]	`e_layers=2`
resnet	CNN-based	✓	ResNet [82]	`num_layers=50`
fact	CNN-based	✓	FACT [83]	`dilation=[1, 2]`, `num_kernels=6`
timesnet	CNN-based	✓	THUML [28]	`top_k=5`, `num_kernels=3`
rnn	RNN-based	✓	MLC [80]	`hidden_size=1024`
gru	RNN-based	✓	MLC [80]	`hidden_size=1024`
lstm	RNN-based	✓	MLC [80]	`num_fc_layers=2`
bilstm	RNN-based	✓	THUML [28]	`hidden_size=1024`
seq2seq	RNN-based	✓	Seq2Seq [84]	`num_kernels=3`
segrnn	RNN-based	✓	THUML [28]	`downsampling_window=1`
mamba	SSM	✓	Mamba [85]	`expand=2`, `d_conv=4`
film	SSM	✗	THUML [28]	`moving_avg=25`
timekan	KAN	✗	TimeKAN [86]	`downsampling_window=2`

Let y_i denote the ground-truth value, $\hat{y}_i$ denote the predicted value, and n denote the number of observations being predicted. MAE can be calculated as follows:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{14}$$

Mean Squared Error (MSE). MSE measures the average squared forecasting error. Because the errors are squared, large mistakes would receive heavy penalties than small mistakes: therefore, MSE is a desired metric when large forecasting errors are strongly discouraged. MSE is computed as follows:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2. \tag{15}$$

Table 7: 19 foundation models included in the benchmark.

Model	Type	CD	Source	Key Hyperparameters
moirai_v1	zero-shot forecaster	✓	MOIRAI v1 [87]	`model_size="base"`, `patch_size=8`
moirai	zero-shot forecaster	✗	MOIRAI v2 [40]	`model_size="small"`, `patch_size=8`
chronos_t5	zero-shot forecaster	✗	Chronos [19]	`quantile_levels=[0.5]`
chronos_bolt	zero-shot forecaster	✗	Chronos [19]	`quantile_levels=[0.5]`
chronos	zero-shot forecaster	✓	Chronos [19]	`quantile_levels=[0.5]`
timer	zero-shot forecaster	✗	Timer [88]	`max_context_length=2800`
sundial	zero-shot forecaster	✗	SunDial [89]	`num_samples=20`
kairos	zero-shot forecaster	✗	Kairos [90]	`preserve_positivity=False`
tirex	zero-shot forecaster	✓	TiRex [91]	–
timesfm_v1	zero-shot forecaster	✗	TimesFM v1 [39]	`num_layers=20`
timesfm_v2	zero-shot forecaster	✗	TimesFM v2 [39]	`num_layers=50`
timesfm	zero-shot forecaster	✗	TimesFM [39]	`max_context=1024`
toto	zero-shot forecaster	✓	Toto [92]	`num_samples=32`
tabpfn	zero-shot forecaster	✓	TabPFN [93]	–
gpt4ts	few-shot learner	✓	GPT4TS [94]	`freeze=False`, `pretrain=True`
moment	few-shot learner	✗	MOMENT [95]	`patch_len=8`
timellm	few-shot learner	✗	TimeLLM [58]	`llm_model="BERT"`
units	few-shot learner	✓	UniTS [96]	`prompt_num=10`
test	few-shot learner	✓	TEST [97]	`use_encoder=True`, `patch_size=11`

Root Mean Squared Error (RMSE). RMSE is the square root of MSE. Like MAE, it is expressed in the same unit as the original series, but it still penalizes large errors more strongly than MAE. A lower RMSE indicates better overall forecasting accuracy, and it can be calculated as below:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{16}$$

Mean Absolute Percentage Error (MAPE). MAPE measures the average absolute error as a percentage of the true value. It is scale-independent and easy to compare across datasets. However, it becomes unstable and less meaningful when the true value is near zero. MAPE is calculated as follows:

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|. \tag{17}$$

Symmetric Mean Absolute Percentage Error (SMAPE). SMAPE is a percentage-based error metric that normalizes the absolute error by the average magnitude of the true and predicted values. Compared with MAPE, it reduces the asymmetry between over-forecasting and under-forecasting. However, it can still be ill-defined when both the true and predicted values are close to zero. SMAPE is calculated as:

$$\text{SMAPE} = \frac{100}{n}\sum_{i=1}^{n}\frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2}. \tag{18}$$

Weighted Absolute Percentage Error (WAPE). WAPE measures the total absolute forecasting error relative to the total magnitude of the observed values. It is often more stable than MAPE because it aggregates errors before normalization, which makes it useful for datasets with multiple small or zero values.

$$\text{WAPE} = \frac{\sum_{i=1}^{n}|y_i - \hat{y}_i|}{\sum_{i=1}^{n}|y_i|}. \tag{19}$$

Modified Symmetric Mean Absolute Percentage Error (MSMAPE). MSMAPE is a modified version of SMAPE designed to improve numerical stability. By adding a small constant ϵ to the denominator, it avoids excessive penalties when both the true and predicted values are very small. This makes it more suitable for sparse, intermittent, or low-volume time series.

$$\text{MSMAPE} = \frac{100}{n}\sum_{i=1}^{n}\frac{|y_i - \hat{y}_i|}{|y_i| + |\hat{y}_i| + \epsilon}. \tag{20}$$

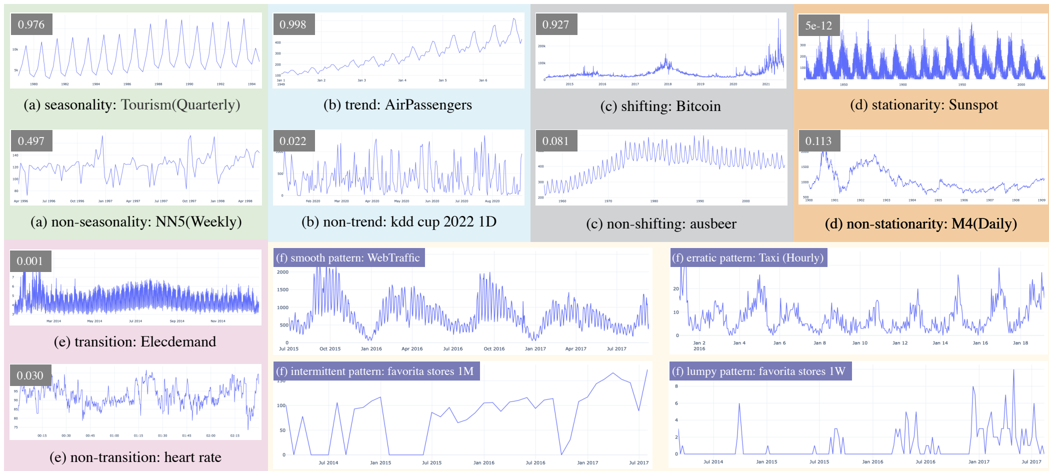

Figure 9: Qualitative examples of common time series patterns. We compare real-world datasets based on a total of 6 time series patterns: seasonality, trend, shifting, stationarity, transition, and 4 different demand patterns.

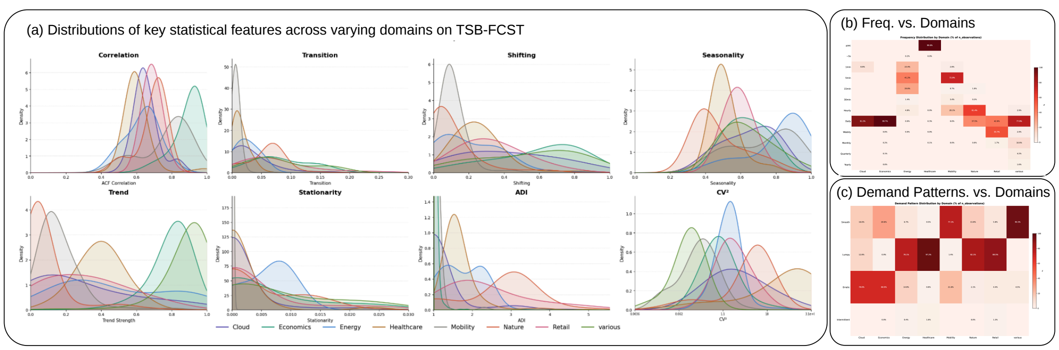

Figure 10: Data diversity and domain-specific characteristics of TSB-FCST Benchmark. (a) Distribution of key temporal features (e.g., trend, seasonality) across eight application domains. (b) Heatmap of data sampling frequencies, representing various granularities from microsecond-level to yearly. (c) Classification of demand patterns (smooth, erratic, intermittent, and lumpy) categorized by domain.

Mean Absolute Scaled Error (MASE). MASE measures the forecasting error relative to the error of a `naive seasonal` forecasting baseline. Unlike percentage-based metrics such as MAPE, MASE is scale-independent and remains well-defined even when the time series contains zeros. A MASE value smaller than 1 indicates that the forecasting model performs better than the naive baseline, while a value greater than 1 indicates worse performance than the baseline. Because of its robustness and comparability across datasets, MASE is widely used in large-scale forecasting benchmarks. Formally, MASE is defined as:

$$\mathrm{MASE} = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{\frac{n}{T-m} \sum_{i=m+1}^{T} |y_i - y_{i-m}|}, \tag{21}$$

T denotes the length of the training series, and m denotes the seasonal period of the time series.

E Details of TSB-FCST Benchmark

E.1 Data Acquisition and Standardization

To facilitate a comprehensive evaluation of time-series forecasting, we initiated an extensive data acquisition process aimed at capturing the vast diversity inherent in real-world temporal dynamics. We initially assembled a massive repository of raw time-series candidates in TSB-FCST, spanning a broad spectrum of domains including energy, meteorology, cloud computing, healthcare, transportation, economics, and retail. These sources encompass a wide range of temporal resolutions—from high-frequency seconds to yearly intervals—challenging a model's adaptability across varying scales and seasonalities.

Following this initial collection, we first conducted and implemented a rigorous preliminary check process to ensure the basic integrity of the benchmark. This stage involved the systematic removal of "low-quality" datasets characterized by an excessive portion of missing values, severe temporal misalignment across multivariate channels, or purely stochastic properties (e.g., *white noise* and *random walk signals*), which are not suitable for model evaluation. After this preliminary filtering, we maintained a high-quality archive of 128 datasets with a total scale of ***over one billion*** observations, balanced between 64 univariate (UTS) and 64 multivariate (MTS) datasets. While this version represents an expansive and diverse archive for broad exploration, it serves as the foundation for the subsequent refined and curated 1B-scale "clean" version used in our primary benchmarking (detailed in Section 3.2).

To streamline the utilization of such a large-scale archive, all datasets have been unified into a standardized tabular format. Considering the heterogeneous original source or file structures, each dataset in TSB-FCST is transformed into a uniform schema with clearly distinguishable fields for Sample ID, Timestamp, and Channel identifiers. This unified representation significantly reduces the engineering overhead for data loading and preprocessing, enabling researchers to efficiently navigate this multi-billion scale archive without the burden of complex data wrangling.

E.2 Data Curation and Quality Control

As outlined in Section E.1, we implement a rigorous, multi-stage curation and cleaning pipeline to ensure the integrity and predictability of the TSB-FCST archive. This process culminates in a ***one-billion-scale*** curated dataset, serving as a comprehensive and ready-to-use archive for large-scale time-series research. Given the inherent heterogeneity of large-scale data sources, we systematically address several critical issues commonly observed in public datasets through an automated screening process, complemented by meticulous human inspection. Our pipeline specifically targets the following four facets: (i) Missing Value and Irregular Sampling Handling, (ii) Anomaly and Artifact Mitigation, (iii) Stochasticity and Random Walk Detection, and (iv) Constant and Inactive Region Filtering. Detailed descriptions of each stage are provided below.

Missing Value and Irregular Sampling Handling: Missing values in time-series data often arise from various real-world constraints, such as sensor malfunctions, network transmission interruptions, or maintenance-related downtime. While certain advanced architectures possess an inherent capacity to process missing data, the majority of conventional and contemporary forecasting methodologies operate under the assumption of continuous and complete input sequences. To bridge this gap, we carefully handle missing entries while maintaining experimental integrity. Specifically, we employ a forward-fill imputation method for datasets with minor gaps, as it effectively preserves the temporal continuity of the signal while strictly preventing temporal data leakage—ensuring that information from the future does not leak to past observations. For datasets with excessive missing proportions that compromise structural integrity, we exclude them to maintain a high signal-to-noise ratio within the benchmark.

Furthermore, we rigorously verify the temporal regularity of the sampled intervals. To ensure a standardized and reliable baseline, the vast majority of the TSB-FCST archive is curated to be regularly sampled, facilitating the evaluation of models that rely on consistent seasonal patterns. However, we acknowledge that irregular sampling is a prevalent challenge in event-driven recording protocols. Instead of forcing these sequences into a uniform grid through aggressive interpolation—which might distort the ground-truth seasonality—we explicitly categorize a specialized subset that retains its original non-uniform interval nature. This subset allows for a targeted assessment of model robustness against irregular temporal dynamics without compromising the overall consistency of the broader benchmark. Finally, for both regular and irregular instances, we ensure cross-channel temporal alignment, ensuring that all multivariate channels are synchronized across the recorded timestamps to ensure a standardized data loading pipeline for all models.

Anomaly and Artifact Mitigation: While the majority of real-world temporal signals exhibit relatively smooth transitions, we observed prevalent anomalies across various source data, often attributable to sensor malfunctions, transmission errors, or data logging artifacts. A recurring debate in the forecasting community is whether models should be trained to be robust against such noise; however, the presence of these anomalies—especially within the test partitions—can significantly distort the overall evaluation of a model's true predictive performance.

With this in mind, we implement a systematic protocol to detect and mitigate these artifacts. To ensure objectivity, we leverage an ensemble of three widely-used anomaly detection algorithms [10] and flag sequences that exhibit high consensus anomaly scores. We acknowledge that the precise identification of anomalies often necessitates substantial domain-specific prior knowledge, as certain "anomalous" fluctuations may represent rare but legitimate physical events. Consequently, our mitigation efforts focus primarily on "spiky" anomalies—extreme deviations (e.g., erroneous values like $-9,999$ or unrealistic spikes) that are highly likely to be the result of sensor failure or a human typo error. Once identified under such automated screening process, these localized artifacts are treated using the same imputation rationale applied to missing values to ensure temporal consistency. Finally, to prevent the unintended removal of critical signals, all flagged anomalies undergo a final round of meticulous human inspection. This hybrid approach ensures that our cleaning actions are both necessary and justified.

Stochasticity and Random Walk Detection: To ensure the benchmark provides meaningful signals for evaluating forecasting efficacy, we systematically exclude datasets that exhibit purely stochastic or unpredictable behavior. Specifically, we target two categories: (1) White Noise, identified via the Ljung-Box test [98]; these sequences lack any temporal autocorrelation, meaning past observations provide zero information about future values. (2) Random Walk processes, characterized by the form $y_t = y_{t-1} + \epsilon_t$, where ϵ_t is a noise term. Such a process is fundamentally unpredictable because its best unbiased estimator is simply the most recent observation (the naive forecast). We employ the Augmented Dickey-Fuller (ADF) test complemented by the Variance Ratio (VR) test.

We validate all flagged candidates by running both naive and SOTA models, ensuring that the final removed series cannot offer genuine predictive challenges for benchmarking purpose. Consequently, we remove datasets with a high proportion of random walk sequences, such as the ExchangeRate dataset, which is widely recognized as an unsuitable benchmark dataset [99].

Constant and Inactive Region Filtering: In contrast to stochastic signals that lack predictability, certain sequences exhibit a lack of temporal variance, rendering them "too simple" to provide meaningful evaluative value. This category primarily includes constant and inactive regions—extended intervals where the signal remains unchanged. Such phenomena are typically artifacts of sensor cold-starts, device malfunctions, or periods of system inactivity rather than legitimate seasonal patterns (e.g., zero-demand during scheduled off-hours). If left unaddressed, these regions can bias evaluation metrics by artificially deflating error rates, as even a naive model can achieve near-zero error on a "flatline" signal.

To mitigate this, we rigorously scan the archive for datasets characterized by long inactive "heads" or "tails." We perform precision trimming on these sequences, removing the stagnant prefixes or suffixes while preserving the high-quality, dynamic segments in the middle. Furthermore, we examine series with a significant portion of inactive intervals in their interior. For these cases, we evaluate the structural integrity of the remaining data: if the "active" segments before or after the gap are of sufficient length and quality, they are retained as independent sub-series; otherwise, the entire flagged series is discarded to prevent models from learning or being evaluated on trivial, non-representative patterns. This curation ensures that the benchmark focuses exclusively on active, dynamic temporal behaviors that challenge a model's forecasting capabilities.

Consistent with the aforementioned quality control philosophy, all sequences processed through the aforementioned automated filters—whether involving imputation, trimming, or removal—undergo a final stage of meticulous human inspection. This manual review ensures that the automated operations accurately preserve the underlying physical semantics of the data.

E.3 Model Search Space and Optimization Protocol

The unprecedented scale and diversity of `TSB-FCST` enable a truly comprehensive evaluation across a wide spectrum of forecasting methodologies. To bridge the performance gap between disparate model classes, we rigorously benchmark **100** widely-adopted models, categorized into four distinct architectural paradigms to capture the evolution of forecasting capabilities. More details can be found in Section C.

Efficient Hyper-parameter Optimization: We acknowledge that optimizing 100 different models on the full 128 `TSB-FCST` archive is computationally infeasible—tuning multiple parameters for each model across 128 datasets would easily exceed 100k experimental runs, with large-scale datasets often requiring days or weeks to converge. Furthermore, per-dataset parameterization risks "overfitting" a model's potential to specific noise patterns rather than generalizable dynamics. To address this, we introduce `TSB-FCST-50M`, a strategically downsampled version from `TSB-FCST-1B` that preserves the original data distributions with proper validation sets while facilitating efficient tuning. This dataset and framework design ensures that each model reach a performance level that is comparable to or exceeds default settings, while maintaining safeguards against dataset-specific over-parameterization, resulting in a more representative assessment of model generalizability.

E.4 Evaluation Pipeline and Metrics

To ensure a robust and statistically sound comparison across 100 disparate models, we implement a standardized evaluation pipeline inspired by the protocols in `TFB` [29] and `TSLib` [28]. This pipeline addresses data partitioning, normalization, and frequency-aware horizon settings.

Data Partitioning and Forecasting Schemes: We employ both fixed and rolling window forecasting schemes to comprehensively assess model stability. In the fixed forecasting setup, models predict a single future window following the training and validation set. For rolling forecasting, which provides a more rigorous evaluation of long-term predictive consistency across more time steps, we ensure sufficient sequence length by adopting a default 6:2:2 split for training, validation, and test sets. We allow overlapping windows for shorter series and enable non-overlapping strategies on longer series for efficiency.

Data Pre-processing and Metrics: Following established practices in large-scale forecasting, all input series are normalized using a Standard Scaler (z-score normalization) to ensure numerical stability during training. For evaluation, we select metrics tailored to the dimensionality of the data: (i) Univariate Time-Series (UTS): We utilize Mean Absolute Scaled Error (MASE) and Mean Symmetric Mean Absolute Percentage Error (MSMAPE). These scale-invariant metrics are particularly suited for UTS as they allow for meaningful aggregation across datasets with vastly different magnitudes; and (ii) Multivariate Time-Series (MTS): we employ the standard Mean Squared Error (MSE) and Mean Absolute Error (MAE) to quantify reconstruction and forecasting accuracy across variables.

Frequency-Aware Evaluation Horizons: Rather than assigning arbitrary forecasting lengths, we define a specific set of horizons (H) for each dataset based on its inherent sampling frequency. These horizons are carefully curated to represent short, medium, and long-term forecasting tasks that align with real-world decision-making requirements (see Table 2 and 3 for details). For instance, hourly datasets may feature horizons ranging from 48, 168 to 336 hours, which match 2 days, 1 week and 2 weeks in practical applications. This intuition-based selection ensures that the benchmark evaluates models on tasks that are both practically relevant and challenging for their respective domains.

F Experimental Settings

Platform and Implementation. We use a cluster equipped with 2xAMD EPYC 7713 64-Core processors on the Ubuntu 22.04.3 LTS operating system with 4 NVIDIA A100 80GB GPU cards to conduct our benchmarking experiments. We implement all forecasting models in Python 3.11 to ensure the fairness and consistency of evaluation. We also list the primary dependencies as follows: torch 2.4.1, numpy 1.26.4, pandas 2.1.4, scikit-learn 1.4.2, statsmodels 0.14.1, and scipy 1.11.4. To promote the reproducibility of our evaluation results and analysis, we release our source code repository and collected datasets [41]. For all model experiments, we run 3 random seeds and take average to reduce the effect of randomness. For each model, we perform hyper-parameter searches across multiple sets, with a limit of 4 sets.

Statistical Test. To more reliably measure each model's overall performance, following [100, 101, 102], we employ Friedman test [103] and post-hoc Nemenyi test [104] with 95% confidence level.

Model Training and Evaluation. For foundation models, we report zero-shot performance when supported and 10% few-shot performance otherwise, while all other task-specific models are evaluated in the full-shot setting.

G Detailed experimental results

In this section, we present the complete forecasting benchmark results for both univariate time series (UTS) and multivariate time series (MTS) settings across all evaluated models and datasets. Following the evaluation protocol of prior forecasting benchmarks such as TFB, we report MASE and MSMAPE for UTS benchmarks, and MAE and MSE for MTS benchmarks. To improve readability and reduce redundancy, results are averaged across all forecasting horizons for each dataset.

For each dataset and evaluation metric, lower values indicate better forecasting performance. We additionally report the average rank of each model, computed as the mean rank across the two reported metrics within the same dataset. The best, second-best, and third-best results are highlighted using boldface, underline, and italic formatting, respectively. Due to the large number of evaluated models, the full benchmark tables are divided into multiple parts in the appendix.

Table 8: **Univariate forecasting results.** Part 1.

Dataset	adida			arch			ardregressor			arima			autoformer			autoregressive			bayesianridge			bilstm			chronos			chronos_bolt		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	1.402	0.284	48.0	4.917	1.947	98.0	0.817	0.279	28.5	1.741	0.342	69.0	1.348	0.304	49.5	*0.433*	*0.099*	3.0	0.674	0.217	14.5	1.682	0.332	67.5	1.013	0.197	21.0	1.252	0.265	38.5
AustralianElectricity	1.938	1.025	89.5	1.980	1.456	96.0	0.787	0.502	48.5	1.716	1.020	85.5	0.975	0.609	67.5	0.820	0.545	60.0	0.782	0.498	45.0	0.767	0.488	41.5	0.906	0.550	62.5	1.011	0.601	69.0
Bitcoin	0.909	0.243	7.5	8.089	1.751	97.5	1.007	0.281	46.0	$> 10^6$	0.287	82.0	1.162	0.289	66.5	1.031	0.246	30.5	1.008	0.281	47.0	1.031	0.261	42.0	0.915	*0.234*	6.0	0.925	0.240	9.0
BrazilianCitiesTemperature	0.822	0.379	14.5	1.318	1.312	92.5	1.154	0.986	73.0	0.766	0.286	8.5	1.262	0.770	74.5	$> 10^5$	0.734	81.5	1.143	1.017	76.5	0.891	0.486	26.5	0.738	0.273	6.5	0.782	0.314	10.5
CIF_2016	0.934	0.489	23.5	2.675	1.793	97.5	1.562	0.590	73.0	0.922	0.489	21.0	1.113	0.571	53.5	*0.793*	0.425	4.5	1.487	0.601	74.0	1.234	0.572	61.0	0.916	0.492	22.0	0.992	0.521	33.5
Car_Parts	1.009	0.934	85.0	0.799	0.778	60.5	0.757	0.693	42.5	0.884	0.910	75.5	0.917	0.913	80.0	0.951	0.864	77.0	0.759	0.703	44.5	0.702	0.593	21.0	*0.469*	0.153	3.5	0.476	0.165	5.0
CoronaVirus_2019_confirmed	0.947	0.080	50.0	8.480	1.935	98.5	1.495	0.140	63.5	0.394	*0.034*	3.5	1.601	0.162	68.5	6.205	0.042	44.5	1.551	0.143	65.0	0.802	0.073	40.0	*0.384*	0.036	3.5	0.558	0.048	12.0
CoronaVirus_2019_deaths	0.945	0.073	43.0	8.418	1.959	98.0	2.607	0.085	67.0	**0.262**	*0.032*	2.0	1.396	0.138	66.0	0.657	0.041	12.0	2.703	0.088	68.0	0.699	0.054	24.0	0.292	0.037	4.5	0.476	0.044	9.0
CoronaVirus_2019_recovered	0.946	0.093	48.5	4.251	1.946	95.5	4.260	0.107	74.5	0.397	0.043	3.0	1.055	0.167	63.5	1.482	0.060	38.0	11.812	0.112	77.0	0.613	0.077	28.5	0.507	0.054	6.0	0.664	0.069	24.5
Elecdemand	1.561	1.063	86.5	1.623	1.375	94.5	0.804	0.612	25.0	1.672	1.072	92.0	1.102	0.770	67.0	0.863	0.716	49.5	0.809	0.614	30.5	0.889	0.701	52.5	0.878	0.657	46.0	1.104	0.770	68.0
Extended_Wikipedia_Web_Traffic_Daily_Dataset	0.898	0.365	33.0	1.628	1.090	96.0	1.023	0.539	78.5	0.908	0.380	43.0	0.994	0.478	69.5	1.398	0.438	76.5	1.016	0.541	78.0	0.919	0.388	47.0	0.822	0.332	5.5	0.808	0.336	5.5
Flood_Modeling_Dataset_1	0.922	0.692	44.0	0.917	1.169	85.5	0.883	1.088	72.0	1.122	0.838	68.5	0.957	1.098	86.5	0.958	1.033	81.0	0.879	1.087	70.5	0.753	0.783	17.0	0.734	0.508	6.0	0.888	0.578	35.5
Flood_Modeling_Dataset_2	0.914	0.643	46.0	0.852	0.875	76.5	0.832	0.845	70.5	0.883	0.879	80.0	0.890	0.893	83.0	39.956	0.885	93.5	0.825	0.831	67.5	0.748	0.671	17.0	0.820	0.519	29.5	0.933	0.568	43.5
Flood_Modeling_Dataset_3	0.977	0.578	35.5	1.041	1.210	86.5	0.997	1.032	72.5	1.094	1.208	88.5	1.007	1.013	74.5	1.121	1.108	88.5	0.958	1.027	67.5	0.801	0.749	13.5	0.932	0.495	27.0	1.006	0.530	39.0
Hospital	0.878	0.961	16.5	1.134	1.573	87.5	1.818	1.281	93.0	0.865	0.993	25.0	0.961	1.048	52.5	1.113	0.984	45.0	1.266	1.241	81.0	0.966	1.001	44.5	0.831	0.933	4.0	0.828	0.942	5.0
M1(Monthly)	0.886	0.666	24.0	2.179	1.689	97.5	1.034	0.721	56.5	0.896	0.691	32.0	0.990	0.720	49.5	*0.700*	0.542	3.5	1.078	0.765	72.5	1.039	0.713	55.0	0.909	0.669	30.5	0.907	0.688	31.5
M1(Quarterly)	0.982	0.657	34.0	2.130	1.813	97.0	1.504	0.817	78.0	0.860	0.621	7.5	1.031	0.718	53.5	1.244	0.695	61.5	1.474	0.832	77.5	1.131	0.667	54.0	0.905	0.610	8.0	0.906	0.615	9.0
M1(Yearly)	1.182	0.734	48.5	2.200	1.835	97.5	1.545	0.950	71.0	1.047	0.715	40.0	1.130	0.749	47.5	1.267	0.842	62.0	1.610	0.953	72.5	0.869	0.615	11.5	0.911	*0.542*	6.5	0.989	0.578	15.5
M3(Monthly)	0.894	0.663	25.5	1.956	1.667	96.5	1.344	0.810	74.5	0.819	0.611	8.5	1.183	0.795	66.5	0.897	0.578	14.0	1.417	0.855	79.5	1.016	0.725	43.5	0.846	0.617	12.5	0.887	0.643	19.5
M3(Quarterly)	1.044	0.421	29.5	3.213	1.832	96.0	3.105	0.696	81.0	0.786	0.384	9.0	1.239	0.501	57.5	0.960	0.426	27.5	2.947	0.725	81.0	1.108	0.417	33.5	0.808	0.369	7.0	0.886	0.391	17.0
M3(Yearly)	1.208	0.564	29.0	3.138	1.786	97.0	2.722	0.874	79.5	1.194	0.486	16.5	1.293	0.638	49.5	1.797	0.680	66.5	2.765	0.903	83.5	2.152	0.716	70.0	1.090	0.481	10.0	1.028	0.496	9.5
M4(Daily)	0.921	0.168	14.5	8.330	1.794	98.5	0.958	0.175	27.0	0.829	0.168	8.5	1.172	0.202	66.0	1.089	0.184	48.0	1.002	0.178	37.5	1.319	0.183	58.5	1.168	0.178	48.0	1.167	0.177	46.5
M4(Hourly)	3.153	1.082	86.0	3.224	1.535	94.5	0.777	0.328	7.0	2.851	1.200	87.5	1.264	0.522	62.0	0.730	0.304	2.0	0.781	0.331	8.0	1.247	0.520	61.0	0.952	0.372	29.0	0.853	0.340	16.5
M4(Monthly)	0.872	0.545	27.5	2.485	1.723	98.0	1.153	0.645	70.0	0.880	0.564	35.5	1.027	0.600	60.0	0.811	0.531	17.0	1.167	0.658	72.0	1.148	0.624	67.5	0.899	0.557	37.0	0.891	0.545	31.0
M4(Quarterly)	1.021	0.369	25.5	4.924	1.806	98.5	2.609	0.556	78.0	0.772	0.337	10.0	1.231	0.444	58.5	1.140	0.421	47.5	2.759	0.582	83.0	1.176	0.405	46.0	0.831	0.301	8.5	0.896	0.325	15.0
M4(Weekly)	0.901	0.435	39.0	9.667	1.728	98.5	0.723	0.409	7.0	0.785	*0.395*	11.5	1.237	0.457	67.0	0.725	**0.380**	3.5	*0.711*	0.414	8.0	1.114	0.509	76.5	0.846	0.447	38.5	0.882	0.445	41.0
M4(Yearly)	1.095	0.455	51.5	8.460	1.872	94.0	39.730	0.726	82.0	0.995	0.400	16.0	1.027	0.504	50.0	0.964	0.475	28.0	39.126	0.742	83.0	1.014	0.438	35.0	0.994	0.331	7.0	0.998	0.361	11.0
M5	0.889	1.206	51.5	0.904	1.481	82.5	0.883	1.250	57.5	0.886	1.226	58.0	0.895	1.215	61.0	0.884	1.224	54.0	0.877	1.306	55.5	0.883	1.210	46.0	**0.763**	**0.791**	1.0	*0.769*	0.826	4.5
NN5	1.400	1.041	87.0	1.648	1.447	97.0	0.830	0.564	38.0	1.398	0.989	84.5	0.852	0.583	50.0	0.830	0.562	35.0	0.830	0.564	39.0	0.794	0.542	9.5	0.881	0.603	57.0	1.076	0.745	80.5
NN5(Weekly)	0.810	1.165	44.0	0.816	1.523	76.5	0.903	1.289	77.0	0.824	1.173	56.5	0.774	1.146	19.0	0.785	1.166	36.0	0.837	1.338	77.0	0.926	1.168	63.0	0.816	1.174	52.5	0.798	1.179	48.5
Pedestrian_Counts	2.360	1.131	88.5	2.207	1.586	93.5	1.099	0.576	50.5	2.043	1.282	90.0	1.218	0.648	72.0	0.952	0.505	27.5	1.116	0.585	55.0	0.965	0.512	30.0	0.748	*0.363*	4.0	0.682	**0.334**	1.5
SaugeenRiverFlow	0.914	0.709	58.5	1.034	1.190	93.0	0.820	0.809	59.0	0.905	0.667	49.0	1.018	0.946	89.5	11.430	0.994	96.0	0.819	0.809	57.5	0.774	0.637	10.5	0.978	0.680	59.0	1.002	0.703	66.5
Solar(5Min.)	1.097	0.806	80.5	1.359	1.641	96.0	0.444	0.395	34.0	1.115	0.999	88.5	0.790	0.623	68.0	0.659	0.584	58.5	0.468	0.424	41.5	0.463	0.413	40.5	*0.276*	*0.236*	3.0	0.271	0.233	2.0
SolarPower	1.082	0.021	42.0	129.063	1.737	97.5	1.240	0.024	54.5	0.706	0.014	13.5	3.070	0.053	83.0	11.231	0.232	92.0	1.243	0.024	55.5	0.904	0.019	24.5	**0.444**	**0.007**	1.0	0.671	0.009	9.0
Sunspot	0.888	0.585	80.5	1.838	1.517	98.0	0.785	0.536	29.0	0.805	0.545	52.5	0.836	0.556	60.5	0.811	0.544	53.5	0.785	0.537	31.5	0.781	0.530	18.0	0.881	0.581	78.0	0.886	0.580	78.5
Taxi(Half_Hourly)	1.367	1.167	82.5	1.224	1.487	88.5	1.022	0.930	33.0	1.211	1.288	84.0	1.214	1.151	74.0	0.929	0.879	8.0	0.983	0.936	28.5	1.070	1.030	48.5	0.861	0.775	3.0	**0.754**	**0.742**	1.0
Taxi(Hourly)	0.804	1.153	57.5	0.680	1.446	64.5	0.583	1.048	14.5	0.805	1.113	52.5	0.819	1.208	74.5	0.643	1.048	20.0	0.586	1.063	18.0	0.745	1.126	46.5	0.810	1.187	66.0	0.799	1.196	62.5
Tourism(Monthly)	1.773	0.595	66.5	3.662	1.773	97.5	1.152	0.434	16.5	1.748	0.601	68.0	1.618	0.563	56.0	1.038	0.398	5.0	1.195	0.450	20.5	1.630	0.554	54.5	1.145	0.423	12.0	1.321	0.474	32.0
Tourism(Quarterly)	1.420	0.484	54.0	4.010	1.858	98.0	1.407	0.361	36.5	1.392	0.480	50.5	1.374	0.449	40.5	1.080	0.316	10.5	1.393	0.374	37.0	1.377	0.470	47.0	1.414	0.460	47.5	1.476	0.493	60.5
Tourism(Yearly)	1.408	0.535	48.5	2.782	1.915	92.0	5.462	1.175	86.5	0.959	0.361	5.0	1.445	0.584	54.5	2.765	0.635	75.0	4.623	1.185	85.5	1.060	0.396	21.5	*0.890*	0.363	4.0	1.003	0.386	15.5
USBirths	2.953	0.913	90.0	4.266	1.724	98.0	0.942	0.293	30.0	2.916	0.896	88.5	1.313	0.511	71.5	0.963	0.297	37.5	0.944	0.294	32.0	0.844	0.273	10.5	0.885	0.283	17.5	1.704	0.595	80.0
UberTLC_daily	1.292	0.633	49.0	2.632	1.767	98.0	2.042	1.164	77.5	1.448	0.717	54.5	1.066	0.529	41.5	0.918	0.541	33.0	2.152	1.219	86.0	0.875	0.364	11.5	1.457	0.783	58.5	1.770	0.982	67.5
UberTLC_hourly	1.973	1.075	90.5	1.809	1.566	93.5	0.810	0.594	48.0	1.585	1.038	85.0	0.952	0.691	67.0	0.793	0.559	44.0	0.818	0.600	49.5	0.907	0.639	61.5	0.519	0.347	2.0	**0.482**	**0.320**	1.0
VehicleTrips	1.386	1.127	66.5	1.776	1.529	92.5	1.382	0.878	41.0	1.209	0.964	37.5	1.191	0.881	27.5	*0.957*	0.716	3.5	1.341	0.888	39.5	1.265	0.935	39.5	1.040	0.745	8.0	1.472	1.095	71.0
Weather	0.825	0.754	52.5	1.173	1.381	97.0	0.810	0.797	58.0	0.823	0.771	58.0	0.819	0.779	57.5	1.082	0.929	92.5	0.811	0.805	62.0	0.768	0.700	14.5	0.771	**0.649**	9.5	0.788	0.686	20.5
WebTraffic	0.868	0.463	14.5	1.512	1.038	94.0	1.226	0.704	75.0	0.989	0.469	36.5	1.123	0.638	68.0	1.759	0.702	87.0	1.255	0.733	81.0	0.893	0.469	23.5	0.836	0.446	5.5	0.839	0.440	6.0
Wike2000	1.088	0.500	52.0	3.095	1.478	98.0	1.130	0.547	69.5	1.145	0.517	64.0	1.113	0.506	57.5	1.198	0.495	64.0	1.131	0.557	72.0	1.139	0.468	56.5	0.870	*0.355*	3.5	0.947	0.412	25.0
Wind_Farms	0.959	0.252	6.0	3.309	1.425	95.0	1.106	0.440	51.5	0.948	0.247	4.5	2.105	0.875	89.0	$> 10^6$	0.503	87.5	1.163	0.473	59.0	1.101	0.318	32.5	0.987	0.261	13.5	1.023	0.263	17.5
ausbeer	3.468	0.759	71.5	3.773	0.827	83.5	**0.542**	0.147	1.5	1.591	0.379	41.0	1.900	0.390	47.0	0.999	0.242	19.5	0.634	0.157	4.5	3.895	0.875	87.0	0.808	0.172	8.5	2.551	0.515	55.5
ecdc_ili	0.920	1.256	56.0	0.805	1.507	62.0	0.848	1.057	18.5	1.198	1.173	63.5	0.928	1.350	65.5	1.956	1.085	57.5	0.802	1.075	17.5	1.339	1.368	85.0	0.776	1.158	25.5	0.816	1.207	36.0
favorita_stores_1D	0.923	0.970	81.5	1.062	1.352	96.5	0.793	0.877	23.5	0.917	0.981	83.5	0.842	0.893	54.0	0.856	0.870	46.0	0.795	0.883	26.5	0.832	0.864	34.0	0.816	0.831	17.0	0.851	0.892	55.0
favorita_stores_1M	0.968	0.688	8.5	1.434	1.624	92.0	1.756	1.034	90.0	0.971	0.685	9.0	1.104	0.873	66.0	1.982	0.816	82.0	1.441	1.029	84.0	1.026	0.712	40.5	0.972	0.696	13.0	0.957	0.689	8.0
favorita_stores_1W	0.783	0.716	36.5	1.242	1.557	96.0	0.756	0.826	54.5	0.734	0.716	22.5	0.747	0.760	40.0	1.326	0.789	77.5	0.750	0.839	52.5	0.728	0.715	16.5	*0.696*	*0.694*	3.0	**0.686**	0.698	3.0
favorita_transactions_1D	1.059	0.783	80.0	1.563	1.278	96.0	0.831	0.619	34.5	1.115	0.813	89.0	0.864	0.608	40.5	0.855	0.596	33.0	0.837	0.623	38.5	0.817	0.598	26.0	0.847	0.617	39.0	1.029	0.740	73.0
favorita_transactions_1M	1.602	0.814	46.5	1.915	1.548	92.0	1.563	0.899	54.5	1.667	0.843	60.5	1.617	0.900	62.0	1.595	0.639	26.5	1.654	1.011	75.5	1.595	0.803	43.0	1.748	0.842	65.0	1.750	0.830	63.0
favorita_transactions_1W	0.958	0.591	28.5	2.400	1.479	97.0	1.155	0.718	67.0	1.065	0.627	54.5	1.182	0.671	63.5	1.019	0.615	48.5	1.179	0.736	74.0	0.968	0.570	23.5	*0.900*	0.557	7.0	**0.883**	*0.545*	2.0
heart_rate	1.043	0.969	14.5	1.283	1.514	93.5	1.029	1.050	23.5	0.992	**0.942**	1.5	1.125	1.072	52.5	1.088	1.176	63.0	1.035	1.056	26.0	1.096	1.125	60.0	1.009	1.010	13.0	1.105	1.025	38.0
kdd_cup_2022_10T	0.783	0.394	34.5	2.501	1.541	97.5	0.939	0.457	74.0	*0.741*	**0.367**	2.0	0.991	0.490	85.5	$> 10^6$	0.458	87.5	0.940	0.458	76.0	0.741	*0.371*	2.5	0.825	0.407	45.5	0.837	0.414	49.0
kdd_cup_2022_1D	0.853	1.286	42.0	0.830	1.522	54.0	0.856	1.353	52.5	0.828	1.410	49.0	0.874	1.260	43.5	0.850	1.386	53.0	0.839	1.441	54.0	0.848	1.199	13.5	0.869	1.217	28.5	0.858	1.203	20.5
kdd_cup_2022_30T	0.847	0.893	20.0	0.958	1.545	93.5	0.881	1.204	71.5	**0.775**	0.960	11.5	0.948	1.122	77.5	0.777	0.953	10.5	0.882	1.215	73.0	0.812	0.991	22.5	0.849	0.884	20.0	0.841	0.915	18.5
physionet_2012	0.728	0.968	8.5	0.901	1.483	92.0	0.758	1.058	39.0	0.730	*0.952*	4.5	0.889	1.058	73.0	0.858	1.047	69.5	0.764	1.078	49.0	1.104	1.081	88.5	0.747	0.946	7.5	0.767	0.958	13.5
rossmann_1D	1.325	1.111	75.5	1.396	1.359	92.0	0.740	0.662	23.5	1.322	1.154	79.0	0.898	0.762	43.5	0.726	0.650	21.0	0.751	0.674	29.5	1.177	0.981	63.0	1.179	0.974	63.0	1.318	1.103	72.0
rossmann_1W	0.869	1.315	50.5	0.849	1.570	55.5	0.962	1.257	36.0	0.802	1.272	26.5	1.136	1.227	38.0	0.903	1.382	57.5	0.845	1.382	48.5	1.176	1.260	59.5	1.272	1.408	92.0	1.273	1.378	88.5
us_gasoline	0.881	0.557	48.5	1.812	1.698	98.5	0.859	0.557	39.0	0.819	0.512	11.5	0.895	0.550	48.0	0.822	0.521	14.5	0.870	0.560	49.0	0.878	0.567	56.0	0.869	0.560	48.0	0.857	0.553	34.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 9: **Univariate forecasting results.** Part 2.

Dataset	chronos_t5			cnn			crossformer			croston			cycleitransformer			cyclenet			decisiontree			dlinear			duet			elasticnet		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	1.228	0.250	35.0	1.522	0.329	61.5	2.601	0.751	84.0	1.399	0.288	48.5	0.803	0.170	10.5	1.494	0.325	60.0	2.365	0.606	80.0	1.625	0.360	69.5	1.412	0.302	53.0	0.661	0.162	4.5
AustralianElectricity	1.032	0.630	71.5	0.712	0.461	27.0	0.642	0.415	12.0	2.371	1.320	96.5	**0.534**	**0.354**	1.0	0.699	0.450	21.5	0.856	0.523	58.0	0.784	0.503	48.5	0.725	0.460	29.5	0.806	0.500	50.0
Bitcoin	0.960	0.247	19.0	1.031	0.260	40.5	7.635	0.827	85.5	0.985	0.280	41.5	1.021	0.263	42.5	0.959	0.250	21.0	7.711	0.947	93.0	1.185	0.312	72.0	0.946	0.253	19.5	1.063	0.290	61.5
BrazilianCitiesTemperature	0.852	0.423	19.5	1.009	0.533	49.0	1.169	1.009	77.0	4.680	1.417	98.0	0.950	0.515	40.5	1.004	0.534	49.5	1.115	0.370	37.0	1.248	0.784	74.0	0.900	0.502	31.5	1.150	1.010	76.0
CIF_2016	1.025	0.543	41.0	1.137	0.589	59.5	1.590	0.776	85.0	1.884	1.044	90.5	0.848	0.437	7.5	1.151	0.599	63.0	1.483	0.691	79.5	1.177	0.612	68.0	1.092	0.569	51.0	1.452	0.614	73.5
Car_Parts	1.444	0.888	85.5	0.701	0.588	19.0	0.774	0.728	52.0	9.452	1.881	98.5	0.764	0.647	41.0	0.723	0.618	33.0	0.872	0.686	55.0	0.892	0.888	74.0	0.686	0.564	13.0	0.775	0.726	52.0
CoronaVirus_2019_confirmed	0.579	0.046	12.0	1.365	0.123	59.0	6.439	0.818	82.0	1.358	0.117	57.5	0.680	0.059	24.0	1.362	0.127	59.5	6.997	1.036	86.5	1.667	0.177	70.5	0.806	0.069	37.5	2.412	0.214	74.0
CoronaVirus_2019_deaths	0.498	0.048	12.0	1.331	0.096	59.5	3.678	0.309	85.0	1.412	0.109	63.0	0.818	0.055	30.5	1.392	0.108	61.5	2.523	0.255	75.0	1.442	0.119	64.5	1.017	0.060	40.0	3.428	0.129	76.5
CoronaVirus_2019_recovered	0.661	0.062	19.0	1.101	0.118	58.0	2.803	0.425	83.5	1.235	0.139	63.0	0.600	0.072	23.0	1.185	0.135	61.0	1.923	0.316	75.5	1.285	0.159	66.0	0.679	0.076	32.5	2.725	0.147	73.5
Elecdemand	0.952	0.713	58.0	0.802	0.603	19.5	0.782	0.595	13.5	2.100	1.360	96.5	0.817	0.620	32.5	0.813	0.621	32.5	0.830	0.604	27.5	0.873	0.647	42.5	0.808	0.605	24.5	0.877	0.631	41.5
Extended_Wikipedia_Web_Traffic_Daily_Dataset	0.968	0.391	54.5	0.838	0.357	15.5	0.923	0.443	57.0	2.268	1.142	98.0	0.864	0.353	21.0	0.836	0.356	14.5	1.215	0.429	74.5	1.021	0.538	77.5	0.841	0.359	18.0	1.066	0.581	84.0
Flood_Modeling_Dataset_1	0.953	0.755	50.0	0.787	0.804	33.0	0.836	0.907	48.0	2.939	1.678	97.5	0.771	0.791	24.0	0.762	0.787	20.5	0.892	0.857	58.0	0.884	1.092	73.5	0.775	0.818	30.0	0.888	1.116	78.5
Flood_Modeling_Dataset_2	0.939	0.652	49.0	0.740	0.662	14.5	0.783	0.758	43.5	2.878	1.604	97.0	0.766	0.698	32.5	0.752	0.676	20.5	0.831	0.734	54.5	0.824	0.827	66.5	0.768	0.710	35.0	0.838	0.856	72.5
Flood_Modeling_Dataset_3	1.000	0.642	43.0	0.842	0.757	23.0	0.879	0.867	45.0	3.747	1.698	98.5	0.855	0.773	30.5	0.835	0.761	21.5	0.920	0.808	47.0	0.933	0.995	62.5	0.821	0.769	21.5	0.967	1.059	73.0
Hospital	0.879	0.991	28.0	0.870	1.009	32.5	1.143	1.256	81.0	1.085	1.089	66.5	0.942	1.008	42.0	0.874	1.002	31.0	1.239	1.244	80.0	0.862	1.000	27.0	0.853	0.990	18.5	1.402	1.252	87.0
M1(Monthly)	0.855	0.647	15.0	0.961	0.727	49.0	1.229	0.841	82.5	1.336	0.944	90.0	0.831	0.598	7.0	0.953	0.718	44.0	1.299	0.835	84.0	0.996	0.735	57.0	0.936	0.713	40.5	1.088	0.735	70.5
M1(Quarterly)	0.900	0.629	12.0	1.061	0.693	51.5	1.436	0.958	79.5	1.214	0.753	67.5	0.933	0.628	14.5	1.073	0.699	56.0	1.508	1.003	85.5	1.080	0.719	59.0	1.031	0.678	48.0	1.290	0.788	71.0
M1(Yearly)	0.998	0.616	22.0	1.261	0.813	59.5	1.776	1.233	81.0	1.500	0.909	69.0	0.978	0.638	21.0	1.269	0.809	60.0	1.782	1.285	84.0	1.281	0.836	64.5	1.239	0.798	55.0	1.519	0.962	71.5
M3(Monthly)	0.864	0.645	18.5	1.073	0.762	57.5	1.286	0.885	77.0	1.361	1.000	86.5	0.891	0.616	17.5	1.065	0.754	55.0	1.445	0.912	86.0	1.094	0.779	61.5	1.041	0.730	50.5	1.314	0.819	72.5
M3(Quarterly)	0.880	0.383	13.0	1.275	0.500	58.5	1.777	0.750	77.0	1.678	0.695	71.0	0.939	0.399	19.5	1.261	0.496	56.5	1.914	0.769	82.0	1.329	0.520	65.0	1.233	0.481	53.5	2.633	0.678	78.5
M3(Yearly)	1.002	0.510	11.0	1.400	0.633	53.5	2.161	0.891	75.5	2.040	0.928	76.5	1.163	0.541	22.5	1.415	0.638	56.5	2.264	0.958	82.0	1.424	0.649	59.5	1.364	0.627	50.0	2.623	0.860	77.0
M4(Daily)	1.063	0.193	54.5	1.186	0.177	47.5	1.799	0.239	78.5	1.580	0.321	84.0	0.881	0.187	29.5	1.132	0.174	39.5	2.048	0.291	87.5	1.296	0.223	74.5	1.120	0.176	40.5	1.134	0.195	60.5
M4(Hourly)	0.849	0.359	17.5	1.097	0.457	51.5	0.857	0.365	21.0	3.454	1.068	88.0	0.800	0.339	13.0	1.117	0.472	53.0	0.959	0.376	32.0	2.239	0.965	78.5	0.984	0.408	42.0	0.857	0.361	20.0
M4(Monthly)	0.922	0.597	49.0	0.990	0.583	55.0	1.309	0.766	80.0	1.485	0.972	90.0	0.797	0.490	7.0	0.988	0.579	54.0	1.383	0.770	83.0	1.037	0.608	61.5	0.932	0.563	42.5	1.074	0.635	66.0
M4(Quarterly)	0.943	0.341	20.0	1.218	0.441	56.5	2.267	0.654	81.5	1.806	0.629	75.0	0.937	0.367	20.5	1.210	0.443	56.5	2.082	0.686	81.5	1.263	0.474	63.5	1.145	0.432	51.5	2.833	0.573	82.5
M4(Weekly)	0.873	0.427	31.5	1.129	0.465	68.5	3.877	0.491	87.0	1.439	0.665	87.0	0.698	0.406	4.5	1.037	0.460	62.0	1.696	0.483	82.0	1.316	0.494	81.0	0.780	0.441	30.0	1.094	0.432	46.0
M4(Yearly)	1.022	0.355	25.5	1.039	0.520	57.0	8.367	0.896	83.0	2.860	0.817	74.0	1.006	0.391	19.5	1.039	0.523	59.5	3.411	0.969	77.5	1.036	0.534	58.0	1.036	0.509	54.5	39.997	0.715	81.5
M5	1.055	1.190	66.0	0.876	1.206	35.5	0.886	1.273	64.0	1.741	1.424	97.0	0.886	1.177	35.0	0.882	1.179	31.5	1.007	1.157	54.5	0.882	1.271	56.5	0.875	1.197	29.0	0.876	1.297	52.0
NN5	0.947	0.643	69.5	0.820	0.561	31.5	0.784	0.535	4.0	2.211	1.362	96.5	0.800	0.548	17.5	0.826	0.563	35.0	0.992	0.627	70.0	0.954	0.671	71.0	0.816	0.558	28.5	0.859	0.586	51.5
NN5(Weekly)	0.775	1.171	36.0	0.754	1.148	14.5	0.797	1.290	59.5	0.932	1.251	76.5	0.795	1.141	23.5	0.760	1.159	24.5	1.096	1.267	82.5	0.760	1.162	25.5	0.753	1.147	11.0	0.995	1.314	85.0
Pedestrian_Counts	*0.722*	0.372	3.5	0.907	0.482	16.5	0.988	0.541	36.5	2.974	1.275	95.5	0.915	0.483	19.5	0.913	0.484	19.5	1.245	0.596	64.5	1.123	0.601	59.5	0.910	0.494	20.5	1.035	0.559	44.5
SaugeenRiverFlow	0.887	0.697	53.5	0.788	0.661	25.0	0.827	0.769	56.0	1.725	1.505	97.0	0.782	0.640	15.5	0.788	0.657	22.5	0.933	0.722	61.5	0.864	0.856	72.5	0.775	0.681	27.0	0.836	0.849	68.5
Solar(5Min.)	0.359	0.312	6.5	0.425	0.376	29.0	0.392	0.349	13.0	2.155	1.496	97.5	0.403	0.352	15.5	0.426	0.379	30.0	0.481	0.372	35.5	0.584	0.546	56.0	0.423	0.372	26.5	0.408	0.365	22.0
SolarPower	4.274	0.025	72.0	1.041	0.021	42.0	1.245	0.019	41.0	1.625	0.038	72.5	0.925	0.018	22.5	1.071	0.023	48.0	1.221	0.014	34.0	2.494	0.038	78.5	1.009	0.022	41.0	8.003	0.075	88.0
Sunspot	0.846	0.566	67.5	0.778	0.529	7.5	0.786	0.540	38.0	1.161	0.888	95.0	0.777	0.528	5.0	0.779	0.529	11.5	0.914	0.596	85.5	0.816	0.554	57.5	0.779	0.531	12.5	0.814	0.555	58.0
Taxi(Half_Hourly)	1.156	1.051	62.5	1.010	0.976	40.0	0.990	0.955	34.5	1.699	1.308	96.0	0.961	0.879	12.0	1.049	1.041	48.0	1.197	0.986	59.0	1.133	1.177	70.5	0.954	0.924	19.0	0.967	0.923	20.5
Taxi(Hourly)	0.799	1.205	64.5	0.814	1.249	76.5	0.572	1.012	9.0	0.942	1.267	90.0	0.662	1.037	18.5	0.803	1.241	71.0	0.727	1.074	36.0	0.840	1.277	86.0	0.694	1.155	44.0	0.577	1.068	16.5
Tourism(Monthly)	1.360	0.484	36.0	1.750	0.595	67.0	1.494	0.529	47.5	1.755	0.600	68.5	1.112	0.419	9.0	1.604	0.565	56.5	1.718	0.627	69.0	1.736	0.608	69.5	1.405	0.512	43.0	1.169	0.428	15.0
Tourism(Quarterly)	1.516	0.435	52.0	1.539	0.509	67.0	2.053	0.524	80.0	1.691	0.522	73.0	1.101	0.345	17.5	1.457	0.495	59.5	1.772	0.566	80.5	1.568	0.525	71.5	1.379	0.463	44.5	1.369	0.366	32.0
Tourism(Yearly)	1.084	0.401	24.5	1.554	0.598	60.0	2.298	1.232	75.0	1.947	0.854	68.5	1.027	0.384	16.0	1.547	0.596	57.5	2.437	1.339	78.0	1.581	0.624	65.0	1.501	0.570	54.5	4.142	1.152	82.5
USBirths	0.976	0.306	41.0	0.922	0.292	28.0	1.191	0.415	65.5	2.591	0.685	83.5	0.827	0.265	6.5	0.882	0.277	15.0	1.556	0.461	73.0	1.869	0.689	82.0	0.929	0.304	34.5	1.075	0.330	49.5
UberTLC_daily	1.409	0.716	52.0	1.043	0.443	35.0	2.205	1.280	89.5	0.977	0.408	26.5	0.723	0.398	12.5	1.070	0.463	40.5	1.704	0.944	64.0	0.971	0.411	27.0	0.895	0.388	15.5	2.060	1.185	81.0
UberTLC_hourly	0.764	0.505	32.5	0.726	0.515	32.0	0.661	0.502	20.0	1.824	0.999	87.0	0.534	0.386	4.5	0.767	0.553	40.0	0.956	0.670	66.0	1.012	0.716	69.5	0.680	0.504	22.5	0.740	0.518	34.0
VehicleTrips	1.064	0.831	15.0	1.386	1.111	64.5	1.509	1.004	64.5	1.900	1.127	85.0	1.204	0.803	18.0	1.348	1.069	53.5	1.642	0.933	65.5	1.407	1.130	71.0	1.280	1.021	43.0	1.337	0.854	32.0
Weather	0.858	0.783	70.0	0.768	0.699	13.0	0.795	0.760	44.5	1.444	1.256	96.5	0.766	0.704	15.0	0.792	0.729	36.5	0.942	0.769	72.0	0.828	0.814	72.5	0.772	0.718	25.0	0.829	0.831	74.5
WebTraffic	0.926	0.462	28.0	0.864	0.464	14.0	1.267	0.615	70.5	2.786	1.300	98.0	0.895	0.476	29.0	0.878	0.473	24.5	1.409	0.600	73.5	0.985	0.597	56.5	0.866	0.470	17.5	1.240	0.725	77.5
Wike2000	0.975	0.398	25.0	0.958	0.407	25.5	1.331	0.494	67.5	3.425	1.402	97.5	1.010	0.386	27.0	1.019	0.460	42.0	1.540	0.536	81.5	1.120	0.553	69.5	0.914	0.380	11.5	1.170	0.591	79.0
Wind_Farms	1.036	0.286	21.5	0.963	0.287	14.0	1.287	0.429	61.0	2.452	0.760	89.5	1.035	0.319	28.0	0.981	0.301	16.5	1.421	0.415	64.0	1.228	0.518	67.5	1.108	0.374	43.0	1.228	0.502	66.5
ausbeer	1.032	0.221	19.5	3.717	0.852	83.5	1.407	0.353	34.0	4.024	0.917	90.0	1.649	0.401	46.5	2.637	0.585	58.5	1.561	0.396	42.0	2.866	0.556	58.5	1.905	0.463	50.0	1.643	0.400	45.5
ecdc_ili	1.021	1.229	68.5	0.990	1.439	86.0	0.731	1.163	20.5	0.945	1.233	59.0	0.813	1.148	30.0	0.989	1.429	83.0	0.790	1.058	12.5	1.012	1.438	90.0	0.940	1.417	74.5	0.816	1.063	18.0
favorita_stores_1D	0.894	0.913	68.5	0.821	0.876	38.5	0.815	0.878	36.0	1.436	1.228	95.5	0.823	0.838	23.5	0.804	0.845	16.5	1.014	0.931	79.5	0.890	0.979	78.5	0.790	0.846	9.0	*0.793*	0.895	31.0
favorita_stores_1M	0.997	0.753	43.5	0.977	0.707	19.0	1.177	0.983	72.0	1.109	0.808	64.0	0.994	0.706	26.0	0.984	0.716	29.5	1.261	1.001	79.5	1.002	0.746	44.5	0.986	0.708	24.5	1.591	0.999	85.5
favorita_stores_1W	0.725	0.751	30.5	0.728	0.717	20.0	0.796	0.843	69.0	0.994	0.893	85.5	0.759	0.736	42.5	0.726	0.724	23.0	1.013	0.908	88.0	0.744	0.742	36.0	0.715	0.708	10.5	0.753	0.822	50.0
favorita_transactions_1D	0.906	0.656	57.5	0.896	0.654	56.0	0.842	0.601	33.5	2.113	1.289	98.0	0.789	0.572	8.0	0.819	0.599	27.5	1.131	0.682	79.0	1.035	0.760	76.0	0.815	0.597	23.5	0.862	0.635	47.0
favorita_transactions_1M	1.636	0.819	53.5	1.412	0.810	25.5	1.461	1.008	54.0	2.116	1.253	92.5	1.573	0.751	26.5	1.416	0.800	20.0	1.793	0.999	80.5	1.441	0.825	33.5	1.443	0.799	23.0	1.615	0.906	63.5
favorita_transactions_1W	0.958	0.585	28.0	0.954	0.566	18.0	1.216	0.750	77.5	2.756	1.192	96.0	0.965	0.534	12.5	0.936	0.556	8.0	1.752	0.762	88.0	1.014	0.618	48.0	0.932	0.559	9.0	1.163	0.717	68.5
heart_rate	1.014	1.029	17.0	1.115	1.082	53.0	1.097	1.135	61.5	1.385	1.097	77.5	1.043	0.989	18.0	1.133	1.103	66.5	1.274	1.106	77.5	1.089	1.087	47.0	1.040	1.075	31.5	1.027	1.075	27.5
kdd_cup_2022_10T	0.850	0.425	53.0	0.752	0.377	19.0	0.899	0.439	61.0	1.828	0.929	96.0	0.747	0.375	15.0	0.749	0.374	15.5	0.967	0.459	80.5	1.065	0.525	89.5	0.766	0.389	29.5	1.042	0.503	88.0
kdd_cup_2022_1D	0.872	1.139	24.0	0.865	1.260	37.5	0.831	1.387	49.5	1.270	1.439	94.0	0.923	1.267	65.0	0.878	1.273	51.5	1.147	1.314	79.5	0.871	1.276	49.5	0.861	1.257	33.0	0.856	1.374	56.5
kdd_cup_2022_30T	0.821	*0.871*	8.0	0.840	1.037	38.5	0.866	1.118	58.0	1.453	1.520	98.0	0.860	0.990	40.0	0.833	0.973	24.0	0.962	1.095	76.0	0.901	1.235	83.0	0.842	1.027	38.0	0.873	1.262	73.5
physionet_2012	0.734	0.984	14.5	0.785	1.034	43.5	0.794	1.094	64.0	0.990	1.143	92.5	0.825	0.992	47.0	0.790	1.026	43.5	0.975	1.118	88.0	0.813	1.048	62.5	0.774	1.005	29.5	0.765	1.068	45.5
rossmann_1D	1.261	0.995	66.5	1.221	1.041	66.0	0.755	0.691	32.5	1.460	1.024	78.0	0.639	0.580	4.5	1.088	0.912	54.0	0.840	0.672	35.0	1.326	1.139	79.5	0.804	0.707	38.0	0.790	0.683	34.0
rossmann_1W	0.981	1.385	64.5	1.125	1.281	59.5	0.857	1.362	49.5	0.977	**1.076**	18.5	1.161	1.216	45.5	1.149	1.292	69.5	1.030	1.093	22.5	1.125	1.281	59.0	1.120	1.276	53.5	0.937	1.248	33.0
us_gasoline	0.900	0.567	62.0	0.949	0.595	74.0	0.847	0.556	31.5	0.931	0.559	60.0	0.803	0.493	6.5	0.935	0.583	70.0	0.980	0.641	83.0	1.035	0.645	90.0	0.863	0.558	43.5	0.882	0.576	61.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 10: **Univariate forecasting results.** Part 3.

Dataset	ets			etsformer			extratrees			fact			fedformer			film			fits			frets			garch			gpt4ts		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	**0.347**	**0.072**	1.0	1.632	0.330	65.0	2.458	0.662	82.5	0.908	0.203	18.5	1.028	0.213	24.5	1.472	0.312	58.0	1.704	0.377	73.0	3.202	0.648	84.5	5.458	1.532	97.5	1.115	0.235	32.0
AustralianElectricity	1.004	0.599	68.0	1.291	0.791	81.0	0.706	0.461	25.5	0.599	0.395	7.0	0.810	0.530	56.5	0.790	0.493	47.0	0.799	0.502	50.0	0.721	0.478	33.5	2.313	1.302	95.5	0.795	0.509	51.5
Bitcoin	**0.897**	0.246	6.0	1.009	0.288	53.0	7.675	0.900	91.0	1.028	0.256	37.0	1.125	0.280	57.5	1.076	0.266	50.5	1.114	0.278	54.5	1.139	0.297	67.5	$> 10^5$	1.572	97.5	1.075	0.263	48.0
BrazilianCitiesTemperature	0.915	0.440	25.5	1.186	0.854	71.0	1.238	0.910	76.5	0.891	0.477	25.5	1.244	0.757	72.5	1.059	0.588	56.5	1.107	0.576	57.5	1.291	0.944	81.0	2.181	1.305	96.0	1.038	0.527	49.5
CIF_2016	**0.687**	**0.352**	1.0	1.023	0.577	47.0	1.474	0.691	79.5	0.877	0.470	15.5	1.107	0.586	56.5	1.176	0.604	66.0	1.163	0.597	63.5	1.416	0.644	75.0	2.984	1.305	95.5	1.184	0.533	51.0
Car_Parts	0.921	0.822	73.0	1.079	0.975	90.5	0.833	0.693	54.0	0.705	0.589	22.0	0.940	0.970	85.0	1.156	1.026	92.5	0.916	0.897	77.0	0.869	0.814	67.0	$> 10^6$	1.384	98.5	0.717	0.598	25.5
CoronaVirus_2019_confirmed	0.374	**0.032**	1.5	0.823	0.093	48.5	6.997	1.036	86.5	0.613	0.056	16.5	1.465	0.130	61.0	1.492	0.143	63.5	1.605	0.157	67.5	4.965	0.379	78.5	16.952	1.586	98.0	0.848	0.075	46.0
CoronaVirus_2019_deaths	*0.268*	**0.029**	2.0	0.596	0.072	30.0	2.523	0.253	74.0	0.804	0.051	26.0	1.177	0.105	58.0	1.539	0.127	66.5	1.706	0.148	69.5	3.795	0.173	81.5	8.931	1.616	97.5	1.186	0.071	49.0
CoronaVirus_2019_recovered	0.394	*0.048*	2.5	0.684	0.102	44.0	1.923	0.316	76.5	0.506	0.064	9.5	1.009	0.136	58.5	1.328	0.162	67.0	1.411	0.184	69.0	2.059	0.198	76.5	6.556	1.613	96.5	0.701	0.087	40.0
Elecdemand	0.976	0.772	64.5	1.359	0.981	80.0	*0.702*	*0.544*	3.0	0.777	0.581	12.0	0.959	0.669	54.5	0.860	0.639	38.0	1.471	1.080	87.5	0.865	0.659	44.0	1.914	1.259	95.5	0.868	0.677	46.5
Extended_Wikipedia_Web_Traffic_Daily_Dataset	0.899	0.374	40.0	1.135	0.578	86.5	0.961	0.465	64.5	0.849	0.349	14.0	0.963	0.466	65.5	0.914	0.419	51.0	0.923	0.422	53.0	0.989	0.482	70.0	2.265	1.094	97.0	0.846	0.362	20.5
Flood_Modeling_Dataset_1	1.182	0.866	71.5	0.947	1.092	83.5	0.856	1.007	62.5	0.784	0.797	30.0	0.889	1.033	72.5	0.852	0.871	49.5	0.887	0.950	62.5	0.870	0.996	64.0	1.312	1.140	94.5	0.764	0.794	24.5
Flood_Modeling_Dataset_2	1.067	0.784	75.0	0.953	0.949	89.0	0.826	0.820	65.0	0.757	0.686	28.0	0.852	0.861	75.5	0.848	0.791	64.0	0.865	0.827	71.5	0.808	0.806	57.5	1.214	0.997	92.5	0.756	0.687	27.5
Flood_Modeling_Dataset_3	1.237	0.802	68.0	1.015	0.954	72.5	0.903	0.918	53.0	0.850	0.776	31.0	0.998	1.048	75.5	0.940	0.830	52.5	0.969	0.952	64.5	1.001	0.958	70.5	1.506	1.143	94.0	0.829	0.759	19.5
Hospital	0.840	0.913	3.5	1.163	1.055	73.0	1.126	1.236	72.5	0.882	0.997	32.5	0.974	1.044	56.0	0.875	0.998	30.0	0.856	0.996	24.0	1.434	1.371	95.0	1.633	1.343	94.5	0.963	0.993	39.0
M1(Monthly)	0.782	*0.540*	3.5	1.019	0.713	53.0	1.228	0.824	80.0	0.802	0.613	6.5	0.964	0.691	38.0	1.005	0.735	60.0	0.972	0.733	53.5	1.226	0.774	78.0	2.764	1.322	96.5	0.938	0.661	27.0
M1(Quarterly)	0.959	0.574	12.0	1.143	0.755	65.5	1.480	0.994	83.5	0.972	0.654	30.5	1.006	0.659	40.5	1.086	0.697	56.5	1.067	0.698	55.0	1.448	0.973	81.0	2.708	1.397	96.0	0.983	0.637	27.0
M1(Yearly)	0.992	0.622	22.0	1.268	0.798	58.0	1.755	1.267	81.5	1.177	0.772	50.5	1.221	0.793	54.0	1.279	0.825	63.5	1.276	0.816	62.0	1.908	1.361	89.0	2.668	1.500	96.5	0.975	0.643	20.5
M3(Monthly)	0.784	**0.528**	1.5	1.102	0.768	61.0	1.285	0.868	74.5	0.930	0.641	23.5	1.072	0.730	52.0	1.082	0.758	58.0	1.079	0.769	59.0	1.563	0.882	85.0	2.479	1.355	97.0	0.934	0.646	26.5
M3(Quarterly)	0.882	0.391	16.5	1.093	0.454	41.5	1.842	0.741	76.0	1.115	0.449	42.5	1.185	0.486	52.5	1.259	0.502	58.5	1.294	0.506	62.0	2.572	0.957	87.5	4.220	1.360	95.0	1.008	0.413	27.0
M3(Yearly)	1.292	0.505	26.5	1.417	0.672	60.5	2.227	0.964	82.0	1.284	0.594	39.0	1.286	0.596	40.5	1.351	0.636	51.5	1.402	0.640	56.5	1.902	0.877	71.0	3.939	1.347	96.0	1.333	0.573	39.5
M4(Daily)	0.863	0.170	11.5	1.114	0.192	56.0	1.933	0.260	82.0	1.005	0.187	45.5	1.036	0.167	26.0	1.138	0.172	39.0	1.245	0.201	67.0	1.166	0.213	67.0	8.622	1.576	98.0	1.125	0.189	55.5
M4(Hourly)	1.083	0.402	45.0	1.604	0.653	70.0	0.753	0.312	4.0	0.852	0.352	17.0	1.166	0.468	54.0	2.110	0.855	75.5	2.757	1.035	81.0	1.014	0.425	45.0	3.833	1.324	95.0	1.272	0.530	63.0
M4(Monthly)	0.788	0.494	7.0	0.904	0.565	41.0	1.312	0.750	79.5	0.832	0.508	16.0	0.972	0.568	49.0	1.003	0.578	55.5	1.030	0.599	60.0	1.601	0.833	89.5	2.852	1.406	97.5	0.826	0.498	12.5
M4(Quarterly)	0.761	0.322	7.5	1.161	0.444	55.5	1.934	0.660	78.0	0.931	0.393	26.5	1.119	0.419	44.0	1.293	0.453	62.5	1.239	0.453	60.5	1.882	0.544	70.0	5.621	1.370	97.5	0.971	0.404	32.0
M4(Weekly)	0.747	0.426	17.5	0.921	0.459	55.0	1.612	0.438	61.0	0.826	0.416	21.0	0.736	0.410	10.0	1.064	0.461	65.0	1.253	0.491	79.0	1.096	0.453	58.5	623.249	1.537	98.0	1.019	0.450	53.0
M4(Yearly)	0.934	0.371	8.5	1.507	0.512	62.0	4.070	0.973	79.0	1.029	0.472	47.0	1.023	0.464	43.5	1.041	0.535	63.0	1.040	0.521	59.0	7.632	0.924	80.5	8.088	1.408	87.0	1.013	0.419	29.0
M5	0.887	1.204	47.5	0.944	1.207	68.5	0.893	1.236	68.0	0.882	1.180	31.5	0.889	1.219	59.0	0.905	1.126	41.5	0.884	1.239	60.0	0.977	1.211	72.0	1.228	1.237	87.0	0.895	1.185	47.0
NN5	0.819	0.549	25.0	0.985	0.685	73.0	**0.761**	**0.521**	1.0	0.789	0.541	6.5	0.847	0.571	46.0	0.916	0.624	66.0	0.983	0.676	72.0	0.806	0.555	23.5	2.235	1.246	96.5	0.837	0.564	39.5
NN5(Weekly)	0.832	1.173	56.5	1.013	1.246	78.5	0.819	1.240	64.0	0.756	1.145	11.5	0.754	**1.094**	4.5	0.777	1.169	36.0	0.750	1.157	19.0	0.908	1.311	81.0	1.062	1.167	66.0	0.759	1.102	6.5
Pedestrian_Counts	1.316	0.645	75.0	1.326	0.718	78.0	1.007	0.551	41.5	0.897	0.482	15.5	1.145	0.617	63.5	1.181	0.629	66.5	1.198	0.670	72.5	1.287	0.629	70.5	2.832	1.328	96.0	1.042	0.560	47.0
SaugeenRiverFlow	0.910	0.698	56.0	0.865	0.740	57.5	0.816	0.783	51.0	0.780	0.654	20.0	0.842	0.780	59.5	0.974	0.816	79.0	1.102	0.897	90.5	0.844	0.817	69.0	1.436	1.139	94.5	0.802	0.671	33.0
Solar(5Min.)	1.250	0.868	88.0	0.771	0.654	68.5	0.374	0.326	9.0	0.409	0.359	21.5	0.839	0.762	73.0	0.692	0.611	63.5	0.918	0.844	78.5	0.720	0.588	63.0	1.529	1.348	95.5	0.728	0.648	67.5
SolarPower	1.002	0.020	34.0	10.148	0.118	91.0	0.832	0.011	12.0	0.934	0.020	32.0	3.677	0.054	84.0	1.511	0.033	69.0	1.190	0.025	56.0	1.670	0.028	67.0	181.813	1.718	97.5	1.220	0.023	50.0
Sunspot	0.989	0.613	89.0	0.855	0.572	70.5	0.790	0.540	40.0	*0.777*	0.528	3.5	0.801	0.535	38.0	0.807	0.545	54.0	0.858	0.567	71.0	0.782	0.538	30.0	$> 10^6$	1.349	98.5	0.793	0.537	37.5
Taxi(Half_Hourly)	1.035	0.897	26.0	1.235	1.150	75.5	0.936	0.903	11.0	0.951	0.896	11.5	1.057	1.009	45.5	1.132	1.123	65.0	1.208	1.235	81.5	1.081	0.950	43.0	1.636	1.284	94.0	1.064	1.030	47.0
Taxi(Hourly)	0.817	1.169	65.0	0.765	1.161	53.0	**0.549**	0.977	3.0	0.663	1.074	26.5	0.758	1.146	50.0	0.860	1.278	87.0	0.866	1.290	89.5	0.611	1.051	20.0	1.187	1.335	96.5	0.818	1.201	71.5
Tourism(Monthly)	**0.898**	0.387	1.5	1.818	0.641	79.0	1.379	0.497	39.5	1.048	0.421	8.5	1.370	0.504	40.0	1.654	0.592	61.0	1.745	0.605	68.0	1.436	0.526	46.0	4.311	1.506	97.0	1.725	0.563	59.5
Tourism(Quarterly)	**0.931**	0.275	1.5	1.576	0.539	74.5	1.713	0.499	71.5	1.051	0.326	11.0	1.380	0.509	56.0	1.504	0.516	66.5	1.575	0.522	71.5	1.592	0.420	54.5	4.584	1.413	97.5	1.145	0.361	21.5
Tourism(Yearly)	0.999	0.365	9.5	1.485	0.609	58.0	2.444	1.350	79.0	1.422	0.542	51.5	1.379	0.516	44.5	1.551	0.581	57.5	1.576	0.605	62.5	3.281	1.586	93.0	3.519	1.475	91.5	1.109	0.419	30.5
USBirths	0.887	0.272	14.0	1.570	0.568	78.0	1.132	0.331	53.0	**0.783**	**0.248**	1.0	1.154	0.456	65.0	1.314	0.433	68.5	1.585	0.564	78.0	0.910	0.286	22.5	4.912	1.336	97.5	0.909	0.282	19.0
UberTLC_daily	1.254	0.608	46.0	1.538	0.776	59.5	2.163	1.254	87.5	0.718	0.365	7.5	0.916	0.432	25.0	1.267	0.574	46.0	1.059	0.450	37.0	1.932	1.101	72.0	3.497	1.532	97.5	0.648	*0.303*	3.5
UberTLC_hourly	1.230	0.763	75.5	1.406	1.010	82.5	0.664	0.507	23.0	0.543	0.400	6.0	0.744	0.558	40.0	1.052	0.718	72.0	1.248	0.889	79.0	0.878	0.684	62.0	2.245	1.406	96.0	0.889	0.639	59.5
VehicleTrips	0.988	0.704	3.5	1.538	1.078	73.0	1.364	0.866	36.5	1.078	0.783	11.5	1.287	0.911	38.0	1.436	1.095	67.5	1.373	1.117	62.5	1.381	0.863	38.0	2.682	1.300	96.0	1.304	0.933	41.5
Weather	0.875	0.793	76.0	0.820	0.806	66.5	0.794	0.764	45.0	0.787	0.716	29.0	0.781	0.757	36.5	0.825	0.739	50.0	0.864	0.789	74.0	0.823	0.788	61.5	$> 10^6$	1.258	98.0	0.785	0.712	26.0
WebTraffic	1.334	0.556	69.5	1.224	0.744	77.5	1.411	0.628	78.0	0.878	0.469	21.5	1.041	0.609	60.0	0.954	0.495	45.0	0.897	0.499	38.0	1.181	0.671	70.0	4.690	1.129	98.5	0.899	0.485	36.0
Wike2000	1.072	0.470	48.5	1.108	0.551	66.0	1.094	0.486	51.5	0.909	0.365	8.0	1.098	0.503	55.0	0.972	0.447	33.0	1.046	0.483	47.0	1.112	0.513	57.5	4.148	1.215	97.5	0.982	0.410	30.0
Wind_Farms	0.985	0.266	14.0	2.230	0.661	87.0	1.308	0.409	60.5	0.994	0.305	19.5	1.471	0.599	81.5	1.453	0.430	68.0	1.281	0.395	55.0	1.125	0.455	54.5	$> 10^6$	$> 10^6$	99.0	1.211	0.337	43.5
ausbeer	0.752	0.162	7.0	2.335	0.494	53.5	1.683	0.388	44.5	0.660	**0.146**	3.0	3.227	0.637	63.5	4.464	0.984	94.0	3.620	0.812	79.0	1.094	0.293	24.5	6.754	1.237	98.0	4.103	0.895	90.0
ecdc_ili	0.944	1.125	44.5	0.948	1.243	62.5	0.718	*1.055*	3.5	0.890	1.234	49.0	0.930	1.387	68.5	0.993	1.442	88.0	1.008	1.436	88.5	0.783	1.131	23.0	1.320	1.347	82.5	0.843	1.263	47.0
favorita_stores_1D	0.877	0.870	49.0	0.820	0.874	35.5	0.800	0.866	22.0	0.797	0.833	6.5	0.837	0.888	48.5	0.870	0.926	64.0	0.890	0.960	74.5	0.835	0.892	49.0	1.733	1.296	97.5	0.829	0.875	39.5
favorita_stores_1M	0.981	0.794	38.5	1.150	0.783	64.0	1.192	0.981	73.5	0.985	0.714	28.5	1.107	0.814	64.0	1.001	0.715	35.5	0.983	0.708	23.5	1.366	1.165	88.0	1.890	1.393	97.0	1.027	0.680	28.0
favorita_stores_1W	0.762	0.746	46.0	0.907	0.811	72.5	0.803	0.847	73.5	0.722	0.710	12.0	0.750	0.753	39.5	0.759	0.727	39.5	0.729	0.717	21.5	0.892	0.866	81.0	1.834	1.365	97.5	0.714	0.708	9.5
favorita_transactions_1D	0.877	0.615	44.5	0.891	0.656	55.5	0.796	0.572	11.5	0.770	*0.554*	2.5	0.909	0.612	48.0	0.916	0.664	60.5	1.031	0.755	74.5	0.792	0.571	9.0	2.257	1.225	97.0	0.875	0.640	51.0
favorita_transactions_1M	1.426	**0.567**	7.5	2.041	0.906	78.0	1.319	0.900	36.5	1.420	0.789	16.5	1.559	0.857	50.5	1.488	0.801	30.5	1.407	0.802	21.5	1.517	1.005	58.5	3.017	1.262	97.0	1.549	0.783	29.0
favorita_transactions_1W	1.016	0.594	42.5	1.184	0.665	63.5	1.301	0.705	74.0	0.940	0.568	16.0	1.054	0.615	50.5	1.009	0.579	36.0	0.948	0.570	18.5	1.282	0.734	79.5	3.692	1.339	98.0	0.986	0.578	30.5
heart_rate	1.000	0.957	4.5	1.059	1.044	29.0	1.054	1.077	36.5	1.175	1.007	47.0	1.088	1.112	54.5	1.121	1.091	60.0	1.188	1.183	83.5	1.255	1.075	65.0	1.497	1.328	96.0	1.143	1.104	68.0
kdd_cup_2022_10T	0.784	0.386	32.0	0.950	0.467	82.5	0.898	0.437	59.5	0.746	0.372	9.5	0.912	0.454	69.0	0.817	0.408	45.0	0.837	0.422	50.0	0.913	0.446	67.5	$> 10^6$	1.308	98.5	0.758	0.378	22.5
kdd_cup_2022_1D	0.830	1.402	50.0	0.985	1.304	77.0	0.892	1.430	76.0	0.890	1.266	51.5	0.868	1.239	29.5	0.891	1.279	57.0	0.867	1.267	44.0	0.938	1.358	83.0	1.131	1.240	56.0	0.876	1.257	41.0
kdd_cup_2022_30T	0.870	0.883	28.5	0.911	1.222	84.0	0.888	1.207	75.0	0.848	0.985	33.0	0.889	1.108	67.0	0.889	1.038	62.0	0.865	1.016	47.5	0.893	1.220	78.5	1.089	1.324	97.0	0.826	1.002	31.0
physionet_2012	0.802	0.994	42.0	0.899	1.092	83.5	0.799	1.070	63.0	0.779	0.996	28.0	0.803	1.011	50.0	0.790	1.011	39.0	0.782	1.034	41.0	0.827	1.068	70.5	1.304	1.311	97.5	0.799	0.980	36.0
rossmann_1D	0.772	0.706	35.5	1.128	0.937	57.0	**0.582**	**0.555**	1.0	0.660	0.606	7.5	0.829	0.720	40.0	1.082	0.971	57.0	1.340	1.164	82.0	0.683	0.629	14.0	2.554	1.432	98.5	1.139	0.952	59.0
rossmann_1W	0.873	1.401	58.5	1.102	1.229	34.5	0.786	1.185	5.0	1.130	1.251	47.0	1.114	1.208	31.0	1.139	1.244	45.5	1.119	1.290	60.0	0.855	1.174	10.5	1.621	1.412	96.0	1.139	1.263	52.5
us_gasoline	0.822	0.530	17.0	0.877	0.556	45.0	0.875	0.580	58.0	0.846	0.543	25.0	0.816	0.514	11.0	0.978	0.613	80.5	1.041	0.643	90.5	0.901	0.574	64.5	1.997	1.400	98.0	0.889	0.565	59.5

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 11: **Univariate forecasting results.** Part 4.

Dataset	gru			holt			huber			imapa			informer			itransformer			kairos			kalmanfilter			kneighbors			lars		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	1.280	0.261	38.5	1.854	0.358	73.5	0.709	0.236	21.0	1.402	0.284	48.0	3.041	0.869	87.0	1.051	0.235	30.0	1.234	0.231	32.0	7.932	0.981	94.0	3.423	1.069	89.5	0.708	0.213	15.0
AustralianElectricity	0.707	0.462	26.5	13.599	1.400	98.5	0.750	0.472	35.0	1.938	1.025	89.0	0.720	0.487	35.5	0.557	0.366	4.0	1.793	0.963	86.5	12.680	1.395	97.5	0.828	0.538	59.0	0.917	0.543	62.0
Bitcoin	0.956	0.252	21.5	0.953	0.279	35.0	0.949	0.256	24.0	0.909	0.243	7.5	7.669	0.888	89.0	0.982	0.260	32.0	0.990	0.247	23.5	2.695	0.633	83.0	7.743	1.011	95.0	1.324	0.366	77.5
BrazilianCitiesTemperature	0.941	0.486	32.5	0.850	0.340	14.5	**0.615**	**0.217**	1.0	0.822	0.379	14.5	1.157	0.931	72.0	0.923	0.497	32.5	0.857	0.421	20.0	2.053	0.491	64.0	1.183	0.920	73.0	1.131	0.996	72.0
CIF_2016	0.999	0.540	37.5	0.798	0.419	4.0	1.797	0.587	74.5	0.934	0.489	23.5	1.557	0.742	83.0	0.930	0.491	24.0	0.852	0.447	10.0	1.096	0.521	41.5	1.647	0.800	88.0	$> 10^6$	0.636	88.0
Car_Parts	0.677	0.545	11.5	1.308	1.069	94.0	**0.435**	**0.094**	1.0	1.009	0.934	85.0	0.779	0.735	56.0	0.758	0.638	39.5	0.873	0.616	50.0	4.577	0.904	89.5	0.723	0.299	18.5	0.772	0.717	49.0
CoronaVirus_2019_confirmed	0.807	0.072	40.0	**0.354**	<u>0.032</u>	1.5	1.283	0.126	57.5	0.947	0.080	50.0	7.036	1.053	91.0	0.690	0.062	26.5	0.466	0.045	8.5	0.407	0.042	6.0	7.025	1.044	90.0	7.461	0.238	85.5
CoronaVirus_2019_deaths	1.010	0.082	49.5	<u>0.265</u>	<u>0.030</u>	2.0	2.527	0.080	62.0	0.945	0.073	43.0	4.193	0.406	89.5	0.806	0.056	31.0	0.425	0.047	10.0	0.333	0.043	7.0	2.576	0.258	79.5	3.793	0.186	81.5
CoronaVirus_2019_recovered	0.568	0.071	18.0	**0.349**	**0.043**	1.0	5.023	0.099	73.5	0.946	0.093	48.5	2.975	0.531	88.0	0.665	0.076	31.5	0.583	0.068	16.5	*0.395*	0.058	4.5	1.951	0.324	79.5	$> 10^6$	0.185	87.0
Elecdemand	0.901	0.718	56.0	14.749	1.563	98.0	0.736	0.556	6.5	1.561	1.063	86.5	1.206	0.882	74.0	0.800	0.608	23.0	1.584	1.046	86.5	22.356	1.570	99.0	0.809	0.606	26.5	1.453	0.792	75.0
Extended_Wikipedia_Web_Traffic_Daily_Dataset	0.833	0.351	10.5	1.167	0.420	72.5	0.865	0.378	34.5	0.898	0.365	33.0	0.929	0.446	58.5	0.864	0.364	26.0	0.897	0.386	42.0	4.302	0.755	96.5	0.948	0.445	61.0	1.125	0.608	87.0
Flood_Modeling_Dataset_1	0.786	0.805	32.5	3.214	0.836	70.5	**0.578**	**0.404**	1.0	0.922	0.692	44.0	0.868	0.962	59.5	0.764	0.786	21.5	0.804	0.694	25.0	9.570	1.149	97.5	0.953	0.903	68.0	0.884	1.106	75.0
Flood_Modeling_Dataset_2	0.735	0.649	11.0	3.767	0.812	82.5	**0.598**	**0.405**	1.0	0.914	0.643	46.0	0.785	0.751	43.5	0.774	0.708	35.0	0.786	0.632	24.5	9.664	1.121	96.5	0.885	0.759	63.5	0.834	0.850	71.5
Flood_Modeling_Dataset_3	0.824	0.783	26.0	2.279	0.822	73.5	0.861	*0.468*	17.0	0.977	0.578	35.5	0.916	0.897	54.0	0.860	0.780	33.5	0.867	0.621	23.5	7.204	1.027	88.0	0.993	0.893	63.0	0.963	1.053	71.5
Hospital	0.963	0.993	39.0	1.007	0.975	37.5	1.266	1.241	82.0	0.878	0.961	16.5	1.145	1.319	85.5	0.938	1.022	45.0	0.984	0.994	44.5	2.582	1.301	94.5	1.122	1.241	73.5	$> 10^6$	1.277	94.0
M1(Monthly)	1.003	0.697	45.5	1.026	0.704	50.0	1.055	0.706	54.0	0.886	0.666	24.0	1.296	0.857	86.0	0.817	0.611	7.5	1.035	0.684	47.0	1.590	0.834	87.0	1.300	0.846	86.5	1.083	0.723	63.5
M1(Quarterly)	1.008	0.652	37.0	0.980	0.618	19.5	1.745	0.891	82.5	0.982	0.657	34.0	1.528	1.011	87.0	0.962	0.657	30.5	1.160	0.634	43.0	1.312	0.721	67.0	1.545	1.046	88.5	$> 10^6$	0.969	91.0
M1(Yearly)	*0.839*	0.597	6.5	**0.806**	0.550	2.5	1.751	0.974	75.5	1.182	0.734	48.5	1.889	1.310	87.5	1.085	0.700	40.0	0.928	0.608	14.0	0.898	0.557	5.5	1.801	1.332	86.5	$> 10^6$	1.009	88.5
M3(Monthly)	0.990	0.694	37.0	1.108	0.656	44.5	1.472	0.834	80.5	0.894	0.663	25.5	1.322	0.911	81.5	0.947	0.657	29.0	0.879	0.659	22.0	1.982	0.943	92.5	1.342	0.891	81.5	$> 10^6$	0.797	83.0
M3(Quarterly)	1.333	0.496	61.0	0.946	0.430	28.0	4.365	0.789	91.5	1.044	0.421	29.5	1.855	0.743	77.0	1.089	0.442	39.0	0.846	0.387	12.5	1.084	0.458	41.5	2.031	0.856	84.0	$> 10^6$	0.863	94.0
M3(Yearly)	1.644	0.673	64.5	1.271	0.521	25.5	3.007	0.916	86.5	1.208	0.564	29.0	2.268	0.944	82.0	1.137	0.558	22.5	1.257	0.533	26.5	1.461	0.662	62.0	2.502	1.101	86.0	$> 10^6$	0.923	91.0
M4(Daily)	1.652	0.214	77.0	**0.792**	0.183	23.0	0.897	0.169	13.0	0.921	0.168	14.5	2.230	0.276	87.5	0.958	0.180	33.5	1.258	0.210	70.5	1.429	0.264	78.5	2.132	0.293	88.5	1.049	0.182	43.0
M4(Hourly)	1.202	0.480	57.0	16.612	1.553	98.0	0.768	0.322	5.5	3.153	1.082	86.0	1.388	0.583	65.5	0.942	0.382	30.5	3.918	1.326	96.0	21.479	1.655	99.0	0.953	0.395	36.5	1.023	0.390	41.0
M4(Monthly)	0.888	0.533	27.5	0.892	0.592	44.5	1.259	0.661	74.5	0.872	0.545	27.5	1.421	0.793	84.0	0.839	0.508	17.0	0.919	0.563	41.0	1.846	0.799	89.0	1.426	0.808	86.5	$> 10^6$	0.683	87.5
M4(Quarterly)	1.059	0.405	37.5	*0.763*	0.304	5.5	2.174	0.538	72.5	1.021	0.369	25.5	2.559	0.700	86.0	0.932	0.392	26.5	0.847	0.309	10.5	1.036	0.394	33.0	2.056	0.678	80.5	3.483	0.567	84.0
M4(Weekly)	0.940	0.455	51.5	0.728	0.414	11.5	0.752	<u>0.394</u>	8.0	0.901	0.435	39.0	4.984	0.534	92.5	0.897	0.453	47.5	0.944	0.458	54.5	2.467	0.528	89.0	1.747	0.436	62.0	0.871	0.439	38.0
M4(Yearly)	1.010	0.431	28.5	1.000	0.408	18.0	2.270	0.747	71.0	1.095	0.455	51.5	8.384	1.008	87.5	1.018	0.440	37.0	0.993	0.313	6.0	1.144	0.351	36.5	4.572	1.010	82.0	$> 10^6$	0.811	88.5
M5	0.875	1.212	37.0	1.001	1.201	68.5	0.789	0.864	6.5	0.889	1.206	51.5	0.918	1.310	85.0	0.887	1.170	35.0	0.903	1.082	38.0	6.751	1.526	99.0	0.860	1.165	14.5	0.880	1.328	59.0
NN5	0.797	0.544	13.5	1.527	1.059	92.5	0.816	0.553	26.5	1.400	1.041	87.0	0.893	0.618	62.5	0.800	0.546	16.0	1.034	0.708	77.0	8.306	1.436	98.0	0.872	0.606	56.0	0.936	0.653	69.5
NN5(Weekly)	0.802	1.150	34.5	0.899	1.171	60.5	1.482	1.352	95.5	0.810	1.165	44.0	0.817	1.300	68.5	0.808	1.157	40.5	1.208	1.273	85.5	2.354	1.438	98.0	0.830	1.372	78.5	0.949	1.307	83.0
Pedestrian_Counts	0.940	0.500	26.0	8.287	1.274	95.5	1.000	0.518	35.5	2.360	1.131	88.5	1.239	0.669	73.5	0.934	0.491	22.0	1.807	0.921	84.5	29.084	1.577	98.5	1.114	0.588	54.5	1.085	0.581	51.5
SaugeenRiverFlow	0.771	0.647	12.5	2.002	0.840	90.0	**0.712**	**0.573**	1.0	0.914	0.709	58.5	0.736	0.646	9.5	0.766	0.630	7.5	1.002	0.757	72.0	2.993	0.965	95.0	0.991	0.797	75.5	0.842	0.830	69.0
Solar(5Min.)	0.435	0.391	32.0	11.886	0.941	93.0	0.313	0.266	4.0	1.097	0.806	80.5	1.081	1.001	85.5	0.413	0.359	22.5	1.221	0.855	86.0	14.806	1.334	97.5	0.460	0.397	36.0	1.334	0.470	71.5
SolarPower	0.880	0.019	21.5	<u>0.457</u>	<u>0.007</u>	2.0	0.981	0.019	31.5	1.082	0.021	42.0	7.556	0.077	88.0	0.921	0.018	23.0	2.978	0.029	74.0	41.929	0.594	94.5	0.746	0.008	9.0	6.113	0.068	86.0
Sunspot	0.780	0.529	11.5	1.385	0.675	95.0	0.779	0.531	16.5	0.888	0.585	80.5	0.791	0.536	33.5	0.781	0.531	16.5	1.039	0.619	92.0	4.328	1.075	97.5	0.840	0.564	66.0	1.200	0.845	95.0
Taxi(Half_Hourly)	1.083	1.032	52.5	1.785	1.204	91.5	1.061	0.932	36.5	1.367	1.167	82.5	1.117	1.046	58.0	1.008	0.925	30.5	1.314	1.164	81.0	5.890	1.437	98.0	0.968	0.919	20.0	0.971	0.952	30.0
Taxi(Hourly)	0.722	1.120	41.5	0.902	1.134	69.5	0.587	1.032	13.5	0.804	1.153	57.5	0.568	0.976	5.0	0.740	1.143	48.0	0.954	1.177	77.5	1.521	1.170	78.5	0.594	1.008	14.5	0.582	1.068	18.0
Tourism(Monthly)	1.634	0.557	56.0	2.029	0.637	84.0	1.169	0.451	20.0	1.773	0.595	66.5	1.786	0.608	72.5	1.230	0.466	25.0	1.253	0.476	30.0	5.152	1.113	96.5	1.492	0.560	51.0	1.160	0.428	15.0
Tourism(Quarterly)	1.357	0.466	43.0	1.318	0.466	40.0	1.316	0.367	27.5	1.420	0.484	54.0	2.309	0.602	89.0	1.038	0.321	9.5	1.074	0.319	10.5	1.962	0.688	87.5	1.911	0.573	83.5	1.950	0.393	58.5
Tourism(Yearly)	0.993	0.384	12.0	**0.822**	<u>0.318</u>	1.5	4.726	1.189	87.5	1.408	0.535	48.5	2.534	1.398	85.0	1.219	0.452	39.5	1.131	0.419	32.0	0.982	0.413	18.5	2.487	1.401	83.5	$> 10^6$	1.214	89.0
USBirths	0.847	0.271	10.5	2.980	0.950	94.5	0.917	0.285	23.5	2.953	0.913	90.0	1.276	0.481	70.0	0.841	0.265	7.0	1.538	0.559	75.5	9.888	1.428	98.5	1.387	0.487	72.5	0.979	0.306	43.0
UberTLC_daily	0.876	0.410	19.5	2.213	1.141	84.0	2.323	1.299	91.5	1.292	0.633	49.0	1.904	1.045	70.5	0.651	0.319	5.0	1.591	0.770	59.0	5.137	1.401	97.5	2.130	1.239	85.5	2.045	1.185	79.5
UberTLC_hourly	0.709	0.525	31.5	37.433	1.682	99.0	0.779	0.555	42.5	1.973	1.075	90.5	1.232	0.899	79.0	0.583	0.420	8.0	1.559	0.957	82.0	34.897	1.650	98.0	0.777	0.638	51.0	0.729	0.503	29.0
VehicleTrips	1.254	0.911	36.0	1.875	1.168	87.5	1.507	0.851	48.5	1.386	1.127	66.5	1.571	1.038	69.5	1.217	0.893	32.5	1.173	0.827	18.5	4.640	1.399	97.5	1.439	0.938	56.0	$> 10^6$	0.871	64.5
Weather	0.774	0.708	20.5	0.909	0.786	75.5	*0.749*	0.667	3.5	0.825	0.754	52.5	0.764	0.732	24.5	0.760	0.700	11.5	0.937	0.764	70.0	2.825	1.094	96.0	0.851	0.800	73.5	0.937	0.985	89.0
WebTraffic	0.868	0.456	12.0	1.109	0.536	57.5	0.925	0.498	43.0	0.868	0.463	14.5	1.326	0.693	79.5	0.911	0.486	38.5	0.851	0.458	10.0	3.545	0.796	95.0	1.288	0.620	72.5	1.254	0.729	80.0
Wike2000	0.925	0.395	16.0	1.223	0.560	80.5	0.963	0.404	25.5	1.088	0.500	52.0	1.567	0.553	85.5	0.980	0.377	20.0	0.985	0.444	37.0	2.280	0.743	93.0	1.302	0.523	74.5	1.160	0.581	77.0
Wind_Farms	1.013	0.350	32.0	3.021	0.416	74.5	*0.914*	0.253	5.5	0.959	0.252	6.0	1.291	0.400	57.5	1.094	0.345	37.5	1.102	0.307	29.0	6.348	0.883	94.0	1.214	0.309	37.5	2.236	0.552	85.5
ausbeer	3.181	0.695	64.0	3.296	0.704	67.0	0.732	0.188	9.0	3.468	0.759	71.5	2.031	0.514	53.0	0.997	0.212	16.0	0.924	0.172	11.5	6.360	1.303	98.0	1.629	0.297	34.5	0.832	0.192	11.5
ecdc_ili	0.880	1.357	58.5	1.905	1.100	59.0	0.907	1.064	29.0	0.920	1.256	56.0	*0.715*	1.072	7.0	0.874	1.257	50.0	1.207	1.177	64.5	1.837	1.154	64.5	0.719	1.088	11.5	0.980	1.090	47.0
favorita_stores_1D	0.805	0.857	21.0	0.958	0.962	83.0	**0.769**	**0.790**	1.0	0.923	0.970	81.5	0.846	0.905	58.0	0.809	0.835	16.0	1.020	0.950	82.5	2.349	1.241	97.5	0.800	0.867	22.5	0.798	0.893	34.0
favorita_stores_1M	1.017	0.715	40.5	1.028	0.711	41.5	1.751	1.085	92.5	0.968	0.688	8.5	1.241	1.083	82.5	1.020	0.727	44.5	1.033	0.692	32.5	1.318	0.842	74.5	1.200	1.028	78.0	$> 10^6$	1.098	95.0
favorita_stores_1W	0.730	0.716	19.5	0.819	0.773	62.0	0.787	0.803	59.5	0.783	0.716	36.5	0.822	0.847	74.5	0.754	0.722	31.0	0.947	0.787	70.5	1.965	0.946	95.5	0.853	0.865	78.5	0.754	0.819	50.5
favorita_transactions_1D	0.814	0.598	24.0	1.089	0.797	84.5	0.805	0.588	17.5	1.059	0.783	80.0	0.970	0.661	63.5	0.789	0.565	6.0	0.996	0.700	69.5	3.077	1.212	97.0	0.837	0.604	34.0	0.842	0.624	41.0
favorita_transactions_1M	1.700	0.803	52.0	2.187	0.888	77.0	2.281	1.003	89.0	1.602	0.814	46.5	1.474	0.972	51.0	1.486	0.797	25.5	1.721	0.813	56.0	2.754	0.994	89.5	1.466	0.972	51.0	2.443	1.017	93.0
favorita_transactions_1W	0.965	0.566	21.0	1.233	0.682	67.5	1.068	0.628	55.5	0.958	0.591	28.5	1.326	0.768	84.5	1.053	0.616	51.0	1.159	0.688	62.0	2.918	1.093	95.0	1.352	0.767	85.5	1.162	0.718	69.0
heart_rate	1.016	*0.956*	6.0	1.425	1.041	61.5	1.062	1.052	33.0	1.043	0.969	14.5	1.070	1.049	33.5	1.120	1.089	58.0	1.296	1.022	57.0	7.197	1.367	97.5	1.112	1.103	60.0	1.128	1.144	71.5
kdd_cup_2022_10T	0.743	0.373	8.5	0.849	0.398	46.5	0.811	0.397	41.0	0.783	0.394	34.5	0.845	0.417	51.0	0.747	0.374	15.0	0.895	0.413	54.0	1.377	0.517	91.5	0.905	0.449	66.5	0.942	0.459	78.0
kdd_cup_2022_1D	0.852	1.215	18.5	0.883	1.253	40.5	0.909	1.288	65.5	0.853	1.286	42.0	0.843	1.353	47.0	0.914	1.274	63.0	1.039	1.241	56.0	5.532	1.515	98.0	0.870	1.458	70.0	0.847	1.372	50.0
kdd_cup_2022_30T	0.792	0.961	13.5	1.088	0.946	56.0	0.799	1.007	25.5	0.847	0.893	20.0	0.983	1.203	86.0	0.846	0.979	29.5	0.911	0.948	49.5	6.928	1.220	91.5	0.979	1.174	82.0	0.878	1.304	76.0
physionet_2012	0.777	0.982	21.5	0.766	0.968	16.0	0.796	1.068	59.0	0.728	0.968	8.5	0.832	1.058	70.5	0.829	1.050	66.5	0.795	0.968	29.5	1.138	1.059	81.5	0.801	1.109	71.5	0.796	1.066	58.5
rossmann_1D	0.858	0.736	42.0	1.353	1.125	81.0	0.707	0.629	16.0	1.325	1.111	75.5	0.800	0.705	36.5	0.711	0.640	18.0	1.399	1.073	79.5	2.523	1.315	95.5	*0.613*	0.584	4.0	0.766	0.671	29.5
rossmann_1W	1.177	1.268	62.5	0.838	1.245	18.5	1.125	1.171	31.0	0.869	1.315	50.5	0.856	1.307	45.5	1.188	1.234	52.0	0.889	1.267	35.0	2.289	1.411	96.5	0.850	1.395	52.0	0.964	1.239	30.0
us_gasoline	0.886	0.557	50.5	0.862	0.559	43.0	0.854	0.553	32.5	0.881	0.557	48.5	0.784	0.495	5.0	0.821	0.501	10.5	1.016	0.570	73.0	1.368	0.679	95.0	0.833	0.540	21.0	0.861	0.563	46.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, <u>underline</u>, and *italic*. The rank column reports the average rank over the two metrics.

Table 12: **Univariate forecasting results.** Part 5.

Dataset	lasso			ldlinear			lightgbm			lightts			linearregression			lstm			mamba			micn			moirai			moment		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	0.717	0.163	8.5	4.859	1.479	95.5	3.747	1.221	90.5	4.381	1.403	94.5	0.706	0.237	20.5	1.484	0.298	55.0	1.444	0.320	57.0	1.588	0.307	61.0	1.446	0.298	54.0	4.917	1.947	97.0
AustralianElectricity	0.858	0.523	58.0	0.724	0.466	32.0	0.680	0.445	17.0	0.774	0.509	47.0	0.812	0.518	55.0	0.757	0.483	39.0	*0.547*	*0.365*	3.0	0.690	0.465	25.5	1.219	0.656	75.5	0.756	0.487	40.0
Bitcoin	1.246	0.321	74.5	1.137	0.294	66.5	7.697	0.935	92.0	1.478	0.383	80.5	1.011	0.283	51.0	1.020	0.256	36.0	0.974	0.253	26.0	1.463	0.308	76.0	0.982	0.235	15.5	2.402	0.302	77.0
BrazilianCitiesTemperature	1.127	1.024	75.0	1.315	1.106	89.0	1.408	1.012	89.0	1.475	1.210	94.5	1.206	1.004	79.5	0.999	0.527	45.5	0.939	0.507	36.0	1.418	0.807	80.0	0.773	0.350	12.0	1.318	1.312	91.5
CIF_2016	1.427	0.636	74.0	2.524	1.385	94.0	1.613	0.765	85.5	2.659	1.424	95.0	1.823	0.592	77.0	0.983	0.544	37.5	1.049	0.540	44.0	1.190	0.584	62.0	0.976	0.506	29.5	2.675	1.793	96.5
Car_Parts	0.773	0.727	51.0	0.846	0.785	64.0	0.775	0.723	51.0	0.990	0.945	85.0	0.779	0.703	50.5	0.691	0.578	15.5	0.769	0.649	42.0	1.002	1.034	89.0	0.622	0.423	9.5	0.890	0.912	77.5
CoronaVirus_2019_confirmed	2.219	0.231	74.5	5.221	0.833	82.0	7.246	1.122	92.5	8.031	1.345	95.0	1.467	0.139	62.0	0.784	0.068	35.0	0.718	0.066	30.5	2.869	0.302	77.0	0.845	0.079	46.5	8.480	1.935	97.5
CoronaVirus_2019_deaths	3.006	0.125	74.5	3.946	0.492	90.0	2.986	0.308	82.5	6.301	1.104	95.5	2.551	0.084	64.5	0.698	0.054	24.0	0.903	0.059	35.0	2.552	0.270	79.0	0.792	0.076	40.0	8.418	1.959	97.0
CoronaVirus_2019_recovered	3.909	0.140	77.5	2.235	0.522	83.5	2.130	0.376	81.0	3.415	1.094	92.0	135.501	0.116	78.0	0.603	0.074	25.0	0.657	0.078	31.5	1.607	0.282	74.5	0.836	0.098	48.0	4.251	1.946	94.5
Elecdemand	0.906	0.651	47.0	0.844	0.629	36.0	0.719	0.552	5.5	0.927	0.689	54.5	0.804	0.613	26.0	0.860	0.662	43.5	0.749	0.577	9.0	1.027	0.683	58.5	0.925	0.653	49.0	0.715	0.576	6.5
Extended_Wikipedia_Web_Traffic_Daily_Dataset	1.120	0.622	87.0	1.046	0.542	81.0	1.118	0.511	80.0	1.142	0.580	87.5	1.016	0.529	76.0	0.831	0.350	9.0	0.860	0.351	18.5	1.090	0.576	83.0	0.798	0.333	4.5	1.628	1.089	95.0
Flood_Modeling_Dataset_1	0.892	1.125	80.5	0.886	1.079	72.5	0.955	0.994	77.0	0.939	1.038	79.0	0.885	1.079	72.5	0.762	0.793	22.0	0.774	0.786	23.0	0.855	0.995	61.0	0.729	0.669	9.0	0.849	0.990	57.0
Flood_Modeling_Dataset_2	0.841	0.860	73.5	0.823	0.826	63.0	1.042	0.839	83.0	0.879	0.864	77.0	0.823	0.826	64.0	0.750	0.675	19.0	0.766	0.696	32.5	0.811	0.801	58.0	0.781	0.636	22.5	0.815	0.797	58.0
Flood_Modeling_Dataset_3	0.973	1.080	75.0	0.990	1.037	72.5	1.077	1.040	81.5	1.057	1.045	80.5	1.346	1.064	90.5	0.818	0.783	23.0	0.895	0.798	40.5	0.947	0.974	63.5	0.793	0.588	8.0	1.041	1.210	85.5
Hospital	1.566	1.257	89.5	1.244	1.350	90.0	1.115	1.249	76.0	1.258	1.366	91.5	1.298	1.244	85.0	0.963	1.007	44.0	0.946	1.034	48.0	1.012	0.988	41.0	0.843	0.962	9.0	1.134	1.573	86.5
M1(Monthly)	1.067	0.733	66.0	2.167	1.361	95.0	1.357	0.936	90.0	2.268	1.357	96.5	1.059	0.719	59.5	0.969	0.710	44.5	0.868	0.666	22.5	1.139	0.703	59.0	0.898	0.666	25.0	2.179	1.689	96.5
M1(Quarterly)	1.210	0.784	69.0	2.211	1.469	97.0	1.828	1.316	92.0	2.172	1.444	96.0	2.095	0.899	85.0	0.997	0.657	38.0	1.015	0.694	47.0	0.937	0.628	15.5	0.979	0.650	29.0	2.130	1.813	96.0
M1(Yearly)	1.568	1.005	75.5	2.229	1.505	96.5	2.013	1.446	92.0	2.129	1.513	94.5	2.140	0.986	84.5	1.086	0.700	41.5	1.205	0.778	53.0	1.000	0.644	26.0	1.058	0.664	34.5	2.200	1.835	96.5
M3(Monthly)	1.222	0.809	68.5	1.904	1.321	93.0	1.358	0.906	83.0	2.011	1.331	95.5	1.564	0.849	82.5	1.027	0.730	46.5	1.016	0.715	42.0	1.072	0.686	45.0	0.865	0.639	17.0	1.956	1.667	95.5
M3(Quarterly)	2.251	0.641	75.5	3.269	1.423	96.0	2.549	1.152	88.0	3.189	1.421	93.5	5.460	0.852	92.5	1.072	0.434	35.5	1.207	0.480	51.5	1.285	0.482	57.0	0.974	0.419	25.5	3.213	1.832	95.0
M3(Yearly)	2.352	0.835	74.5	3.078	1.401	94.0	2.785	1.285	91.5	3.142	1.431	96.5	3.028	0.906	86.5	1.187	0.585	31.5	1.296	0.617	44.0	1.212	0.559	29.0	1.128	0.535	18.5	3.138	1.786	96.0
M4(Daily)	1.223	0.212	69.5	2.585	0.398	92.5	2.046	0.273	85.5	2.764	0.424	94.5	3.564	0.188	74.5	0.955	0.176	28.5	0.970	0.171	25.0	1.211	0.201	66.5	1.013	0.167	24.5	8.330	1.793	97.5
M4(Hourly)	0.911	0.376	26.0	1.057	0.454	49.0	0.814	0.331	11.0	2.334	0.949	78.5	0.786	0.332	9.5	1.466	0.596	66.5	0.920	0.371	25.0	1.174	0.437	52.0	1.570	0.539	67.0	3.224	1.535	93.5
M4(Monthly)	1.062	0.639	66.0	1.932	1.146	93.5	1.430	0.824	87.5	2.299	1.271	95.0	1.282	0.673	76.0	0.894	0.544	31.5	0.898	0.539	30.5	0.981	0.563	47.0	0.912	0.543	33.5	2.485	1.723	97.0
M4(Quarterly)	2.675	0.574	81.0	4.507	1.357	94.5	2.532	0.769	86.5	4.746	1.432	96.5	2.778	0.552	79.0	1.066	0.412	39.5	1.091	0.425	44.5	1.317	0.434	60.0	1.140	0.369	34.5	4.924	1.806	97.5
M4(Weekly)	0.788	0.434	27.0	1.550	0.588	87.0	1.647	0.458	73.0	1.457	0.541	85.5	0.750	0.412	12.0	1.031	0.455	57.5	0.794	0.434	28.5	1.088	0.400	33.5	0.757	0.433	23.0	9.667	1.728	97.5
M4(Yearly)	40.856	0.720	83.0	9.798	1.448	93.0	6.212	1.276	83.5	7.824	1.475	87.5	40.942	0.760	86.5	1.022	0.453	39.0	1.031	0.482	51.5	**0.607**	0.460	22.0	1.014	0.394	26.5	8.460	1.872	93.0
M5	0.880	1.336	61.0	0.917	1.225	73.5	0.918	1.198	61.5	1.006	1.222	78.5	0.910	1.225	72.0	0.880	1.217	44.0	0.902	1.152	39.0	0.904	1.190	53.0	0.815	1.056	9.0	0.904	1.481	81.5
NN5	0.890	0.608	59.5	0.863	0.589	52.5	0.807	0.555	24.5	0.928	0.640	68.0	0.833	0.567	41.0	0.795	0.541	10.5	0.814	0.556	27.0	0.822	0.562	33.0	0.867	0.594	54.0	1.648	1.447	96.0
NN5(Weekly)	0.896	1.294	77.0	0.924	1.339	85.0	0.788	1.347	61.5	0.917	1.337	83.5	1.462	1.365	95.5	0.791	1.155	33.0	0.819	1.182	57.5	0.865	1.152	48.0	0.779	1.147	23.0	0.816	1.523	75.5
Pedestrian_Counts	1.039	0.560	46.0	1.126	0.598	59.0	0.995	0.532	35.5	1.289	0.693	76.5	1.142	0.597	59.5	1.057	0.598	54.0	0.939	0.498	24.5	1.194	0.637	69.0	0.867	0.445	9.0	1.101	0.580	52.5
SaugeenRiverFlow	0.842	0.874	72.5	0.823	0.807	60.0	0.824	0.739	50.0	0.871	0.836	72.5	0.819	0.806	55.0	0.761	*0.621*	5.5	0.779	0.649	17.5	0.872	0.861	75.0	0.841	0.689	47.0	0.837	0.802	62.0
Solar(5Min.)	0.436	0.380	31.5	0.527	0.483	52.0	0.409	0.334	18.0	0.672	0.620	63.0	0.496	0.444	49.0	0.539	0.479	52.0	0.407	0.362	21.0	0.514	0.410	45.5	0.513	0.396	42.5	0.662	0.587	59.5
SolarPower	13.745	0.115	91.5	1.513	0.029	67.5	0.928	0.012	19.5	1.437	0.023	57.5	1.249	0.024	57.0	1.179	0.024	51.5	0.908	0.018	23.0	2.963	0.044	80.0	0.648	0.009	7.5	2.186	0.036	74.0
Sunspot	0.845	0.575	70.0	0.786	0.537	34.0	0.798	0.543	47.5	0.809	0.551	55.5	0.785	0.536	30.5	0.781	0.530	14.0	0.782	0.532	23.0	0.801	0.541	46.0	0.867	0.557	69.0	0.807	0.554	55.5
Taxi(Half_Hourly)	0.968	0.940	25.0	1.064	1.031	48.5	0.972	0.927	25.0	1.308	1.215	87.0	1.071	0.955	44.5	1.113	1.072	60.5	1.002	0.921	26.5	1.120	1.007	54.0	0.868	0.823	5.0	1.224	1.487	87.5
Taxi(Hourly)	0.576	1.090	20.0	0.685	1.200	50.5	0.581	0.992	9.0	0.718	1.250	61.5	0.592	1.051	19.0	0.813	1.221	73.5	0.712	1.123	41.0	0.837	1.100	58.0	0.802	1.170	59.0	0.680	1.446	63.5
Tourism(Monthly)	1.144	0.415	9.5	2.555	1.006	92.0	1.570	0.542	50.5	3.284	1.234	94.5	1.249	0.466	27.0	1.840	0.596	71.5	1.255	0.488	34.0	1.818	0.583	67.5	1.432	0.516	44.5	3.662	1.773	96.5
Tourism(Quarterly)	1.321	0.361	26.5	3.805	1.420	96.0	2.272	0.643	89.0	3.772	1.358	94.0	1.467	0.384	45.0	1.315	0.468	40.0	1.196	0.374	25.5	1.210	0.373	25.5	1.400	0.485	53.0	4.010	1.858	97.0
Tourism(Yearly)	3.638	1.122	80.5	2.926	1.577	92.0	2.534	1.458	86.5	2.809	1.574	90.5	4.726	1.189	86.5	1.211	0.446	37.5	1.414	0.551	51.5	1.226	0.425	36.5	1.057	0.388	19.5	2.782	1.915	91.0
USBirths	1.167	0.351	58.5	1.047	0.328	48.0	1.113	0.334	52.0	1.100	0.365	54.5	0.944	0.294	32.5	0.903	0.284	20.0	0.831	0.266	7.5	1.227	0.442	67.5	0.961	0.308	42.0	0.949	0.298	35.5
UberTLC_daily	1.984	1.139	75.0	2.463	1.306	93.0	1.760	0.962	66.5	2.610	1.358	95.0	2.055	1.130	77.5	0.943	0.392	21.0	0.923	0.444	28.5	0.876	0.388	15.0	1.452	0.776	57.0	2.632	1.767	97.0
UberTLC_hourly	0.748	0.509	33.5	0.862	0.626	55.0	0.686	0.526	31.0	1.199	0.852	76.5	0.821	0.600	51.0	0.879	0.623	56.0	*0.531*	0.384	3.5	0.607	0.477	14.5	0.741	0.484	28.0	0.808	0.575	46.0
VehicleTrips	1.347	0.887	39.5	1.933	1.283	92.5	1.527	1.051	68.5	1.899	1.263	90.5	1.511	0.872	54.5	1.373	1.082	56.5	1.203	0.913	34.0	1.417	0.885	48.0	<u>0.947</u>	0.741	4.5	1.776	1.529	91.5
Weather	0.853	0.870	81.5	0.829	0.810	74.0	0.829	0.787	65.5	0.873	0.851	84.5	0.814	0.797	61.0	0.771	0.710	19.0	0.764	0.697	10.5	0.851	0.841	78.5	0.801	0.705	30.0	1.173	1.381	96.0
WebTraffic	1.276	0.758	84.5	1.195	0.751	77.5	2.190	0.672	86.5	1.450	0.868	91.5	1.230	0.704	75.0	0.870	0.471	20.0	0.921	0.493	41.0	1.450	0.727	87.5	**0.819**	0.433	2.5	1.512	1.038	93.0
Wike2000	1.244	0.644	84.5	1.391	0.709	88.0	1.296	0.561	82.0	1.532	0.781	91.0	1.116	0.535	65.0	0.946	0.410	24.0	0.957	0.383	17.5	1.453	0.539	80.5	0.888	0.362	5.0	3.095	1.478	97.0
Wind_Farms	1.295	0.529	71.5	1.225	0.517	66.5	1.247	0.397	55.0	1.318	0.533	74.0	$> 10^6$	0.686	93.0	1.084	0.316	29.5	1.032	0.333	30.5	1.256	0.552	72.0	<u>0.912</u>	<u>0.239</u>	2.0	1.670	0.358	62.5
ausbeer	1.581	0.374	40.0	4.075	0.885	89.0	1.542	0.369	38.0	5.190	0.976	94.5	<u>0.605</u>	0.157	3.0	3.781	0.859	86.0	1.421	0.316	32.0	3.523	0.619	67.0	1.217	0.273	26.5	3.773	0.827	82.5
ecdc_ili	0.760	1.074	13.0	0.900	1.329	57.0	0.737	1.079	12.0	0.904	1.349	60.5	0.945	1.080	40.5	0.927	1.389	68.5	0.949	1.335	70.0	0.995	1.281	72.0	0.796	1.231	36.0	0.805	1.507	61.0
favorita_stores_1D	0.800	0.911	38.5	0.879	0.959	71.5	0.859	0.903	60.0	0.910	0.947	73.5	0.797	0.875	24.5	0.824	0.868	36.0	0.816	0.844	21.5	0.877	0.881	54.5	0.807	0.846	20.0	1.062	1.352	95.5
favorita_stores_1M	1.588	0.990	84.0	1.535	1.362	93.0	1.228	1.084	82.5	1.482	1.345	91.0	1.741	1.084	91.5	1.004	0.708	32.0	1.026	0.724	45.0	1.144	0.782	63.0	0.998	0.710	31.5	1.434	1.624	91.0
favorita_stores_1W	0.759	0.833	58.0	1.209	1.204	92.5	0.854	0.899	81.0	1.252	1.242	94.5	0.798	0.847	71.5	0.738	0.719	24.5	0.785	0.729	46.0	1.023	0.832	80.0	0.710	0.722	18.0	1.242	1.557	95.0
favorita_transactions_1D	0.899	0.662	58.5	1.013	0.763	73.5	0.931	0.626	55.5	1.062	0.779	80.5	0.836	0.621	37.5	0.870	0.633	48.0	0.804	0.574	13.0	1.015	0.637	62.0	0.818	0.590	22.5	1.563	1.278	95.0
favorita_transactions_1M	1.453	0.901	44.5	2.052	1.312	92.5	1.630	1.143	75.5	2.106	1.307	93.0	2.395	1.032	93.0	1.618	0.810	48.5	1.427	0.798	20.0	2.190	0.928	82.0	1.730	0.841	63.5	1.915	1.548	91.0
favorita_transactions_1W	1.161	0.727	70.0	2.112	1.134	92.5	1.470	0.804	89.0	2.210	1.152	93.5	1.268	0.711	74.5	0.955	0.564	16.5	1.037	0.610	47.5	1.698	0.724	83.5	0.917	0.561	8.5	2.400	1.479	96.0
heart_rate	1.028	1.097	36.5	1.099	1.078	47.5	1.090	1.063	40.0	1.131	1.129	70.5	1.049	1.061	30.5	1.095	1.102	53.0	1.123	1.094	61.0	1.170	1.119	73.5	1.058	1.048	29.0	1.302	1.505	94.5
kdd_cup_2022_10T	1.106	0.531	90.5	0.932	0.454	71.0	0.907	0.442	65.5	0.949	0.463	81.5	0.939	0.457	73.5	0.743	0.372	5.0	0.742	0.373	7.5	0.890	0.439	58.5	0.797	0.396	38.5	1.314	0.650	93.0
kdd_cup_2022_1D	0.839	1.382	50.0	0.920	1.346	73.0	0.887	1.290	59.0	0.909	1.335	68.5	0.926	1.349	78.0	0.855	1.249	25.5	0.918	1.280	66.5	0.897	1.265	53.0	0.831	*1.156*	7.5	0.830	1.522	54.0
kdd_cup_2022_30T	0.876	1.317	75.5	0.883	1.201	72.5	0.906	1.154	74.0	0.907	1.226	83.0	0.883	1.200	70.5	0.818	0.996	24.0	0.859	0.975	36.5	0.915	1.255	87.5	0.799	0.890	6.5	0.823	1.032	34.0
physionet_2012	0.763	1.074	46.5	0.982	1.272	92.5	0.794	1.054	52.5	0.981	1.274	92.5	0.788	1.062	52.5	0.781	0.996	29.5	0.797	1.003	43.0	1.085	1.072	85.0	**0.721**	0.966	6.0	0.901	1.483	91.0
rossmann_1D	0.818	0.694	37.5	1.321	1.126	76.5	0.691	0.651	18.0	1.467	1.214	91.5	0.748	0.671	25.5	1.246	1.047	67.0	0.689	0.620	12.5	0.698	0.604	11.0	1.025	0.897	50.0	1.396	1.359	91.0
rossmann_1W	0.910	1.261	34.5	1.073	1.286	56.0	0.867	1.271	34.5	1.078	1.284	55.0	1.148	1.229	42.5	1.147	1.273	58.0	1.234	1.277	72.0	1.312	1.214	53.0	1.065	1.380	65.5	0.849	1.570	54.5
us_gasoline	0.875	0.578	59.0	1.053	0.650	92.0	0.849	0.561	40.5	0.962	0.601	76.0	0.862	0.557	39.0	0.938	0.587	71.5	<u>0.764</u>	0.494	3.5	0.861	0.534	28.5	0.895	0.578	65.0	1.812	1.698	97.5

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, <u>underline</u>, and *italic*. The rank column reports the average rank over the two metrics.

Table 13: **Univariate forecasting results.** Part 6.

Dataset	naivedrift			naivemean			naivemovingaverage			naiveseasonal			nbeats			nlinear			nonstationary-transformer			patch_tst			patchmlp			prophet		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	1.944	0.371	75.5	3.863	1.240	91.5	1.357	0.281	44.0	1.000	0.229	26.5	0.917	0.220	23.5	2.126	0.418	77.5	1.383	0.281	46.0	0.956	0.203	20.5	0.983	0.188	17.5	0.665	0.138	4.5
AustralianElectricity	2.050	1.041	93.5	1.774	1.090	90.5	1.940	1.032	91.0	1.000	0.594	67.0	0.685	0.454	21.5	0.717	0.450	25.0	0.621	0.411	9.5	0.677	0.445	17.0	1.092	0.730	75.5	1.554	0.820	84.0
Bitcoin	0.947	0.270	32.5	2.019	0.375	80.5	0.968	0.249	21.5	1.000	0.253	30.0	1.251	0.329	76.0	0.938	0.269	29.5	1.168	0.286	65.5	0.949	0.248	18.5	1.185	0.283	64.5	1.041	0.281	55.0
BrazilianCitiesTemperature	0.753	0.272	7.0	1.365	1.071	90.0	0.964	0.450	32.0	1.000	0.319	29.0	1.210	0.852	73.0	1.105	0.628	59.5	0.977	0.602	51.5	0.917	0.506	33.5	1.012	0.520	47.5	1.460	0.895	82.5
CIF_2016	0.887	0.473	16.5	2.223	1.252	92.0	1.026	0.532	39.0	1.000	0.504	31.5	1.166	0.596	63.5	1.016	0.523	36.0	0.943	0.466	21.0	0.906	0.481	19.0	0.955	0.476	23.5	1.091	0.567	49.5
Car_Parts	1.658	0.425	54.0	0.889	0.942	79.0	1.017	0.844	80.0	1.000	0.230	44.0	0.831	0.716	57.5	1.525	0.960	93.0	0.687	0.569	14.5	0.792	0.707	53.5	0.734	0.613	32.0	1.119	0.970	90.5
CoronaVirus_2019_confirmed	0.640	0.050	16.0	5.089	0.750	80.0	1.156	0.099	55.5	1.000	0.085	53.5	1.964	0.188	72.5	0.699	0.068	30.0	0.729	0.064	30.5	0.627	0.057	19.0	0.662	0.057	21.0	0.885	0.075	46.5
CoronaVirus_2019_deaths	0.630	0.051	18.0	4.686	0.701	93.5	1.192	0.091	57.5	1.000	0.078	47.5	1.752	0.117	66.5	0.755	0.053	25.5	1.031	0.052	35.0	0.750	0.051	22.5	0.781	0.052	25.0	0.611	0.064	27.5
CoronaVirus_2019_recovered	0.711	0.061	24.0	2.930	0.746	88.5	1.126	0.118	59.5	1.000	0.102	52.5	1.458	0.149	68.0	0.616	0.068	21.5	0.684	0.072	30.0	0.541	0.072	17.5	0.520	0.063	10.0	1.012	0.105	54.5
Elecdemand	1.709	1.092	94.5	1.478	1.089	89.0	1.544	1.070	88.0	1.000	0.715	61.0	0.852	0.648	39.0	0.805	0.596	20.0	0.966	0.717	60.5	0.798	0.601	16.5	1.471	1.066	84.5	1.366	0.904	78.5
Extended_Wikipedia_Web_Traffic_Daily_Dataset	1.290	0.419	72.5	0.980	0.484	70.0	0.899	0.363	33.0	1.000	0.369	49.0	1.006	0.486	72.5	1.099	0.442	71.0	0.907	0.393	47.0	0.890	0.365	32.0	0.853	0.369	28.0	0.946	0.425	57.0
Flood_Modeling_Dataset_1	1.190	0.652	51.0	0.915	1.127	83.0	0.920	0.719	46.0	1.000	0.627	48.0	0.865	0.953	58.0	0.920	0.962	71.0	0.795	0.830	38.5	0.782	0.806	31.5	0.791	0.838	40.0	1.416	1.160	96.0
Flood_Modeling_Dataset_2	1.009	0.615	48.5	0.889	0.921	83.0	0.941	0.673	53.5	1.000	0.616	48.5	0.784	0.751	42.5	0.889	0.843	76.0	0.774	0.729	37.5	0.759	0.686	28.0	0.770	0.710	36.0	1.379	1.112	93.5
Flood_Modeling_Dataset_3	1.106	0.580	48.5	1.103	1.279	91.5	0.933	0.600	32.0	1.000	0.567	38.0	0.934	0.935	59.5	1.013	0.917	69.5	0.896	0.848	45.5	0.848	0.769	27.5	0.836	0.791	29.5	1.730	1.260	96.5
Hospital	1.124	1.008	59.5	1.012	1.277	73.5	0.860	0.966	13.5	1.000	0.976	37.5	1.078	1.233	67.0	1.151	1.067	73.5	0.971	0.990	39.5	0.963	1.063	56.5	1.069	1.036	60.0	0.894	*0.927*	16.5
M1(Monthly)	0.936	0.668	31.0	1.587	1.089	92.5	0.993	0.733	54.5	1.000	0.670	42.0	1.120	0.727	66.0	1.129	0.760	73.5	1.057	0.708	56.0	0.867	0.668	22.5	0.943	0.630	23.5	1.042	0.663	42.0
M1(Quarterly)	0.937	0.598	10.5	1.797	1.308	90.5	1.108	0.713	59.5	1.000	0.663	40.5	1.179	0.769	68.0	1.042	0.670	48.5	1.147	0.655	50.0	0.948	0.651	26.0	1.030	0.641	35.0	1.029	0.653	39.5
M1(Yearly)	0.869	**0.495**	2.5	1.943	1.420	90.5	1.456	0.976	71.5	1.000	0.565	17.0	1.492	1.145	74.0	1.099	0.653	36.0	1.018	0.655	30.5	1.145	0.756	49.0	1.096	0.713	43.0	1.086	0.670	37.0
M3(Monthly)	0.875	0.582	10.5	1.620	1.198	91.5	0.959	0.698	36.0	1.000	0.668	35.0	1.303	0.824	72.5	1.020	0.703	41.0	1.030	0.707	43.5	0.930	0.660	28.0	0.959	0.629	25.5	1.100	0.719	53.0
M3(Quarterly)	0.852	0.377	11.0	2.527	1.157	88.0	1.348	0.522	66.5	1.000	0.423	29.0	1.491	0.586	69.0	0.977	0.419	26.5	1.219	0.455	48.5	1.086	0.450	40.0	0.973	0.402	22.5	1.225	0.464	50.5
M3(Yearly)	**0.930**	**0.459**	1.0	2.585	1.218	87.0	1.647	0.695	66.0	1.000	0.475	4.5	2.459	0.889	79.0	1.144	0.568	26.0	1.619	0.617	55.0	1.196	0.575	31.0	1.268	0.556	29.0	1.404	0.643	57.5
M4(Daily)	<u>0.793</u>	0.161	3.0	2.729	0.413	93.5	1.075	0.185	48.0	1.000	0.188	46.0	1.106	0.198	58.0	0.937	0.182	32.5	0.951	0.181	32.5	0.954	0.176	25.5	0.960	0.200	48.5	1.501	0.263	79.0
M4(Hourly)	3.175	1.128	90.0	2.842	1.236	87.5	3.468	1.075	89.0	1.000	0.386	37.5	0.799	0.337	12.0	1.340	0.555	64.5	1.568	0.675	70.0	0.953	0.396	36.5	0.930	0.364	25.0	1.163	0.456	52.0
M4(Monthly)	0.885	0.549	31.5	1.787	1.071	92.0	0.928	0.554	40.0	1.000	0.587	56.5	1.124	0.685	72.5	0.981	0.611	58.0	0.836	0.493	12.5	0.844	0.517	20.0	0.790	0.487	5.0	1.047	0.599	60.5
M4(Quarterly)	0.783	0.289	5.0	3.362	1.022	92.5	1.330	0.472	65.5	1.000	0.347	23.0	2.247	0.564	75.5	1.075	0.403	38.0	1.281	0.423	54.0	0.930	0.387	24.5	1.040	0.409	37.5	1.522	0.544	68.5
M4(Weekly)	0.850	0.432	30.5	2.581	0.574	91.5	1.194	0.467	71.0	1.000	0.437	46.0	0.994	0.397	28.0	0.858	0.441	38.0	0.839	0.437	33.5	0.776	0.426	21.0	0.888	0.422	28.5	0.916	0.469	60.0
M4(Yearly)	<u>0.679</u>	**0.308**	1.5	6.736	1.265	83.5	1.061	0.661	64.5	1.000	0.341	9.5	63.735	0.731	85.0	1.015	0.425	34.0	1.012	0.431	29.5	1.027	0.473	46.0	1.004	0.386	17.5	2.451	0.475	58.5
M5	1.065	0.909	51.5	0.896	1.276	73.5	0.905	1.192	54.5	1.000	0.885	48.5	0.974	1.181	58.5	0.971	1.175	55.5	0.892	1.235	67.0	0.903	1.174	44.0	0.879	1.188	29.0	0.915	1.198	61.0
NN5	1.839	1.067	95.0	1.433	1.059	92.0	1.417	1.041	88.0	1.000	0.630	71.0	0.839	0.581	46.0	0.905	0.621	64.5	0.827	0.563	36.0	0.811	0.558	26.5	<u>0.779</u>	<u>0.532</u>	2.0	0.823	0.552	28.0
NN5(Weekly)	1.237	1.271	85.5	0.768	1.318	51.5	0.767	1.146	15.0	1.000	1.205	75.0	0.904	1.224	71.0	1.172	1.258	82.5	0.781	*1.114*	16.0	0.773	1.147	19.5	0.835	1.171	56.0	0.766	1.131	10.5
Pedestrian_Counts	2.892	1.139	93.0	2.066	1.307	91.5	2.253	1.141	90.0	1.000	0.477	26.0	1.022	0.565	44.5	0.979	0.513	31.5	1.123	0.551	50.0	0.892	0.482	15.5	1.410	0.875	81.5	1.193	0.602	65.0
SaugeenRiverFlow	0.949	0.701	61.5	1.181	1.214	95.0	0.993	0.759	71.0	1.000	0.735	68.0	0.865	0.821	70.5	0.940	0.794	71.5	0.747	0.657	15.0	0.804	0.668	33.0	0.794	0.649	21.5	1.451	1.025	94.5
Solar(5Min.)	1.231	0.772	83.5	1.122	1.089	91.0	1.110	0.864	85.0	1.000	0.666	75.0	0.461	0.405	37.5	0.479	0.411	42.0	0.667	0.581	59.0	0.417	0.368	25.5	1.104	1.051	89.0	1.654	0.990	92.5
SolarPower	0.528	*0.007*	3.5	6.165	0.098	88.5	1.189	0.023	49.5	1.000	0.020	33.0	2.282	0.030	72.5	1.015	0.022	42.5	0.833	0.012	14.0	0.939	0.020	32.5	1.310	0.036	68.0	0.636	0.008	5.0
Sunspot	1.032	0.625	92.0	0.876	0.583	78.0	0.893	0.602	84.5	1.000	0.641	92.0	0.817	0.557	61.0	0.782	0.529	16.0	0.781	0.536	23.5	0.781	0.529	12.0	<u>0.776</u>	**0.528**	1.5	0.885	0.571	75.0
Taxi(Half_Hourly)	1.592	1.182	87.0	1.237	1.376	88.5	1.287	1.185	82.0	1.000	0.862	18.5	0.975	0.907	20.5	1.178	1.060	65.5	1.092	1.055	57.0	0.968	0.907	18.0	1.005	0.912	25.5	1.404	1.304	93.0
Taxi(Hourly)	0.861	1.091	58.5	0.681	1.249	56.5	0.878	1.257	86.5	1.000	1.338	96.0	0.560	<u>0.973</u>	3.0	0.818	1.129	60.5	0.739	1.085	38.0	0.728	1.122	43.5	0.676	1.042	20.5	0.714	1.165	47.5
Tourism(Monthly)	2.224	0.704	89.5	2.232	0.782	91.0	1.642	0.576	59.5	1.000	0.407	5.5	1.335	0.503	37.5	2.105	0.711	89.5	1.671	0.554	57.0	1.135	0.452	16.5	1.246	0.467	27.5	1.279	0.477	32.5
Tourism(Quarterly)	1.458	0.498	61.5	2.952	0.973	93.0	1.581	0.520	71.5	1.000	0.297	4.0	1.383	0.422	40.5	1.697	0.600	80.5	1.452	0.465	52.0	1.364	0.418	36.5	1.222	0.387	28.5	1.352	0.452	39.0
Tourism(Yearly)	<u>0.839</u>	**0.314**	1.5	2.418	1.370	78.5	1.908	0.819	67.5	1.000	0.367	11.0	2.205	1.158	71.5	1.121	0.448	34.5	1.052	0.411	24.0	1.375	0.529	46.0	1.084	0.398	24.5	1.108	0.376	20.5
USBirths	3.099	0.845	91.5	2.970	0.880	90.5	2.968	0.921	93.0	1.000	0.306	43.0	0.897	0.306	31.5	1.137	0.375	59.0	0.844	0.275	11.0	0.889	0.289	22.5	<u>0.800</u>	<u>0.251</u>	2.0	1.124	0.348	54.5
UberTLC_daily	2.027	0.955	70.5	1.959	1.024	71.5	1.035	0.435	33.0	1.000	0.519	37.0	1.705	0.946	65.0	1.889	0.885	65.0	0.949	0.416	26.0	0.719	0.382	8.5	0.724	0.343	7.5	0.734	0.400	14.5
UberTLC_hourly	2.701	1.195	96.0	1.610	1.090	88.0	1.773	1.119	91.5	1.000	0.613	61.0	0.683	0.554	31.5	0.827	0.581	49.5	1.025	0.765	73.5	0.594	0.446	11.0	1.102	0.756	74.0	0.893	0.643	62.0
VehicleTrips	2.133	1.196	92.0	1.651	1.324	90.0	1.292	1.083	51.5	1.000	*0.713*	4.5	1.299	0.835	28.5	2.191	1.199	93.0	1.309	0.975	45.5	1.173	0.832	19.5	1.284	0.905	37.0	1.095	0.753	11.5
Weather	0.992	0.753	67.0	1.249	1.220	95.5	0.811	0.751	45.0	1.000	0.776	75.0	0.853	0.801	75.0	0.864	0.778	69.5	**0.735**	0.680	3.0	0.796	0.731	38.5	0.791	0.718	33.0	1.048	0.885	91.0
WebTraffic	1.116	0.506	54.5	1.927	0.808	94.0	0.900	0.478	34.0	1.000	0.486	47.0	1.263	0.637	74.5	1.058	0.575	58.5	0.931	0.508	47.0	0.893	0.480	31.5	0.890	0.477	29.0	1.676	0.622	80.0
Wike2000	1.128	0.521	62.5	1.507	0.613	87.5	1.026	0.467	44.0	1.000	0.396	29.0	1.112	0.520	59.0	1.195	0.545	75.0	0.985	0.425	33.5	0.903	0.374	9.0	0.950	0.397	21.5	1.220	0.523	71.5
Wind_Farms	1.134	0.276	28.0	2.754	0.716	89.5	0.975	0.256	10.5	1.000	0.261	14.5	1.130	0.456	55.5	1.165	0.379	47.5	0.974	0.307	17.0	1.044	0.329	32.5	1.369	0.423	64.5	1.943	0.437	74.0
ausbeer	3.672	0.795	78.5	4.739	1.010	95.0	3.531	0.784	75.5	1.000	0.211	16.5	1.638	0.365	41.0	2.567	0.633	59.0	3.087	0.723	64.5	1.208	0.231	22.5	2.485	0.603	57.0	1.513	0.345	35.0
ecdc_ili	1.002	1.129	55.0	0.817	1.439	63.5	1.007	1.369	80.5	1.000	1.241	68.0	0.739	1.058	8.0	1.074	1.179	64.0	0.906	1.221	48.0	0.902	1.243	51.0	0.941	1.325	65.0	**0.712**	**0.986**	1.0
favorita_stores_1D	1.275	1.028	93.5	0.954	1.032	88.5	0.939	0.977	84.0	1.000	0.869	61.0	0.883	0.893	61.0	0.981	0.949	80.5	0.861	0.896	59.5	0.798	0.838	9.5	0.801	0.843	14.0	0.873	0.867	46.0
favorita_stores_1M	0.964	0.679	6.0	1.259	1.157	85.5	0.984	0.715	28.5	1.000	0.739	41.5	1.179	0.993	74.5	1.106	0.791	62.0	1.042	0.710	42.0	1.028	0.731	47.5	1.072	0.752	56.5	1.074	0.744	54.5
favorita_stores_1W	1.094	0.832	79.5	0.888	0.919	84.0	0.757	0.715	29.0	1.000	0.825	76.5	0.885	0.915	83.0	1.132	0.844	83.5	0.765	0.754	49.0	0.755	0.723	33.5	0.787	0.738	49.5	0.851	0.803	67.5
favorita_transactions_1D	1.582	0.975	94.0	1.080	0.798	84.5	1.059	0.786	80.5	1.000	0.700	69.5	0.875	0.592	34.5	1.092	0.774	81.0	0.892	0.647	54.5	0.790	0.568	7.0	0.806	0.570	12.5	0.892	0.605	44.5
favorita_transactions_1M	1.792	0.774	45.0	1.651	1.154	77.0	1.322	0.774	10.0	**1.000**	0.692	3.5	1.528	0.864	48.5	1.860	0.877	71.5	1.590	0.809	42.5	1.410	0.781	13.0	1.726	0.816	58.0	<u>1.088</u>	0.677	3.5
favorita_transactions_1W	1.252	0.683	69.0	1.400	0.772	87.0	1.001	0.615	44.5	1.000	0.557	24.5	1.326	0.735	82.0	1.275	0.702	72.5	0.945	0.564	14.0	0.966	0.579	26.5	1.062	0.609	49.0	1.466	0.647	72.5
heart_rate	1.045	0.964	15.0	1.344	1.378	95.0	1.120	1.009	38.5	*1.000*	0.957	3.5	1.112	1.108	61.5	1.042	0.994	17.5	1.067	1.016	27.5	1.196	0.964	43.5	1.160	1.151	77.5	1.463	1.140	86.5
kdd_cup_2022_10T	0.790	0.378	29.5	1.814	0.814	95.0	0.883	0.441	59.0	1.000	0.482	85.5	0.878	0.433	55.0	0.758	0.379	23.0	0.777	0.393	33.5	0.756	0.379	23.0	0.744	0.372	7.0	1.312	0.579	92.0
kdd_cup_2022_1D	1.490	1.348	84.0	0.826	1.453	50.5	0.908	1.249	44.0	1.000	**1.081**	46.0	0.923	1.358	80.0	1.351	1.349	84.0	0.860	1.209	22.5	0.899	1.262	52.5	0.921	1.250	51.0	0.828	1.350	40.0
kdd_cup_2022_30T	0.929	0.897	48.5	0.880	1.122	64.5	0.876	0.958	39.0	1.000	0.975	61.0	0.906	1.159	75.0	0.838	0.991	29.0	0.878	1.084	60.5	0.853	1.022	43.0	0.831	0.992	28.5	1.026	1.108	79.5
physionet_2012	0.853	0.988	52.0	0.820	1.139	79.5	0.782	1.020	37.5	1.000	1.125	91.5	0.805	1.048	60.5	0.947	1.033	69.0	0.800	1.015	48.5	0.802	0.985	39.5	0.833	1.016	60.0	0.823	0.996	49.5
rossmann_1D	1.506	1.114	85.5	1.399	1.190	89.5	1.315	1.097	71.0	1.000	0.805	46.5	0.680	0.622	12.0	1.558	1.181	91.0	1.096	0.951	56.0	0.712	0.644	19.5	0.667	0.593	7.5	0.784	0.693	34.5
rossmann_1W	1.094	1.220	31.5	0.915	1.487	64.0	1.016	1.291	55.0	1.000	*1.167*	21.5	0.912	1.277	43.0	1.346	1.315	86.0	1.362	1.274	73.5	1.161	1.237	50.5	1.255	1.233	55.0	1.062	1.368	63.5
us_gasoline	0.829	0.536	20.0	1.005	0.598	80.0	1.021	0.632	86.0	1.000	0.601	80.5	0.885	0.576	62.0	0.914	0.585	69.0	0.785	<u>0.489</u>	4.0	0.859	0.549	32.5	0.855	0.544	29.5	**0.731**	**0.446**	1.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, <u>underline</u>, and *italic*. The rank column reports the average rank over the two metrics.

Table 14: **Univariate forecasting results.** Part 7.

Dataset	pyraformer			randomforest			reformer			resnet			ridge			rmlp			rnn			sarima			scinet			segrnn		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	2.246	0.763	82.0	2.537	0.661	82.5	2.051	0.485	77.5	1.235	0.243	35.5	0.690	0.218	15.5	1.212	0.251	35.0	1.305	0.267	40.5	0.417	0.091	2.0	2.734	0.804	85.5	1.758	0.344	70.0
AustralianElectricity	0.616	0.416	10.0	0.709	0.462	27.5	0.815	0.539	58.5	0.736	0.488	39.0	0.802	0.512	53.0	0.632	0.413	10.5	0.977	0.632	69.5	0.949	0.554	64.0	0.642	0.418	12.5	0.789	0.525	53.0
Bitcoin	5.006	0.699	84.0	7.673	0.895	90.0	7.624	0.879	86.5	1.124	0.281	57.5	1.009	0.282	49.0	0.973	0.248	21.5	1.099	0.279	55.0	$> 10^6$	0.236	52.0	7.639	0.834	86.5	0.905	0.246	8.5
BrazilianCitiesTemperature	0.988	0.742	54.5	1.179	0.869	70.5	1.258	1.084	86.5	0.963	0.527	43.0	1.205	1.005	79.5	0.895	0.484	27.0	0.924	0.489	32.0	34.562	0.690	80.5	1.132	0.995	72.0	0.738	0.320	9.5
CIF_2016	0.927	0.453	17.0	1.473	0.685	78.0	1.362	0.638	74.0	1.040	0.561	46.5	1.595	0.568	67.5	0.924	0.508	25.0	0.987	0.517	32.0	$> 10^6$	*0.406*	51.0	1.623	0.797	87.0	0.834	0.446	8.5
Car_Parts	0.720	0.606	28.0	0.829	0.712	56.0	0.807	0.794	62.5	0.735	0.613	33.0	0.777	0.702	48.0	0.703	0.587	19.5	0.704	0.592	21.5	0.932	0.825	74.0	0.791	0.754	58.0	1.375	0.844	83.0
CoronaVirus_2019_confirmed	2.950	0.608	78.5	7.011	1.041	89.0	6.721	0.992	85.0	0.646	0.064	23.5	1.780	0.161	70.0	0.638	0.055	17.0	0.774	0.072	37.5	0.702	0.039	17.0	6.482	0.833	83.0	0.682	0.050	20.0
CoronaVirus_2019_deaths	2.205	0.335	78.0	2.552	0.258	78.5	4.262	0.424	90.5	0.825	0.061	35.0	2.954	0.097	70.5	0.729	0.053	24.0	1.080	0.089	54.5	$> 10^6$	0.042	53.0	3.809	0.331	87.0	0.649	0.045	14.5
CoronaVirus_2019_recovered	1.592	0.397	77.5	1.938	0.320	78.5	2.964	0.526	87.0	0.545	0.061	10.0	4.546	0.118	77.0	0.556	0.068	15.0	0.898	0.123	55.0	1.093	0.067	36.0	2.852	0.447	85.0	0.734	0.058	24.0
Elecdemand	0.868	0.655	43.0	0.701	0.539	2.0	1.485	0.988	84.0	0.938	0.696	56.0	0.807	0.614	29.0	0.822	0.609	30.5	0.912	0.717	57.0	0.937	0.728	60.5	0.768	0.569	8.5	1.271	0.955	77.5
Extended_Wikipedia_Web_Traffic_Daily_Dataset	0.937	0.457	61.5	1.164	0.481	81.5	0.959	0.476	65.5	0.931	0.389	49.5	1.015	0.529	75.0	0.838	0.345	10.5	0.839	0.354	15.0	1.471	0.441	77.5	0.933	0.449	60.0	1.156	0.393	68.5
Flood_Modeling_Dataset_1	0.839	0.962	54.5	0.849	0.953	54.0	0.850	1.015	61.5	0.839	0.838	44.5	0.885	1.079	72.5	0.776	0.793	26.5	0.754	0.774	15.5	0.978	1.013	81.5	0.830	0.891	46.5	0.881	0.959	60.5
Flood_Modeling_Dataset_2	0.801	0.783	50.0	0.806	0.783	52.0	0.807	0.797	55.0	0.787	0.731	42.5	0.823	0.826	65.0	0.741	0.660	15.0	0.747	0.676	18.5	1.890	0.874	89.0	0.781	0.759	44.0	0.803	0.797	53.5
Flood_Modeling_Dataset_3	0.906	0.904	52.0	0.893	0.881	47.5	0.931	0.993	60.5	0.842	0.750	22.0	1.154	1.055	88.0	0.838	0.766	24.5	0.820	0.765	19.0	1.130	1.115	90.5	0.892	0.890	47.5	0.909	0.917	53.5
Hospital	0.933	1.006	38.0	1.113	1.244	73.0	0.967	1.039	54.0	0.873	0.989	22.5	1.258	1.240	79.0	0.864	0.986	19.0	0.993	0.984	38.0	1.092	0.989	45.5	1.143	1.256	80.0	1.012	0.992	45.5
M1(Monthly)	0.930	0.657	25.0	1.212	0.835	80.5	1.170	0.805	77.5	0.981	0.728	52.0	1.066	0.726	62.0	0.877	0.649	19.5	1.005	0.669	42.0	**0.680**	**0.508**	1.0	1.292	0.880	86.5	0.867	0.633	15.0
M1(Quarterly)	0.987	0.629	24.5	1.510	1.015	87.0	1.379	0.903	76.5	0.939	0.640	20.5	1.475	0.827	77.5	0.991	0.664	39.5	1.152	0.661	54.0	0.902	0.595	5.5	1.427	0.947	78.5	0.906	0.601	8.5
M1(Yearly)	0.903	0.607	10.5	1.778	1.279	83.0	1.805	1.212	82.5	1.037	0.682	36.0	1.622	0.956	73.5	1.181	0.749	49.5	1.012	0.663	30.5	0.824	0.670	18.0	1.783	1.227	82.0	0.918	0.521	6.5
M3(Monthly)	0.863	0.589	10.5	1.291	0.871	76.0	1.145	0.778	63.5	1.021	0.734	48.0	1.451	0.826	79.0	0.935	0.645	26.5	1.150	0.727	57.0	0.848	*0.560*	6.5	1.337	0.932	83.0	0.823	0.583	6.5
M3(Quarterly)	0.957	0.390	19.0	1.890	0.763	81.0	1.758	0.699	72.5	1.099	0.441	40.0	3.161	0.757	86.0	1.121	0.447	42.5	1.075	0.427	34.0	*0.770*	0.366	5.0	1.777	0.755	77.0	0.822	0.365	6.5
M3(Yearly)	1.200	0.528	23.0	2.233	0.961	82.0	2.093	0.873	71.0	1.136	0.576	27.5	2.784	0.878	82.0	1.253	0.608	38.0	1.351	0.574	41.0	1.603	0.488	35.5	2.161	0.891	75.5	*0.945*	*0.468*	3.0
M4(Daily)	0.948	0.184	35.0	1.919	0.265	83.0	2.488	0.324	91.0	1.127	0.188	55.5	0.974	0.174	28.5	0.916	0.178	26.5	1.130	0.188	55.5	0.841	0.173	13.0	1.831	0.240	79.5	*0.815*	0.172	10.5
M4(Hourly)	0.929	0.386	31.0	0.768	0.317	5.5	2.176	0.884	77.0	0.958	0.388	35.5	0.790	0.333	10.5	0.879	0.355	20.0	1.186	0.502	57.5	**0.714**	**0.294**	1.0	0.894	0.379	26.5	2.495	1.003	80.0
M4(Monthly)	0.754	0.471	2.0	1.319	0.752	80.5	1.232	0.690	76.0	0.840	0.526	21.0	1.164	0.651	71.0	0.832	0.499	13.5	0.999	0.577	54.0	0.815	0.513	15.0	1.309	0.763	80.0	0.837	0.541	22.5
M4(Quarterly)	0.962	0.374	25.0	1.934	0.655	78.0	2.119	0.579	77.5	1.031	0.417	37.0	2.805	0.566	81.0	1.037	0.401	34.0	1.071	0.374	32.0	**0.733**	0.310	6.0	2.263	0.637	80.5	0.804	0.300	7.0
M4(Weekly)	0.860	0.431	31.0	1.630	0.458	73.0	5.164	0.593	95.0	1.042	0.470	68.0	0.721	0.409	6.0	0.819	0.441	34.5	1.174	0.487	74.0	0.765	0.418	16.0	4.067	0.488	87.0	**0.696**	0.404	3.5
M4(Yearly)	1.001	0.369	14.0	3.933	0.985	79.5	8.361	0.839	81.0	1.013	0.426	30.5	40.588	0.742	84.0	1.028	0.477	48.5	1.010	0.419	26.0	0.952	0.418	16.0	8.365	0.876	82.0	0.992	*0.310*	5.0
M5	0.890	1.281	71.0	0.891	1.223	61.5	0.888	1.303	68.0	0.924	1.234	78.0	0.910	1.225	72.0	0.881	1.183	31.5	0.879	1.210	40.0	0.880	1.206	40.0	0.888	1.257	65.0	0.881	1.211	43.5
NN5	0.808	0.550	22.5	0.793	0.541	8.5	0.893	0.612	61.0	0.845	0.570	44.0	0.833	0.567	40.0	0.795	0.546	13.0	0.824	0.559	32.5	0.795	0.535	9.0	0.785	0.535	5.0	0.853	0.592	52.0
NN5(Weekly)	0.777	1.127	16.0	0.863	1.282	73.0	0.749	1.141	5.5	0.754	1.152	17.0	1.343	1.350	94.0	0.761	1.152	19.5	0.790	1.146	24.5	0.811	1.154	40.0	0.784	1.255	51.0	1.031	1.224	77.0
Pedestrian_Counts	1.115	0.610	59.5	1.038	0.553	44.5	1.593	0.824	82.0	0.959	0.506	29.0	1.141	0.596	58.5	0.848	0.450	9.0	1.032	0.544	41.5	0.888	0.471	12.5	0.996	0.544	38.5	1.061	0.580	50.0
SaugeenRiverFlow	0.715	0.605	2.0	0.826	0.761	55.0	0.735	0.651	11.0	0.793	0.658	25.5	0.819	0.806	56.0	0.797	0.661	27.5	0.763	0.629	6.5	22.154	0.949	95.5	0.820	0.759	51.5	0.805	0.747	46.0
Solar(5Min.)	0.726	0.631	66.5	0.375	0.328	10.5	0.986	0.763	76.5	0.491	0.430	46.0	0.493	0.442	48.0	0.400	0.357	15.5	0.683	0.603	62.5	0.953	0.540	66.5	0.401	0.357	16.5	0.477	0.432	44.0
SolarPower	2.151	0.026	68.0	0.834	0.010	12.0	32.020	0.272	93.5	1.366	0.023	57.5	1.278	0.025	59.0	0.958	0.022	38.0	1.042	0.020	37.0	$> 10^6$	0.025	80.0	1.252	0.019	42.5	1.008	0.021	39.5
Sunspot	0.786	0.535	29.5	0.790	0.541	41.0	0.895	0.595	83.5	0.801	0.539	44.0	0.785	0.537	31.5	**0.776**	0.528	1.5	0.781	0.530	16.0	0.797	0.538	41.5	0.787	0.539	38.0	0.792	0.538	39.5
Taxi(Half_Hourly)	1.008	0.941	33.5	0.952	0.918	16.0	1.194	1.241	80.5	0.999	0.955	34.0	1.055	0.952	39.0	0.940	0.888	10.0	1.108	1.073	60.0	0.963	0.866	11.5	1.006	0.969	38.0	1.229	1.138	74.0
Taxi(Hourly)	0.697	1.069	30.0	0.553	0.981	4.0	0.611	1.092	27.0	0.725	1.109	40.0	0.592	1.051	18.0	0.732	1.133	46.0	0.814	1.210	73.0	$> 10^6$	1.000	53.5	0.575	1.004	8.5	0.750	1.118	45.0
Tourism(Monthly)	1.380	0.478	37.0	1.370	0.487	37.5	1.880	0.642	80.5	1.496	0.507	45.5	1.231	0.462	25.0	1.189	0.433	18.0	1.596	0.543	51.5	0.965	0.400	4.5	1.529	0.540	49.5	1.891	0.640	80.5
Tourism(Quarterly)	1.129	0.356	19.5	1.734	0.495	70.5	2.082	0.571	85.5	1.309	0.463	36.5	1.426	0.376	42.0	1.065	0.320	10.5	1.273	0.431	32.0	1.057	0.308	7.0	2.131	0.551	84.0	1.333	0.458	38.5
Tourism(Yearly)	0.995	0.374	11.5	2.467	1.375	82.0	2.461	1.297	78.5	1.193	0.451	38.0	4.198	1.150	82.5	1.413	0.526	47.5	1.146	0.421	34.0	1.642	0.433	51.0	2.298	1.232	75.0	0.994	0.369	10.0
USBirths	1.139	0.396	60.5	1.137	0.340	55.5	1.515	0.520	74.5	0.873	0.295	23.5	0.944	0.294	31.5	*0.806*	0.256	3.5	0.959	0.308	42.0	0.920	0.285	23.5	1.182	0.405	64.0	1.043	0.369	53.0
UberTLC_daily	0.928	0.423	25.5	1.941	1.076	72.0	1.702	0.906	63.0	1.076	0.458	40.5	2.064	1.139	79.5	**0.601**	**0.292**	1.0	0.797	0.355	9.5	0.860	0.423	20.5	2.246	1.310	92.0	2.047	0.984	74.0
UberTLC_hourly	0.678	0.518	25.5	0.685	0.525	29.5	0.954	0.679	66.0	0.637	0.478	16.0	0.820	0.600	49.5	0.603	0.453	12.5	0.904	0.642	62.0	0.776	0.545	40.0	0.682	0.512	25.5	1.071	0.778	75.0
VehicleTrips	1.258	0.872	30.0	1.378	0.894	44.0	1.498	1.056	66.5	1.384	1.083	60.5	1.448	0.859	46.0	1.143	0.830	17.5	1.566	1.076	73.0	0.986	**0.689**	2.5	1.579	1.057	72.0	1.810	1.166	86.5
Weather	0.752	0.710	13.0	0.803	0.759	45.5	0.762	0.735	24.5	0.788	0.712	27.0	0.814	0.797	61.0	0.783	0.713	26.0	0.782	0.714	26.5	0.879	0.780	73.0	0.788	0.762	42.5	0.823	0.764	54.5
WebTraffic	1.082	0.633	65.0	1.345	0.623	75.5	1.344	0.690	80.0	1.016	0.523	54.5	1.230	0.704	75.0	0.871	0.464	18.0	0.861	0.461	11.5	1.522	0.634	82.0	1.245	0.619	68.5	0.957	0.507	48.5
Wike2000	0.925	0.389	14.0	1.200	0.508	67.5	1.644	0.619	90.5	1.131	0.450	52.5	1.116	0.535	64.0	0.898	0.364	6.0	0.945	0.395	18.5	1.148	0.469	58.0	1.372	0.497	69.5	0.940	0.418	24.0
Wind_Farms	1.217	0.362	49.0	1.324	0.407	61.0	1.653	0.548	81.0	1.042	0.324	31.0	1.174	0.478	61.0	0.952	0.283	11.0	1.086	0.318	30.5	$> 10^6$	0.475	86.0	1.302	0.450	65.5	1.014	0.369	34.5
ausbeer	1.322	0.312	29.0	1.635	0.389	43.0	3.087	0.797	69.5	3.557	0.791	76.5	*0.632*	*0.155*	3.0	1.193	0.269	24.5	1.698	0.354	42.0	0.767	0.172	8.5	3.267	0.741	67.5	2.586	0.517	57.0
ecdc_ili	0.805	1.127	24.5	0.733	1.076	11.0	0.742	1.138	21.5	0.990	1.436	85.5	0.908	1.066	30.0	0.886	1.281	54.0	0.888	1.332	56.5	2.574	1.243	77.5	0.762	1.244	37.0	0.946	1.117	45.0
favorita_stores_1D	0.831	0.861	33.0	0.819	0.875	36.5	0.848	0.916	60.5	1.036	0.866	59.5	0.797	0.875	23.5	0.794	*0.823*	4.0	0.814	0.857	23.5	0.846	0.866	39.0	0.815	0.890	39.5	0.838	0.885	48.0
favorita_stores_1M	0.987	0.718	32.5	1.186	1.007	76.5	1.103	0.866	65.0	0.983	0.709	23.5	1.586	1.045	88.5	0.981	0.693	14.5	0.982	0.703	17.5	0.982	0.750	36.0	1.177	0.983	72.0	**0.941**	0.673	2.5
favorita_stores_1W	0.756	0.737	40.5	0.792	0.832	65.0	0.771	0.807	56.0	0.753	0.725	33.0	0.795	0.845	69.5	0.704	0.698	6.5	0.991	0.773	70.0	0.740	0.762	39.0	0.800	0.831	66.5	0.930	0.766	67.5
favorita_transactions_1D	0.803	0.592	18.0	0.834	0.585	23.0	0.942	0.673	64.5	0.806	0.584	16.5	0.836	0.621	36.5	0.791	0.574	11.0	0.810	0.589	19.5	0.835	0.583	22.0	0.852	0.607	38.0	0.863	0.624	44.0
favorita_transactions_1M	1.520	0.786	27.0	1.518	0.962	54.0	1.390	0.895	36.0	1.470	0.800	27.5	2.088	1.009	88.0	1.460	0.783	20.0	1.853	0.802	57.0	*1.241*	*0.585*	3.0	1.461	1.008	54.0	1.782	0.834	64.5
favorita_transactions_1W	0.992	0.566	27.0	1.320	0.693	73.5	1.198	0.692	67.5	0.997	0.601	40.5	1.251	0.709	73.0	0.937	0.557	10.5	0.933	0.553	5.5	0.978	0.596	35.5	1.176	0.723	70.5	1.106	0.646	57.0
heart_rate	1.033	1.013	18.5	1.062	1.064	35.0	1.114	1.179	72.0	1.205	1.195	86.5	1.049	1.062	30.5	1.149	1.003	43.5	1.019	0.992	12.5	1.000	0.947	3.5	1.078	1.145	58.0	**0.987**	0.959	3.5
kdd_cup_2022_10T	0.746	0.374	12.0	0.903	0.439	62.0	0.922	0.454	69.5	0.768	0.385	28.5	0.939	0.457	74.5	0.744	0.373	9.0	0.743	0.372	5.0	$> 10^3$	0.429	76.0	0.894	0.437	58.0	0.801	0.398	40.5
kdd_cup_2022_1D	0.891	1.337	63.0	0.926	1.448	88.5	0.816	1.261	21.0	0.866	1.256	34.5	0.921	1.351	76.5	0.869	1.246	31.5	0.854	1.218	21.0	0.844	1.382	50.5	0.833	1.391	50.5	0.896	1.352	68.5
kdd_cup_2022_30T	0.876	1.242	73.5	0.880	1.189	68.5	0.917	1.307	90.0	0.870	1.051	55.0	0.883	1.200	71.5	0.856	0.984	37.0	0.809	0.958	14.0	*0.779*	0.961	14.0	0.856	1.116	54.5	0.864	1.130	58.5
physionet_2012	0.793	0.979	29.0	0.783	1.041	44.5	0.782	1.012	35.0	0.796	1.021	48.0	0.785	1.062	50.0	0.783	0.994	29.5	0.969	1.093	86.0	$> 10^6$	1.136	95.0	0.796	1.090	64.5	0.803	0.960	33.5
rossmann_1D	0.740	0.646	22.0	0.605	*0.579*	2.5	1.101	0.967	58.5	0.749	0.672	28.0	0.748	0.671	26.5	0.664	0.607	8.5	0.928	0.795	45.0	0.736	0.675	27.0	0.805	0.733	39.5	1.055	0.892	51.0
rossmann_1W	1.137	1.252	48.5	*0.794*	1.194	6.5	0.856	1.266	29.0	1.147	1.269	55.0	1.118	1.228	34.0	1.124	1.270	51.0	1.124	1.234	38.0	0.860	1.281	39.5	0.858	1.377	51.5	0.879	1.277	40.5
us_gasoline	0.798	*0.490*	5.5	0.865	0.573	52.0	0.795	0.522	11.5	0.984	0.590	77.0	0.863	0.557	40.5	0.875	0.550	41.0	0.874	0.558	47.0	0.805	0.509	9.5	0.848	0.559	38.0	0.841	0.542	23.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 15: **Univariate forecasting results.** Part 8.

Dataset	seq2seq			sgdregressor			sim_mtm			simpleexpsmoothing			simpletm			sparse_tsf			sundial			svr			tabpfn			tcn		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	1.276	0.262	38.5	0.747	0.207	16.0	0.720	0.174	10.0	1.767	0.347	71.0	0.879	0.193	14.0	1.399	0.286	48.5	1.434	0.330	58.5	4.107	1.387	93.0	1.063	0.234	30.0	1.455	0.305	56.0
AustralianElectricity	1.179	0.747	77.5	0.744	0.477	35.0	0.691	0.452	22.5	1.893	1.023	88.0	0.617	0.404	8.5	1.096	0.665	74.0	0.777	0.478	40.0	0.666	0.430	14.0	0.983	0.602	68.0	0.960	0.612	67.5
Bitcoin	1.196	0.282	65.0	1.192	0.323	73.5	0.998	0.265	38.5	0.911	0.257	20.0	0.985	0.255	28.0	1.183	0.293	68.5	1.020	0.241	25.5	7.898	1.199	96.0	0.922	0.237	8.0	1.208	0.284	67.5
BrazilianCitiesTemperature	1.003	0.513	44.5	1.214	1.004	80.0	0.945	0.546	44.5	*0.694*	0.259	4.0	0.907	0.484	27.5	1.205	0.561	64.0	0.841	0.453	19.5	0.857	0.493	27.0	0.856	0.287	13.5	1.130	0.554	57.5
CIF_2016	1.030	0.555	44.5	1.463	0.585	66.5	1.041	0.458	30.0	0.892	0.466	16.0	0.871	0.459	13.5	1.218	0.618	71.0	1.041	0.531	40.5	2.005	0.999	90.5	0.852	0.425	7.0	1.234	0.612	70.5
Car_Parts	0.705	0.601	24.5	0.785	0.714	53.5	0.737	0.617	35.0	1.084	0.807	78.5	0.720	0.599	27.0	1.065	0.906	84.5	0.778	0.628	43.5	0.591	0.386	8.5	0.669	0.317	9.0	0.963	0.800	73.5
CoronaVirus_2019_confirmed	0.842	0.072	42.5	2.611	0.215	75.0	0.584	0.052	15.0	0.805	0.068	36.5	0.689	0.061	25.5	1.702	0.160	69.0	0.875	0.077	47.5	8.050	1.583	96.0	0.507	0.046	10.0	1.592	0.154	66.0
CoronaVirus_2019_deaths	1.045	0.071	46.0	3.574	0.126	76.0	0.993	0.050	30.0	0.784	0.061	32.5	0.758	0.055	28.0	1.750	0.150	70.5	0.917	0.074	43.0	4.167	0.559	91.0	0.431	0.044	9.0	1.557	0.143	68.5
CoronaVirus_2019_recovered	0.658	0.084	33.5	4.117	0.147	78.5	0.599	0.068	17.5	0.810	0.076	39.0	0.649	0.073	26.5	1.457	0.190	71.0	0.919	0.095	49.5	2.821	0.639	87.0	0.608	0.065	17.5	1.328	0.185	69.0
Elecdemand	1.122	0.830	72.0	0.913	0.681	52.5	0.805	0.606	23.5	1.585	1.060	87.5	0.791	0.595	14.5	1.034	0.781	67.0	0.841	0.607	30.5	0.769	0.578	11.0	0.988	0.724	63.0	0.902	0.697	53.0
Extended_Wikipedia_Web_Traffic_Daily_Dataset	0.851	0.369	27.5	$> 10^6$	0.735	96.5	0.881	0.367	31.0	1.012	0.378	53.5	0.851	0.352	17.5	1.045	0.451	71.5	0.884	0.393	41.5	0.872	0.410	41.5	0.811	0.336	6.5	0.974	0.407	57.0
Flood_Modeling_Dataset_1	0.780	0.824	32.5	0.908	1.072	75.5	0.825	0.862	44.0	0.915	0.717	44.0	0.766	0.779	20.5	0.974	0.992	78.0	0.896	0.909	63.0	0.689	0.668	7.5	0.716	0.565	6.0	0.811	0.873	45.0
Flood_Modeling_Dataset_2	0.756	0.694	29.5	$> 10^4$	1.584	98.5	0.777	0.709	36.5	0.897	0.600	42.5	0.754	0.685	23.5	0.950	0.879	85.0	0.882	0.805	69.0	0.703	0.617	8.0	0.756	0.564	13.0	0.791	0.737	45.0
Flood_Modeling_Dataset_3	0.832	0.805	30.0	1.082	1.035	81.0	1.187	0.852	72.0	0.952	0.594	34.5	0.875	0.771	33.0	1.073	0.949	74.0	0.938	0.805	49.5	*0.767*	0.655	10.5	0.820	0.538	8.5	0.914	0.863	49.5
Hospital	0.878	0.963	19.0	1.235	1.274	85.0	1.103	1.031	61.0	0.871	0.958	13.5	0.944	1.015	44.0	0.906	1.013	40.0	0.856	0.984	16.0	1.115	1.246	75.0	0.876	0.941	14.5	0.936	1.053	49.0
M1(Monthly)	0.960	0.706	40.5	1.138	0.754	73.5	0.849	0.625	11.5	0.868	0.653	18.0	0.814	0.610	6.5	1.038	0.749	65.0	0.864	0.664	18.5	1.478	0.946	91.5	0.844	0.633	12.5	1.075	0.763	71.5
M1(Quarterly)	0.996	0.646	31.5	1.378	0.824	73.5	1.006	0.647	34.0	0.947	0.618	15.0	0.929	0.636	16.5	1.126	0.754	64.0	0.937	0.629	16.0	1.816	1.269	90.5	0.930	0.621	13.0	1.099	0.726	61.0
M1(Yearly)	1.166	0.739	48.0	1.523	0.968	72.5	0.908	0.606	10.5	0.999	0.568	16.5	1.037	0.679	36.0	1.298	0.853	67.0	1.135	0.724	45.5	1.979	1.419	90.5	0.959	0.612	16.0	1.278	0.802	60.5
M3(Monthly)	1.062	0.737	52.5	1.424	0.832	78.5	0.935	0.638	24.0	0.923	0.681	29.0	0.888	0.615	16.0	1.160	0.803	67.0	0.956	0.682	33.0	1.505	0.999	89.0	0.823	0.573	5.5	1.094	0.786	61.5
M3(Quarterly)	1.173	0.457	47.5	2.969	0.747	83.5	0.982	0.399	23.0	0.857	0.373	11.0	0.945	0.409	21.5	1.369	0.538	67.5	1.046	0.424	32.0	2.500	1.103	86.5	0.833	0.380	9.5	1.293	0.510	62.0
M3(Yearly)	1.278	0.609	40.5	2.774	0.864	79.5	1.196	0.531	23.0	1.001	0.484	6.5	1.078	0.549	18.0	1.458	0.629	55.5	1.133	0.562	22.5	2.701	1.218	88.5	1.039	0.509	12.5	1.390	0.651	56.5
M4(Daily)	1.047	0.169	28.5	1.024	0.176	34.5	0.890	0.181	25.0	0.860	0.170	11.5	0.903	0.179	26.0	1.285	0.204	71.0	1.285	0.193	66.0	2.484	0.330	91.0	0.885	0.162	7.5	1.249	0.208	69.5
M4(Hourly)	2.104	0.828	74.5	0.848	0.362	18.0	1.013	0.411	44.0	3.148	1.088	86.5	0.947	0.381	31.0	3.066	1.092	87.0	1.016	0.409	44.5	0.928	0.370	25.0	0.958	0.379	32.5	2.998	1.088	86.0
M4(Monthly)	0.946	0.552	40.5	1.131	0.660	70.5	0.801	*0.477*	5.5	0.887	0.560	34.5	0.807	0.494	9.5	1.080	0.623	65.5	0.948	0.567	46.5	1.616	0.887	90.5	0.806	0.503	11.0	1.080	0.616	64.5
M4(Quarterly)	1.120	0.427	47.0	2.864	0.579	84.0	0.961	0.381	25.5	0.875	0.308	12.0	0.906	0.368	18.5	1.348	0.471	65.5	1.037	0.384	31.0	3.071	0.842	91.5	0.821	**0.286**	5.0	1.283	0.472	64.0
M4(Weekly)	0.994	0.460	58.5	0.901	0.422	29.5	0.726	0.427	16.5	0.825	0.425	22.5	0.781	0.425	20.5	1.227	0.492	78.5	1.097	0.447	56.5	1.983	0.452	70.0	0.831	0.446	37.0	1.193	0.476	73.5
M4(Yearly)	1.029	0.481	50.0	38.737	0.762	84.0	1.000	0.372	14.0	1.003	0.348	12.0	1.014	0.413	28.0	1.039	0.523	59.5	1.022	0.424	35.0	8.413	1.285	90.0	1.001	0.349	11.5	1.040	0.535	63.0
M5	0.876	1.202	32.5	0.910	1.221	69.5	0.954	1.198	64.0	0.889	1.149	32.0	0.885	1.167	30.5	0.999	1.175	57.0	0.861	1.138	12.5	0.826	1.079	10.0	0.791	0.819	5.5	0.933	1.195	61.5
NN5	1.025	0.722	77.5	0.845	0.576	47.0	0.806	0.559	25.5	1.424	1.034	87.5	*0.782*	*0.533*	3.0	1.129	0.779	82.0	0.897	0.624	64.0	0.813	0.567	34.0	0.851	0.572	47.0	1.039	0.721	78.0
NN5(Weekly)	0.754	1.150	15.0	1.113	1.327	89.0	0.847	1.154	47.0	0.904	1.199	69.0	0.793	1.154	33.0	0.818	1.177	56.0	*0.750*	1.150	12.5	**0.745**	1.195	30.5	0.864	1.175	61.5	0.797	1.166	41.0
Pedestrian_Counts	1.275	0.732	76.5	1.145	0.601	62.0	0.904	0.496	20.0	2.577	1.112	89.5	0.866	0.461	10.5	1.387	0.744	79.5	0.743	0.383	4.5	1.160	0.637	67.5	0.925	0.449	15.0	1.491	0.864	82.0
SaugeenRiverFlow	0.773	0.678	24.5	0.826	0.790	57.5	0.821	0.693	42.5	0.886	0.678	50.0	0.799	0.666	30.5	1.101	0.881	89.5	0.933	0.754	65.0	*0.726*	0.631	4.5	0.895	0.700	55.0	0.916	0.785	69.0
Solar(5Min.)	0.896	0.799	75.5	0.477	0.428	42.5	0.440	0.389	32.5	1.094	0.839	81.0	0.406	0.358	18.5	0.898	0.765	75.0	0.360	0.306	6.0	0.424	0.374	28.0	0.372	0.304	6.0	0.791	0.689	72.0
SolarPower	1.414	0.029	66.5	2.296	0.034	74.5	0.900	0.019	23.5	1.009	0.020	35.5	0.904	0.018	20.5	1.643	0.036	72.5	1.274	0.028	62.0	5.963	0.077	87.0	1.311	0.033	66.5	1.757	0.041	75.0
Sunspot	0.795	0.534	34.0	0.802	0.543	49.0	0.789	0.535	30.5	0.865	0.579	75.0	0.779	0.530	12.5	0.871	0.572	74.5	0.819	0.547	57.5	0.805	0.544	51.5	0.904	0.591	84.0	0.847	0.562	66.0
Taxi(Half_Hourly)	1.200	1.192	77.5	1.029	0.946	36.5	1.080	0.955	44.5	1.306	1.187	83.5	0.977	0.908	21.5	1.360	1.241	90.0	0.883	0.861	6.0	0.947	0.928	19.0	1.142	0.978	54.5	1.281	1.212	84.5
Taxi(Hourly)	0.800	1.238	69.0	0.592	1.051	18.5	0.677	1.073	27.0	0.792	1.100	44.5	0.701	1.100	36.0	0.903	1.295	93.5	0.835	1.253	82.0	*0.556*	*0.973*	3.0	0.819	1.189	71.0	0.885	1.292	91.5
Tourism(Monthly)	1.998	0.629	81.0	1.204	0.453	23.0	1.265	0.481	32.5	1.923	0.629	79.0	1.149	0.433	15.0	2.021	0.669	86.0	1.249	0.452	25.5	1.933	0.673	85.0	*0.955*	*0.388*	3.0	1.933	0.670	84.0
Tourism(Quarterly)	1.402	0.479	51.5	1.439	0.393	44.5	1.088	0.331	15.0	1.486	0.498	63.0	1.071	0.324	12.0	1.721	0.566	80.0	1.205	0.342	20.5	2.663	0.760	92.0	1.100	0.307	10.5	1.516	0.539	70.5
Tourism(Yearly)	1.398	0.512	44.5	3.972	1.157	82.5	0.969	0.366	7.0	1.048	0.379	16.5	1.126	0.426	33.5	1.574	0.622	63.0	1.302	0.503	41.5	2.532	1.454	85.5	0.936	0.389	11.5	1.550	0.603	59.5
USBirths	1.104	0.373	56.0	0.957	0.299	37.0	0.872	0.309	29.5	3.021	0.950	95.5	0.806	*0.253*	3.5	2.164	0.723	83.5	0.978	0.320	45.0	1.545	0.468	73.0	0.945	0.303	36.5	2.396	0.775	84.5
UberTLC_daily	0.969	0.401	24.0	2.138	1.202	85.0	0.993	0.454	34.0	1.180	0.537	43.5	0.732	0.388	11.0	1.538	0.728	57.0	1.032	0.439	33.0	2.078	1.189	83.0	1.970	0.853	67.0	1.058	0.426	33.0
UberTLC_hourly	1.349	0.884	79.0	0.832	0.610	52.5	0.708	0.505	27.0	1.836	1.078	91.0	0.606	0.449	12.5	1.719	1.072	87.5	0.625	0.432	12.0	0.718	0.571	38.5	0.862	0.560	48.5	1.464	1.034	83.5
VehicleTrips	1.398	1.113	66.5	1.342	0.839	32.0	1.304	0.883	36.0	1.919	1.176	89.5	1.171	0.823	16.5	1.471	1.095	71.0	1.046	0.745	9.0	1.392	0.887	47.0	1.033	0.720	6.0	1.508	1.117	76.0
Weather	0.759	0.698	9.0	0.819	0.802	64.0	0.787	0.721	31.0	0.817	0.738	45.5	0.791	0.717	32.0	0.938	0.826	85.5	0.844	0.777	65.0	0.776	0.720	27.5	0.809	0.685	25.0	0.859	0.802	77.0
WebTraffic	0.874	0.469	19.0	1.244	0.716	77.5	0.941	0.504	46.0	0.915	0.478	35.5	0.899	0.481	34.0	1.020	0.545	56.0	0.871	0.471	21.0	0.953	0.550	51.5	0.839	0.449	7.0	0.960	0.514	50.5
Wike2000	1.001	0.438	38.0	1.169	0.540	73.0	0.954	0.395	19.0	1.133	0.522	65.0	0.930	0.369	11.5	1.195	0.530	71.0	0.902	0.371	8.0	1.220	0.418	55.0	*0.863*	0.360	3.5	1.054	0.464	45.0
Wind_Farms	1.147	0.321	37.0	$> 10^6$	0.871	94.0	1.410	0.467	71.0	0.962	0.253	8.0	0.989	0.309	21.0	1.469	0.416	67.0	1.153	0.346	42.5	1.624	0.440	72.5	1.225	0.324	43.0	1.338	0.425	64.5
ausbeer	3.980	0.908	89.0	1.014	0.232	19.5	0.855	0.215	14.5	3.528	0.749	73.0	1.016	0.247	21.5	3.678	0.816	80.5	0.717	0.173	8.0	1.457	0.331	33.5	0.906	0.206	13.0	3.584	0.766	76.0
ecdc_ili	0.933	1.429	75.5	0.862	1.071	23.5	0.870	1.196	39.5	0.915	1.224	50.0	0.877	1.215	42.5	1.020	1.403	85.5	0.854	1.294	50.0	<u>0.714</u>	<u>1.052</u>	2.0	0.932	1.190	49.5	1.032	1.420	88.5
favorita_stores_1D	0.882	0.950	70.5	0.801	0.877	29.5	0.821	0.854	28.0	0.960	0.964	84.0	0.802	0.837	12.5	1.039	1.002	91.0	0.837	0.892	49.5	0.805	0.850	19.5	0.904	0.845	43.5	0.913	0.974	80.5
favorita_stores_1M	0.991	0.707	25.0	1.471	1.043	86.0	1.015	0.717	41.0	*0.947*	<u>0.660</u>	2.5	0.999	0.694	23.5	0.990	0.754	41.5	0.974	0.705	16.0	1.203	1.033	79.5	0.994	0.704	24.0	1.054	0.755	57.5
favorita_stores_1W	0.726	0.715	15.5	0.782	0.841	63.5	0.816	0.741	55.5	0.904	0.733	59.5	0.755	0.730	37.5	0.837	0.766	63.0	0.702	0.716	11.5	0.898	0.906	83.5	0.734	0.713	18.0	0.779	0.774	53.5
favorita_transactions_1D	1.033	0.746	74.5	0.844	0.626	42.5	0.785	0.572	7.0	1.093	0.812	86.5	**0.764**	**0.546**	1.0	1.114	0.776	83.5	0.890	0.645	53.0	0.785	<u>0.548</u>	3.0	0.954	0.666	64.5	1.104	0.769	82.0
favorita_transactions_1M	1.540	0.798	31.5	1.687	0.980	72.5	1.611	0.779	35.5	1.733	0.817	59.5	1.435	0.767	13.5	1.589	0.846	52.0	1.530	0.801	34.0	1.484	0.984	54.0	1.284	0.745	6.0	1.606	0.836	53.5
favorita_transactions_1W	0.970	0.565	22.0	1.200	0.707	70.0	1.011	0.584	37.5	0.995	0.605	41.0	0.979	0.573	27.5	1.088	0.624	55.0	0.943	0.575	19.0	1.219	0.690	68.5	<u>0.899</u>	**0.532**	1.5	1.035	0.610	47.5
heart_rate	1.171	1.176	80.5	1.067	1.072	38.0	1.294	1.089	72.0	1.011	0.971	9.0	1.060	1.020	26.5	1.239	1.170	84.0	1.110	1.111	61.5	1.096	1.078	46.0	1.106	1.084	51.0	1.252	1.196	89.5
kdd_cup_2022_10T	0.770	0.390	31.0	1.024	0.497	87.0	0.765	0.383	26.5	**0.739**	<u>0.369</u>	1.5	0.746	0.374	13.0	0.933	0.462	76.5	0.903	0.446	65.5	0.827	0.405	45.5	0.804	0.397	40.0	0.887	0.442	60.5
kdd_cup_2022_1D	0.850	1.230	20.0	0.909	1.369	74.0	0.956	1.260	63.0	0.863	1.263	39.0	0.909	1.268	59.5	0.936	1.276	70.5	0.865	1.260	37.0	0.819	1.266	25.0	0.912	1.201	40.0	0.938	1.281	73.0
kdd_cup_2022_30T	0.855	1.030	45.5	0.898	1.184	75.0	0.854	1.039	47.0	0.855	0.917	26.0	0.853	0.996	37.5	0.910	1.059	70.0	0.857	0.984	37.5	0.888	1.042	62.0	0.819	0.917	12.5	0.890	1.038	62.5
physionet_2012	0.778	1.009	31.5	0.777	1.076	50.0	0.842	1.002	56.0	<u>0.726</u>	0.958	3.5	0.795	0.994	37.0	0.852	1.041	67.5	0.743	0.986	17.0	0.832	1.122	82.0	0.760	0.965	13.0	0.808	1.024	55.5
rossmann_1D	1.350	1.188	85.5	0.751	0.674	30.5	0.711	0.637	18.0	1.334	1.105	76.0	0.674	0.614	10.5	1.586	1.211	93.0	1.138	0.963	59.5	0.629	<u>0.570</u>	3.0	1.125	0.931	56.0	1.467	1.173	89.0
rossmann_1W	1.149	1.253	52.5	1.007	1.229	29.0	1.285	1.243	60.0	0.849	1.343	45.0	1.157	1.239	50.0	1.249	1.286	77.0	1.157	1.308	75.5	**0.735**	1.194	5.0	0.868	1.302	46.5	1.241	1.286	76.0
us_gasoline	1.016	0.632	84.5	0.876	0.561	53.0	0.835	0.549	24.0	0.917	0.573	65.0	0.847	0.543	25.5	1.213	0.727	95.0	0.955	0.612	77.5	0.785	0.518	9.0	0.849	0.553	32.5	1.000	0.615	82.5

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, <u>underline</u>, and *italic*. The rank column reports the average rank over the two metrics.

Table 16: **Univariate forecasting results.** Part 9.

Dataset	test			tft			theta			tide			timekan			timellm			timemixer			timer			timesfm			timesfm_v1		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	1.508	0.305	58.5	0.919	0.192	16.5	1.019	0.214	25.0	0.889	0.185	13.0	1.528	0.351	66.0	1.542	0.291	56.5	1.360	0.295	48.5	1.634	0.357	69.5	0.913	0.197	18.0	1.113	0.219	28.5
AustralianElectricity	1.744	1.054	89.5	0.783	0.501	46.5	1.994	1.039	92.5	0.717	0.451	26.0	0.753	0.479	37.5	1.214	0.684	76.5	<u>0.546</u>	<u>0.361</u>	2.0	1.323	0.784	81.0	0.581	0.381	6.0	1.220	0.715	78.0
Bitcoin	1.156	0.286	64.0	1.068	0.262	47.0	*0.903*	0.252	12.5	0.986	0.270	39.5	1.038	0.269	48.5	1.404	0.363	77.5	0.995	0.255	30.5	1.032	0.259	41.5	0.905	<u>0.233</u>	3.0	0.972	0.247	20.5
BrazilianCitiesTemperature	1.073	0.550	54.5	0.975	0.569	48.5	0.849	0.378	15.5	0.993	0.509	42.5	1.024	0.578	55.0	1.468	0.592	76.5	0.944	0.502	36.0	0.924	0.453	28.5	0.702	0.259	4.0	0.889	0.418	21.0
CIF_2016	1.034	0.555	44.5	0.975	0.534	34.5	<u>0.774</u>	<u>0.393</u>	2.0	0.903	0.441	12.0	1.070	0.549	46.0	1.158	0.588	60.5	1.100	0.560	50.0	1.138	0.584	57.0	0.860	0.434	8.5	1.125	0.578	55.5
Car_Parts	0.729	0.631	35.0	0.736	0.653	39.0	0.897	0.896	76.0	0.712	0.585	21.0	0.721	0.603	29.0	1.482	0.893	86.5	0.740	0.626	36.5	0.778	0.636	44.0	0.484	0.173	6.0	0.855	0.783	64.0
CoronaVirus_2019_confirmed	0.964	0.085	52.5	0.744	0.066	32.0	0.706	0.063	28.5	0.839	0.075	44.0	0.858	0.080	48.0	1.079	0.084	53.0	0.780	0.071	37.0	1.294	0.175	64.0	0.503	0.044	8.5	0.662	0.058	23.0
CoronaVirus_2019_deaths	0.832	0.068	38.5	0.985	0.059	37.0	0.675	0.050	18.0	1.053	0.076	49.0	1.166	0.066	46.5	1.228	0.081	55.0	1.066	0.062	43.5	1.286	0.167	66.5	0.566	0.045	12.0	0.710	0.058	27.0
CoronaVirus_2019_recovered	0.894	0.092	46.5	0.669	0.078	34.0	0.759	0.075	36.5	0.592	0.087	30.0	0.703	0.090	41.0	0.891	0.094	47.5	0.665	0.079	34.0	1.202	0.239	69.0	0.706	0.070	30.5	0.793	0.085	42.5
Elecdemand	1.474	1.067	86.0	0.886	0.659	47.0	1.639	1.080	92.0	0.804	0.602	20.5	0.808	0.621	31.5	1.185	0.800	71.0	0.756	0.572	8.5	1.230	0.898	75.0	**0.362**	**0.259**	1.0	1.287	0.917	77.5
Extended_Wikipedia_Web_Traffic_Daily_Dataset	0.891	0.382	39.5	0.876	0.361	26.5	0.910	0.371	41.0	0.941	0.384	49.5	0.851	0.369	26.5	1.131	0.472	77.5	0.842	0.356	16.5	0.888	0.383	39.0	**0.774**	**0.309**	1.0	0.877	0.376	35.0
Flood_Modeling_Dataset_1	0.788	0.830	37.0	0.787	0.836	37.0	1.057	0.702	53.0	0.764	0.778	18.5	0.784	0.796	30.0	1.050	0.918	74.0	3.862	0.911	78.0	0.977	1.072	84.0	0.666	0.518	4.5	0.792	0.755	26.0
Flood_Modeling_Dataset_2	0.810	0.760	52.5	0.755	0.677	22.5	0.964	0.658	52.5	0.788	0.730	42.5	0.754	0.678	22.0	2.361	1.166	95.5	0.760	0.687	30.5	0.951	0.925	88.0	*0.694*	0.515	3.5	0.815	0.671	38.0
Flood_Modeling_Dataset_3	0.849	0.809	36.5	0.837	0.783	29.0	1.101	0.657	51.5	1.061	0.812	64.0	0.857	0.776	31.5	1.455	0.892	77.0	0.859	0.787	35.0	1.126	1.115	89.5	**0.749**	0.490	2.5	0.877	0.664	27.0
Hospital	0.977	1.021	53.0	0.961	0.990	35.0	0.856	**0.902**	5.5	1.056	1.045	62.0	0.932	1.020	42.0	1.993	1.265	91.5	0.977	1.039	55.0	0.893	0.996	32.5	**0.826**	0.937	3.5	0.852	0.970	12.0
M1(Monthly)	0.967	0.730	51.0	0.999	0.691	43.0	<u>0.689</u>	<u>0.532</u>	2.0	0.842	0.627	11.0	1.000	0.708	47.5	1.328	0.815	83.5	0.953	0.703	38.0	1.008	0.735	60.0	0.808	0.618	7.5	0.869	0.669	25.5
M1(Quarterly)	1.054	0.689	50.5	0.973	0.630	23.0	0.884	*0.578*	3.5	0.952	0.651	26.0	1.044	0.695	51.5	1.339	0.750	69.0	1.038	0.696	51.5	1.012	0.659	42.0	**0.755**	**0.522**	1.0	*0.880*	0.583	3.5
M1(Yearly)	1.249	0.799	56.5	0.924	0.609	14.0	0.923	0.573	10.0	0.995	0.647	25.5	1.075	0.709	40.5	0.901	0.610	11.5	1.031	0.684	36.0	1.292	0.844	66.0	0.970	0.598	13.5	1.066	0.672	37.0
M3(Monthly)	1.065	0.730	51.5	1.034	0.709	44.5	**0.747**	<u>0.540</u>	1.5	0.831	0.585	8.0	1.053	0.751	52.5	1.825	0.860	84.5	1.040	0.721	47.0	0.972	0.701	38.0	0.821	0.604	8.5	0.971	0.686	35.0
M3(Quarterly)	1.278	0.497	58.5	1.095	0.429	37.0	<u>0.769</u>	<u>0.342</u>	2.0	0.896	0.388	16.5	1.154	0.474	49.5	1.408	0.517	66.5	1.152	0.473	48.5	1.149	0.466	47.5	0.845	0.359	7.5	0.841	*0.355*	6.0
M3(Yearly)	1.379	0.636	53.0	1.461	0.618	54.5	<u>0.940</u>	<u>0.466</u>	2.0	1.093	0.557	20.0	1.311	0.618	45.5	1.687	0.608	54.5	1.330	0.591	42.0	1.345	0.571	39.5	0.984	0.477	4.5	1.053	0.523	14.5
M4(Daily)	0.977	0.191	47.0	0.929	0.187	36.5	0.816	0.165	5.5	0.935	0.177	26.0	1.056	0.174	33.5	1.450	0.253	77.5	0.995	0.176	31.5	1.281	0.197	67.0	0.897	<u>0.156</u>	7.5	0.955	*0.160*	15.0
M4(Hourly)	2.856	1.111	86.5	1.163	0.448	51.5	3.234	1.151	92.0	1.228	0.481	58.5	1.235	0.509	60.0	2.149	0.804	75.0	1.058	0.430	48.0	1.848	0.785	72.0	0.943	0.391	34.0	0.962	0.382	35.5
M4(Monthly)	0.922	0.555	39.5	0.879	0.520	23.5	*0.772*	0.484	3.5	0.793	0.489	6.0	0.953	0.571	48.0	1.257	0.672	74.5	0.907	0.547	35.5	0.967	0.575	50.0	0.844	0.505	18.0	0.959	0.565	46.5
M4(Quarterly)	1.204	0.435	55.0	1.127	0.418	45.0	0.789	*0.288*	5.0	0.909	0.383	22.5	1.095	0.429	47.0	1.305	0.427	57.5	1.140	0.433	50.5	1.153	0.415	46.5	0.875	0.329	14.5	0.849	0.300	8.5
M4(Weekly)	1.316	0.468	75.0	0.887	0.449	43.5	0.727	0.425	14.5	0.748	0.426	17.5	1.025	0.460	62.0	1.361	0.503	82.0	0.856	0.457	45.0	1.182	0.477	73.5	1.013	0.452	53.0	1.011	0.467	63.5
M4(Yearly)	1.038	0.522	57.5	1.012	0.424	28.5	*0.834*	<u>0.310</u>	2.5	1.010	0.389	20.5	1.024	0.465	44.5	1.030	0.461	46.5	1.021	0.454	39.0	8.324	0.530	72.5	1.014	0.389	26.0	1.013	0.383	23.0
M5	0.889	1.168	36.5	0.872	1.212	36.0	0.904	1.223	66.5	0.981	1.169	55.0	0.883	1.227	55.5	1.285	1.230	85.5	0.882	1.183	32.0	0.875	1.193	27.5	0.782	0.822	4.5	0.889	1.162	34.0
NN5	1.393	1.040	85.0	0.792	0.542	9.0	1.401	1.041	89.0	0.845	0.573	47.0	0.841	0.572	45.0	1.085	0.691	78.0	0.794	0.543	11.0	1.501	1.057	91.5	0.804	0.545	16.5	1.008	0.708	75.5
NN5(Weekly)	0.776	1.137	16.5	0.775	1.142	17.5	0.854	1.168	55.5	0.801	1.142	27.5	0.773	1.158	28.0	1.190	1.240	80.5	0.802	1.149	33.0	0.796	1.187	48.0	0.789	1.156	32.5	0.798	1.190	50.5
Pedestrian_Counts	2.044	1.250	89.0	1.418	0.638	76.5	2.625	1.138	91.5	1.030	0.535	40.0	0.986	0.533	35.0	2.462	0.898	88.0	0.927	0.502	25.0	0.980	0.521	33.0	0.811	0.417	7.0	0.900	0.456	13.0
SaugeenRiverFlow	0.947	0.811	77.5	0.778	0.662	22.5	0.941	0.727	63.0	0.778	0.652	18.0	0.796	0.663	28.5	0.994	0.772	73.5	0.776	0.637	11.5	0.979	0.810	79.0	0.789	0.652	22.0	0.906	0.735	60.0
Solar(5Min.)	1.102	1.042	88.0	0.708	0.676	68.0	1.269	0.876	89.0	0.945	0.591	69.0	0.461	0.405	38.0	1.906	0.949	92.5	0.457	0.405	36.5	0.586	0.486	54.5	0.396	0.309	9.5	0.407	0.331	15.5
SolarPower	2.331	0.049	80.0	1.156	0.021	45.0	613.184	1.060	97.0	1.091	0.023	48.5	0.860	0.018	19.0	2.325	0.045	79.0	1.017	0.019	34.5	70.477	1.109	96.0	*0.498*	0.008	4.0	0.937	0.017	23.0
Sunspot	0.857	0.567	70.0	0.784	0.531	23.0	0.997	0.616	90.0	0.778	*0.528*	4.0	0.781	0.531	18.5	0.855	0.563	68.0	0.779	0.528	7.0	0.837	0.556	62.0	0.837	0.556	62.0	0.899	0.579	80.0
Taxi(Half_Hourly)	1.245	1.222	85.0	1.151	1.122	66.5	1.452	1.172	85.0	1.164	0.997	57.5	1.104	1.079	60.0	1.546	1.196	88.5	0.983	0.946	30.5	1.074	1.068	55.0	<u>0.800</u>	*0.784*	2.5	1.090	1.038	53.5
Taxi(Hourly)	0.885	1.281	53.0	0.678	1.069	27.0	0.809	1.112	53.0	0.830	1.186	70.5	0.788	1.205	62.0	1.092	1.285	94.0	0.790	1.194	60.0	0.807	1.235	72.0	0.838	1.201	76.0	0.803	1.210	68.5
Tourism(Monthly)	1.945	0.635	81.5	1.381	0.481	38.5	1.080	0.430	11.5	1.365	0.535	42.5	1.735	0.595	64.5	4.600	1.020	95.5	1.599	0.564	55.5	1.972	0.645	84.5	<u>0.946</u>	**0.386**	1.5	1.146	0.436	16.0
Tourism(Quarterly)	1.483	0.499	63.5	1.326	0.452	37.0	1.087	0.320	12.0	1.092	0.350	17.0	1.339	0.467	42.5	1.569	0.489	64.5	1.422	0.474	53.5	1.826	0.563	81.0	1.151	0.312	14.5	*0.965*	**0.273**	2.0
Tourism(Yearly)	1.498	0.574	54.5	1.084	0.405	25.5	0.913	*0.324*	3.5	1.046	0.399	20.5	1.322	0.523	43.5	1.630	0.529	56.0	1.344	0.530	46.0	1.574	0.614	63.0	1.066	0.405	24.5	1.079	0.410	25.5
USBirths	2.900	0.880	87.5	0.819	0.277	9.0	2.961	0.913	92.0	0.916	0.287	25.0	0.924	0.292	28.0	1.645	0.512	77.0	0.847	0.274	11.5	2.919	0.827	87.0	0.912	0.287	24.0	1.038	0.338	49.0
UberTLC_daily	0.965	0.403	24.5	*0.631*	<u>0.299</u>	2.5	2.083	0.991	77.0	0.859	0.454	25.0	1.028	0.440	33.0	1.618	0.614	55.0	0.891	0.390	17.0	1.447	0.611	50.5	1.065	0.475	40.0	1.284	0.552	46.0
UberTLC_hourly	1.636	1.096	89.5	0.557	*0.377*	4.5	2.232	1.117	94.0	1.012	0.681	68.5	0.717	0.524	32.0	1.618	0.968	84.0	0.643	0.494	18.0	0.891	0.577	53.0	0.647	0.429	13.0	0.725	0.475	24.0
VehicleTrips	1.458	1.117	72.0	1.111	0.804	14.0	1.511	1.140	80.0	1.225	0.856	26.0	1.390	1.034	56.0	2.665	1.182	92.5	1.440	1.090	67.0	1.401	1.070	61.5	**0.932**	0.725	3.5	1.065	0.825	14.0
Weather	0.839	0.764	61.5	0.772	0.699	17.0	0.890	0.807	82.0	0.812	0.740	45.0	0.777	0.714	25.0	1.061	0.846	90.0	<u>0.743</u>	0.683	4.0	0.870	0.784	72.5	0.788	*0.663*	17.5	0.851	0.769	64.0
WebTraffic	0.924	0.500	43.5	0.883	0.472	25.0	0.974	0.510	50.5	0.983	0.514	52.0	0.910	0.493	39.0	1.151	0.584	61.5	0.899	0.483	34.5	0.908	0.482	36.0	*0.824*	**0.427**	2.0	0.843	0.446	7.5
Wike2000	1.058	0.458	44.5	0.991	0.425	35.0	1.161	0.536	71.0	0.991	0.395	26.5	0.985	0.427	34.5	2.289	0.761	94.0	0.980	0.403	28.0	1.031	0.460	43.5	**0.840**	**0.338**	1.0	0.927	0.399	19.5
Wind_Farms	1.456	0.436	69.0	1.213	0.396	51.0	14.096	0.946	95.0	3.015	0.388	70.0	1.006	0.315	23.5	2.874	0.454	79.0	1.547	0.355	60.5	1.862	0.966	89.5	**0.901**	**0.235**	1.0	1.205	0.341	44.0
ausbeer	3.311	0.731	68.5	1.358	0.342	31.5	1.184	0.211	19.0	2.091	0.489	52.5	2.021	0.466	51.0	6.282	0.956	94.5	3.743	0.844	83.0	7.194	1.142	98.0	1.363	0.298	29.5	1.544	0.363	37.5
ecdc_ili	0.966	1.411	78.5	0.923	1.398	69.0	0.949	1.117	47.0	0.937	1.220	53.5	0.995	1.407	83.0	1.493	1.381	86.5	0.974	1.423	81.5	0.809	1.231	38.0	0.792	1.181	29.5	0.858	1.277	49.0
favorita_stores_1D	0.887	0.944	70.0	0.806	0.858	22.5	0.950	0.995	86.0	0.817	0.839	21.0	0.835	0.888	47.0	1.693	1.169	95.5	0.819	0.866	31.0	0.912	0.932	73.0	0.802	<u>0.819</u>	10.0	0.915	0.955	77.0
favorita_stores_1M	1.007	0.710	34.0	1.011	0.724	42.0	<u>0.942</u>	**0.642**	1.5	1.031	0.733	49.0	1.081	0.745	55.5	1.320	0.836	74.5	1.024	0.748	49.5	0.994	0.705	24.5	0.975	0.697	14.5	0.947	*0.673*	3.5
favorita_stores_1W	0.764	0.726	39.5	0.709	0.705	8.0	0.752	0.726	33.5	0.800	0.752	56.5	0.743	0.722	26.5	1.487	0.997	95.0	0.781	0.724	40.5	0.752	0.731	35.0	<u>0.688</u>	<u>0.688</u>	2.0	0.725	0.703	11.0
favorita_transactions_1D	1.027	0.739	72.0	0.792	0.585	14.5	1.096	0.797	85.5	0.827	0.596	25.5	0.875	0.634	50.0	2.034	1.068	95.0	0.824	0.599	28.5	1.132	0.793	88.0	0.805	0.583	15.0	0.944	0.689	67.0
favorita_transactions_1M	1.432	0.795	18.5	1.508	0.798	28.0	1.253	<u>0.580</u>	3.0	1.531	0.756	24.0	1.688	0.823	58.0	2.203	0.864	76.5	1.607	0.821	51.0	1.664	0.892	63.5	1.684	0.801	49.0	1.550	0.778	28.0
favorita_transactions_1W	0.988	0.593	37.5	0.935	0.555	6.5	1.039	0.618	51.5	0.981	0.581	31.5	0.972	0.584	31.0	1.980	0.841	91.0	0.988	0.577	30.5	1.351	0.751	84.0	0.920	0.556	6.5	0.970	0.589	31.5
heart_rate	1.230	1.182	85.0	1.158	1.162	77.5	1.030	0.974	14.0	1.304	1.071	66.0	1.157	1.149	76.0	1.532	1.188	95.0	1.191	1.102	71.0	1.121	1.088	57.5	1.022	0.972	11.5	1.103	1.035	38.5
kdd_cup_2022_10T	0.974	0.481	84.0	0.754	0.380	22.5	0.942	0.410	63.5	0.745	0.372	8.5	0.751	0.376	18.0	0.888	0.430	56.0	0.747	0.375	16.5	1.291	0.727	92.5	0.789	0.391	34.5	0.937	0.459	76.0
kdd_cup_2022_1D	0.877	1.253	38.5	0.860	1.198	19.5	0.819	1.259	19.5	0.923	1.255	54.0	0.874	1.265	45.5	1.158	1.299	79.0	0.880	1.271	52.0	*0.816*	1.174	3.5	0.881	1.198	29.5	0.916	1.252	49.5
kdd_cup_2022_30T	0.896	1.042	66.0	0.823	1.009	30.0	0.851	0.872	19.5	0.824	0.972	22.0	0.867	1.017	49.0	1.036	1.073	78.0	0.847	0.961	27.0	0.855	1.001	41.5	0.793	0.891	6.5	0.871	0.946	35.0
physionet_2012	0.787	1.016	39.5	0.825	1.030	60.5	0.743	0.956	8.0	0.798	0.985	37.0	0.798	1.021	48.5	1.009	1.078	86.5	0.858	1.029	57.0	0.805	0.998	47.5	0.734	0.959	7.5	0.735	0.976	13.0
rossmann_1D	1.366	1.173	85.5	0.745	0.659	24.0	1.346	1.108	78.0	0.923	0.762	43.5	1.166	0.991	63.0	2.754	1.325	97.0	1.047	0.887	50.0	1.672	1.390	97.0	1.027	0.865	48.5	1.248	1.051	68.5
rossmann_1W	1.154	1.270	59.5	1.158	1.172	40.5	0.828	1.283	34.5	1.212	1.235	54.0	1.149	1.281	63.0	1.981	1.306	85.5	1.166	1.277	66.5	1.196	1.245	57.0	0.977	1.366	59.5	0.921	1.276	42.0
us_gasoline	1.034	0.643	88.5	0.837	0.550	26.0	0.816	0.522	13.5	0.866	0.549	36.0	0.971	0.608	78.5	1.360	0.718	95.0	0.850	0.531	24.5	0.970	0.594	75.5	0.867	0.526	30.5	0.970	0.609	78.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, <u>underline</u>, and *italic*. The rank column reports the average rank over the two metrics.

Table 17: **Univariate forecasting results.** Part 10.

Dataset	timesfm_v2			timesnet			timexer			tirex			toto			transformer			tsmixer			units			xgboost		
	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank	mase	msmape	rank
AirPassengers	1.608	0.331	65.0	1.325	0.281	43.0	0.905	0.196	16.0	0.926	0.196	18.0	1.797	0.366	74.0	3.029	0.905	87.0	4.262	1.317	93.0	1.344	0.272	42.0	2.344	0.611	80.0
AustralianElectricity	1.343	0.795	82.5	0.685	0.433	17.5	0.577	0.380	5.0	1.122	0.669	75.0	1.405	0.780	81.5	0.669	0.437	15.5	0.712	0.468	29.5	0.768	0.485	40.5	0.684	0.447	18.5
Bitcoin	0.932	0.247	13.0	1.034	0.264	46.0	1.015	0.261	39.5	0.902	**0.230**	1.5	0.934	0.258	23.0	7.665	0.869	87.5	1.421	0.393	80.0	1.042	0.270	50.0	7.727	0.967	94.0
BrazilianCitiesTemperature	0.867	0.391	20.0	0.985	0.534	46.0	0.943	0.514	38.5	0.716	*0.252*	4.0	0.674	0.235	2.0	1.187	0.979	75.5	1.327	1.197	91.0	1.072	0.540	53.0	1.284	0.901	78.5
CIF_2016	0.970	0.523	31.5	1.185	0.608	68.0	0.960	0.456	20.5	0.824	0.446	7.5	0.920	0.515	24.5	1.466	0.694	79.0	2.497	1.427	94.5	1.020	0.555	41.5	1.512	0.708	82.0
Car_Parts	0.713	0.581	21.0	0.721	0.597	26.5	0.700	0.579	16.5	0.473	*0.149*	3.5	0.455	0.121	2.0	0.796	0.761	59.5	0.886	0.834	70.5	0.685	0.568	13.0	0.779	0.691	49.0
CoronaVirus_2019_confirmed	0.528	0.049	12.0	0.774	0.070	35.5	0.658	0.057	21.0	0.420	0.038	5.5	0.655	0.056	19.5	7.219	1.124	92.5	6.578	1.174	89.0	0.822	0.071	39.5	7.003	1.038	88.0
CoronaVirus_2019_deaths	0.522	0.049	13.0	1.084	0.069	47.0	0.819	0.050	26.0	0.284	0.039	4.5	0.523	0.049	14.0	4.270	0.426	91.5	5.779	0.957	94.5	0.926	0.063	38.5	2.538	0.256	76.5
CoronaVirus_2019_recovered	0.576	0.072	20.5	0.722	0.082	40.0	0.578	0.066	14.5	0.488	0.053	4.5	0.589	0.069	19.0	3.022	0.560	89.5	2.986	0.962	90.5	0.677	0.077	34.0	1.930	0.317	77.5
Elecdemand	1.305	0.969	79.0	0.787	0.604	17.0	0.798	0.607	21.5	1.061	0.802	69.0	1.091	0.806	70.0	1.244	0.844	74.5	0.873	0.642	41.5	0.789	0.602	16.0	0.719	0.550	4.5
Extended_Wikipedia_Web_Traffic_Daily_Dataset	0.792	0.324	2.0	0.852	0.367	26.5	0.836	0.358	15.0	*0.798*	*0.330*	3.0	0.846	0.353	16.0	0.896	0.431	49.0	1.040	0.563	80.5	0.852	0.371	29.5	1.164	0.490	83.0
Flood_Modeling_Dataset_1	0.759	0.700	13.5	0.794	0.822	37.0	0.777	0.804	29.0	0.653	0.475	2.0	*0.662*	*0.487*	3.0	0.853	0.926	54.5	0.842	0.966	56.0	0.754	0.783	17.5	0.860	0.919	55.5
Flood_Modeling_Dataset_2	0.810	0.652	33.0	0.756	0.682	24.5	0.756	0.686	26.0	0.667	0.470	2.0	0.697	*0.490*	3.5	0.791	0.757	46.0	0.803	0.790	52.0	0.740	0.659	14.0	0.818	0.774	55.0
Flood_Modeling_Dataset_3	0.823	0.622	14.0	0.847	0.762	25.5	0.838	0.760	23.0	0.816	0.464	4.5	0.759	**0.459**	1.5	0.909	0.870	50.5	1.002	1.012	73.0	0.809	0.742	13.5	0.908	0.868	49.0
Hospital	0.848	0.969	11.0	0.953	1.018	46.0	0.937	1.008	40.0	*0.828*	0.937	4.0	0.884	1.041	44.0	1.045	1.195	65.0	1.261	1.329	90.5	0.944	0.972	28.0	1.183	1.244	80.0
M1(Monthly)	0.870	0.639	18.0	1.026	0.753	63.5	0.881	0.623	16.0	0.923	0.666	28.5	0.983	0.772	60.0	1.278	0.845	83.5	2.216	1.374	97.0	1.056	0.727	61.0	1.289	0.871	85.5
M1(Quarterly)	1.022	0.644	35.0	1.079	0.704	57.0	0.970	0.643	25.0	0.924	0.598	8.5	0.958	0.741	43.0	1.409	0.911	77.5	2.128	1.413	94.0	1.001	0.644	32.5	1.481	0.982	83.5
M1(Yearly)	1.030	0.655	30.5	1.255	0.812	58.5	0.977	0.645	22.0	0.980	0.627	21.0	0.986	0.620	20.5	1.852	1.291	86.5	2.175	1.457	94.0	1.011	0.671	32.0	1.754	1.266	80.5
M3(Monthly)	0.857	0.633	14.5	1.107	0.753	59.5	0.890	0.602	15.0	*0.802*	0.591	6.5	0.903	0.715	34.5	1.251	0.874	75.0	2.023	1.362	97.0	1.029	0.697	41.0	1.320	0.887	79.5
M3(Quarterly)	0.771	0.358	4.0	1.294	0.511	63.5	1.005	0.403	25.5	**0.753**	**0.340**	1.0	0.883	0.433	26.0	1.769	0.727	74.0	3.244	1.407	94.5	1.140	0.440	41.5	1.879	0.760	80.0
M3(Yearly)	1.028	0.505	11.0	1.393	0.630	52.5	1.153	0.564	26.5	1.018	0.497	9.5	1.147	0.622	35.5	2.127	0.888	73.5	3.183	1.416	96.5	1.285	0.581	38.0	2.225	0.968	82.0
M4(Daily)	0.901	**0.152**	7.5	1.103	0.183	47.0	0.905	0.186	32.5	0.964	0.165	19.0	0.914	0.191	37.5	1.920	0.264	83.0	3.201	0.522	95.5	0.980	0.178	34.5	2.028	0.280	86.0
M4(Hourly)	*0.744*	*0.312*	3.0	1.218	0.476	57.0	0.963	0.404	40.5	0.870	0.348	18.5	1.046	0.406	44.5	1.552	0.659	69.0	1.873	0.785	73.0	1.492	0.626	67.5	0.817	0.335	13.0
M4(Monthly)	0.864	0.534	24.0	0.975	0.573	50.5	**0.743**	**0.459**	1.0	0.822	0.512	15.0	0.924	0.624	54.5	1.439	0.805	87.0	2.262	1.317	95.0	0.873	0.524	23.5	1.376	0.797	83.5
M4(Quarterly)	0.850	0.312	12.5	1.247	0.450	60.0	1.053	0.389	33.5	0.779	0.287	3.5	0.959	0.386	26.5	2.483	0.708	85.5	4.501	1.369	94.5	1.120	0.426	47.0	2.005	0.664	79.5
M4(Weekly)	0.977	0.454	52.0	0.978	0.460	59.0	0.788	0.413	17.0	0.893	0.425	30.0	0.818	0.433	28.0	4.689	0.532	91.5	2.273	0.665	92.5	1.195	0.457	65.5	1.703	0.467	79.0
M4(Yearly)	1.013	0.413	26.5	1.037	0.506	54.5	1.007	0.407	22.0	1.007	0.374	17.5	1.376	0.491	59.5	8.370	0.914	84.0	7.947	1.445	87.0	1.015	0.441	36.5	4.159	0.974	80.0
M5	1.471	1.303	93.5	0.884	1.188	37.5	0.877	1.167	20.0	0.765	0.793	2.0	0.785	*0.815*	4.0	0.913	1.274	80.0	0.883	1.311	64.0	0.878	1.203	35.0	0.906	1.177	50.0
NN5	1.068	0.739	79.5	0.806	0.548	19.5	0.798	0.547	16.0	0.905	0.616	63.0	1.213	0.887	83.0	0.873	0.596	55.5	0.885	0.609	59.5	0.879	0.607	57.5	0.802	0.552	20.0
NN5(Weekly)	0.802	1.173	47.5	0.817	1.159	46.0	0.769	1.115	11.0	0.887	1.201	67.0	0.824	1.197	60.5	0.808	1.229	57.0	0.987	1.375	90.0	0.769	1.137	13.0	0.890	1.262	73.0
Pedestrian_Counts	0.801	0.412	6.0	0.957	0.500	26.5	0.875	0.476	12.5	**0.678**	0.344	1.5	0.979	0.493	26.0	1.238	0.636	70.0	1.172	0.634	66.5	1.001	0.523	37.0	1.086	0.578	50.5
SaugeenRiverFlow	0.804	0.644	23.5	0.788	0.665	27.0	0.775	0.656	17.5	0.783	0.639	15.5	0.820	0.669	37.0	0.727	0.640	7.5	0.835	0.801	60.5	0.803	0.674	34.0	0.823	0.744	50.0
Solar(5Min.)	0.411	0.323	16.0	0.485	0.433	46.5	0.405	0.353	16.5	**0.269**	**0.222**	1.0	0.483	0.366	34.5	0.776	0.662	69.5	0.542	0.500	54.0	0.601	0.516	56.0	0.375	0.329	10.5
SolarPower	0.670	0.010	9.5	1.041	0.021	39.5	0.862	0.016	17.0	0.971	0.017	25.0	0.598	0.008	5.5	3.061	0.028	74.0	1.923	0.026	67.0	1.118	0.021	42.5	0.927	0.012	18.5
Sunspot	0.943	0.598	87.0	0.778	0.529	7.0	0.783	0.531	23.5	0.859	0.563	69.0	0.917	0.610	87.5	0.783	0.533	24.0	0.793	0.543	44.5	0.794	0.538	39.5	0.801	0.542	47.5
Taxi(Half_Hourly)	1.083	1.039	53.0	1.108	1.052	58.0	0.964	0.904	15.5	*0.819*	0.785	3.5	1.297	1.161	79.0	1.144	1.149	67.5	1.177	1.200	76.5	1.132	1.109	63.5	0.969	0.922	22.0
Taxi(Hourly)	0.811	1.208	71.5	0.793	1.173	58.0	0.708	1.108	37.0	0.834	1.184	70.5	0.771	1.153	51.5	0.602	1.042	17.0	0.684	1.273	60.0	0.875	1.258	86.5	0.563	**0.964**	3.0
Tourism(Monthly)	1.270	0.467	30.0	1.880	0.608	75.0	1.230	0.444	21.0	1.061	0.420	8.5	1.621	0.606	63.5	1.818	0.623	74.0	3.255	1.257	94.5	2.040	0.635	83.5	1.432	0.522	44.5
Tourism(Quarterly)	0.937	*0.286*	2.5	1.552	0.527	71.5	1.117	0.314	13.0	1.134	0.305	12.0	1.918	0.745	87.0	2.275	0.633	89.0	4.132	1.381	96.5	1.353	0.444	38.0	1.775	0.543	79.5
Tourism(Yearly)	0.970	0.371	9.0	1.520	0.596	57.5	1.134	0.429	35.0	1.040	0.392	18.5	1.005	0.459	28.5	2.503	1.423	84.5	2.887	1.524	90.5	1.123	0.415	30.5	2.464	1.371	81.0
USBirths	1.218	0.403	64.5	0.885	0.286	19.5	0.822	0.282	11.0	1.059	0.356	53.0	2.002	0.633	81.5	1.162	0.398	62.0	1.118	0.351	55.0	0.955	0.301	37.0	1.379	0.409	68.0
UberTLC_daily	1.466	0.699	55.0	0.883	0.401	19.0	0.616	0.304	3.0	1.439	0.753	54.0	2.429	1.258	90.5	1.445	0.690	52.0	2.548	1.280	92.0	0.927	0.389	19.0	2.047	1.173	79.0
UberTLC_hourly	0.713	0.481	24.0	0.685	0.499	23.0	0.593	0.442	10.0	0.568	0.389	6.5	0.950	0.666	64.0	0.863	0.632	56.5	1.014	0.743	71.5	0.892	0.631	58.5	0.702	0.544	32.0
VehicleTrips	1.244	0.973	39.5	1.446	1.068	63.5	1.191	0.814	17.5	1.116	0.801	13.5	1.481	1.143	78.0	1.467	1.027	62.0	1.978	1.290	93.5	1.383	1.052	55.0	1.450	0.912	55.5
Weather	0.823	0.724	45.0	0.762	0.697	9.0	0.753	0.701	11.0	0.776	0.654	12.0	0.829	0.735	52.0	0.768	0.722	24.5	0.831	0.839	76.5	0.770	0.691	12.5	0.828	0.771	59.5
WebTraffic	0.832	0.428	3.0	0.886	0.476	27.5	0.880	0.473	25.5	0.822	*0.431*	2.5	0.860	0.450	9.5	1.259	0.640	75.0	1.346	0.857	89.0	0.875	0.477	25.5	1.421	0.623	78.0
Wike2000	0.912	0.384	12.0	1.056	0.441	42.0	0.977	0.397	24.5	0.847	0.348	2.0	1.089	0.521	57.0	1.571	0.566	88.0	1.979	0.915	94.0	0.978	0.432	32.5	1.350	0.526	76.0
Wind_Farms	1.161	0.338	41.5	1.090	0.325	34.0	0.983	0.299	16.5	1.038	0.275	20.5	0.944	*0.246*	3.5	1.447	0.481	74.0	1.307	0.548	74.0	1.059	0.310	28.0	1.751	0.603	84.5
ausbeer	1.818	0.407	48.5	3.146	0.698	64.0	0.932	0.239	18.0	1.128	0.272	24.0	3.322	0.693	66.5	3.281	0.849	75.5	3.471	0.748	71.5	4.139	0.921	92.0	1.383	0.326	31.5
ecdc_ili	0.810	1.213	35.0	0.969	1.388	76.5	0.861	1.214	40.0	0.747	1.133	21.0	0.739	1.136	20.0	0.731	1.098	13.5	0.966	1.395	77.0	0.878	1.333	55.5	0.761	1.103	18.5
favorita_stores_1D	0.885	0.931	68.0	0.823	0.866	32.0	0.828	0.854	29.5	0.809	0.824	14.0	0.915	0.957	78.0	0.856	0.903	59.5	0.910	1.024	82.0	0.848	0.900	57.0	0.839	0.891	50.5
favorita_stores_1M	1.045	0.739	51.5	1.001	0.737	41.5	1.011	0.721	41.0	0.967	0.677	6.0	1.051	0.815	60.5	1.142	0.952	68.5	1.498	1.356	92.0	1.007	0.700	28.5	1.171	0.974	70.5
favorita_stores_1W	0.701	0.695	4.5	0.794	0.745	52.5	0.731	0.726	27.5	0.697	**0.686**	2.5	0.806	0.882	76.0	0.783	0.789	56.5	1.263	1.266	95.5	0.720	0.713	12.0	0.831	0.859	76.5
favorita_transactions_1D	0.930	0.679	63.5	0.851	0.625	43.0	*0.776*	0.559	3.5	0.817	0.602	27.5	1.268	0.940	92.0	0.929	0.670	62.0	1.127	0.851	90.0	0.939	0.682	65.0	0.870	0.597	36.0
favorita_transactions_1M	1.895	0.928	77.5	1.523	0.833	43.5	1.505	0.768	21.0	1.715	0.810	54.5	1.848	0.937	76.5	1.441	0.972	47.5	1.957	1.227	89.5	1.603	0.800	40.0	1.587	0.982	62.5
favorita_transactions_1W	0.956	0.585	27.0	1.016	0.601	44.5	0.949	0.556	11.0	0.936	0.572	16.5	1.124	0.708	64.5	1.188	0.650	63.5	2.480	1.265	96.0	0.981	0.579	31.0	1.450	0.730	83.0
heart_rate	1.074	1.085	43.5	1.231	1.149	81.0	1.102	1.049	40.0	1.053	1.091	41.5	1.120	1.100	60.5	1.096	1.113	58.5	1.134	1.174	76.5	1.238	1.186	87.0	1.089	1.085	46.0
kdd_cup_2022_10T	0.819	0.404	44.0	0.755	0.377	20.5	0.763	0.384	26.5	0.796	0.392	36.0	0.760	0.400	34.0	0.840	0.416	50.0	0.941	0.459	79.0	0.777	0.388	30.5	0.906	0.441	64.0
kdd_cup_2022_1D	0.916	1.213	43.5	0.908	1.266	57.0	0.878	1.256	40.5	0.856	1.205	20.0	0.905	1.257	48.5	**0.812**	1.259	17.5	0.892	1.343	64.0	0.850	1.221	19.0	0.931	1.397	86.5
kdd_cup_2022_30T	0.820	0.938	13.5	0.842	0.998	33.5	0.835	1.011	34.0	0.821	**0.863**	7.5	0.843	0.866	15.0	0.979	1.233	90.0	0.895	1.228	80.5	0.824	0.999	29.0	0.915	1.166	78.5
physionet_2012	*0.728*	0.961	6.0	0.831	1.037	64.5	0.792	0.993	33.0	0.747	**0.938**	7.5	0.732	1.077	42.5	0.831	1.087	77.5	0.978	1.285	92.5	0.788	1.002	35.5	0.812	1.071	69.5
rossmann_1D	1.285	1.050	69.0	0.999	0.859	46.5	0.701	0.626	15.0	1.084	0.885	51.5	1.346	1.147	81.5	1.081	0.959	55.0	1.376	1.220	89.0	1.320	1.130	76.5	0.643	0.611	8.0
rossmann_1W	0.978	1.385	63.5	1.218	1.248	59.0	1.149	1.253	54.0	1.197	1.414	90.0	0.819	1.308	40.0	0.855	1.300	42.0	1.088	1.303	60.0	1.157	1.245	51.5	0.827	1.177	6.5
us_gasoline	0.902	0.563	61.0	0.884	0.558	52.0	0.828	0.517	14.5	0.844	0.535	22.0	0.883	0.664	75.0	*0.781*	0.503	5.5	1.029	0.670	90.5	1.040	0.643	89.0	0.875	0.589	60.5

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 18: **Multivariate forecasting results.** Part 1.

Dataset	arima			autoformer			autoregressive			bilstm			chronos			chronos_bolt			chronos_t5			cnn			crossformer			cycleitransformer		
	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.718	1.634	60.5	0.810	1.282	64.0	0.663	1.249	50.0	0.651	0.886	35.0	*0.565*	0.715	6.0	0.617	0.850	26.0	0.654	0.947	39.0	0.738	1.141	60.0	0.848	1.323	65.0	0.576	0.695	6.0
Bike_Sharing	0.953	2.242	71.0	0.662	0.895	51.5	0.648	0.925	52.5	0.501	0.652	6.0	0.491	0.808	20.0	**0.489**	0.761	14.5	0.611	0.992	51.5	0.556	0.741	27.0	0.496	0.637	3.0	0.527	0.695	14.0
COVID-19_Deaths	**1.235**	55.007	1.5	5.158	252.143	59.0	143.124	$> 10^6$	73.0	2.579	124.780	39.0	1.322	55.543	4.5	1.964	101.515	29.0	2.135	95.992	31.0	2.387	107.636	36.0	12.459	$> 10^3$	67.0	1.788	81.797	22.0
Czelan	0.275	0.290	54.0	0.362	0.312	65.5	0.281	0.349	60.0	0.231	0.191	9.0	**0.190**	**0.175**	1.0	0.255	0.262	45.5	0.237	0.220	27.0	0.248	0.221	31.5	0.325	0.271	61.0	0.224	0.206	10.5
DailyDelhiClimate	1.387	45.574	13.5	1.673	47.044	23.0	1.815	60.436	36.5	2.907	91.758	60.0	1.041	43.806	6.5	1.288	56.871	23.5	1.151	44.794	9.5	2.176	65.802	44.0	**0.954**	**42.733**	1.0	2.795	120.657	64.5
Dominick	0.663	1.384	9.5	0.886	1.710	53.5	0.798	2.299	59.5	0.698	1.431	22.0	0.559	1.375	4.5	0.601	*1.344*	5.0	0.796	1.981	55.0	0.722	1.479	33.0	0.905	1.819	56.0	0.675	1.412	15.0
ETTH1	0.662	1.198	70.0	0.480	0.485	44.0	0.412	0.414	4.5	0.455	0.452	33.0	0.418	0.462	21.0	0.414	0.449	15.5	0.455	0.547	44.0	0.488	0.528	48.0	0.523	0.557	54.0	0.438	0.427	25.5
ETTH2	0.481	0.553	63.0	0.422	0.377	49.0	0.362	**0.303**	1.5	0.388	0.353	27.5	*0.363*	0.339	8.5	0.393	0.401	43.0	0.424	0.476	57.5	0.413	0.391	47.5	0.716	1.028	67.0	0.388	0.347	24.5
ETTM1	0.804	1.561	70.0	0.608	0.739	57.0	0.605	0.797	60.0	0.482	0.511	27.5	0.495	0.674	40.5	0.536	0.769	52.5	0.637	0.931	64.5	0.476	0.505	22.5	*0.447*	*0.436*	3.0	0.458	0.480	11.0
ETTM2	0.326	0.243	64.0	0.295	0.178	36.5	0.294	0.181	37.0	0.273	0.166	7.0	0.278	0.186	26.0	0.290	0.194	37.5	0.310	0.212	54.0	0.276	0.172	13.0	0.323	0.223	60.0	0.279	0.174	19.5
FRED-MD	1.246	85.306	26.0	2.014	114.204	58.0	$> 10^3$	$> 10^6$	72.0	1.666	69.881	37.0	1.297	75.680	26.5	1.214	51.492	4.5	1.287	56.686	11.0	1.539	65.336	31.0	4.244	330.662	66.0	1.332	67.852	24.5
ILY	1.269	4.133	46.5	1.422	3.762	50.0	1.191	3.296	33.5	1.150	3.291	30.0	1.121	3.235	26.5	1.218	3.743	40.0	1.161	3.170	28.5	1.122	3.002	23.0	1.510	4.861	60.5	1.010	2.361	4.5
KDDCup2018	0.753	1.282	70.0	0.632	0.871	53.5	0.596	0.967	40.5	0.593	0.790	15.0	0.582	0.790	10.5	**0.478**	**0.590**	1.0	0.656	0.928	62.5	0.620	0.858	49.5	0.648	0.873	57.5	0.611	0.845	42.5
METR-LA	0.735	1.637	62.0	0.825	1.465	63.0	$> 10^4$	$> 10^6$	72.0	0.610	1.076	10.0	0.637	1.376	35.5	0.711	1.515	56.5	0.727	1.669	61.0	0.654	1.125	29.5	0.605	1.043	5.0	0.648	1.144	27.5
MIT-BIH	0.858	3.618	72.0	0.663	1.757	45.5	0.612	*1.363*	26.0	0.562	2.088	26.5	0.533	1.442	5.5	0.609	1.717	34.5	0.627	1.800	44.0	0.613	2.549	54.5	0.536	1.415	5.0	0.599	2.843	49.5
Metro_Interstate_Traffic_Volume	0.514	0.781	71.0	0.357	0.310	29.0	0.339	0.330	27.0	0.340	0.318	23.0	0.306	0.383	30.5	*0.300*	0.338	21.5	0.322	0.361	31.5	0.350	0.326	30.5	0.345	0.300	19.5	0.333	0.312	15.5
MexicoCityBikes	0.849	1.280	32.5	0.849	1.280	30.5	0.856	1.303	54.5	0.848	1.278	12.0	**0.845**	1.300	24.5	0.849	1.310	44.0	0.930	1.614	69.0	0.848	1.278	14.5	0.848	*1.274*	12.0	0.848	1.280	27.0
NASDAQ	*1.216*	3.644	6.0	1.715	5.380	62.5	1.839	18.869	71.0	1.334	3.904	11.5	1.354	4.399	23.0	1.278	3.837	10.5	1.342	3.912	12.5	1.678	5.542	61.5	**1.138**	**2.363**	1.0	1.409	4.135	22.5
NYSE	**0.354**	0.451	3.0	0.468	0.585	54.0	0.377	0.599	34.0	0.360	**0.442**	1.5	0.372	0.461	9.0	0.372	0.461	9.0	0.380	0.513	28.0	0.399	0.489	30.5	0.900	2.586	68.0	0.387	0.482	23.5
National_Stock_Exchange	0.583	*0.651*	3.5	0.890	1.544	59.5	0.717	1.128	23.0	0.690	1.056	16.5	0.642	0.873	9.5	0.650	0.906	10.5	0.618	0.852	7.0	0.792	1.317	43.0	0.806	1.340	47.0	0.711	1.231	25.0
Oikolab	0.989	1.741	72.0	0.634	0.673	50.0	0.588	0.612	11.0	0.606	0.633	28.0	0.622	0.731	53.5	0.648	0.781	62.5	0.627	0.730	53.5	0.603	0.634	26.5	0.554	0.535	2.0	0.609	0.648	35.0
PEMS-BAY	0.702	3.808	68.0	0.617	1.138	53.0	$> 10^6$	$> 10^6$	73.0	0.413	0.818	14.0	0.584	1.384	56.0	0.657	1.618	63.5	0.627	1.487	58.5	0.449	0.851	22.5	**0.330**	0.579	1.5	0.456	0.891	27.0
PEMS03	0.677	0.807	61.5	0.413	0.315	43.0	$> 10^6$	$> 10^6$	73.0	0.254	0.161	5.5	0.507	0.542	55.5	0.678	0.873	63.5	0.664	0.770	60.0	0.288	0.194	18.0	0.227	0.131	2.0	0.300	0.205	24.0
PEMS04	0.854	1.430	68.5	0.365	0.239	35.0	0.774	1.653	67.5	0.247	0.137	16.0	0.558	0.611	56.0	0.810	1.249	65.5	0.752	0.998	62.0	0.248	0.133	16.0	**0.213**	**0.106**	1.0	0.291	0.189	26.5
PEMS07	0.656	0.883	63.5	0.350	0.227	33.0	$> 10^6$	$> 10^6$	73.0	0.240	0.145	10.0	0.597	0.693	58.0	0.720	0.951	66.0	0.650	0.768	60.0	0.241	0.146	11.0	**0.201**	**0.106**	1.0	0.280	0.175	22.5
PEMS08	0.835	1.308	66.0	0.482	0.485	49.0	0.784	1.445	66.0	0.265	0.237	7.0	0.578	0.677	58.5	0.824	1.365	66.5	0.789	1.187	64.0	0.337	0.300	27.0	**0.228**	0.222	1.5	0.328	0.290	25.5
Power_Consumption_of_Tetouan_City	0.609	0.856	67.0	0.501	0.447	55.5	0.509	0.512	59.0	0.293	0.248	8.5	0.351	0.418	38.5	0.463	0.541	57.0	0.497	0.556	59.5	0.310	0.276	19.5	0.324	0.248	16.5	0.298	0.257	11.5
Residential_Power_and_Battery_Data	0.859	2.330	63.0	0.898	1.899	59.5	0.969	9.539	71.0	0.767	1.743	24.0	0.786	2.115	52.5	0.802	2.092	54.0	0.838	2.284	59.0	0.770	1.749	28.5	0.743	*1.621*	3.5	0.753	1.768	23.0
Rideshare	0.268	*0.224*	2.5	0.709	0.881	49.0	0.988	$> 10^3$	71.5	0.546	0.631	28.5	0.316	0.310	12.0	0.304	0.278	8.5	0.303	0.260	6.5	0.658	0.928	49.0	0.738	1.103	58.0	0.361	0.309	13.0
Sleep_EDFX	1.253	8.675	71.0	0.844	5.897	65.0	0.685	6.431	46.5	0.708	4.982	32.5	0.719	4.599	23.0	0.816	5.745	62.0	0.714	4.742	26.0	0.736	4.929	34.0	0.653	5.016	24.0	0.644	4.716	10.0
Spanish_energy	0.628	1.077	70.0	0.461	0.483	57.5	0.400	0.414	30.0	0.394	0.394	20.0	0.411	0.476	46.5	0.376	0.415	20.0	0.445	0.502	57.5	0.414	0.422	42.5	0.408	0.371	19.5	0.391	0.388	13.5
Spanish_weather	0.523	0.725	71.5	0.361	0.328	46.5	0.327	0.313	17.5	0.322	0.316	16.0	0.321	0.352	29.5	*0.316*	0.344	25.5	0.364	0.399	59.0	0.332	0.333	38.5	0.331	**0.297**	16.5	0.319	0.314	11.5
Temperature-Rain	0.960	79.047	66.5	0.916	4.419	52.0	$> 10^6$	$> 10^6$	71.0	0.819	4.253	23.0	0.727	4.961	32.0	0.725	4.965	32.0	0.874	5.433	58.5	0.804	4.194	10.0	0.803	*4.161*	7.5	0.850	4.392	42.5
Traffic	1.076	3.956	70.0	0.386	0.642	43.5	$> 10^6$	$> 10^6$	73.0	0.332	0.638	32.5	0.278	0.519	10.5	0.300	0.689	30.5	0.340	0.822	41.5	0.377	0.663	44.5	0.295	0.553	16.0	0.296	0.433	7.5
Traffic_Flow_Forecasting	1.066	1.873	69.0	0.699	0.789	51.0	0.756	0.865	54.0	0.401	0.314	22.5	0.685	1.006	53.5	0.674	0.965	51.5	0.709	0.915	54.0	0.432	0.365	27.5	**0.294**	**0.172**	1.0	0.354	0.266	11.0
WIND	0.779	1.535	66.0	0.791	1.314	65.0	$> 10^6$	$> 10^6$	72.5	0.663	1.048	17.5	0.693	1.207	43.5	0.711	1.244	52.0	0.706	1.235	49.0	0.711	1.181	47.5	**0.627**	0.955	1.5	0.685	1.088	30.5
Zafnoo_env	0.801	1.471	72.0	0.493	0.525	52.0	0.405	0.449	4.5	0.433	0.463	18.0	0.420	0.555	31.5	*0.409*	0.509	23.0	0.418	0.532	28.0	0.470	0.522	48.0	0.441	0.468	28.0	0.447	0.489	40.0
Zafnoo_sapf	0.538	0.868	72.0	0.370	0.371	59.5	0.291	0.302	31.5	0.293	0.307	36.5	**0.248**	0.270	1.5	*0.251*	0.277	4.0	0.294	0.378	48.0	0.296	0.304	37.5	0.282	0.288	18.5	0.287	0.299	27.5
australian_tourism	0.780	0.960	17.5	1.009	1.566	64.5	1.542	26.204	73.0	0.829	1.057	33.0	**0.624**	**0.725**	1.0	0.734	0.961	13.5	0.669	0.811	4.0	0.843	1.089	41.0	0.762	0.916	10.5	0.827	1.091	37.0
beijing_multisite_air_quality	0.564	1.080	68.5	0.493	0.773	56.0	46.158	$> 10^6$	72.0	0.443	0.702	10.0	0.449	0.772	36.0	0.452	0.837	46.0	0.511	0.893	66.0	0.462	0.737	40.0	0.436	*0.669*	3.5	0.449	0.709	21.5
boomlet_1062	0.772	1.053	60.5	0.699	0.925	36.5	0.690	0.958	37.0	0.681	0.898	18.5	**0.602**	1.006	25.5	0.636	0.999	27.5	0.772	1.817	67.5	0.689	0.922	31.5	0.718	0.990	47.0	0.678	0.893	12.5
boomlet_1209	0.420	1.869	49.0	0.490	1.838	51.5	$> 10^6$	$> 10^6$	73.0	0.403	1.773	37.5	0.365	1.722	21.5	0.374	1.753	27.5	0.464	1.856	50.5	0.401	1.782	38.0	0.634	4.970	65.5	0.360	1.606	12.0
boomlet_1225	0.293	0.781	19.0	0.417	0.934	50.0	0.363	11.309	60.0	0.352	0.916	44.5	0.255	0.743	5.0	0.267	0.734	6.5	0.272	0.790	14.5	0.425	1.342	55.0	0.719	6.393	64.0	0.290	0.751	13.0
boomlet_1230	4.103	580.211	12.0	5.647	640.093	55.5	$> 10^6$	$> 10^6$	72.0	4.224	624.300	32.5	4.069	602.028	19.0	4.140	624.740	28.0	4.247	698.867	48.5	4.284	635.406	45.0	9.246	$> 10^3$	66.0	4.020	589.971	12.5
boomlet_1282	0.471	7.869	57.5	0.468	1.191	47.0	$> 10^3$	$> 10^6$	72.0	0.437	1.096	28.5	0.425	1.221	33.0	0.435	1.137	30.0	0.443	1.289	47.0	0.439	1.160	36.5	1.013	8.678	66.0	0.408	0.997	3.0
boomlet_1487	0.254	0.793	53.0	0.243	0.755	46.5	16.201	$> 10^6$	72.0	0.217	0.735	23.5	0.190	0.735	9.0	0.192	0.736	12.0	0.249	1.004	58.5	0.233	0.750	42.5	0.287	0.766	52.0	0.207	0.736	19.0
boomlet_1631	0.595	0.864	45.5	0.683	0.894	56.5	0.551	0.687	21.5	0.618	0.818	49.0	0.548	0.673	17.5	0.557	0.720	29.5	0.572	0.770	36.5	0.640	0.865	52.5	0.874	1.477	67.0	0.538	0.644	9.0
boomlet_1676	0.575	1.698	7.0	1.005	7.778	45.5	566.905	$> 10^6$	71.5	0.901	9.606	42.5	0.671	4.474	30.0	*0.554*	**1.519**	2.0	0.576	2.437	16.0	0.995	13.016	50.0	3.694	265.842	61.0	0.598	2.191	18.0
boomlet_1855	0.629	92.036	26.5	0.728	92.141	35.5	0.717	369.330	55.0	0.884	96.260	57.5	0.486	91.274	3.0	0.495	91.256	3.0	0.562	91.998	16.5	1.062	100.243	65.5	0.896	93.119	49.5	0.644	93.164	38.5
boomlet_1975	0.568	0.820	52.5	0.508	0.567	22.0	$> 10^6$	$> 10^6$	72.0	0.493	0.535	19.5	0.552	1.169	59.0	0.530	0.938	48.0	0.533	1.037	50.5	0.590	0.798	55.5	0.402	0.482	10.5	*0.317*	*0.309*	3.0
boomlet_2187	0.779	**22.071**	19.0	0.961	46.352	63.0	1.092	624.010	71.0	0.782	43.032	47.5	0.655	34.123	15.0	*0.629*	28.167	4.0	0.708	26.486	10.0	0.801	41.549	50.5	0.741	40.057	34.5	0.686	32.449	13.5
boomlet_285	0.321	0.335	52.5	0.386	0.402	65.0	0.306	0.315	44.0	0.269	0.301	25.5	**0.190**	0.286	10.0	0.239	0.469	36.0	0.386	0.499	66.5	0.285	0.300	32.0	0.280	0.284	25.0	0.258	0.285	17.0
boomlet_619	0.853	1.432	69.5	0.445	0.462	48.0	0.629	0.711	60.0	0.319	0.319	11.0	0.365	0.463	41.5	0.690	1.011	65.5	0.827	1.658	70.0	0.363	0.343	31.0	0.310	0.303	2.0	0.329	0.315	12.5
boomlet_772	0.382	1.561	54.5	0.469	1.164	53.5	$> 10^6$	$> 10^6$	72.0	0.458	1.505	60.5	0.294	1.039	7.5	0.321	1.072	21.0	0.344	1.213	40.5	0.437	1.228	56.5	0.413	1.162	49.5	0.326	1.039	15.0
boomlet_963	0.816	3.962	67.5	0.740	1.799	58.5	$> 10^6$	$> 10^6$	72.0	0.651	1.601	38.0	**0.583**	1.495	7.0	0.701	2.429	60.5	0.678	1.895	55.5	0.791	2.174	64.0	0.653	1.603	39.5	0.650	1.660	42.0
electricity_2016	1.005	1.833	71.5	0.314	0.204	46.5	$> 10^4$	$> 10^6$	73.0	0.284	0.184	33.5	0.258	0.195	27.5	0.246	0.175	11.5	0.277	0.223	40.0	0.314	0.206	47.0	*0.245*	0.150	2.5	0.250	0.156	7.5
electricityloaddiagrams20112014	6.076	$> 10^3$	57.0	5.715	$> 10^3$	50.0	$> 10^6$	$> 10^6$	73.0	4.541	$> 10^3$	42.5	2.173	715.810	27.0	1.738	425.363	19.5	1.914	351.480	19.5	4.563	$> 10^3$	47.0	15.636	$> 10^4$	63.0	1.650	318.589	7.0
gasrate_co2	1.060	1.707	67.0	0.879	1.143	46.5	0.544	0.477	2.0	0.935	1.345	63.0	0.807	1.019	23.5	0.830	1.077	32.0	0.691	0.786	10.0	0.889	1.190	53.5	0.707	0.735	9.5	0.770	0.917	15.5
gefcom2012_load	1.596	7.693	33.0	1.532	7.588	24.5	1.403	7.178	6.0	2.522	13.596	72.5	1.458	8.048	31.5	1.450	7.902	27.0	1.577	7.600	29.0	1.426	7.205	10.5	1.461	7.743	24.0	1.704	8.906	55.5
gefcom2012_wind	1.106	2.162	71.0	0.917	1.202	56.5	0.875	*1.077*	16.5	0.867	1.105	15.5	0.885	1.281	52.5	0.941	1.435	67.0	0.900	1.429	59.5	0.880	1.158	42.0	0.855	1.057	2.0	0.875	1.126	33.5
gvar	**0.226**	**0.188**	1.0	0.413	0.370	57.5	0.260	0.248	23.0	0.286	0.250	36.0	*0.235*	0.206	3.5	0.240	0.212	5.0	0.265	0.232	18.0	0.290	0.247	35.5	0.754	1.175	63.5	0.262	0.236	19.5
italy_air_quality	0.674	1.590	64.0	0.676	1.330	59.5	785.128	$> 10^6$	72.0	0.599	1.242	43.0	0.535	1.251	26.5	*0.530*	1.322	28.0	0.519	1.186	17.0	0.618	1.238	46.0	0.611	1.216	43.0	0.568	1.109	15.0
vessel_ais	0.792	1.051	57.0	0.783	1.041	53.0	0.819	1.112	67.0	0.756	0.973	39.0	0.769	1.009	42.0	0.800	1.079	61.0	0.812	1.111	66.0	0.755	0.970	38.0	0.679	0.802	12.0	0.679	0.803	11.0
weather_Wetterstation	0.294	0.337	52.5	0.388	0.363	65.0	0.366	16.174	68.0	0.282	0.227	31.0	0.241	0.203	3.5	0.274	0.249	32.0	0.281	0.242	34.5	0.314	0.310	55.5	0.279	0.206	16.5	0.267	0.220	16.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 19: **Multivariate forecasting results.** Part 2.

Dataset	cyclenet			dlinear			duet			elasticnet			ets			etsformer			fact			fedformer			film			fits		
	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.598	0.740	13.5	0.694	0.917	43.5	0.607	0.756	17.0	0.692	0.933	44.0	3.155	98.746	72.5	0.636	0.783	24.0	0.579	0.713	8.0	0.737	1.050	56.0	0.629	0.809	24.5	0.624	0.800	22.0
Bike_Sharing	0.559	0.758	30.5	0.593	0.799	39.5	0.516	0.684	11.0	0.518	0.661	10.0	1.015	15.005	72.0	0.660	0.900	51.5	0.533	0.690	16.0	0.644	0.822	46.5	0.645	0.890	49.0	0.833	1.339	67.5
COVID-19_Deaths	4.558	230.801	55.0	4.910	256.349	58.5	1.545	82.136	17.5	6.872	768.704	63.0	1.291	61.642	5.0	6.794	232.567	58.5	*1.309*	*55.312*	3.0	5.102	253.453	59.0	4.434	224.171	53.5	4.654	247.538	56.5
Czelan	0.234	0.211	19.0	0.303	0.235	48.0	0.230	0.215	17.5	0.317	0.242	53.0	0.847	5.904	72.5	0.278	0.203	30.0	*0.216*	*0.188*	3.0	0.381	0.326	66.5	0.277	0.269	52.0	0.245	0.219	29.5
DailyDelhiClimate	1.672	51.726	26.5	1.405	44.915	13.5	2.502	100.360	57.0	1.563	56.440	27.0	2.462	60.927	45.0	3.137	79.408	58.0	2.089	71.904	44.0	1.641	46.712	20.5	2.115	72.298	45.0	1.499	45.815	16.0
Dominick	0.713	1.459	31.0	0.778	1.489	40.5	0.705	1.445	27.5	1.007	2.704	68.0	0.739	1.846	50.5	0.968	2.035	60.5	0.674	1.389	13.5	0.871	1.696	52.0	0.727	1.516	40.0	0.725	1.458	34.5
ETTH1	0.416	0.413	7.0	0.423	0.416	13.5	0.414	0.402	4.0	0.422	0.416	12.0	0.802	5.675	73.0	0.525	0.563	55.0	0.437	0.433	25.5	0.474	0.474	41.0	0.433	0.440	25.5	0.573	0.719	63.0
ETTH2	0.379	0.344	18.5	0.416	0.370	43.0	0.371	*0.322*	6.0	0.524	0.605	64.0	1.053	7.890	70.5	0.447	0.404	56.5	0.373	0.340	13.5	0.410	0.364	39.5	0.371	0.335	9.5	0.404	0.368	38.5
ETTM1	0.461	0.516	20.0	0.483	0.555	33.5	0.452	0.475	8.0	0.468	0.505	19.5	1.748	27.711	73.0	0.577	0.674	51.0	0.448	0.455	5.0	0.532	0.555	40.5	0.499	0.612	38.0	0.510	0.633	40.5
ETTM2	0.276	0.170	11.0	0.305	0.202	51.5	0.280	0.177	25.5	0.323	0.226	61.0	0.526	1.304	73.0	0.324	0.209	58.5	0.263	0.157	2.0	0.274	*0.160*	6.0	0.279	0.174	22.0	0.281	0.172	19.5
FRED-MD	1.599	83.129	41.5	2.010	115.657	58.5	1.411	85.178	35.5	1.781	105.125	52.0	1.800	254.456	58.0	3.163	73.263	48.0	1.333	72.258	28.0	1.951	113.883	56.0	1.624	84.613	43.0	1.706	93.130	49.0
ILY	1.243	3.173	34.5	1.529	4.374	57.0	1.085	2.753	17.0	1.067	2.794	16.0	3.005	36.912	73.0	1.339	3.599	46.0	1.026	2.486	7.5	1.244	3.101	34.0	1.479	4.075	52.0	1.563	4.606	62.0
KDDCup2018	0.595	0.801	18.5	0.611	0.771	21.5	0.588	0.787	10.5	0.604	0.764	17.5	0.892	6.356	72.0	0.648	0.854	54.0	0.601	0.807	28.0	0.611	0.828	39.5	0.622	0.835	47.0	0.603	0.787	21.0
METR-LA	0.650	1.113	24.5	0.672	*1.023*	24.5	0.665	1.158	36.0	0.672	1.047	26.5	0.861	2.337	67.5	0.753	1.290	57.0	0.631	1.110	19.0	0.761	1.214	54.5	0.651	1.105	23.5	0.665	1.094	29.5
MIT-BIH	0.617	2.909	60.0	0.588	2.080	35.5	0.604	2.523	47.5	0.568	1.644	16.0	7.249	$> 10^3$	73.0	0.612	1.527	33.0	0.598	3.113	52.0	0.643	2.011	52.0	0.607	1.857	39.0	0.595	1.846	32.5
Metro_Interstate_Traffic_Volume	0.353	0.337	36.5	0.352	0.311	24.0	0.348	0.329	31.0	0.356	0.314	31.5	0.844	13.672	72.5	0.466	0.412	64.5	0.335	0.312	16.0	0.340	0.296	14.5	0.376	0.372	55.5	0.442	0.470	65.0
MexicoCityBikes	0.850	1.287	40.5	0.849	1.276	21.5	0.848	1.278	10.5	0.850	1.276	24.5	0.865	1.334	62.0	0.858	1.306	56.5	0.848	1.278	8.0	0.848	1.279	24.5	0.858	1.301	54.0	0.848	1.278	18.0
NASDAQ	1.633	5.049	51.5	1.674	5.290	57.5	1.509	4.726	42.5	1.471	4.199	30.5	1.442	4.941	39.5	1.451	4.409	33.5	1.401	4.132	21.5	1.748	5.481	64.0	1.517	4.211	34.5	1.652	5.256	54.0
NYSE	0.410	0.498	38.0	0.488	0.605	57.5	0.393	0.490	29.5	0.443	0.557	47.0	0.386	0.539	32.5	0.410	0.493	35.5	0.376	0.460	11.0	0.456	0.566	50.0	0.451	0.565	49.0	0.462	0.572	51.5
National_Stock_Exchange	0.800	1.317	43.5	0.831	1.368	50.0	0.728	1.278	29.5	1.418	5.533	72.0	0.668	0.848	9.5	0.945	1.528	60.0	0.732	1.199	25.5	0.851	1.383	53.5	0.832	1.616	56.5	0.805	1.337	46.0
Oikolab	0.611	0.652	39.5	0.599	0.614	15.0	0.597	0.625	17.5	0.600	0.620	18.0	1.718	18.645	73.0	0.602	0.601	16.0	0.592	0.612	12.0	0.630	0.664	47.0	0.617	0.662	43.0	0.614	0.658	41.5
PEMS-BAY	0.483	0.999	38.0	0.508	0.964	40.5	0.419	0.755	11.5	0.516	1.000	43.0	1.907	55.678	71.0	0.573	0.991	46.5	0.410	0.731	7.0	0.618	1.212	55.0	0.512	1.065	43.5	0.537	1.083	45.5
PEMS03	0.374	0.316	41.0	0.388	0.291	40.5	0.281	0.181	16.0	0.417	0.321	44.5	1.233	4.586	71.0	0.370	0.243	35.0	0.309	0.213	27.0	0.702	1.083	66.0	0.425	0.399	48.0	0.428	0.390	47.0
PEMS04	0.386	0.326	45.0	0.368	0.274	38.5	0.264	0.143	18.5	0.407	0.319	45.5	1.483	8.093	72.0	0.301	0.174	26.5	0.278	0.162	21.5	0.360	0.247	35.0	0.424	0.394	50.0	0.438	0.406	51.0
PEMS07	0.388	0.336	42.5	0.394	0.321	42.5	0.234	*0.117*	4.5	0.531	0.541	53.5	1.241	4.563	72.0	0.318	0.190	29.5	0.295	0.186	27.0	0.476	0.474	50.0	0.430	0.388	47.5	0.446	0.393	48.5
PEMS08	0.408	0.425	41.5	0.415	0.412	42.5	0.270	**0.198**	4.0	0.437	0.444	44.5	1.266	4.558	72.0	0.381	0.317	32.5	0.286	0.231	8.5	0.380	0.348	35.5	0.478	0.533	52.5	0.476	0.526	51.0
Power_Consumption_of_Tetouan_City	0.314	0.278	22.0	0.375	0.313	35.5	0.301	0.249	11.0	0.362	0.301	33.0	2.873	52.368	73.0	0.438	0.345	42.5	0.272	0.226	2.0	0.467	0.396	49.0	0.399	0.387	41.5	0.407	0.394	43.0
Residential_Power_and_Battery_Data	0.770	1.784	32.5	0.763	1.654	12.5	0.792	1.820	46.0	0.760	1.635	9.5	1.043	3.250	71.0	0.814	1.761	42.5	0.765	1.779	28.5	0.879	1.892	57.0	0.812	1.843	49.0	0.845	1.823	50.5
Rideshare	0.616	0.813	41.0	0.593	0.663	32.5	0.538	0.685	31.0	0.601	0.972	44.0	0.323	0.275	10.0	0.464	0.420	21.0	0.437	0.428	19.0	0.726	1.032	55.0	0.560	0.683	32.5	0.591	0.730	34.5
Sleep_EDFX	0.716	5.009	35.5	0.692	4.880	24.0	0.628	4.674	7.5	0.697	4.958	27.0	2.243	50.627	72.0	0.779	5.126	48.0	0.647	4.709	10.0	0.798	5.696	60.0	0.768	5.340	51.5	0.746	4.989	39.5
Spanish_energy	0.398	0.403	24.0	0.400	0.379	17.0	*0.379*	0.367	2.5	0.406	0.388	23.0	0.933	14.514	71.5	0.465	0.449	54.0	0.379	0.370	4.0	0.418	0.413	41.0	0.409	0.420	38.0	0.401	0.407	30.5
Spanish_weather	0.326	0.321	24.5	0.337	*0.300*	20.5	0.315	0.302	3.0	0.348	0.305	25.5	0.431	0.638	69.5	0.387	0.342	55.0	0.318	0.312	7.0	0.354	0.332	45.5	0.341	0.337	43.0	0.332	0.321	33.0
Temperature-Rain	0.830	4.263	28.0	0.851	4.197	28.0	0.831	4.482	43.0	0.848	4.872	50.5	1.104	182.200	69.5	1.107	5.933	67.5	0.832	4.321	32.0	0.900	4.369	48.0	0.894	4.511	56.5	0.855	4.240	35.0
Traffic	0.347	0.534	27.0	0.382	0.606	40.0	0.292	0.449	6.0	0.511	1.017	59.5	1.430	41.251	71.0	0.517	0.881	58.5	0.297	*0.432*	7.5	0.400	0.647	46.0	0.348	0.572	29.5	0.356	0.579	34.0
Traffic_Flow_Forecasting	0.530	0.565	37.5	0.756	0.882	55.0	0.331	0.215	5.0	0.409	0.303	21.0	2.175	12.841	73.0	0.636	0.662	42.5	0.345	0.240	6.5	0.522	0.479	35.0	0.838	1.092	61.0	0.962	1.361	65.0
WIND	0.685	1.084	29.5	0.656	1.007	10.0	0.661	1.033	13.5	0.651	1.005	6.5	2.112	35.340	71.0	0.695	1.040	27.0	0.651	1.023	10.0	0.713	1.120	45.0	0.715	1.145	46.5	0.709	1.118	41.5
Zafnoo_env	0.435	0.478	29.0	0.426	0.447	10.5	0.431	0.469	21.5	0.423	0.442	6.5	0.832	12.700	73.0	0.472	0.493	45.0	0.431	0.469	20.5	0.456	0.490	41.0	0.476	0.515	48.5	0.699	0.822	68.5
Zafnoo_sapf	0.281	0.288	18.5	0.279	0.283	12.5	0.272	0.282	8.5	0.270	*0.275*	5.0	0.351	0.357	56.5	0.365	0.359	58.0	0.279	0.285	15.5	0.363	0.354	55.5	0.313	0.303	42.0	0.492	0.571	69.0
australian_tourism	0.800	0.994	25.5	0.841	1.101	41.0	0.801	0.981	25.0	1.067	2.547	71.0	0.822	1.210	39.5	1.065	1.809	70.0	0.769	0.958	15.5	1.023	1.558	64.5	0.763	0.928	13.0	0.821	1.041	28.5
beijing_multisite_air_quality	0.450	0.709	21.5	0.455	0.694	20.5	0.444	0.691	8.0	0.464	0.788	49.0	0.797	2.802	70.5	0.506	0.753	54.5	0.445	0.703	12.5	0.480	0.739	47.0	0.473	0.752	46.0	0.504	0.806	61.5
boomlet_1062	0.684	0.914	23.5	0.692	0.924	34.0	0.674	*0.887*	7.5	0.695	0.943	37.0	0.791	1.355	67.0	0.710	0.953	41.0	0.673	0.887	7.0	0.697	0.909	30.0	0.718	0.949	42.0	0.686	0.916	27.5
boomlet_1209	0.365	1.603	15.0	0.556	2.221	61.0	0.362	1.599	11.5	0.595	2.512	63.5	0.490	2.101	57.5	0.568	1.810	53.5	0.364	1.575	9.5	0.479	1.828	50.5	0.380	1.602	20.0	0.390	1.598	21.5
boomlet_1225	0.292	0.760	15.5	0.476	1.715	57.5	0.314	0.798	29.0	0.522	2.533	59.0	0.362	0.991	49.5	0.444	0.935	51.5	0.331	0.847	35.0	0.361	0.880	43.5	0.317	0.811	31.0	0.314	0.804	29.0
boomlet_1230	4.003	588.011	11.0	4.216	650.800	41.5	3.971	591.183	10.5	4.348	642.287	50.5	4.422	656.893	53.5	7.530	617.572	45.5	3.987	591.576	12.0	5.513	638.608	54.0	4.162	585.433	17.5	4.358	632.910	45.5
boomlet_1282	0.427	1.051	17.5	0.840	4.393	61.0	0.419	1.091	19.5	1.011	7.108	63.5	0.591	34.574	64.5	0.485	1.122	41.5	0.411	1.032	8.5	0.461	1.171	44.5	0.430	1.073	21.5	0.443	1.174	41.0
boomlet_1487	0.214	0.737	26.5	0.352	0.803	63.0	0.213	0.742	28.5	0.426	1.852	70.5	0.315	1.003	63.0	0.326	0.808	61.5	0.212	0.735	18.0	0.237	0.740	37.5	0.232	0.746	39.5	0.217	0.736	27.5
boomlet_1631	0.539	0.649	10.5	0.665	0.906	56.0	0.554	0.663	19.0	0.688	0.950	59.5	0.689	1.213	62.0	0.664	0.909	56.0	0.593	0.766	40.0	0.609	0.754	41.5	0.545	0.653	12.5	*0.533*	*0.623*	3.0
boomlet_1676	0.585	1.999	15.5	3.200	284.418	63.5	0.673	3.034	27.5	3.173	273.549	61.5	0.964	14.362	49.5	1.314	5.407	45.0	0.790	7.450	36.0	0.931	7.748	40.5	0.584	1.887	13.5	0.552	*1.585*	2.5
boomlet_1855	0.593	92.472	25.5	0.753	92.395	40.0	0.620	92.685	32.5	82.192	$> 10^6$	72.0	0.752	93.856	50.0	1.228	95.758	64.5	0.659	94.206	44.5	0.687	92.029	29.5	0.825	99.841	57.5	0.605	91.834	20.0
boomlet_1975	0.551	0.719	44.5	0.582	0.656	45.0	0.351	0.352	4.5	0.537	0.618	33.5	0.585	0.858	57.5	0.508	0.545	22.0	0.456	0.537	17.5	0.398	0.418	9.0	0.588	0.796	54.0	0.578	0.785	51.5
boomlet_2187	0.735	36.764	28.5	0.774	32.344	24.5	0.705	30.617	11.5	0.966	42.718	58.5	0.928	92.683	65.0	1.502	40.345	58.5	0.720	35.128	24.5	0.974	46.100	64.0	0.798	39.343	43.0	0.798	38.587	41.0
boomlet_285	0.258	0.277	15.0	0.321	0.286	36.5	0.261	0.275	15.5	0.292	0.356	50.5	0.331	0.830	65.0	0.322	0.311	46.0	0.250	0.267	5.0	0.311	0.310	43.0	0.272	0.284	20.5	0.252	0.271	8.0
boomlet_619	0.337	0.324	23.0	0.324	0.316	12.5	0.324	0.323	17.0	0.348	0.386	33.0	0.977	1.860	72.0	0.527	0.554	55.5	0.319	*0.305*	5.0	0.415	0.415	44.5	0.383	0.376	40.0	0.317	0.312	6.0
boomlet_772	0.324	1.032	11.0	0.385	1.087	38.5	0.326	1.032	11.0	0.492	3.878	65.5	0.517	4.791	67.5	0.499	1.230	60.5	0.351	1.083	31.5	0.419	1.117	48.5	0.345	1.057	23.5	0.344	1.057	23.5
boomlet_963	0.597	1.471	8.0	0.635	1.520	24.5	0.588	**1.424**	1.5	0.643	1.545	30.0	1.152	16.149	69.0	0.736	1.703	54.0	0.602	1.461	8.0	0.669	1.673	48.0	0.619	1.536	23.0	0.597	1.479	9.0
electricity_2016	0.278	0.191	34.5	0.280	0.188	35.0	0.261	0.161	14.0	0.369	0.333	59.5	0.437	0.858	67.0	0.361	0.258	55.5	0.244	*0.151*	2.5	0.330	0.220	50.5	0.267	0.188	29.5	0.266	0.187	26.5
electricityloaddiagrams20112014	1.719	349.611	13.0	4.538	$> 10^3$	40.5	2.550	$> 10^3$	31.0	16.170	$> 10^4$	65.0	12.654	$> 10^4$	62.0	21.342	$> 10^3$	63.5	1.639	*313.913*	4.5	6.169	$> 10^3$	51.5	1.708	342.804	9.0	1.736	361.628	15.0
gasrate_co2	0.864	1.138	41.0	0.882	1.157	49.5	0.785	0.939	17.0	0.613	0.595	4.0	1.425	3.764	72.5	0.857	1.185	44.0	0.809	0.985	23.0	0.986	1.467	66.0	0.873	1.146	46.0	0.868	1.130	41.5
gefcom2012_load	1.440	7.121	9.5	1.424	7.190	9.0	2.065	9.538	66.0	2.165	12.938	70.0	2.044	8.788	62.0	2.133	10.022	67.0	1.482	7.757	27.5	1.562	7.568	24.5	1.483	**6.867**	13.0	1.416	*7.115*	5.0
gefcom2012_wind	0.881	1.135	40.5	0.874	1.099	21.5	0.871	1.116	22.0	0.860	1.078	5.0	2.756	43.716	73.0	0.888	1.174	47.0	0.875	1.120	31.5	0.894	1.133	44.5	0.956	1.242	62.5	0.919	1.189	56.0
gvar	0.288	0.247	35.5	0.359	0.299	55.0	0.294	0.264	46.5	0.413	0.492	59.0	0.250	0.220	8.0	0.413	0.420	59.0	0.262	0.224	13.5	0.392	0.351	56.0	0.291	0.263	43.5	0.308	0.271	52.0
italy_air_quality	0.544	1.091	9.5	0.653	1.255	53.0	0.537	**1.061**	4.0	0.609	1.166	35.0	0.690	1.778	67.5	0.658	1.210	47.0	0.554	1.133	17.5	0.667	1.292	56.0	0.571	1.176	27.5	0.653	1.331	56.5
vessel_ais	0.684	0.814	18.5	0.690	0.827	21.0	0.676	0.797	5.5	0.685	0.813	18.5	0.832	1.191	71.5	0.778	1.019	46.5	0.676	0.798	5.5	0.748	0.954	35.0	0.706	0.860	25.0	0.677	0.798	7.5
weather_Wetterstation	0.272	0.218	15.5	0.297	0.215	28.5	0.272	0.219	17.5	0.288	0.224	30.0	0.971	6.167	70.0	0.349	0.273	55.5	0.264	0.218	12.0	0.363	0.303	60.5	0.309	0.256	46.5	0.290	0.236	38.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 20: **Multivariate forecasting results.** Part 3.

Dataset	frets			gpt4ts			gru			informer			itransformer			kairos			ldlinear			lightts			linearregression			lstm		
	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.772	1.620	64.5	0.599	0.771	16.5	0.646	0.847	30.0	1.211	2.488	71.0	0.586	0.710	8.5	0.692	1.075	50.0	0.675	0.885	38.0	0.747	1.110	60.0	0.640	0.867	30.5	0.657	0.877	34.5
Bike_Sharing	0.564	0.729	29.0	0.620	0.894	47.0	0.508	0.649	6.5	0.685	1.088	58.0	0.540	0.722	20.5	0.833	1.626	70.0	0.580	0.772	36.0	0.590	0.793	38.5	0.505	**0.633**	4.5	*0.495*	*0.642*	3.0
COVID-19_Deaths	2.030	75.199	24.0	1.474	57.639	7.0	2.573	130.913	39.0	13.437	$> 10^3$	70.0	1.614	66.582	14.5	1.341	**54.763**	3.5	8.730	643.421	63.0	11.325	$> 10^3$	65.0	15.480	$> 10^3$	72.0	2.859	186.948	47.0
Czelan	0.307	0.242	52.0	0.251	0.234	35.5	0.238	0.202	18.5	0.426	0.415	69.5	0.222	0.199	6.5	0.291	0.279	57.5	0.316	0.279	62.0	0.300	0.240	49.0	0.264	0.200	24.0	0.232	0.192	11.5
DailyDelhiClimate	1.765	57.784	34.0	1.726	51.897	29.0	4.544	490.364	72.5	1.015	42.845	4.0	2.747	114.114	63.0	2.044	98.595	49.5	2.076	63.287	41.0	2.535	88.792	54.5	2.130	81.909	47.5	2.903	84.660	57.5
Dominick	0.846	1.676	51.0	0.672	<u>1.344</u>	8.5	0.694	1.425	20.0	1.019	2.156	65.0	0.715	1.522	36.5	0.841	2.171	59.0	1.107	2.441	67.5	1.128	2.627	69.0	$> 10^6$	$> 10^6$	72.0	0.700	1.444	25.5
ETTH1	0.444	0.445	29.5	0.450	0.465	34.0	0.481	0.489	46.0	0.769	1.027	71.0	0.428	0.424	21.0	0.596	0.851	66.5	0.417	0.411	7.5	0.466	0.474	40.0	0.494	0.543	50.0	0.466	0.472	39.5
ETTH2	0.417	0.373	46.0	0.366	0.324	5.5	0.397	0.371	37.5	1.594	3.933	72.5	0.384	0.342	21.0	0.432	0.447	57.0	0.392	0.340	24.0	0.467	0.473	60.5	0.550	0.666	65.0	0.390	0.355	29.0
ETTM1	0.469	0.502	18.5	0.470	0.529	25.5	0.510	0.559	37.0	0.611	0.709	56.5	0.455	0.485	11.0	0.815	1.647	71.5	0.466	0.520	22.0	0.471	0.507	22.5	0.462	0.503	16.5	0.482	0.507	26.0
ETTM2	0.298	0.190	44.5	0.281	0.176	25.0	0.279	0.174	20.5	0.474	0.494	72.0	0.278	0.174	18.0	0.340	0.245	65.5	0.284	0.178	29.5	0.304	0.195	46.5	0.293	0.196	43.0	*0.270*	0.165	4.0
FRED-MD	1.868	74.861	44.5	1.306	62.727	18.0	1.569	77.058	40.0	4.918	339.432	69.0	1.370	76.204	31.0	1.322	65.350	23.0	1.993	98.333	54.5	2.758	154.639	61.5	1.663	105.686	50.0	1.504	63.383	25.5
ILY	1.265	3.409	40.0	1.189	3.017	27.5	1.070	2.825	17.5	1.551	5.086	65.0	1.029	2.427	7.0	1.316	4.566	52.0	1.552	4.620	62.5	1.888	6.682	69.5	1.028	2.661	10.0	1.151	3.049	26.0
KDDCup2018	0.617	0.806	35.5	0.595	0.801	18.5	0.597	0.802	22.5	0.739	1.061	69.0	0.605	0.824	33.5	0.673	0.984	65.5	0.598	0.761	14.0	0.629	0.814	41.0	0.614	0.794	30.5	0.589	0.780	8.0
METR-LA	0.651	1.072	19.0	0.646	1.050	14.5	0.651	1.212	34.5	0.667	1.259	44.0	0.656	1.180	37.0	0.729	1.416	58.0	0.675	1.037	26.0	0.665	1.056	23.5	1.539	5.908	69.5	0.613	1.117	17.5
MIT-BIH	0.609	2.132	47.0	0.571	1.711	18.5	0.583	1.990	31.5	0.570	<u>1.358</u>	9.0	0.583	2.395	39.0	0.622	1.681	38.0	0.616	2.542	55.0	0.595	2.476	43.0	0.626	2.727	58.5	0.577	1.946	29.0
Metro_Interstate_Traffic_Volume	0.382	0.360	54.5	0.370	0.359	52.5	0.334	0.314	17.0	0.415	0.406	59.5	0.347	0.329	31.0	0.447	0.582	68.5	0.359	0.336	42.5	0.369	0.357	51.5	0.344	0.308	19.5	0.340	0.320	22.5
MexicoCityBikes	0.851	1.282	40.5	0.865	1.341	63.5	0.848	1.278	11.0	<u>0.847</u>	1.274	3.0	0.848	1.279	24.0	0.884	1.432	67.0	0.853	1.287	44.5	0.868	1.340	63.5	0.864	1.325	60.0	0.848	1.277	9.5
NASDAQ	1.597	4.913	48.5	1.576	4.854	46.5	1.849	6.563	71.0	1.429	3.416	16.0	1.456	4.334	31.0	1.431	4.547	33.5	1.691	5.561	62.5	1.662	5.260	56.0	1.274	3.704	9.0	1.505	4.481	38.0
NYSE	0.459	0.597	53.0	0.405	0.497	35.5	0.367	0.448	5.0	1.068	3.257	71.0	0.388	0.483	25.5	0.416	0.583	48.0	0.525	0.705	61.5	0.743	1.463	66.0	0.390	0.489	28.5	0.371	0.451	6.5
National_Stock_Exchange	1.072	2.145	66.5	0.696	1.179	20.5	0.715	1.142	23.0	0.830	1.456	52.5	0.747	1.289	36.0	0.744	1.238	29.5	1.144	2.077	67.0	1.188	2.204	69.0	1.548	6.869	73.0	0.774	1.258	33.5
Oikolab	0.605	0.628	25.5	0.637	0.676	52.0	0.584	0.599	7.5	0.655	0.689	57.0	0.608	0.647	33.0	0.772	1.073	70.0	0.607	0.637	29.5	0.601	0.619	19.0	0.601	0.628	21.0	0.588	0.607	10.5
PEMS-BAY	0.467	0.903	30.0	0.469	0.949	34.0	0.426	0.829	19.0	0.401	0.781	7.5	0.441	0.867	22.5	0.675	1.577	63.0	0.507	0.960	39.5	0.459	0.808	22.0	1.180	4.850	70.0	0.410	0.805	11.0
PEMS03	0.301	0.201	23.0	0.346	0.275	35.5	0.270	0.183	14.0	0.261	0.158	7.0	0.295	0.198	21.0	0.855	1.244	69.0	0.368	0.278	37.5	0.303	0.194	22.0	$> 10^6$	$> 10^6$	72.0	0.258	0.179	9.5
PEMS04	0.314	0.211	30.0	0.381	0.318	43.5	0.232	0.128	9.5	0.238	0.128	11.0	0.298	0.201	27.5	0.866	1.316	67.0	0.362	0.269	36.5	0.273	0.152	20.0	0.610	0.884	60.5	0.219	0.117	4.0
PEMS07	0.315	0.216	30.5	0.380	0.329	41.0	0.222	0.131	4.5	0.248	0.182	18.5	0.276	0.176	22.0	0.806	1.142	69.5	0.394	0.320	41.5	0.256	0.141	12.5	0.660	0.812	61.5	*0.212*	0.129	3.5
PEMS08	0.355	0.325	31.0	0.389	0.389	39.0	0.307	0.269	17.0	0.322	0.328	27.0	0.323	0.282	21.5	0.859	1.345	67.0	0.412	0.408	41.0	0.324	0.263	18.5	0.469	0.500	49.0	0.264	0.239	7.0
Power_Consumption_of_Tetouan_City	0.363	0.290	32.5	0.357	0.319	35.0	0.293	0.242	7.5	0.700	0.770	69.0	0.296	0.256	10.5	0.612	0.800	67.0	0.354	0.289	30.0	0.357	0.278	28.0	0.349	0.305	31.0	0.304	0.258	13.5
Residential_Power_and_Battery_Data	0.769	1.670	18.0	0.774	1.761	33.5	0.789	1.800	44.0	<u>0.732</u>	1.648	4.0	0.757	1.737	18.0	0.898	2.326	67.5	0.763	1.659	12.5	0.784	1.701	28.0	0.770	1.674	20.0	0.768	1.730	23.5
Rideshare	1.284	5.742	71.5	0.782	1.361	64.5	0.438	0.414	19.0	0.745	1.037	57.5	0.449	0.447	22.0	0.437	0.442	20.0	0.483	0.410	20.5	0.770	0.892	54.5	0.925	2.620	70.0	0.564	0.669	32.0
Sleep_EDFX	0.702	5.060	33.5	0.726	5.060	38.5	0.709	5.103	36.5	0.796	5.518	57.5	0.648	4.607	8.0	0.898	5.927	67.0	0.756	5.125	44.5	0.654	4.806	16.0	**0.327**	**0.543**	1.0	0.706	4.955	29.5
Spanish_energy	0.405	0.384	21.0	0.403	0.405	30.0	0.405	0.409	32.5	0.551	0.652	67.0	0.392	0.390	15.5	0.527	0.709	67.5	0.410	0.398	31.0	0.412	0.393	31.0	$> 10^6$	$> 10^6$	73.0	0.409	0.419	37.5
Spanish_weather	0.344	0.319	35.5	0.326	0.321	25.5	0.328	0.324	30.0	0.406	0.398	63.5	0.321	0.316	14.5	0.427	0.517	68.0	0.341	0.314	29.0	0.348	0.330	43.5	0.342	0.312	27.0	0.327	0.320	25.5
Temperature-Rain	0.878	4.441	50.0	0.823	4.233	23.5	0.817	4.225	17.0	0.789	4.214	11.0	0.861	4.437	48.5	0.879	5.264	58.5	0.885	4.466	53.0	0.989	4.908	62.0	$> 10^6$	$> 10^6$	72.5	0.806	4.187	10.0
Traffic	0.316	0.498	17.5	0.314	0.478	16.5	0.361	0.666	43.5	0.452	0.852	55.5	0.293	<u>0.432</u>	5.5	0.885	1.867	68.0	0.358	0.575	34.5	0.390	0.584	40.5	0.666	1.399	64.5	0.329	0.633	31.5
Traffic_Flow_Forecasting	0.352	0.232	7.0	0.614	0.659	41.0	0.360	0.260	11.0	0.375	0.253	12.0	0.398	0.313	21.0	1.209	2.300	72.0	0.427	0.320	25.5	0.508	0.427	32.5	0.401	0.297	19.5	0.437	0.360	27.5
WIND	0.663	1.005	11.5	0.690	1.082	31.0	0.660	1.045	15.5	0.714	1.115	44.0	0.692	1.101	35.5	0.784	1.438	66.0	0.667	1.007	15.0	0.658	1.007	11.5	<u>0.627</u>	*0.967*	2.5	0.669	1.064	23.0
Zafnoo_env	0.427	0.449	12.5	0.447	0.485	39.0	0.443	0.473	31.0	0.651	0.804	66.0	0.440	0.481	33.5	0.665	1.013	69.0	0.426	0.449	10.0	0.441	0.465	25.0	0.412	**0.441**	4.0	0.441	0.485	34.5
Zafnoo_sapf	0.306	0.303	40.0	0.293	0.298	31.5	0.302	0.316	42.0	0.414	0.529	65.0	0.290	0.298	28.5	0.481	0.698	68.0	0.273	0.282	8.5	0.334	0.335	52.0	0.284	0.295	22.0	0.302	0.317	41.5
australian_tourism	0.874	1.224	50.0	0.768	0.907	12.0	0.825	1.052	30.5	0.895	1.221	51.5	0.791	1.090	32.0	0.864	1.741	58.5	0.901	1.292	54.5	1.038	1.575	67.0	1.170	2.863	72.0	0.830	1.060	34.0
beijing_multisite_air_quality	0.459	0.697	23.0	0.456	0.720	34.0	0.444	0.706	13.5	0.489	0.775	56.5	0.451	0.711	24.0	0.507	0.873	65.0	0.464	0.701	25.5	0.466	0.717	37.0	0.749	4.365	70.5	0.447	0.713	22.0
boomlet_1062	0.741	1.058	57.5	0.696	0.923	34.5	0.679	0.892	14.0	0.836	1.271	68.0	0.680	0.897	17.0	0.663	0.989	26.5	0.733	0.991	50.5	0.734	1.050	55.5	$> 10^6$	$> 10^6$	73.0	0.679	0.891	14.0
boomlet_1209	0.476	1.901	53.5	0.363	1.584	10.0	0.398	1.739	34.0	1.183	7.337	70.5	0.360	1.582	6.0	0.446	1.986	52.0	0.472	1.705	39.0	0.492	1.998	57.5	0.550	2.174	60.0	0.391	1.796	35.5
boomlet_1225	0.748	17.141	68.5	0.305	0.765	19.5	0.371	0.984	50.0	1.029	7.748	70.0	0.294	0.760	16.5	0.284	0.785	15.5	0.451	1.542	56.5	0.664	5.151	62.5	0.706	2.831	61.5	0.362	0.967	48.0
boomlet_1230	4.532	755.684	59.0	**3.947**	**574.572**	1.0	4.236	626.521	36.0	9.734	$> 10^3$	70.0	*3.963*	584.105	4.5	4.532	748.923	59.0	4.151	634.749	34.5	5.577	864.123	62.5	4.230	594.296	27.5	4.300	641.533	48.0
boomlet_1282	0.706	3.463	59.5	0.416	1.037	12.5	0.442	1.120	33.5	1.323	10.787	70.5	*0.409*	*0.996*	3.0	0.479	1.253	51.5	0.541	1.372	54.0	0.752	3.459	59.5	0.942	6.224	62.0	0.437	1.125	30.5
boomlet_1487	0.318	0.790	57.5	0.220	0.742	35.0	0.231	0.744	38.0	0.344	0.817	63.5	0.207	0.735	17.0	0.201	0.735	10.5	0.363	0.816	64.5	0.418	0.917	67.5	0.513	1.484	70.5	0.213	0.738	26.5
boomlet_1631	0.779	1.777	66.0	<u>0.532</u>	0.637	3.5	0.624	0.821	50.5	1.225	2.443	72.0	0.545	0.659	13.0	0.605	0.812	45.5	0.620	0.813	49.0	0.856	1.536	66.5	0.712	1.038	61.5	0.609	0.788	45.5
boomlet_1676	10.033	$> 10^3$	70.0	0.629	2.962	23.0	1.011	17.459	53.0	4.073	274.523	64.5	0.659	3.660	27.5	0.623	2.244	20.5	1.068	15.331	53.0	6.692	387.433	68.0	4.465	512.062	68.0	0.895	8.643	40.5
boomlet_1855	0.868	93.988	53.5	0.592	92.030	20.0	0.994	99.940	63.0	1.024	93.497	57.5	0.592	91.992	18.0	0.542	91.476	10.5	0.729	92.772	41.5	1.064	111.141	67.5	56.016	$> 10^5$	71.0	1.009	102.260	65.0
boomlet_1975	0.445	0.500	15.5	0.545	0.717	42.5	0.530	0.610	30.5	0.584	0.829	56.0	0.375	0.398	7.0	0.623	1.265	68.5	0.542	0.640	37.5	0.499	0.573	22.0	0.508	0.607	25.5	0.486	0.597	22.0
boomlet_2187	0.720	30.684	15.0	0.723	33.205	20.0	0.823	42.610	54.0	0.987	41.082	57.5	0.681	32.355	11.5	0.795	34.442	35.0	0.769	31.525	21.0	0.799	32.336	30.0	0.849	33.552	40.0	0.795	41.474	47.0
boomlet_285	0.324	0.308	44.5	0.273	0.288	25.0	0.276	0.309	33.0	0.413	0.373	66.0	0.269	0.295	23.5	0.284	0.348	46.5	0.312	0.285	33.5	0.331	0.335	56.5	$> 10^6$	$> 10^6$	72.0	0.272	0.302	27.5
boomlet_619	0.356	0.351	31.5	0.352	0.338	28.0	0.323	0.323	15.5	0.465	0.538	50.5	0.335	0.322	21.0	0.641	0.893	63.5	0.376	0.367	37.0	0.359	0.347	31.0	0.370	0.394	38.0	*0.312*	0.316	7.5
boomlet_772	0.406	2.371	59.0	0.320	1.032	8.5	0.405	1.161	47.5	0.613	1.521	66.5	0.319	1.033	10.0	0.336	1.102	31.5	0.367	1.067	33.0	0.493	1.331	61.0	2.717	217.400	71.0	0.406	1.189	51.5
boomlet_963	2.253	100.738	70.0	0.619	1.553	24.5	0.646	1.607	37.5	0.748	1.745	56.0	0.632	1.522	24.5	0.708	2.259	60.0	0.605	<u>1.441</u>	7.0	0.655	1.544	34.0	4.750	263.347	71.0	0.670	1.750	51.0
electricity_2016	0.262	0.172	17.0	0.266	0.178	22.5	0.295	0.192	40.0	0.461	0.435	65.5	0.250	0.160	11.0	0.612	0.824	67.5	0.269	0.184	28.0	0.306	0.205	45.5	0.621	4.698	70.5	0.291	0.193	39.5
electricityloaddiagrams20112014	3.542	$> 10^3$	34.5	1.681	327.388	8.0	4.560	$> 10^3$	47.0	16.790	$> 10^4$	67.5	*1.623*	315.620	4.0	5.096	$> 10^3$	52.0	1.784	379.098	20.0	67.578	$> 10^3$	66.0	16.352	$> 10^4$	68.0	4.540	$> 10^3$	43.0
gasrate_co2	*0.608*	*0.585*	3.0	0.859	1.113	36.0	0.919	1.274	58.5	0.664	0.649	6.5	0.821	1.032	27.5	1.074	2.096	69.0	0.843	1.085	34.5	0.875	1.215	52.0	0.618	0.633	5.0	1.110	1.862	68.5
gefcom2012_load	1.907	11.315	65.0	1.718	8.197	51.0	1.469	7.306	17.0	*1.334*	7.410	9.5	1.581	7.980	39.5	2.476	11.668	70.5	1.466	8.211	35.0	1.425	7.658	19.0	1.930	10.807	65.5	1.423	7.189	8.0
gefcom2012_wind	0.864	1.091	10.5	0.881	1.121	36.0	0.868	1.109	18.0	0.883	1.120	35.5	0.875	1.124	32.0	0.944	1.419	66.5	0.862	1.081	8.0	0.877	1.117	30.0	0.940	1.360	63.5	0.861	1.088	9.0
gvar	0.826	1.262	65.0	0.270	0.235	22.5	0.283	0.249	34.0	0.910	1.674	68.5	0.281	0.246	29.0	0.269	0.235	20.5	1.637	4.398	70.0	1.910	6.040	73.0	0.347	0.431	57.0	0.293	0.260	43.5
italy_air_quality	0.617	1.162	36.0	0.574	1.170	28.0	0.606	1.220	42.0	0.992	2.104	70.0	0.564	1.144	20.0	0.673	1.610	64.0	0.599	1.090	22.5	0.625	1.166	38.5	0.636	1.226	47.5	0.593	1.200	34.0
vessel_ais	0.689	0.825	20.0	0.734	0.925	32.0	0.754	0.970	37.0	0.822	1.120	68.0	0.681	0.808	17.0	0.823	1.146	70.0	*0.675*	<u>0.794</u>	2.5	0.712	0.871	26.0	0.680	0.803	13.5	0.758	0.977	40.0
weather_Wetterstation	0.295	0.227	36.5	0.280	0.229	30.5	0.289	0.233	37.0	0.565	0.672	68.5	0.266	0.219	14.5	0.314	0.293	52.5	0.284	0.214	22.0	0.290	0.226	33.5	0.286	0.221	28.0	0.276	0.221	23.5

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, <u>underline</u>, and *italic*. The rank column reports the average rank over the two metrics.

Table 21: **Multivariate forecasting results.** Part 4.

Dataset	mamba			micn			moirai			moment			naivemovingaverage			naiveseasonal			nbeats			nlinear			nonstationary-transformer			patch_tst		
	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.605	0.770	17.0	0.721	0.962	50.5	0.629	0.884	30.0	**0.557**	**0.623**	1.0	0.704	1.085	53.0	0.731	1.195	58.5	0.726	1.016	53.0	0.693	1.102	52.5	0.687	0.959	44.5	0.592	0.744	13.0
Bike_Sharing	0.541	0.704	18.5	0.556	0.713	23.5	0.558	0.845	37.5	0.561	0.736	29.0	0.833	1.569	68.5	0.722	1.399	64.0	0.546	0.721	21.0	0.593	0.810	42.0	0.725	1.260	61.5	0.548	0.737	24.5
COVID-19_Deaths	1.675	69.230	18.0	3.797	161.162	48.5	2.759	117.465	39.0	13.906	$> 10^3$	71.0	3.780	174.102	48.5	3.145	131.907	43.5	3.561	142.194	46.5	2.074	82.451	26.5	2.811	136.938	43.0	1.433	59.054	7.0
Czelan	0.231	0.212	17.5	0.351	0.283	64.0	0.231	0.248	30.0	0.220	0.176	3.0	0.269	0.256	45.0	0.276	0.279	53.0	0.275	0.222	37.0	0.237	0.214	21.5	0.256	0.218	31.5	0.237	0.225	29.0
DailyDelhiClimate	2.184	78.045	47.0	1.709	48.927	27.0	1.532	60.148	28.5	1.167	43.160	8.5	1.420	47.291	18.5	1.552	84.441	37.0	2.631	95.989	58.0	2.905	116.339	65.0	3.062	111.443	65.5	2.488	103.946	57.5
Dominick	0.694	1.443	24.0	0.936	1.878	59.0	0.597	**1.251**	3.5	1.184	2.589	69.5	0.685	1.448	22.0	0.666	2.037	36.0	0.873	1.877	56.5	0.608	1.821	31.0	0.724	1.616	42.5	0.685	1.406	15.5
ETTH1	0.464	0.474	39.0	0.500	0.538	50.0	0.440	0.485	37.5	0.432	0.422	20.5	0.619	0.855	67.5	0.466	0.570	48.5	0.457	0.461	34.0	*0.413*	*0.408*	3.0	0.514	0.586	55.0	0.432	0.423	21.5
ETTH2	0.380	0.334	14.5	0.413	0.372	43.0	0.398	0.402	45.5	0.384	0.342	22.0	0.449	0.499	60.5	0.445	0.525	60.0	0.466	0.438	59.0	0.371	0.323	6.0	0.422	0.397	52.0	0.380	0.342	18.0
ETTM1	0.468	0.501	16.5	0.521	0.552	37.5	0.544	0.768	53.5	**0.401**	**0.363**	1.0	0.791	1.463	69.0	0.826	1.589	71.5	0.473	0.525	26.0	0.473	0.550	29.0	0.490	0.505	26.5	0.461	0.505	18.0
ETTM2	0.281	0.177	26.5	0.312	0.195	49.5	0.310	0.229	57.5	0.271	0.161	4.5	0.343	0.247	67.0	0.364	0.278	68.0	0.294	0.195	44.0	0.276	0.171	12.5	0.290	0.185	33.5	0.285	0.173	23.0
FRED-MD	1.319	66.032	23.0	2.104	115.757	60.0	1.259	55.771	9.0	5.751	351.699	71.0	1.440	61.329	21.0	1.484	63.563	25.0	3.332	178.391	63.0	1.188	48.241	2.0	1.529	70.080	33.0	1.245	54.017	7.0
ILY	1.041	2.530	9.5	1.317	3.267	40.0	1.116	2.923	20.5	1.964	7.570	71.0	1.547	5.070	64.0	2.008	8.276	72.0	1.346	3.920	49.0	1.238	3.628	41.0	1.109	2.715	17.0	1.006	2.470	5.0
KDDCup2018	0.606	0.825	34.5	0.649	0.854	54.5	0.590	0.796	15.0	0.587	0.777	6.0	0.700	1.054	67.5	0.755	1.290	71.0	0.630	0.831	47.0	0.596	0.789	17.0	0.660	0.920	62.5	0.596	0.798	20.0
METR-LA	0.638	1.163	26.0	0.693	1.145	42.5	0.611	1.191	24.0	0.663	1.032	19.5	0.733	1.581	60.5	0.761	1.726	64.5	0.653	1.072	21.0	0.668	1.105	33.5	0.627	1.161	23.0	0.655	1.073	22.5
MIT-BIH	0.601	2.350	44.5	0.574	2.094	32.5	0.567	1.562	15.0	0.587	1.833	28.5	0.677	2.093	55.5	0.686	2.149	58.5	0.563	2.219	31.0	0.625	2.952	62.0	0.577	1.956	29.0	0.607	2.970	55.5
Metro_Interstate_Traffic_Volume	0.359	0.345	46.5	0.354	0.297	22.5	0.312	0.343	26.0	0.355	0.325	33.0	0.459	0.595	69.5	0.364	0.524	59.0	0.337	0.308	16.0	0.357	0.345	45.5	0.318	*0.283*	6.5	0.358	0.340	44.5
MexicoCityBikes	0.848	1.279	25.0	0.959	1.642	71.0	0.858	1.334	60.5	0.851	1.282	41.0	0.871	1.361	65.0	1.132	2.488	72.0	0.854	1.292	49.5	0.852	1.290	44.5	0.855	1.300	52.0	0.851	1.288	42.5
NASDAQ	1.468	4.384	33.5	1.594	5.743	58.0	1.253	3.333	6.0	1.925	5.680	68.5	1.547	4.559	43.5	1.505	4.482	39.0	1.440	4.030	23.5	1.370	3.969	15.0	1.426	4.244	26.0	1.374	3.975	16.0
NYSE	0.384	0.477	19.0	0.759	1.177	65.5	0.381	0.469	16.0	2.188	9.047	73.0	0.404	0.497	35.5	0.421	0.664	53.5	0.486	0.680	59.5	0.380	0.654	36.5	0.400	0.493	32.5	0.384	0.467	15.5
National_Stock_Exchange	0.798	1.369	47.0	1.098	1.835	64.5	0.589	0.738	5.0	1.148	1.912	67.0	0.864	1.497	57.5	0.863	1.830	60.0	0.946	1.861	64.0	0.572	0.644	2.0	0.698	1.109	19.5	0.742	1.284	32.0
Oikolab	0.609	0.647	33.5	0.644	0.704	57.0	0.642	0.764	60.5	0.603	0.609	19.0	0.812	1.102	71.0	0.741	1.032	68.5	0.630	0.663	47.0	0.611	0.654	39.5	0.610	0.630	30.0	0.619	0.645	39.5
PEMS-BAY	0.417	0.833	16.5	0.496	0.903	33.5	0.556	1.267	52.5	*0.363*	**0.552**	2.0	0.690	1.731	66.5	0.716	1.882	68.5	0.452	0.872	25.5	0.546	1.091	47.0	0.411	0.805	12.0	0.463	0.915	31.0
PEMS03	0.272	0.176	11.5	0.308	0.202	24.5	0.482	0.490	53.0	0.279	0.176	13.5	0.718	0.855	64.0	0.738	0.901	66.0	0.275	0.173	11.5	0.410	0.392	46.0	*0.244*	0.145	3.5	0.340	0.261	33.0
PEMS04	0.263	0.156	19.0	0.246	0.129	13.0	0.516	0.568	55.0	0.288	0.173	23.5	0.905	1.369	68.0	0.926	1.426	69.5	0.287	0.187	24.5	0.417	0.381	48.5	0.227	0.125	7.0	0.370	0.294	41.5
PEMS07	0.251	0.149	14.0	0.259	0.149	16.5	0.520	0.547	53.5	0.270	0.158	18.0	0.684	0.824	63.0	0.707	0.871	64.5	0.285	0.188	26.5	0.421	0.384	45.5	0.232	0.140	6.5	0.363	0.287	36.5
PEMS08	0.320	0.282	21.0	0.294	0.232	10.0	0.528	0.622	56.0	0.304	0.281	17.5	0.892	1.415	68.5	0.911	1.464	70.0	0.309	0.244	14.0	0.456	0.496	47.0	*0.259*	0.228	3.5	0.382	0.367	37.5
Power_Consumption_of_Tetouan_City	0.281	0.237	4.0	0.447	0.417	48.0	0.442	0.514	53.5	**0.267**	**0.206**	1.0	0.635	0.874	69.0	0.642	0.902	70.0	0.345	0.282	27.0	0.323	0.299	26.5	0.378	0.353	38.0	0.304	0.261	13.5
Residential_Power_and_Battery_Data	0.760	1.714	15.0	0.748	1.649	7.5	0.744	1.865	27.0	0.812	1.703	33.0	0.856	2.315	61.5	0.895	2.559	68.0	0.767	1.730	21.5	0.836	1.814	48.0	0.750	1.644	7.0	0.768	1.721	21.5
Rideshare	0.536	0.644	27.5	0.861	1.409	67.0	*0.271*	0.217	2.5	0.771	0.764	50.5	**0.262**	**0.206**	1.0	0.316	0.310	12.0	0.829	1.707	67.0	0.289	0.274	6.5	0.748	1.242	61.0	0.497	0.555	25.5
Sleep_EDFX	0.675	4.673	17.0	0.822	5.364	57.0	0.738	5.176	44.5	0.735	5.686	50.0	0.923	6.108	69.5	0.921	5.961	68.5	0.667	4.807	20.5	0.753	5.226	47.5	0.735	5.158	43.0	0.650	4.751	13.0
Spanish_energy	0.389	0.388	12.0	0.413	0.388	29.5	0.414	0.472	48.5	0.392	*0.368*	9.0	0.564	0.792	69.0	0.485	0.660	65.5	0.420	0.426	45.0	0.401	0.407	30.5	0.411	0.420	39.0	0.398	0.393	21.0
Spanish_weather	0.323	0.321	21.5	0.358	0.313	34.0	0.357	0.406	57.5	0.324	0.306	11.5	0.455	0.554	69.5	0.386	0.513	64.5	0.334	0.310	23.0	0.329	0.322	31.0	0.351	0.359	51.0	0.327	0.321	25.0
Temperature-Rain	0.850	4.449	46.0	0.938	4.509	58.0	0.739	4.223	10.5	0.861	4.461	49.5	0.828	4.370	33.5	0.970	7.230	66.0	0.828	4.428	37.5	0.995	7.989	67.5	0.810	4.205	14.5	0.839	4.322	36.0
Traffic	0.273	**0.407**	1.5	0.312	0.531	18.0	0.324	0.579	27.5	0.661	1.179	63.0	0.954	1.854	68.0	0.463	1.134	59.5	0.316	0.511	19.0	0.349	0.572	31.0	0.348	0.660	39.0	0.301	0.468	13.0
Traffic_Flow_Forecasting	0.367	0.279	14.5	0.502	0.458	32.0	0.689	0.997	53.5	0.863	1.029	60.0	1.097	1.846	69.5	1.143	1.993	71.0	*0.306*	0.191	3.5	0.595	0.668	41.5	0.393	0.304	19.0	0.376	0.287	17.0
WIND	0.718	1.203	52.0	0.656	1.000	7.5	0.707	1.234	49.0	0.718	1.091	43.0	0.821	1.607	69.0	0.845	1.709	70.0	0.667	1.043	18.0	0.684	1.082	28.0	0.733	1.189	53.5	0.699	1.119	39.0
Zafnoo_env	0.444	0.491	39.5	0.433	0.452	17.5	0.411	0.506	23.5	0.439	0.465	24.0	0.783	1.082	71.0	0.517	0.812	62.5	0.437	0.474	28.0	0.616	0.941	67.0	0.494	0.557	56.5	0.425	0.467	15.0
Zafnoo_sapf	0.284	0.296	22.5	0.333	0.354	53.0	0.260	0.281	5.5	0.270	**0.262**	4.5	0.586	0.770	72.0	0.338	0.497	59.5	0.302	0.318	42.5	0.532	1.033	72.0	0.298	0.319	42.0	0.277	0.284	11.5
australian_tourism	0.861	1.156	46.0	1.041	1.801	69.0	0.756	0.932	12.0	0.798	0.959	21.0	0.828	1.053	32.0	0.875	1.490	57.0	0.871	1.301	51.5	0.831	1.287	44.5	0.914	1.341	57.5	0.786	0.980	20.5
beijing_multisite_air_quality	0.452	0.711	26.5	0.487	0.733	47.5	0.440	0.721	20.0	0.453	0.718	32.5	0.554	1.030	67.0	0.559	1.142	68.5	0.457	0.725	36.5	0.460	0.725	37.0	0.477	0.774	52.5	0.449	0.705	17.5
boomlet_1062	0.674	**0.884**	7.0	0.769	1.085	61.0	0.661	0.984	25.5	0.681	0.900	20.0	0.777	1.141	64.0	0.824	1.684	70.5	0.799	1.467	68.0	0.687	0.914	27.5	0.724	0.994	49.0	0.684	0.914	24.5
boomlet_1209	0.367	1.596	15.0	0.505	1.724	44.0	0.370	1.585	15.0	0.411	1.647	32.5	0.408	1.714	35.0	0.446	2.305	55.5	0.455	2.233	55.5	0.405	1.864	46.0	0.408	1.760	38.0	0.367	1.573	10.5
boomlet_1225	0.305	0.776	20.5	0.544	1.869	59.5	0.266	0.745	7.5	0.280	**0.695**	5.0	0.302	0.829	27.0	0.354	1.104	49.5	0.542	3.757	60.5	0.333	0.856	37.5	0.373	0.997	51.5	0.285	0.730	9.0
boomlet_1230	4.041	602.682	18.0	4.272	625.839	39.0	4.151	607.321	25.0	4.452	612.791	40.5	4.216	636.597	39.0	4.952	891.362	61.5	5.497	$> 10^3$	62.5	4.130	*576.887*	13.0	4.382	659.791	53.5	3.957	582.746	3.5
boomlet_1282	0.413	1.012	7.0	0.548	1.420	55.0	0.443	1.201	42.0	0.485	1.447	54.5	0.439	1.145	34.5	0.563	1.649	56.5	0.614	2.220	58.0	0.430	1.074	22.5	0.451	1.209	47.0	0.410	1.025	5.5
boomlet_1487	0.206	0.736	20.0	0.345	0.802	61.5	0.193	**0.730**	3.0	0.214	0.734	18.5	0.233	0.773	47.0	0.270	1.056	61.0	0.289	0.769	53.5	0.269	0.778	52.5	0.278	0.779	54.0	0.227	0.751	41.0
boomlet_1631	0.578	0.754	36.0	0.745	1.159	62.5	0.552	0.699	24.0	0.537	0.618	4.5	0.598	0.859	46.0	0.651	0.961	57.5	0.675	0.924	58.0	0.537	0.635	5.0	0.614	0.809	46.5	0.549	0.674	18.5
boomlet_1676	0.641	3.309	25.0	1.668	58.589	58.0	0.561	1.753	6.5	0.648	2.577	23.5	0.572	1.618	6.0	0.650	2.201	22.5	1.658	49.933	57.0	0.565	1.746	6.5	0.923	9.717	43.5	0.696	5.076	32.5
boomlet_1855	0.666	93.050	38.5	0.840	93.194	49.5	**0.484**	**91.229**	1.0	0.613	146.164	48.0	0.536	91.404	9.0	0.588	91.770	15.5	1.025	94.663	60.5	0.717	92.546	37.5	0.809	95.353	54.0	0.596	91.615	17.0
boomlet_1975	0.295	0.295	2.0	0.357	0.333	4.5	0.541	0.910	51.0	**0.289**	**0.192**	1.0	0.589	0.811	56.0	0.592	0.853	59.0	0.435	0.512	15.0	0.555	0.820	51.0	0.380	0.368	6.5	0.520	0.672	32.5
boomlet_2187	0.683	31.559	10.0	0.798	32.905	31.0	0.685	33.453	16.0	1.014	61.853	69.5	0.825	41.355	53.0	0.856	47.073	63.0	0.718	36.947	26.5	1.368	29.324	39.0	0.820	43.245	55.0	0.702	33.475	18.5
boomlet_285	0.258	0.284	16.0	0.390	0.421	66.5	0.259	0.335	36.0	0.287	0.314	40.0	0.311	0.333	49.5	0.325	0.548	62.5	0.342	0.308	47.5	0.334	0.580	65.0	0.323	0.327	51.0	0.253	0.270	8.5
boomlet_619	0.322	0.308	7.5	0.347	0.355	29.5	0.601	0.885	61.5	0.349	0.335	25.5	0.788	1.186	68.0	0.902	1.458	70.5	0.329	0.336	21.0	0.372	0.362	35.5	0.338	0.343	26.0	0.334	0.321	19.5
boomlet_772	0.343	1.058	23.0	0.471	1.158	52.0	0.310	1.061	13.5	0.332	1.038	16.0	0.379	1.366	51.5	0.413	1.984	60.0	0.360	1.088	34.0	0.349	1.066	27.5	0.398	1.167	48.5	0.313	**1.023**	4.5
boomlet_963	0.632	1.578	29.5	0.647	1.513	27.5	0.637	1.681	41.0	0.609	1.523	16.5	0.778	2.687	65.0	0.865	3.060	67.5	0.600	*1.442*	6.0	0.612	1.492	14.0	0.691	1.806	56.5	0.609	1.487	13.0
electricity_2016	**0.241**	**0.149**	1.0	0.287	0.177	29.0	0.293	0.230	46.5	0.262	0.159	13.5	0.813	1.160	70.5	0.307	0.286	52.5	0.264	0.171	17.0	0.266	0.187	27.5	0.280	0.181	30.0	0.265	0.175	19.5
electricityloaddiagrams20112014	**1.590**	288.494	1.5	2.625	873.497	31.0	2.724	$> 10^3$	33.5	1.594	**279.587**	1.5	5.919	$> 10^3$	54.5	6.117	$> 10^3$	57.0	2.403	782.936	28.5	1.711	348.006	11.5	4.556	$> 10^3$	41.5	1.711	343.028	10.0
gasrate_co2	0.835	1.072	32.0	1.280	2.366	71.0	0.807	1.055	25.5	0.726	0.816	12.5	0.923	1.257	59.0	0.962	1.305	62.5	0.821	1.066	28.5	0.917	1.330	60.5	0.870	1.172	48.0	0.820	1.009	25.5
gefcom2012_load	1.599	8.010	42.0	1.629	7.820	38.5	1.570	7.640	28.5	**1.201**	7.019	1.5	1.633	7.851	41.0	2.239	10.151	68.5	1.669	9.505	55.5	1.962	8.688	60.5	1.609	7.950	42.0	1.517	7.535	22.5
gefcom2012_wind	0.872	1.118	26.5	0.872	1.108	21.5	0.913	1.347	60.0	0.904	1.127	46.0	1.049	1.832	70.0	1.140	2.170	72.0	0.872	1.117	24.5	0.881	1.131	39.0	0.901	1.178	50.5	0.869	1.098	15.0
gvar	0.274	0.243	25.5	1.086	1.622	68.5	0.287	0.238	31.0	1.879	4.911	72.0	0.291	0.247	36.5	0.292	0.294	49.0	0.540	0.720	61.5	0.251	0.225	11.5	0.286	0.269	43.0	0.264	0.232	17.0
italy_air_quality	0.553	1.112	12.5	0.637	1.207	44.5	0.535	1.241	24.5	0.539	1.067	6.0	0.605	1.414	51.5	0.599	1.469	50.5	0.608	1.179	36.5	0.575	1.122	22.5	0.636	1.333	55.5	0.547	*1.067*	8.0
vessel_ais	0.681	0.807	16.0	0.778	1.028	48.5	0.806	1.091	63.5	0.716	0.884	27.0	0.806	1.087	63.5	1.090	1.970	73.0	0.705	0.857	24.0	0.676	0.796	4.0	0.746	0.953	34.0	0.677	0.798	7.5
weather_Wetterstation	0.276	0.229	29.0	0.334	0.264	53.0	0.270	0.233	24.5	0.261	*0.194*	4.0	0.295	0.285	47.5	0.308	0.335	55.0	0.280	0.215	20.5	0.275	0.224	24.0	0.329	0.318	58.0	0.272	0.225	21.5

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 22: **Multivariate forecasting results.** Part 5.

Dataset	patchmlp			pyraformer			reformer			resnet		
	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.706	1.014	50.5	1.046	1.889	68.0	1.140	2.209	69.0	0.734	1.087	57.5
Bike_Sharing	0.695	1.045	58.0	0.565	0.768	34.5	0.797	1.380	66.5	0.582	0.830	41.0
COVID-19_Deaths	2.133	104.389	32.0	2.298	105.617	34.5	13.054	$> 10^3$	68.5	2.848	136.813	43.0
Czelan	0.277	0.250	46.5	0.401	0.346	68.0	1.074	1.963	72.0	0.256	0.231	36.0
DailyDelhiClimate	2.402	79.722	49.5	1.221	47.274	16.0	1.144	43.176	8.0	4.480	403.399	71.0
Dominick	0.724	1.558	40.0	0.725	1.583	42.0	1.012	2.113	64.0	0.729	1.503	39.5
ETTH1	0.577	0.723	64.5	0.676	0.778	67.5	0.738	0.953	69.5	0.551	0.636	61.5
ETTH2	0.396	0.353	30.5	1.366	3.192	70.5	1.229	2.335	69.5	0.436	0.407	56.0
ETTM1	0.597	0.776	57.5	0.537	0.579	44.0	0.610	0.646	53.5	0.532	0.607	43.5
ETTM2	0.286	0.175	28.0	0.396	0.330	69.0	0.446	0.392	71.0	0.298	0.187	42.0
FRED-MD	1.371	64.530	23.5	3.621	230.115	64.5	4.858	337.736	68.0	1.614	76.383	40.5
ILY	*0.953*	*2.235*	3.0	1.363	4.309	52.5	1.536	4.788	62.0	0.911	**1.935**	1.5
KDDCup2018	0.598	0.811	26.5	0.670	0.878	60.5	0.726	1.000	67.5	0.637	0.884	57.0
METR-LA	0.667	1.165	39.5	0.655	1.171	35.5	0.652	1.295	40.0	0.671	1.179	42.0
MIT-BIH	0.641	1.761	44.5	*0.536*	**1.256**	2.0	0.604	1.366	23.0	0.615	2.179	51.0
Metro_Interstate_Traffic_Volume	0.398	0.407	59.0	0.368	0.325	39.0	0.310	**0.259**	4.5	0.418	0.438	63.5
MexicoCityBikes	0.848	1.278	15.0	0.849	1.275	20.0	0.849	**1.274**	15.0	0.849	1.279	27.5
NASDAQ	1.515	4.706	42.5	1.188	2.439	2.0	1.360	3.062	8.5	1.657	5.599	59.5
NYSE	0.387	0.486	25.5	0.734	2.036	66.0	1.096	3.037	71.0	0.421	0.537	44.0
National_Stock_Exchange	0.744	1.322	38.5	0.679	0.992	14.5	0.784	1.318	42.5	0.781	1.284	37.0
Oikolab	0.676	0.755	65.0	**0.552**	**0.527**	1.0	0.572	0.566	4.5	0.632	0.681	51.0
PEMS-BAY	0.550	1.139	50.5	0.421	0.804	14.5	0.424	0.798	14.5	0.408	0.771	7.5
PEMS03	0.652	0.794	60.0	0.263	0.170	8.5	0.270	0.178	11.5	0.289	0.198	19.5
PEMS04	0.631	0.741	60.5	*0.217*	*0.112*	3.0	0.236	0.131	11.5	0.243	0.131	13.0
PEMS07	0.639	0.824	61.0	0.243	0.173	15.0	0.263	0.197	24.0	0.235	0.137	7.0
PEMS08	0.639	0.846	61.0	0.286	0.286	16.5	0.290	0.311	19.5	0.271	0.237	8.0
Power_Consumption_of_Tetouan_City	0.470	0.461	54.5	0.549	0.514	60.0	1.050	1.709	72.0	0.316	0.278	21.0
Residential_Power_and_Battery_Data	0.878	2.049	59.0	**0.728**	**1.617**	1.0	*0.739*	1.751	16.0	0.774	1.718	27.0
Rideshare	0.619	0.933	47.0	0.387	0.325	15.0	0.673	0.811	45.0	0.736	1.155	58.5
Sleep_EDFX	0.718	4.989	35.5	0.768	5.316	50.0	0.805	5.416	57.5	0.764	5.145	46.5
Spanish_energy	0.460	0.498	58.0	0.464	0.467	56.0	0.486	0.509	63.5	0.462	0.504	60.0
Spanish_weather	0.374	0.388	59.5	0.356	0.313	32.0	0.369	0.332	50.5	0.405	0.428	65.0
Temperature-Rain	0.831	4.272	29.0	0.776	4.148	5.0	0.809	4.247	18.5	0.835	4.339	35.5
Traffic	0.649	1.134	62.0	0.395	0.733	51.0	0.382	0.721	49.0	0.339	0.628	32.0
Traffic_Flow_Forecasting	0.616	0.677	43.0	0.308	*0.189*	3.5	0.509	0.504	35.0	0.349	0.265	9.5
WIND	0.725	1.182	52.0	0.652	0.995	6.5	0.690	1.080	30.0	0.721	1.154	49.0
Zafnoo_env	0.601	0.693	63.5	0.458	0.466	31.0	0.478	0.516	49.5	0.465	0.536	50.0
Zafnoo_sapf	0.488	0.570	68.0	0.311	0.332	47.5	0.326	0.379	56.0	0.325	0.330	49.0
australian_tourism	0.901	1.411	57.5	0.922	1.363	59.0	0.875	1.167	49.0	0.837	1.079	36.5
beijing_multisite_air_quality	0.469	0.750	44.0	0.466	0.713	35.0	0.473	0.741	44.5	0.486	0.796	56.5
boomlet_1062	0.731	0.967	45.0	0.679	0.918	22.5	0.803	1.138	65.0	0.688	0.909	26.0
boomlet_1209	0.368	1.566	11.0	1.213	6.850	70.0	1.184	7.132	70.5	0.396	1.776	36.0
boomlet_1225	0.330	0.874	37.0	0.985	7.116	67.5	0.934	7.375	67.0	0.335	0.906	42.0
boomlet_1230	4.125	598.922	20.0	9.690	$> 10^3$	68.0	9.744	$> 10^3$	69.5	4.258	630.994	38.5
boomlet_1282	0.422	1.032	13.0	1.244	10.006	67.5	1.262	10.261	68.5	0.445	1.238	47.5
boomlet_1487	0.203	0.733	8.0	0.325	0.794	59.5	0.400	0.849	66.5	0.221	0.741	34.0
boomlet_1631	0.587	0.731	35.0	1.127	2.161	69.0	1.156	2.222	70.0	0.593	0.747	37.0
boomlet_1676	0.968	11.135	49.0	4.028	271.795	63.0	4.092	271.556	63.5	0.936	9.255	43.5
boomlet_1855	0.740	92.509	39.5	0.968	93.392	55.5	0.966	93.373	54.5	0.728	92.314	36.5
boomlet_1975	0.406	0.439	11.0	0.567	0.775	48.5	0.466	0.709	28.5	0.578	0.770	49.5
boomlet_2187	0.763	39.131	35.0	0.989	41.077	57.5	0.967	41.101	56.5	0.803	43.506	54.5
boomlet_285	0.280	0.296	30.0	0.375	0.386	63.5	0.404	0.356	63.5	0.276	0.303	31.5
boomlet_619	0.524	0.573	56.0	0.373	0.369	37.0	0.443	0.474	48.5	0.413	0.414	43.5
boomlet_772	0.356	1.120	38.0	0.548	1.312	62.0	0.603	1.432	64.5	0.367	1.101	37.5
boomlet_963	0.625	1.553	26.5	0.765	1.762	58.0	0.742	1.780	58.5	0.679	1.760	52.5
electricity_2016	0.501	0.498	66.5	0.398	0.335	62.5	0.394	0.334	61.5	0.316	0.226	51.0
electricityloaddiagrams20112014	4.079	$> 10^3$	38.0	16.856	$> 10^4$	69.5	16.663	$> 10^4$	68.0	4.564	$> 10^3$	45.0
gasrate_co2	0.891	1.173	52.5	0.659	0.680	6.5	0.737	0.825	13.5	0.881	1.162	49.5
gefcom2012_load	1.743	8.851	57.5	1.575	7.826	34.0	1.343	7.292	8.5	1.439	7.218	12.5
gefcom2012_wind	0.884	1.140	44.0	*0.855*	1.079	5.0	0.866	1.106	15.5	0.903	1.202	53.5
gvar	0.281	0.261	36.5	0.281	0.266	40.5	0.811	1.413	65.0	0.284	0.252	36.0
italy_air_quality	0.595	1.242	41.0	0.888	1.843	69.0	0.815	1.631	67.5	0.687	1.399	62.5
vessel_ais	0.749	0.959	36.0	0.785	1.038	53.0	0.831	1.143	70.0	0.773	1.010	44.0
weather_Wetterstation	0.339	0.294	57.5	0.476	0.483	66.5	0.997	1.817	70.0	0.318	0.305	55.5

Dataset	rmlp			rnn			scinet			segrnn		
	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.587	0.716	10.5	0.627	0.803	23.0	0.848	1.323	66.0	0.584	0.721	10.0
Bike_Sharing	0.532	0.712	17.0	0.569	0.766	34.5	0.502	0.657	7.0	0.587	0.807	39.5
COVID-19_Deaths	1.719	74.278	20.0	2.524	124.208	37.5	12.459	$> 10^3$	66.0	2.308	91.433	32.5
Czelan	0.229	0.209	13.0	0.282	0.256	51.5	0.309	0.259	56.5	0.263	0.212	29.0
DailyDelhiClimate	1.790	55.015	31.0	3.891	235.091	69.0	0.966	42.748	2.0	2.592	250.144	63.5
Dominick	0.667	1.370	9.5	0.688	1.430	18.5	0.906	1.804	56.0	0.689	1.566	32.0
ETTH1	0.422	0.418	13.5	0.449	0.437	28.5	0.544	0.600	59.5	0.417	**0.397**	5.0
ETTH2	0.380	0.352	22.0	0.418	0.382	49.5	0.703	0.963	66.0	0.375	0.325	10.0
ETTM1	0.460	0.509	18.5	0.687	1.052	67.0	0.440	0.431	2.0	0.487	0.533	31.0
ETTM2	0.273	0.167	7.0	0.298	0.186	41.5	0.316	0.219	58.0	0.282	0.174	25.5
FRED-MD	1.288	63.024	16.0	1.540	70.152	34.5	4.244	330.662	67.0	*1.190*	*48.245*	3.0
ILY	1.064	2.635	12.5	1.117	2.885	20.5	1.401	4.293	52.5	1.132	2.995	23.0
KDDCup2018	0.608	0.827	35.5	0.589	0.784	10.5	0.619	0.788	30.5	0.600	0.855	38.0
METR-LA	0.648	1.093	20.0	0.652	1.137	29.0	*0.602*	1.046	5.0	0.642	1.089	18.5
MIT-BIH	0.647	3.401	68.0	0.565	1.883	23.5	0.543	1.423	7.5	0.560	1.843	19.5
Metro_Interstate_Traffic_Volume	0.343	0.327	28.5	0.355	0.337	40.0	0.343	0.300	17.5	0.353	0.326	32.0
MexicoCityBikes	0.849	1.280	28.5	0.848	1.278	20.0	0.848	1.274	6.0	0.849	1.281	30.0
NASDAQ	1.429	4.249	27.0	1.540	4.927	46.0	1.237	*2.707*	4.0	1.217	3.535	5.5
NYSE	0.388	0.480	23.5	0.365	0.443	3.0	0.914	2.629	69.0	0.374	0.619	34.5
National_Stock_Exchange	0.672	1.063	15.0	0.744	1.239	30.5	0.837	1.448	53.0	**0.557**	**0.628**	1.0
Oikolab	0.611	0.650	37.5	0.594	0.625	16.5	*0.557*	*0.549*	3.0	0.602	0.630	24.0
PEMS-BAY	0.474	0.947	34.0	0.425	0.854	20.0	0.335	*0.597*	2.5	0.484	0.917	34.5
PEMS03	0.344	0.276	35.5	0.277	0.186	16.0	**0.226**	**0.130**	1.0	0.321	0.220	29.0
PEMS04	0.366	0.298	41.0	0.231	0.122	7.5	0.215	0.109	2.0	0.338	0.237	32.0
PEMS07	0.364	0.300	38.0	0.276	0.174	21.5	0.203	0.107	2.0	0.350	0.250	34.5
PEMS08	0.376	0.376	36.0	0.280	0.248	10.5	0.233	*0.227*	2.5	0.369	0.318	31.0
Power_Consumption_of_Tetouan_City	0.308	0.278	19.5	0.328	0.280	25.5	0.322	0.239	13.5	0.345	0.266	23.0
Residential_Power_and_Battery_Data	0.773	1.795	35.0	0.774	1.765	33.5	0.745	1.619	4.0	0.755	1.676	12.0
Rideshare	0.745	1.353	61.0	0.553	0.652	30.0	0.667	0.898	48.0	0.328	0.286	11.5
Sleep_EDFX	0.667	4.875	20.5	0.699	4.969	29.0	0.653	4.981	21.0	0.655	4.966	21.5
Spanish_energy	0.391	0.391	15.0	0.400	0.403	26.5	0.407	0.372	19.5	0.398	0.387	15.0
Spanish_weather	0.324	0.318	19.5	0.355	0.356	51.0	0.328	0.297	14.5	0.330	0.309	19.0
Temperature-Rain	0.817	4.195	13.5	0.809	4.169	10.5	0.803	4.161	7.5	0.887	6.709	62.0
Traffic	0.307	0.473	14.5	0.342	0.648	36.0	0.288	0.557	14.0	0.341	0.657	36.0
Traffic_Flow_Forecasting	0.393	0.316	21.5	0.544	0.510	37.0	0.305	0.181	2.0	0.650	0.682	45.0
WIND	0.690	1.096	33.5	0.651	1.006	7.0	*0.632*	**0.948**	2.0	0.666	1.033	16.0
Zafnoo_env	0.427	0.478	25.0	0.425	0.456	12.0	0.437	0.457	21.5	0.421	*0.446*	6.5
Zafnoo_sapf	0.279	0.285	14.5	0.315	0.322	47.5	0.287	0.289	21.5	0.289	0.297	26.5
australian_tourism	0.777	0.919	14.5	0.843	1.267	46.5	0.762	0.916	10.5	0.790	1.149	33.5
beijing_multisite_air_quality	0.448	0.706	18.0	0.442	0.701	9.0	0.437	0.668	3.5	0.445	0.681	9.0
boomlet_1062	0.678	0.895	14.0	0.678	0.887	10.0	0.715	0.990	46.0	0.684	0.894	19.5
boomlet_1209	0.365	1.558	8.0	0.393	1.753	32.5	0.634	4.970	66.5	0.388	1.691	25.5
boomlet_1225	0.289	0.743	11.0	0.340	0.897	41.5	0.720	6.396	65.0	0.312	0.706	15.0
boomlet_1230	3.972	585.826	7.5	4.268	633.150	41.5	9.230	$> 10^3$	66.0	4.228	666.732	45.5
boomlet_1282	0.411	1.012	5.5	0.435	1.093	27.0	1.013	8.677	65.0	0.422	1.061	17.5
boomlet_1487	0.210	0.738	24.5	0.215	0.742	31.0	0.287	0.766	51.0	0.259	0.750	46.5
boomlet_1631	0.534	0.643	5.5	0.600	0.757	42.0	0.873	1.474	66.0	0.559	0.676	24.5
boomlet_1676	0.608	2.437	20.0	0.947	10.738	47.0	3.694	265.841	61.0	0.619	1.770	15.0
boomlet_1855	0.604	92.038	23.5	0.750	93.328	47.5	0.896	93.119	49.5	0.569	91.354	10.0
boomlet_1975	0.522	0.685	34.0	0.525	0.623	31.0	0.402	0.482	11.5	0.546	0.708	42.0
boomlet_2187	0.698	33.449	16.0	0.794	44.622	51.0	0.741	40.057	34.5	0.714	23.492	9.5
boomlet_285	0.258	0.278	14.5	0.280	0.309	34.5	0.288	0.284	28.0	0.307	0.296	35.0
boomlet_619	0.321	0.320	13.0	0.325	0.318	14.5	0.313	0.306	4.0	0.331	0.318	16.5
boomlet_772	0.317	*1.027*	6.5	0.390	1.152	44.5	0.414	1.177	53.0	0.343	1.067	25.0
boomlet_963	0.594	1.469	6.0	0.659	1.639	44.0	0.650	1.558	34.5	0.609	1.500	14.5
electricity_2016	0.256	0.173	15.5	0.316	0.217	48.5	0.247	0.152	6.0	0.274	0.182	28.0
electricityloaddiagrams20112014	1.630	315.396	4.0	4.568	$> 10^3$	49.0	15.646	$> 10^4$	64.0	5.374	$> 10^3$	54.0
gasrate_co2	0.827	1.054	29.0	1.171	2.059	69.5	0.691	0.740	9.0	0.860	1.151	41.5
gefcom2012_load	1.472	7.529	19.5	1.428	7.209	11.5	1.461	7.743	24.0	1.884	8.051	53.5
gefcom2012_wind	0.875	1.133	34.0	0.871	1.114	22.0	**0.850**	**1.052**	1.0	0.864	1.105	13.5
gvar	0.269	0.233	20.5	0.279	0.245	27.5	0.737	1.111	62.5	0.250	0.219	7.5
italy_air_quality	0.550	1.111	11.5	0.574	1.151	26.0	0.594	1.180	33.0	0.576	1.121	22.5
vessel_ais	0.681	0.806	15.0	0.744	0.945	33.0	0.679	0.802	10.5	0.723	0.902	30.0
weather_Wetterstation	0.272	0.224	21.5	0.313	0.263	48.0	0.258	0.184	3.0	0.269	0.201	8.5

Dataset	seq2seq			sim_mtm		
	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.701	1.021	50.5	0.595	0.738	12.5
Bike_Sharing	0.698	1.032	58.0	0.530	0.683	13.5
COVID-19_Deaths	2.676	134.719	41.0	1.330	55.340	4.5
Czelan	0.281	0.259	52.0	0.237	0.215	23.5
DailyDelhiClimate	2.085	53.122	34.0	4.871	196.495	70.5
Dominick	0.712	1.440	27.0	0.701	1.413	21.0
ETTH1	0.522	0.594	56.5	0.425	0.414	13.5
ETTH2	0.414	0.373	44.5	0.391	0.357	30.0
ETTM1	0.536	0.616	45.5	0.448	0.453	4.0
ETTM2	0.291	0.177	32.0	0.271	0.165	5.0
FRED-MD	1.549	65.227	31.0	1.293	62.467	15.5
ILY	1.292	3.678	45.0	1.051	2.713	13.0
KDDCup2018	0.608	0.832	38.5	0.612	0.822	38.0
METR-LA	1.301	3.952	68.5	0.667	1.141	35.5
MIT-BIH	0.572	1.646	18.5	0.607	2.731	51.0
Metro_Interstate_Traffic_Volume	0.391	0.391	57.5	0.369	0.351	50.0
MexicoCityBikes	0.848	1.278	18.0	0.853	1.292	48.0
NASDAQ	1.740	5.710	66.0	1.464	4.518	36.0
NYSE	0.442	0.559	47.0	0.390	0.482	26.0
National_Stock_Exchange	0.783	1.290	39.0	0.784	1.316	41.5
Oikolab	0.659	0.720	60.0	0.617	0.639	38.0
PEMS-BAY	0.446	0.924	28.0	0.461	0.889	27.5
PEMS03	0.466	0.423	51.5	0.328	0.237	30.5
PEMS04	0.247	0.135	16.0	0.353	0.276	36.5
PEMS07	0.280	0.163	20.5	0.339	0.244	33.0
PEMS08	0.269	0.230	5.5	0.358	0.322	31.0
Power_Consumption_of_Tetouan_City	0.432	0.402	45.0	0.292	0.245	7.0
Residential_Power_and_Battery_Data	0.789	1.743	35.5	0.774	1.727	27.5
Rideshare	0.618	0.809	40.5	0.652	1.087	52.0
Sleep_EDFX	0.758	5.335	49.5	0.659	4.719	16.0
Spanish_energy	0.442	0.471	53.5	0.399	0.394	22.0
Spanish_weather	0.380	0.394	60.5	0.334	0.326	36.0
Temperature-Rain	0.818	4.201	15.5	0.854	4.483	50.0
Traffic	0.677	1.231	64.5	0.292	0.456	8.0
Traffic_Flow_Forecasting	0.751	0.852	53.0	0.375	0.273	14.5
WIND	0.704	1.093	37.5	0.680	1.051	24.0
Zafnoo_env	0.541	0.605	61.0	0.433	0.476	26.0
Zafnoo_sapf	0.383	0.420	63.0	0.278	0.287	14.0
australian_tourism	0.837	1.080	37.5	0.788	1.040	24.0
beijing_multisite_air_quality	0.477	0.759	50.0	0.456	0.716	32.5
boomlet_1062	0.735	0.982	49.0	0.673	0.897	12.0
boomlet_1209	0.422	1.859	48.5	0.363	1.573	8.0
boomlet_1225	0.360	1.025	49.0	0.301	0.779	20.0
boomlet_1230	4.270	631.702	40.5	4.053	586.120	13.5
boomlet_1282	0.440	1.153	36.5	0.411	1.030	7.5
boomlet_1487	0.203	*0.733*	6.5	0.239	0.758	46.0
boomlet_1631	0.598	0.746	38.0	0.538	0.647	9.0
boomlet_1676	1.043	17.341	53.0	0.671	3.946	29.0
boomlet_1855	0.784	92.827	44.5	0.631	92.155	29.5
boomlet_1975	0.431	0.617	20.0	0.523	0.670	32.5
boomlet_2187	0.802	46.744	57.5	0.726	33.722	24.0
boomlet_285	0.359	0.412	64.0	0.250	**0.264**	5.0
boomlet_619	0.407	0.436	44.5	0.352	0.335	26.5
boomlet_772	0.391	1.158	46.0	0.321	1.035	11.5
boomlet_963	0.657	1.655	44.0	0.594	1.453	5.0
electricity_2016	0.429	0.362	63.5	0.274	0.179	27.5
electricityloaddiagrams20112014	7.147	$> 10^3$	59.5	1.738	358.023	15.0
gasrate_co2	0.902	1.232	56.0	0.807	0.968	21.5
gefcom2012_load	1.421	7.159	6.5	1.919	9.227	61.5
gefcom2012_wind	0.877	1.111	29.0	0.865	1.093	12.0
gvar	0.291	0.249	37.5	0.257	0.229	13.5
italy_air_quality	0.622	1.278	49.5	0.568	1.123	19.5
vessel_ais	0.781	1.028	49.0	0.679	0.803	13.0
weather_Wetterstation	0.327	0.282	52.0	0.280	0.232	31.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 23: **Multivariate forecasting results.** Part 6.

Dataset	simpletm			sparse_tsf			sundial			tabpfn			tcn			test			tft			tide			timekan			timellm		
	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.565	*0.679*	3.5	0.691	0.932	42.0	0.647	0.888	35.0	0.610	0.821	22.0	0.712	0.979	50.0	0.793	1.205	62.5	0.732	1.075	55.5	0.611	0.769	19.0	0.568	0.684	4.5	0.640	0.835	29.0
Bike_Sharing	0.529	0.713	16.5	0.740	1.133	61.5	0.553	0.764	28.0	0.555	0.967	40.5	0.759	1.207	62.5	0.830	1.338	66.0	0.733	1.297	62.5	0.586	0.803	38.5	0.559	0.747	29.5	0.621	0.902	49.0
COVID-19_Deaths	1.511	63.828	9.5	3.752	197.851	49.5	3.046	140.988	45.0	1.531	64.166	10.5	4.504	211.315	53.0	2.067	89.356	28.0	2.290	99.110	32.5	1.647	73.673	18.0	1.588	64.942	13.5	1.812	86.061	25.0
Czelan	0.239	0.236	34.5	0.277	0.257	49.5	0.235	0.216	23.0	0.225	0.235	24.5	0.282	0.262	54.0	0.302	0.277	58.0	0.256	0.237	38.5	0.238	0.218	27.5	0.233	0.209	17.5	0.269	0.262	48.5
DailyDelhiClimate	1.962	62.173	38.5	2.419	103.759	56.0	1.429	49.841	21.0	1.219	70.152	27.5	1.699	61.253	35.0	1.564	47.142	21.0	3.048	91.995	61.0	2.648	109.712	61.5	1.657	47.637	24.0	4.538	309.972	71.0
Dominick	0.672	1.387	12.5	0.784	1.737	51.0	0.670	1.352	9.0	0.585	1.593	23.5	0.756	1.637	47.0	0.713	1.484	32.0	0.739	1.609	45.5	0.688	1.433	20.0	0.722	1.484	34.0	0.710	1.497	32.0
ETTH1	0.427	0.419	17.5	0.505	0.588	54.5	**0.408**	0.415	5.0	0.423	0.477	28.0	0.538	0.604	59.0	0.579	0.722	64.5	0.511	0.538	50.5	0.419	0.421	13.5	0.429	0.423	21.0	0.459	0.467	36.5
ETTH2	0.383	0.349	23.0	0.406	0.374	42.5	0.364	0.324	5.0	0.399	0.417	47.5	0.418	0.379	48.5	0.409	0.373	42.0	0.407	0.384	45.5	0.376	0.331	11.5	0.364	0.310	3.5	0.390	0.353	27.0
ETTM1	0.454	0.491	11.0	0.591	0.832	59.0	0.522	0.723	46.5	0.530	0.750	49.5	0.648	0.932	65.5	0.725	1.118	68.0	0.523	0.583	41.0	0.473	0.544	29.0	0.453	0.466	7.5	0.526	0.644	44.0
ETTM2	0.277	0.174	17.5	0.314	0.206	55.0	0.291	0.185	34.5	0.291	0.198	42.5	0.305	0.188	45.0	0.320	0.203	56.0	0.294	0.187	39.0	0.278	0.173	16.0	0.273	0.171	10.5	0.336	0.237	63.5
FRED-MD	1.296	61.878	15.0	1.821	102.811	52.5	1.517	89.694	40.5	1.300	65.283	20.5	1.908	114.972	56.5	1.834	97.664	52.0	1.519	55.908	22.0	1.288	62.176	14.0	1.279	56.960	11.5	1.489	70.205	31.0
ILY	1.069	2.694	14.5	1.665	5.382	67.5	1.271	3.537	42.0	1.308	4.193	49.0	1.496	4.705	58.5	1.531	4.465	58.0	1.184	3.313	33.0	1.237	3.329	36.5	1.236	3.279	34.0	1.509	4.612	58.0
KDDCup2018	0.611	0.830	40.0	0.638	0.865	54.5	*0.548*	*0.714*	3.0	0.643	0.920	59.0	0.627	0.878	53.0	0.616	0.837	44.5	0.598	0.810	25.5	0.595	0.803	19.0	0.590	0.784	11.0	0.643	0.913	59.0
METR-LA	0.638	1.048	12.5	0.723	1.199	49.0	0.560	**0.946**	1.5	0.628	1.359	33.0	0.689	1.219	48.0	0.724	1.266	53.0	0.694	1.306	53.0	0.668	1.142	37.0	0.641	1.082	17.0	0.662	1.079	25.0
MIT-BIH	0.589	2.507	42.0	0.616	1.940	45.0	**0.528**	1.468	6.0	0.603	2.007	39.0	0.612	1.756	37.0	0.637	1.749	42.5	0.571	1.866	25.5	0.633	3.131	65.0	0.594	2.894	47.5	0.848	2.833	66.0
Metro_Interstate_Traffic_Volume	0.342	0.322	24.5	0.435	0.480	65.5	0.301	0.295	4.0	0.375	0.522	61.5	0.418	0.424	62.5	0.443	0.472	66.0	0.325	0.297	9.0	0.356	0.345	43.5	0.354	0.332	36.5	1.531	10.821	72.5
MexicoCityBikes	0.848	1.278	15.5	0.855	1.302	52.5	0.853	1.296	49.0	0.849	1.320	44.5	0.856	1.308	56.0	0.849	1.282	35.5	0.850	1.282	39.0	0.852	1.291	45.5	0.848	1.278	17.0	0.893	1.438	68.0
NASDAQ	1.418	4.053	21.5	1.757	5.858	68.0	1.416	3.979	19.5	1.246	3.728	8.5	1.699	5.346	60.5	1.598	4.792	48.0	1.787	6.059	69.0	1.440	4.390	31.0	1.674	5.303	58.5	1.708	6.355	67.0
NYSE	0.387	0.473	20.5	0.488	0.660	59.0	0.411	0.514	41.5	0.373	0.468	11.5	0.474	0.611	56.5	0.463	0.580	52.5	0.399	0.496	32.5	0.398	0.498	34.0	0.413	0.503	40.5	0.570	1.021	63.0
National_Stock_Exchange	0.740	1.262	30.0	0.935	1.867	63.5	0.746	1.231	30.0	0.677	0.957	13.0	0.841	1.480	55.0	0.736	1.268	30.0	0.705	1.123	20.5	0.815	1.392	50.0	0.783	1.294	40.0	1.218	2.996	70.5
Oikolab	0.609	0.651	35.0	0.665	0.745	63.0	0.559	0.570	4.5	0.672	0.839	66.0	0.634	0.687	53.5	0.741	0.861	68.5	0.598	0.617	15.0	0.613	0.658	41.5	0.611	0.638	33.5	0.644	0.694	56.5
PEMS-BAY	0.430	0.863	21.5	0.642	1.368	58.5	0.549	1.198	50.5	0.616	1.518	58.0	0.545	1.136	48.0	0.689	1.527	63.5	0.414	0.793	11.5	0.495	1.023	40.5	0.486	0.904	34.0	0.525	1.091	46.0
PEMS03	0.344	0.271	34.0	0.523	0.538	55.5	0.483	0.498	54.0	0.762	1.061	67.5	0.444	0.358	46.5	1.003	1.375	70.0	0.265	0.179	11.5	0.406	0.391	45.0	0.370	0.288	38.5	0.458	0.414	50.5
PEMS04	0.335	0.248	33.5	0.558	0.614	57.0	0.501	0.567	54.0	0.767	1.070	63.0	0.391	0.281	42.5	0.998	1.382	69.5	0.229	0.121	6.5	0.413	0.380	47.0	0.378	0.292	41.5	0.449	0.413	52.0
PEMS07	0.370	0.287	38.5	0.512	0.505	51.5	0.518	0.581	54.0	0.722	0.990	67.0	0.456	0.362	47.0	0.978	1.418	70.5	0.244	0.141	11.0	0.423	0.385	46.5	0.370	0.276	37.0	0.560	0.556	55.0
PEMS08	0.353	0.326	31.0	0.554	0.658	57.0	0.519	0.619	55.0	0.753	1.091	62.0	0.492	0.491	50.0	1.004	1.504	71.0	0.312	0.268	17.0	0.457	0.497	48.0	0.415	0.392	41.0	0.495	0.526	53.5
Power_Consumption_of_Tetouan_City	*0.275*	*0.229*	3.0	0.447	0.466	52.0	0.418	0.436	47.0	0.394	0.458	46.5	0.435	0.428	47.5	0.617	0.694	66.5	0.348	0.315	32.0	0.339	0.312	29.5	0.309	0.272	18.5	0.380	0.357	39.0
Residential_Power_and_Battery_Data	0.777	1.760	34.0	0.873	1.908	56.5	0.851	1.916	55.0	0.867	2.385	64.0	0.879	2.054	61.0	0.878	2.047	59.0	0.778	1.716	28.5	0.789	1.803	43.5	0.779	1.757	35.0	1.055	2.846	71.0
Rideshare	0.417	0.391	16.5	0.610	0.838	41.5	0.441	0.474	22.5	0.276	0.238	4.5	0.607	0.803	38.0	0.685	1.086	53.5	0.858	1.652	67.5	0.719	1.871	62.0	0.632	0.875	44.5	0.602	0.975	45.0
Sleep_EDFX	*0.638*	4.926	14.0	0.823	5.630	61.0	0.745	5.481	50.5	0.785	5.413	54.5	0.852	5.847	65.0	0.861	5.943	67.0	0.732	5.150	41.5	0.750	5.217	46.5	0.652	4.525	7.5	0.777	5.452	54.5
Spanish_energy	0.385	0.381	8.0	0.454	0.492	57.0	0.386	0.402	16.5	0.428	0.507	55.5	0.429	0.451	50.0	0.509	0.572	65.5	0.428	0.458	50.0	0.400	0.406	27.0	0.396	0.392	18.5	0.422	0.432	46.5
Spanish_weather	0.324	0.323	24.5	0.364	0.371	57.0	0.326	0.326	27.5	0.333	0.371	45.5	0.397	0.417	64.0	0.417	0.436	66.5	0.340	0.326	39.0	0.326	0.319	23.0	0.322	0.314	14.5	0.499	0.764	71.5
Temperature-Rain	0.839	4.474	45.5	1.009	5.172	64.5	0.825	4.339	30.5	**0.702**	**3.158**	1.0	0.885	4.410	49.0	0.837	4.321	34.0	0.820	4.228	21.5	0.831	4.381	36.0	0.833	4.219	25.5	0.838	4.302	33.0
Traffic	0.292	0.437	6.0	0.378	0.613	40.0	0.328	0.570	24.5	0.398	0.864	53.5	0.445	0.777	54.0	0.804	1.431	66.5	0.375	0.710	46.0	0.351	0.574	32.5	0.342	0.532	26.0	0.607	1.000	60.0
Traffic_Flow_Forecasting	0.362	0.275	13.5	1.088	1.694	68.5	0.683	0.952	51.5	0.769	1.140	61.0	1.056	1.577	67.0	1.037	1.510	66.0	0.452	0.384	29.0	0.665	0.782	47.0	0.479	0.410	30.5	0.689	0.901	52.0
WIND	0.685	1.089	30.0	0.774	1.264	62.0	0.712	1.171	46.5	0.739	1.308	61.0	0.794	1.308	65.0	0.802	1.319	66.5	0.734	1.151	51.0	0.683	1.077	26.0	0.668	1.047	20.0	0.726	1.171	51.0
Zafnoo_env	0.447	0.493	41.0	0.522	0.608	60.0	0.412	0.472	14.0	0.428	0.554	37.0	0.524	0.620	61.0	0.696	0.823	68.5	0.483	0.577	56.0	0.438	0.478	32.0	0.434	0.473	26.0	0.487	0.533	52.5
Zafnoo_sapf	0.280	0.291	19.0	0.350	0.362	57.0	0.278	0.288	15.0	0.370	0.469	62.5	0.344	0.344	53.5	0.488	0.569	67.0	0.382	0.411	62.0	0.297	0.304	38.5	0.291	0.296	26.5	0.312	0.316	43.5
australian_tourism	0.783	1.010	21.5	0.919	1.568	62.5	0.689	0.830	5.0	*0.668*	*0.803*	3.0	0.888	1.321	54.5	0.844	1.120	43.5	1.024	1.698	67.0	0.765	0.980	16.5	0.856	1.196	46.5	0.820	1.044	28.5
beijing_multisite_air_quality	0.451	0.713	25.0	0.495	0.799	60.0	*0.433*	0.706	10.5	0.474	0.819	56.0	0.503	0.812	61.5	0.495	0.799	59.0	0.453	0.728	34.5	0.451	0.713	25.0	0.444	0.700	9.0	0.474	0.757	47.5
boomlet_1062	0.677	0.891	10.0	0.716	0.969	43.5	0.681	0.917	25.0	0.629	1.097	32.5	0.705	0.939	38.0	0.767	1.022	57.5	0.690	0.919	32.0	0.686	0.916	27.5	0.682	0.899	20.5	0.701	0.941	38.0
boomlet_1209	0.362	*1.564*	4.5	0.462	1.917	53.0	0.403	1.683	31.5	0.383	1.843	37.0	0.396	1.805	39.0	0.390	1.815	37.5	0.394	1.797	37.5	0.376	1.639	21.5	0.362	**1.556**	4.0	0.382	1.665	23.5
boomlet_1225	0.289	0.754	13.0	0.335	0.850	37.0	0.308	0.811	27.5	0.285	0.819	20.5	0.341	0.893	41.5	0.329	0.854	35.0	0.350	0.901	43.0	0.300	0.777	19.0	0.313	0.841	31.5	0.306	0.791	24.0
boomlet_1230	3.967	587.148	8.0	4.492	680.533	56.5	4.225	606.691	29.5	4.178	649.548	40.0	4.345	635.920	47.5	4.255	632.823	39.0	4.314	633.681	44.5	3.997	576.213	5.5	4.034	586.013	11.5	4.130	621.376	25.5
boomlet_1282	**0.405**	**0.983**	1.0	0.462	1.234	49.0	0.431	1.068	22.5	0.434	1.204	36.5	0.440	1.155	36.5	0.444	1.175	43.5	0.441	1.115	32.0	0.430	1.071	22.0	0.418	1.036	12.5	0.438	1.115	30.0
boomlet_1487	0.211	0.736	23.5	0.234	0.767	46.5	0.209	0.734	12.5	0.196	0.736	15.0	0.205	0.734	11.0	0.203	0.734	8.0	0.208	0.735	16.0	0.220	0.747	36.5	0.217	0.741	30.5	0.229	0.748	39.5
boomlet_1631	0.535	0.641	5.5	0.577	0.706	31.0	0.551	0.677	20.0	0.553	0.689	24.0	0.584	0.732	35.0	0.580	0.714	32.5	0.646	0.912	55.5	**0.524**	**0.614**	1.0	0.551	0.669	18.5	0.547	0.663	14.5
boomlet_1676	0.580	1.955	13.5	0.588	1.771	14.0	0.575	1.754	9.0	0.956	21.761	51.0	0.932	9.965	45.0	0.812	7.091	36.0	1.243	30.682	55.5	0.585	2.170	15.5	0.709	4.050	32.0	0.649	3.213	25.5
boomlet_1855	0.610	92.160	27.5	0.716	101.542	52.5	0.544	95.598	34.5	0.516	91.358	7.0	0.629	91.739	21.5	0.607	92.076	25.5	0.898	94.539	56.5	0.559	91.949	15.0	0.651	92.575	34.5	0.668	93.164	40.5
boomlet_1975	0.387	0.390	7.5	0.674	0.931	65.5	0.558	0.887	55.0	0.541	1.061	54.0	0.665	0.860	62.5	0.684	0.887	64.5	0.512	0.592	25.0	0.541	0.735	41.5	0.531	0.681	36.5	0.600	0.779	55.5
boomlet_2187	0.675	32.062	9.5	0.937	48.010	64.5	0.770	33.451	27.5	0.660	37.271	20.0	0.769	43.449	45.5	0.782	44.796	50.5	0.852	44.029	58.5	0.749	38.196	33.5	0.748	37.280	32.5	0.777	40.956	41.0
boomlet_285	0.246	*0.268*	5.0	0.304	0.312	42.0	0.300	0.315	43.5	0.195	0.325	24.0	0.276	0.303	30.5	0.320	0.341	53.0	0.279	0.333	41.5	0.252	0.274	10.0	0.262	0.271	14.5	0.291	0.309	37.5
boomlet_619	0.320	0.315	9.5	0.518	0.545	52.5	0.522	0.586	56.0	0.454	0.588	53.0	0.429	0.431	46.0	0.646	0.733	62.0	0.379	0.402	40.5	0.379	0.370	39.0	0.332	0.314	14.0	0.410	0.421	44.0
boomlet_772	0.311	1.025	4.0	0.369	1.099	39.0	0.346	1.081	30.0	0.312	1.073	18.0	0.363	1.107	38.0	0.353	1.099	34.0	0.414	1.215	54.5	0.334	1.056	20.0	0.326	1.040	16.5	0.334	1.055	19.0
boomlet_963	*0.594*	1.456	4.0	0.636	1.588	32.0	0.638	1.660	39.0	0.615	1.603	26.5	0.637	1.621	36.0	0.644	1.627	38.5	0.631	1.547	26.0	0.594	1.467	6.0	0.604	1.496	12.5	0.618	1.523	21.5
electricity_2016	0.245	0.157	5.0	0.364	0.298	58.0	0.249	0.168	11.0	0.280	0.235	44.5	0.350	0.266	55.5	0.763	0.854	69.0	0.279	0.184	30.5	0.266	0.187	28.0	0.267	0.179	25.0	0.345	0.252	54.0
electricityloaddiagrams20112014	1.634	318.509	5.5	1.950	458.180	25.0	1.854	459.152	24.0	3.170	$> 10^3$	35.5	6.239	$> 10^3$	56.0	4.929	$> 10^3$	45.5	4.577	$> 10^3$	48.0	1.713	345.682	11.5	1.770	388.287	19.0	2.477	743.016	28.5
gasrate_co2	0.808	0.987	23.0	0.979	1.384	65.0	0.786	1.005	22.0	0.839	1.212	44.0	0.824	0.947	23.0	0.879	1.166	49.5	0.923	1.276	59.5	0.864	1.141	42.0	0.867	1.132	42.0	1.520	3.753	72.5
gefcom2012_load	1.686	8.328	50.5	1.641	8.524	50.5	1.484	7.594	24.0	1.954	9.257	63.0	1.601	7.938	40.5	1.605	7.650	33.0	1.567	7.911	35.5	1.854	9.370	61.0	1.587	7.846	36.5	2.539	13.242	72.5
gefcom2012_wind	0.869	1.121	24.5	0.938	1.257	61.0	0.904	1.219	55.5	0.959	1.482	69.0	0.913	1.199	55.5	0.911	1.186	53.0	0.883	1.204	48.5	0.872	1.126	27.5	0.867	1.087	12.0	0.879	1.134	38.5
gvar	0.267	0.234	19.5	0.304	0.265	49.5	0.259	0.219	10.5	0.255	0.224	11.5	0.298	0.267	49.0	0.299	0.262	46.5	0.287	0.265	42.5	0.279	0.252	32.5	0.281	0.245	27.5	0.347	0.388	55.5
italy_air_quality	0.542	1.081	7.5	0.661	1.363	59.0	0.554	1.218	29.0	0.534	1.213	20.5	0.680	1.406	62.5	0.643	1.329	55.0	0.581	1.215	35.5	0.570	1.119	19.0	0.585	1.131	25.5	0.568	1.144	22.0
vessel_ais	0.677	0.800	9.0	0.770	1.006	42.0	0.796	1.063	59.0	0.810	1.121	67.0	0.782	1.034	51.0	0.790	1.046	55.0	0.774	1.012	45.0	0.675	*0.795*	2.5	0.691	0.830	22.0	0.732	0.916	31.0
weather_Wetterstation	0.263	0.214	8.5	0.332	0.288	55.5	0.274	0.226	24.5	0.282	0.267	40.0	0.330	0.286	54.5	0.347	0.303	58.5	0.327	0.323	58.0	0.271	0.219	16.5	0.266	0.213	9.0	0.355	0.382	64.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 24: **Multivariate forecasting results.** Part 7.

Dataset	timemixer			timer			timesfm			timesfm_v1			timesfm_v2			timesnet			timexer			tirex			toto			transformer		
	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.659	0.951	40.5	0.672	0.932	40.0	0.607	0.830	22.5	0.675	0.992	45.0	0.605	0.827	20.0	0.663	0.870	35.5	0.561	0.658	2.0	0.605	0.833	22.0	0.635	0.933	35.0	1.163	2.332	70.0
Bike_Sharing	0.517	0.680	10.5	0.638	0.909	50.5	0.527	0.821	26.5	0.631	0.975	52.5	0.610	0.934	49.5	0.547	0.723	22.5	0.500	0.650	5.0	0.531	0.846	30.5	0.555	0.933	40.0	0.765	1.297	64.5
COVID-19_Deaths	1.949	83.612	25.0	4.000	175.754	50.0	1.639	69.038	15.5	1.984	82.960	25.5	1.569	64.206	12.0	4.399	220.431	52.5	1.653	69.119	17.0	1.511	64.469	10.5	1.786	70.672	19.5	13.125	$> 10^3$	68.5
Czelan	0.228	0.209	13.5	0.280	0.255	49.0	0.199	0.195	4.0	0.225	0.215	15.0	0.221	0.228	19.0	0.245	0.204	22.0	0.223	0.201	8.0	0.231	0.250	31.0	0.233	0.233	27.5	0.487	0.451	70.5
DailyDelhiClimate	2.395	91.211	51.5	1.588	55.923	27.0	1.110	48.257	14.5	1.551	56.102	25.5	1.221	46.822	14.0	2.637	94.833	58.0	1.992	56.775	35.0	1.081	46.640	9.5	*1.014*	42.858	4.0	1.083	*42.825*	5.0
Dominick	0.718	1.551	37.5	0.996	2.222	64.0	*0.582*	1.502	18.0	0.647	1.366	7.5	0.635	1.418	11.5	0.774	1.734	49.5	0.693	1.436	21.5	0.593	1.561	22.5	**0.552**	1.508	18.0	0.996	2.093	62.5
ETTH1	0.420	0.409	8.5	0.427	0.471	27.5	0.413	0.450	15.5	0.443	0.507	39.0	0.426	0.485	31.0	0.474	0.478	42.5	0.424	0.420	16.0	0.416	0.461	18.5	0.437	0.513	37.5	0.786	1.017	71.0
ETTH2	0.382	0.344	21.5	0.386	0.375	35.0	0.377	0.358	24.0	0.381	0.358	26.5	0.374	0.357	21.0	0.404	0.365	37.5	0.379	0.334	13.0	**0.357**	0.337	7.0	0.392	0.398	41.5	1.449	3.431	71.5
ETTM1	0.451	0.477	7.5	0.585	0.787	57.0	0.547	0.764	53.5	0.618	0.926	63.5	0.515	0.701	44.0	0.476	0.496	20.5	0.452	0.466	7.0	0.516	0.725	46.0	0.607	0.937	63.0	0.573	0.648	49.5
ETTM2	0.275	0.169	9.5	0.298	0.200	48.5	0.292	0.200	44.0	0.314	0.215	57.0	0.296	0.196	46.0	0.278	0.174	21.0	**0.260**	**0.153**	1.0	0.285	0.191	36.0	0.314	0.219	57.0	0.410	0.360	70.0
FRED-MD	1.229	56.926	8.5	1.509	78.560	36.5	**0.933**	**34.977**	1.0	1.522	92.871	42.5	1.400	85.686	36.0	1.716	87.416	47.5	1.364	70.991	28.0	1.199	53.090	5.0	1.246	52.491	7.0	5.002	341.661	70.0
ILY	1.049	2.612	10.5	1.237	3.221	33.5	**0.852**	1.992	1.5	1.240	3.566	40.0	1.199	3.500	36.0	1.159	3.075	27.0	1.015	2.482	6.5	1.057	2.851	16.0	1.132	3.571	33.5	1.534	4.915	62.5
KDDCup2018	0.587	0.783	8.0	0.620	0.821	40.0	0.597	0.829	30.0	0.633	0.881	56.0	0.599	0.821	29.5	0.609	0.834	40.0	0.601	0.815	29.5	0.492	0.645	2.0	0.609	0.862	43.5	0.697	0.933	65.0
METR-LA	73.740	$> 10^4$	71.0	0.711	1.287	52.5	**0.502**	0.978	1.5	0.779	1.625	64.0	0.745	1.579	61.0	0.678	1.233	48.0	0.629	1.096	17.0	0.625	1.362	32.5	0.788	17.034	68.0	0.615	1.225	27.5
MIT-BIH	0.699	2.784	65.0	0.604	1.726	31.5	0.540	1.507	10.0	0.684	1.798	47.5	0.562	1.415	9.0	0.607	2.933	55.0	0.591	3.221	50.5	0.537	1.397	5.0	0.545	1.502	10.5	0.586	1.489	19.5
Metro_Interstate_Traffic_Volume	0.337	0.313	18.5	0.332	0.331	23.5	0.302	0.366	29.0	0.306	0.337	22.5	**0.298**	0.318	10.5	0.350	0.332	34.5	0.336	0.311	16.5	0.299	0.356	25.0	0.335	0.414	39.0	0.329	0.279	7.5
MexicoCityBikes	3.146	98.013	73.0	0.853	1.292	47.0	*0.847*	1.311	30.0	0.941	1.617	70.0	0.850	1.303	45.0	0.848	1.278	13.0	0.848	1.279	22.5	0.852	1.328	53.0	0.882	1.417	66.0	0.848	1.274	5.0
NASDAQ	1.666	5.361	58.5	1.468	4.240	30.5	1.492	4.700	40.0	1.369	4.020	16.0	1.577	5.012	49.0	1.660	5.252	54.5	1.401	4.072	20.0	1.375	4.341	24.0	1.381	4.114	20.0	1.473	3.554	22.5
NYSE	0.396	0.481	26.0	0.408	0.505	38.5	0.366	0.453	6.0	0.385	0.481	21.0	0.374	0.466	11.0	0.413	0.507	41.0	0.386	0.473	19.0	*0.364*	*0.447*	3.0	0.374	0.470	13.5	1.067	3.295	71.0
National_Stock_Exchange	0.776	1.285	37.0	0.756	1.344	41.5	0.633	0.846	6.5	0.684	0.961	14.5	*0.578*	0.662	3.5	0.825	1.451	51.5	0.707	1.212	23.5	0.641	0.867	8.5	0.653	1.044	13.0	0.846	1.530	57.0
Oikolab	0.599	0.620	18.0	0.631	0.703	53.0	0.588	0.675	28.5	0.673	0.826	66.0	0.610	0.684	43.0	0.603	0.631	26.0	0.600	0.618	17.5	0.601	0.686	36.5	0.631	0.735	56.5	0.586	0.583	7.5
PEMS-BAY	0.483	0.958	35.5	0.639	1.381	58.5	0.502	1.134	44.0	0.712	1.728	67.0	0.680	1.609	64.0	0.448	0.903	26.5	0.418	0.754	10.5	0.563	1.308	53.5	0.631	1.492	59.5	0.399	0.723	4.5
PEMS03	0.296	0.211	24.0	0.621	0.691	57.5	0.401	0.375	43.5	0.711	0.904	65.5	0.318	0.250	30.0	0.294	0.203	22.0	0.321	0.228	29.0	0.449	0.452	51.0	0.576	0.723	57.5	0.259	0.154	6.0
PEMS04	0.300	0.203	28.5	0.591	0.666	59.0	0.414	0.389	48.5	0.810	1.140	64.5	0.332	0.278	35.0	0.269	0.166	21.0	0.288	0.177	25.0	0.487	0.534	53.0	0.571	0.618	58.0	0.224	0.118	5.0
PEMS07	0.280	0.186	25.0	0.648	0.816	60.5	0.386	0.356	42.5	0.787	1.106	68.5	0.307	0.260	32.0	0.258	0.146	15.0	0.287	0.171	22.5	0.499	0.582	53.5	0.573	0.672	57.0	0.239	0.173	14.0
PEMS08	0.326	0.307	25.5	0.616	0.733	60.0	0.429	0.468	44.5	0.817	1.174	64.0	0.317	0.268	18.0	0.302	0.259	13.5	0.325	0.265	19.5	0.471	0.518	50.0	0.576	0.715	58.5	0.328	0.341	31.0
Power_Consumption_of_Tetouan_City	0.319	0.269	20.0	0.557	0.632	64.0	0.383	0.418	43.5	0.508	0.582	61.5	0.407	0.441	46.5	0.307	0.262	15.0	0.284	0.240	5.5	0.448	0.524	55.5	0.497	0.588	61.0	0.653	0.682	67.5
Residential_Power_and_Battery_Data	0.778	1.760	34.5	0.894	2.155	64.5	0.760	2.026	34.0	0.840	2.112	58.0	0.773	1.864	39.5	0.770	1.770	32.0	0.764	1.745	22.5	0.786	2.106	52.0	0.777	2.089	47.5	0.748	1.680	10.5
Rideshare	0.625	0.902	46.0	0.511	0.462	24.5	0.408	0.531	20.5	0.544	0.673	30.5	0.610	0.836	40.5	0.645	0.916	47.5	0.700	1.144	55.5	0.272	0.242	4.5	0.314	0.301	10.0	0.848	1.417	66.5
Sleep_EDFX	0.661	4.645	13.5	0.831	5.710	63.0	0.670	*4.418*	12.0	0.798	4.570	33.0	0.705	4.270	16.0	0.698	4.877	24.5	0.652	4.674	11.5	0.665	4.945	22.5	0.700	4.517	16.5	0.787	5.409	54.5
Spanish_energy	0.385	0.377	6.0	0.420	0.460	49.0	0.389	0.442	27.0	0.434	0.512	58.5	0.411	0.456	44.0	0.392	0.387	13.5	0.389	0.379	9.5	**0.337**	**0.344**	1.0	0.416	0.477	50.0	0.475	0.517	63.5
Spanish_weather	0.321	0.314	14.5	0.363	0.342	50.5	0.320	0.351	27.5	0.339	0.372	48.5	0.331	0.356	42.0	0.329	0.317	25.0	0.320	0.313	10.5	**0.305**	0.332	21.0	0.328	0.359	40.5	0.373	0.352	54.5
Temperature-Rain	0.819	4.305	24.5	0.822	4.323	29.0	0.712	4.360	18.0	0.856	4.396	45.5	0.821	5.287	44.5	0.841	4.393	41.5	0.818	4.225	19.0	*0.713*	4.365	19.0	0.739	4.514	31.0	0.797	4.234	15.0
Traffic	$> 10^3$	$> 10^6$	72.0	0.404	0.718	51.5	0.299	0.531	16.0	*0.276*	0.468	6.5	**0.259**	0.451	4.0	0.356	0.660	41.5	0.304	0.463	12.5	0.312	0.600	26.0	0.459	0.957	57.5	0.372	0.700	45.0
Traffic_Flow_Forecasting	0.416	0.339	25.0	0.875	1.178	63.0	0.720	1.045	57.5	0.584	0.716	42.0	0.469	0.516	33.5	0.504	0.462	33.0	0.355	0.254	9.5	0.695	1.054	56.5	0.888	1.330	64.0	0.367	0.255	12.0
WIND	0.664	1.038	16.0	0.740	1.254	60.0	0.710	1.254	52.5	0.754	1.366	64.0	0.708	1.219	49.0	0.686	1.076	28.0	0.670	1.049	22.0	0.695	1.216	45.0	0.703	1.208	46.0	0.734	1.179	52.5
Zafnoo_env	0.425	0.471	16.5	0.458	0.540	49.0	0.410	0.498	22.5	0.473	0.581	55.0	0.462	0.570	52.0	0.444	0.485	37.0	0.426	0.466	16.0	**0.391**	0.476	14.0	0.426	0.533	34.0	0.531	0.664	62.0
Zafnoo_sapf	0.288	0.298	26.5	0.304	0.299	37.5	0.259	0.292	12.0	0.293	0.332	40.5	0.275	0.320	28.0	0.291	0.302	33.0	0.275	0.284	11.5	0.250	0.276	3.0	0.265	0.299	18.0	0.315	0.356	52.0
australian_tourism	0.935	1.378	60.0	0.800	1.064	29.5	0.649	0.766	2.0	0.705	0.835	6.5	0.700	0.841	6.5	0.956	1.451	62.5	0.787	0.981	22.0	0.705	0.899	8.0	0.784	1.076	26.5	0.842	1.085	39.5
beijing_multisite_air_quality	0.445	0.703	14.0	0.470	0.758	46.5	**0.406**	**0.665**	1.0	0.485	0.850	59.5	0.452	0.773	41.5	0.452	0.718	29.5	0.447	0.707	18.0	0.429	0.733	20.5	0.448	0.761	33.5	0.481	0.766	52.0
boomlet_1062	0.881	1.611	71.0	0.822	1.282	68.0	*0.611*	1.008	27.0	0.759	1.186	62.0	0.655	0.942	20.5	0.685	0.901	23.5	0.677	0.890	10.0	0.610	1.016	27.5	0.634	1.070	32.0	0.742	1.016	55.0
boomlet_1209	0.362	1.586	10.0	0.536	3.107	62.5	0.353	1.725	16.0	0.414	1.908	49.5	0.370	1.768	27.0	0.391	1.795	35.5	0.371	1.604	19.5	**0.352**	1.708	13.5	0.389	1.808	35.5	1.168	7.114	69.0
boomlet_1225	600.413	$> 10^6$	73.0	0.312	0.796	26.5	**0.254**	*0.717*	2.0	0.331	0.860	37.5	0.275	0.823	19.5	0.374	1.085	53.0	0.292	0.758	15.0	*0.255*	0.717	3.5	0.260	0.722	4.5	0.964	7.559	68.0
boomlet_1230	4.023	588.494	12.5	4.223	620.095	30.5	4.046	622.076	23.0	4.336	668.977	52.5	4.087	634.622	30.5	4.267	626.664	38.5	4.152	603.129	24.5	4.033	605.792	18.0	4.214	704.073	44.5	9.672	$> 10^3$	68.5
boomlet_1282	0.414	1.031	9.5	0.444	1.089	33.0	0.415	1.061	14.0	0.447	1.143	39.5	0.426	1.057	17.5	0.443	1.149	38.5	0.412	0.988	5.0	0.429	1.145	27.5	0.431	1.158	32.0	1.278	10.520	69.5
boomlet_1487	0.223	0.749	39.0	0.218	0.735	26.0	**0.190**	0.736	11.0	0.242	0.793	52.0	0.194	0.732	4.0	0.200	0.734	8.0	0.215	0.735	23.5	*0.191*	0.734	7.0	0.210	0.745	29.0	0.350	0.817	64.0
boomlet_1631	0.554	0.683	24.0	0.566	0.688	27.5	0.563	0.717	30.0	0.616	0.867	51.0	0.554	0.701	26.0	0.600	0.787	43.0	0.556	0.670	22.0	0.537	0.664	12.0	0.573	0.756	35.5	1.210	2.383	71.0
boomlet_1676	$> 10^4$	$> 10^6$	73.0	0.578	1.802	11.5	0.557	1.585	4.0	0.706	4.066	32.0	0.586	1.816	14.5	0.937	7.803	43.0	0.857	9.553	41.0	**0.545**	1.563	1.5	0.688	6.484	33.0	4.124	275.602	66.0
boomlet_1855	1.140	96.894	65.0	0.578	91.909	16.0	0.506	91.392	6.5	0.593	91.702	16.5	0.513	91.298	5.5	0.719	92.655	39.0	0.649	92.871	36.5	*0.488*	*91.268*	3.0	0.523	91.426	9.0	0.943	93.197	53.0
boomlet_1975	0.518	0.671	31.5	0.637	0.882	62.5	0.547	1.107	57.0	0.598	1.283	68.0	0.530	1.063	51.0	0.516	0.581	25.0	0.436	0.491	14.0	0.559	1.160	60.0	0.539	1.041	52.0	0.485	0.624	24.5
boomlet_2187	0.717	35.773	24.0	0.805	36.209	41.0	0.623	34.956	15.0	0.810	46.752	59.5	0.702	40.167	28.5	0.791	44.689	51.0	0.746	39.597	34.5	**0.608**	*26.416*	2.0	0.664	29.182	6.0	0.985	40.987	56.0
boomlet_285	0.253	0.271	10.5	0.339	0.330	55.5	0.212	0.316	24.5	0.296	0.368	51.5	0.255	0.302	22.0	0.270	0.315	35.0	0.254	0.269	9.0	*0.199*	0.314	21.5	0.224	2.744	38.0	0.405	0.352	63.5
boomlet_619	**0.306**	**0.299**	1.0	0.668	0.810	63.5	0.520	0.734	58.0	0.742	1.112	66.5	0.520	0.691	57.0	0.335	0.348	26.5	0.329	0.315	14.0	0.630	1.158	64.5	0.493	0.676	55.0	0.488	0.567	52.5
boomlet_772	0.581	1.960	66.5	0.367	1.100	38.0	**0.292**	1.032	4.0	0.367	1.169	44.0	0.314	1.065	16.0	0.369	1.092	37.5	0.319	1.042	13.0	*0.301*	1.073	15.5	0.304	1.058	12.5	0.627	1.503	66.0
boomlet_963	0.632	1.572	29.5	0.651	1.680	44.5	0.617	1.763	37.0	0.715	1.901	59.0	0.666	2.378	56.5	0.662	1.665	46.5	0.612	1.517	17.5	0.612	1.766	36.0	0.617	1.605	28.5	0.788	1.908	63.0
electricity_2016	0.245	0.158	6.5	0.303	0.225	47.5	0.273	0.201	35.0	0.256	0.177	17.0	0.250	0.166	11.5	0.303	0.204	43.5	0.252	0.158	10.0	0.246	0.178	14.0	0.320	0.267	54.5	0.392	0.325	60.0
electricityloaddiagrams20112014	1.829	401.436	22.0	3.582	$> 10^3$	36.0	3.267	$> 10^3$	35.5	2.664	858.512	31.0	1.760	370.981	17.0	4.539	$> 10^3$	39.5	1.779	395.752	20.5	1.901	526.913	25.0	4.599	$> 10^3$	50.5	16.567	$> 10^4$	67.0
gasrate_co2	0.861	1.122	37.5	0.724	0.868	13.0	0.785	0.996	21.0	0.835	1.068	32.0	0.907	1.289	58.5	0.942	1.326	62.5	0.861	1.126	38.5	0.764	0.963	17.0	0.774	0.953	17.5	0.713	0.693	9.5
gefcom2012_load	1.611	7.759	37.5	1.336	7.325	9.0	1.688	8.369	51.5	1.784	8.444	56.0	1.733	8.725	55.5	1.567	7.833	33.0	1.704	8.245	51.5	1.673	8.099	48.0	1.741	7.981	49.5	1.310	7.325	8.5
gefcom2012_wind	0.873	1.109	24.5	0.880	1.154	41.0	0.912	1.363	60.5	0.936	1.414	64.0	0.884	1.216	49.5	0.890	1.143	46.0	0.860	1.085	6.5	0.894	1.316	55.0	0.900	1.376	57.5	0.873	1.131	31.5
gvar	0.294	0.258	43.0	0.269	0.224	16.0	0.242	*0.205*	4.5	0.241	0.207	5.0	0.271	0.258	32.5	0.302	0.272	51.5	0.261	0.236	19.0	0.233	0.202	2.0	0.255	0.225	11.5	0.884	1.617	67.0
italy_air_quality	0.553	1.128	16.0	0.574	1.160	25.5	0.538	1.279	29.5	0.610	1.544	55.0	0.570	1.417	43.5	0.600	1.213	38.5	0.565	1.112	16.0	**0.485**	1.086	3.0	0.537	1.302	30.0	1.074	2.551	71.0
vessel_ais	0.796	1.501	65.0	0.797	1.062	59.0	0.778	1.029	49.5	0.802	1.072	61.0	0.784	1.038	53.0	0.721	0.895	29.0	**0.674**	**0.792**	1.0	0.776	1.027	46.5	0.803	1.092	63.5	0.791	1.050	56.0
weather_Wetterstation	$> 10^3$	$> 10^6$	72.0	0.303	0.249	43.5	*0.254*	0.244	21.0	0.288	0.262	40.5	**0.128**	**0.098**	1.0	0.306	0.254	45.0	0.263	0.211	7.5	0.267	0.232	22.5	0.276	0.248	32.0	0.529	0.518	67.5

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, underline, and *italic*. The rank column reports the average rank over the two metrics.

Table 25: **Multivariate forecasting results.** Part 8.

Dataset	tsmixer			units			var		
	mae	mse	rank	mae	mse	rank	mae	mse	rank
Appliances_Energy_Prediction	0.682	0.855	36.5	0.571	0.705	6.0	2.101	334.570	72.5
Bike_Sharing	0.537	0.674	14.5	0.557	0.763	30.0	$> 10^6$	$> 10^6$	73.0
COVID-19_Deaths	10.636	609.196	62.5	2.257	90.893	30.5	5.064	637.376	60.0
Czelan	0.305	0.231	47.0	0.239	0.217	27.5	0.394	2.376	70.0
DailyDelhiClimate	2.529	70.824	49.5	2.201	59.539	41.5	1.990	57.012	35.5
Dominick	1.160	2.655	70.0	0.702	1.439	25.0	$> 10^6$	$> 10^6$	73.0
ETTH1	0.544	0.574	57.0	0.434	0.439	25.5	0.546	0.644	61.5
ETTH2	1.153	1.891	68.5	0.373	0.329	9.0	0.397	0.357	33.5
ETTM1	0.503	0.536	33.5	0.466	0.501	16.5	0.598	0.788	59.0
ETTM2	0.342	0.242	65.0	0.278	0.170	14.0	0.291	0.179	34.5
FRED-MD	3.478	123.765	62.5	1.705	90.524	47.5	$> 10^4$	$> 10^6$	73.0
ILY	1.716	5.833	68.5	1.469	4.192	52.5	1.605	7.437	68.5
KDDCup2018	0.617	0.782	26.5	0.596	0.807	22.0	$> 10^6$	$> 10^6$	73.0
METR-LA	0.682	1.114	38.0	0.648	1.085	19.5	$> 10^6$	$> 10^6$	73.0
MIT-BIH	0.582	2.031	32.5	0.597	2.099	40.5	0.579	1.464	16.5
Metro_Interstate_Traffic_Volume	0.364	0.325	38.0	0.354	0.340	40.5	0.410	0.414	61.0
MexicoCityBikes	0.850	1.275	23.0	0.848	1.278	18.0	0.849	1.280	35.0
NASDAQ	1.544	4.400	40.0	1.635	5.121	52.5	4.068	119.589	73.0
NYSE	0.702	1.235	64.5	0.445	0.549	47.0	0.400	0.734	49.5
National_Stock_Exchange	1.249	2.415	70.5	0.697	1.105	18.5	0.893	1.671	61.0
Oikolab	0.582	0.574	6.0	0.617	0.644	38.0	0.662	0.713	60.0
PEMS-BAY	0.410	0.713	6.5	0.450	0.828	21.5	334.550	$> 10^6$	72.0
PEMS03	0.258	*0.141*	4.5	0.350	0.256	34.5	0.639	0.859	61.0
PEMS04	0.243	0.126	11.0	0.347	0.243	33.5	$> 10^6$	$> 10^6$	73.0
PEMS07	0.253	0.135	10.5	0.327	0.213	31.0	0.773	2.424	69.5
PEMS08	0.328	0.270	22.5	0.377	0.361	35.5	$> 10^6$	$> 10^6$	73.0
Power_Consumption_of_Tetouan_City	0.452	0.358	46.5	0.308	0.263	16.5	0.470	0.491	56.0
Residential_Power_and_Battery_Data	0.765	1.670	15.0	0.785	1.824	44.0	1.859	953.879	73.0
Rideshare	0.714	0.753	45.0	0.755	1.257	62.0	$> 10^6$	$> 10^6$	73.0
Sleep_EDFX	0.663	4.467	10.5	0.657	4.524	10.0	30.048	$> 10^5$	73.0
Spanish_energy	0.429	0.403	41.0	0.400	0.402	25.5	0.763	16.481	71.5
Spanish_weather	0.344	0.306	25.5	0.323	0.318	18.5	$> 10^6$	$> 10^6$	73.0
Temperature-Rain	0.918	4.660	59.0	0.820	4.199	17.0	$> 10^6$	$> 10^6$	72.0
Traffic	0.405	0.573	40.5	0.323	0.508	19.0	0.793	2.256	67.5
Traffic_Flow_Forecasting	0.422	0.309	23.0	0.639	0.739	45.5	0.590	0.617	39.5
WIND	0.650	0.973	4.0	0.673	1.050	23.0	$> 10^6$	$> 10^6$	72.0
Zafnoo_env	0.433	0.455	17.0	0.436	0.476	29.0	0.639	0.712	65.0
Zafnoo_sapf	0.293	0.301	33.5	0.286	0.296	24.0	0.493	0.560	68.0
australian_tourism	0.911	1.314	56.0	0.844	1.081	41.0	0.951	1.447	61.5
beijing_multisite_air_quality	0.446	0.675	9.5	0.452	0.711	25.5	$> 10^6$	$> 10^6$	73.0
boomlet_1062	0.753	1.276	62.5	0.733	0.971	47.0	0.731	1.042	53.5
boomlet_1209	0.786	2.848	66.0	*0.358*	1.589	7.5	0.651	19.856	69.5
boomlet_1225	0.990	4.618	66.0	0.307	0.796	25.5	5.324	$> 10^4$	72.0
boomlet_1230	10.540	849.034	66.5	3.967	593.226	10.5	$> 10^6$	$> 10^6$	73.0
boomlet_1282	1.116	6.277	64.5	0.418	1.046	14.0	$> 10^5$	$> 10^6$	73.0
boomlet_1487	0.290	0.777	55.0	0.207	0.734	11.0	$> 10^6$	$> 10^6$	73.0
boomlet_1631	0.831	1.265	64.5	0.543	0.656	12.0	121.862	$> 10^6$	73.0
boomlet_1676	7.003	147.865	64.0	0.826	7.555	37.5	$> 10^3$	$> 10^6$	71.5
boomlet_1855	1.085	98.106	65.0	0.595	92.051	23.0	$> 10^6$	$> 10^6$	73.0
boomlet_1975	0.487	0.497	17.5	0.573	0.807	52.0	$> 10^6$	$> 10^6$	72.5
boomlet_2187	1.007	33.356	43.5	0.739	36.676	28.5	$> 10^6$	$> 10^6$	73.0
boomlet_285	0.329	0.330	53.0	0.269	0.298	24.0	$> 10^6$	$> 10^6$	73.0
boomlet_619	0.360	0.347	31.0	0.326	0.310	10.5	$> 10^6$	$> 10^6$	73.0
boomlet_772	0.486	1.190	57.0	0.329	1.038	15.0	$> 10^6$	$> 10^6$	73.0
boomlet_963	0.683	1.614	46.0	0.622	1.561	27.0	$> 10^6$	$> 10^6$	73.0
electricity_2016	0.295	0.189	39.5	0.289	0.196	40.0	0.388	0.392	62.0
electricityloaddiagrams20112014	72.102	$> 10^4$	67.0	1.771	371.993	18.5	9.360	$> 10^4$	66.0
gasrate_co2	0.870	1.184	48.5	0.865	1.131	41.0	**0.445**	**0.308**	1.0
gefcom2012_load	1.479	7.582	21.5	1.746	7.917	48.5	1.560	7.656	27.5
gefcom2012_wind	0.860	1.079	6.0	0.869	1.104	17.0	0.872	1.077	14.0
gvar	1.676	4.624	71.0	0.281	0.252	34.0	0.473	1.548	63.5
italy_air_quality	0.630	1.181	41.5	0.563	1.141	19.0	$> 10^6$	$> 10^6$	73.0
vessel_ais	0.697	0.838	23.0	0.717	0.888	28.0	0.771	1.005	42.0
weather_Wetterstation	0.290	0.209	24.0	0.273	0.215	17.0	$> 10^3$	$> 10^6$	73.0

Note: Results are averaged over all forecasting horizons/windows for each dataset. Lower values indicate better performance. For each metric, best/second/third are marked by **bold**, <u>underline</u>, and *italic*. The rank column reports the average rank over the two metrics.

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